



IEEE Standard for Ethernet Structure of Management Information version 2 (SMIPv2) Data Model Definitions

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

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IEEE Standard for Ethernet Structure of Management Information version 2 (SMIPv2) Data Model Definitions

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LAN/MAN Standards Committee
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IEEE Computer Society

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Abstract: This standard defines Structure of Management Information version 2 (SMIv2) MIB module specifications for IEEE Std 802.3 Ethernet and associated managed object branch and leaf assignments used in the variable descriptors in IEEE Std 802.3 Variable Request operations, administration, and maintenance protocol data units (OAMPDU).

Keywords: Ethernet, IEEE 802.3.1™, Management Information Base, MIB, network management, Simple Network Management Protocol, SMIv2, SNMP, Structure of Management Information version 2

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Introduction

This introduction is not part of IEEE Std 802.3.1-2024, IEEE Standard for Ethernet Structure of Management Information version 2 (SMIPv2) Data Model Definitions.

The initial version of this standard provided Management Information Base (MIB) module definitions that superseded and made obsolete the Guidelines for the definition of Managed Objects (GDMO) MIB modules defined in Annex 30A and Annex 30B of IEEE Std 802.3™-2008 [B5]⁶, and Structure of Management Information version 2 (SMIPv2) MIB modules defined in Annex F.6 of IEEE Std 802.1AB™-2009 [B4], IETF RFC 2108 [B6], IETF RFC 3621 [B8], IETF RFC 3635 [B9], IETF RFC 3637 [B11], IETF RFC 4836 [B14], IETF RFC 4837 [B15], IETF RFC 4878 [B16], and IETF RFC 5066 [B17].

The first revision of this standard updated the SMIPv2 MIB modules to support IEEE Std 802.3 amendments published since IEEE Std 802.3.1 was first published in 2011 and removed the GDMO MIB modules except for the associated branch and leaf assignments, which are used in IEEE Std 802.3 for Variable Request operations, administration, and maintenance protocol data units (OAMPDU).

The second revision of this standard updates the SMIPv2 MIB modules to support IEEE Std 802.3 amendments published since IEEE Std 802.3.1 was revised in 2013.

Ethernet technology, as defined by the IEEE 802.3 Working Group, continues to evolve, with scalable increases in speed, new types of cabling and interfaces, and new features. The evolution may require changes in the managed objects to reflect this new functionality. This document, as with other documents issued by this working group, reflects a certain stage in the evolution of Ethernet technology. In the future, this document might be revised, or new documents might be issued, to reflect the evolution of Ethernet technology.

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IEEE Standard for Ethernet Structure of Management Information version 2 (SMIv2) Data Model Definitions

1. Overview

This standard defines Structure of Management Information version 2 (SMIv2) Management Information Base (MIB) module specifications for IEEE Std 802.3 Ethernet that can be accessed by a Network Management System using the Simple Network Management Protocol (SNMP). This standard also defines associated managed object branch and leaf assignments used in the variable descriptors in IEEE Std 802.3 Variable Request operations, administration, and maintenance protocol data units (OAMPDU).

The term “Ethernet-like interfaces” was historically used because the interfaces defined by the IEEE 802.3 Working Group were not considered “Ethernet” per se, but “Ethernet-like”, because “Ethernet” was taken to mean “Ethernet version 2” according to the (DEC, Intel, Xerox) DIX “blue book”. Today and in the context of SNMP management and SMIv2 MIB modules, “Ethernet”, “Ethernet-like”, and “IEEE 802.3” are synonymous and interchangeable in the marketplace. The term “Ethernet-like” is retained in this document because of its common usage in the SNMP-based network management community.

1.1 Scope

This standard defines Structure of Management Information version 2 (SMIv2) Management Information Base (MIB) module specifications for IEEE Std 802.3 Ethernet and associated managed object branch and leaf assignments used in the variable descriptors in IEEE Std 802.3 Variable Request operations, administration, and maintenance protocol data unit (OAMPDU).

1.2 Purpose

The purpose of the standard is to define Structure of Management Information version 2 (SMIv2) MIB module specifications for IEEE Std 802.3 and associated managed object branch and leaf assignments and publish them in a machine-readable format.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).^{8,9}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

1.4 Internet-Standard Management Framework

For a detailed overview of the documents that describe the current Internet-Standard Management Framework, please refer to section 7 of IETF RFC 3410.¹⁰

Managed objects are accessed via a virtual information store, termed the Management Information Base or MIB. MIB objects are generally accessed through the SNMP.

Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This standard specifies MIB modules that are compliant to the SMIv2, which is described in IETF RFC 2578, IETF RFC 2579, and IETF RFC 2580.

1.5 Security considerations

SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for example by using IPSec), there is no control as to who on the secure network is allowed to access and GET/SET (read/change/create/delete) the objects in a MIB module.

Implementers should consider the security features as provided by the SNMPv3 framework (see section 8 of IETF RFC 3410), including full support for the SNMPv3 cryptographic mechanisms (for authentication and privacy).

SNMPv3 should be deployed, rather than previous versions of SNMP, and cryptographic security should be enabled. It is then a customer/operator responsibility to ensure that the SNMP entity giving access to an instance of this MIB module is properly configured to give access to the objects only to those principals (users) that have legitimate rights to indeed GET or SET (change/create/delete) them.

Throughout this standard, there are a number of accessible management objects that could be considered sensitive or vulnerable in some network environments. The support for some operations in a non-secure environment without proper protection can have a negative effect on network operations. Such management objects are detailed in the clauses that define them.

⁸ The use of the word *must* is deprecated and cannot be used when stating mandatory requirements, *must* is used only to describe unavoidable situations.

⁹ The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact.

¹⁰ Information on references can be found in Clause 2.

The user of these MIB modules should therefore be aware that support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

The readable objects in these MIB modules (i.e., those with MAX-ACCESS other than not-accessible) could be considered sensitive in some environments since, collectively, they provide information about the performance of network interfaces and can reveal some aspects of their configuration. In such environments, it is important to control GET and NOTIFY access to these objects and possibly encrypt their values when sending them over the network via SNMP.

1.6 Conformance

Specific conformance statements and compliance statements, written in accordance with IETF RFC 2580, are included in each MIB module. They can be found by searching for the text strings “Conformance statements” and “Compliance statements”.

1.7 SMIv2 syntax validation and formatting

All SMIv2 modules included in this standard are SMIv2 (see IETF RFC 2578) compliant and pass automated checks using tools available at the time of publication, including the open source and/or free versions of SMIv2 validation ‘smilint’ (see <https://linux.die.net/man/1/smilint>), as well as other SMIv2 validation tools listed at <https://wiki.ietf.org/group/ops/mib-review-tools>. When using “smilint” tool, “smilint: level 6” and “smilint: hide namelength-32” configuration options were used. The “smidump” utility was used to produce a common structure/format for SMIv2 modules included in this standard.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

ANSI T1.231-1997, Layer 1 In-Service Digital Transmission Performance Monitoring.¹¹

ANSI T1.424-2004, Interface Between Networks and Customer Installation—Very-high-bit-rate Digital Subscriber Lines (VDSL)—Metallic Interface (DMT Based).

IEEE Std 802[®], IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.^{12, 13}

IEEE Std 802.1Q[™], IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks.

IEEE Std 802.1AB[™], IEEE Standard for Local and Metropolitan Area Networks—Station and Media Access Control Connectivity Discovery.

IEEE Std 802.3[™], IEEE Standard for Ethernet.

IEEE Std 802.9a[™]-1995, IEEE Standards for Local and Metropolitan Area Networks—Supplement to Integrated Services (IS) LAN Interface at the Medium Access Control (MAC) and Physical (PHY) Layers—Specification of IsLAN16-T.

IETF RFC 1213, Management Information Base for Network Management of TCP/IP-based internets: MIB-II, K. McCloghrie and M. Rose, Mar. 1991.¹⁴

IETF RFC 1516, Definitions of Managed Objects for IEEE 802.3 Repeater Devices, D. McMaster and K. McCloghrie, Sep. 1993.

IETF RFC 2119, Key words for use in RFCs to Indicate Requirement Levels, S. Bradner, Mar. 1997.

IETF RFC 2434, Guidelines for Writing an IANA Considerations Section in RFCs, T. Narten and H. Alvestrand, Oct. 1998.

IETF RFC 2578 (STD 58), Structure of Management Information Version 2 (SMIv2), K. McCloghrie, D. Perkins, and J. Schoenwaelder, Apr. 1999.

IETF RFC 2579 (STD 58), Textual Conventions for SMIv2, K. McCloghrie, D. Perkins, and J. Schoenwaelder, Apr. 1999.

IETF RFC 2580 (STD 58), Conformance Statements for SMIv2, K. McCloghrie, D. Perkins, and J. Schoenwaelder, Apr. 1999.

IETF RFC 2863, The Interfaces Group MIB, K. McCloghrie and F. Kastholz, Jun. 2000.

¹¹ ANSI publications are available from the American National Standards Institute (<https://www.ansi.org/>).

¹² IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

¹³ The IEEE standards or products referred to in Clause 2 are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

¹⁴ IETF RFCs are available from the Internet Engineering Task Force <https://www.ietf.org/>.

IETF RFC 2864, The Inverted Stack Table Extension to the Interfaces Group MIB, K. McCloghrie and G. Hanson, Jun. 2000.

IETF RFC 3410, Introduction and Applicability Statements for Internet Standard Management Framework, J. Case, R. Mundy, D. Partain, and B. Stewart, Dec. 2002.

IETF RFC 3592, Definitions of Managed Objects for the Synchronous Optical Network/Synchronous Digital Hierarchy (SONET/SDH) Interface Type, K. Tesink, Sep. 2003.

ITU-T Recommendation G.983.1, 1998—Optical line systems for local and access networks—Broadband optical access systems based on Passive Optical Networks (PON).¹⁵

ITU-T Recommendation G.991.2, 2003—Single-pair high-speed digital subscriber line (SHDSL) transceivers.

ITU-T Recommendation G.993.1, 2004—Very high speed digital subscriber line transceivers.

¹⁵ ITU-T publications are available from the International Telecommunications Union (<https://www.itu.int/>).

3. Definitions

For the purposes of this document, the following terms and definitions apply. *The IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.¹⁶

agent: An entity, typically implemented in software, which provides remote access to management instrumentation, via the Simple Network Management Protocol (SNMP).

jack type: The jack connector type, as it appears on the outside of the system. The type of mechanical interface to the transmission medium.

Loss of Code-group Delineation: See 50.3.5.3 of IEEE Std 802.3.

managed object: An abstract representation of network resources that are managed. A managed object is defined according to the *Structure of Management Information version 2* (SMIv2) defined in IETF RFC 2578.

managed repeater: A repeater as defined by IEEE Std 802.3 incorporating a management entity that complies with the Management Information Base (MIB) module definition contained in Clause 7 of IEEE Std 802.3.1-2024.

module: In the context of the MIB definitions, a specification of management capabilities related to the system.

non-trivial repeater: A repeater as defined by IEEE Std 802.3 having multiple ports.

path coding violations: In IEEE Std 802.3, the path layer coding violations count is based on block errors and not on BIP-8 errors; that is, it is incremented only once for each B3 byte that indicates incorrect parity, regardless of the number of bits in error. Note that section 8.4.5.1 of ANSI T1.231-1997 allows either path BIP-8 errors or path block errors to be used for the path layer error count.

repeater system: A managed entity compliant with this standard and incorporating at least one managed IEEE 802.3 repeater.

repeater-unit: The portion of a repeater that is inboard of its Physical Medium Attachment (PMA)/Physical Signaling Sublayer (PLS) or PMA/Physical Coding Sublayer (PCS).

Signal Label Mismatch: This defect is called Path Label Mismatch (PLM) in IEEE Std 802.3. It is reported by setting both the sonetPathSignalLabelMismatch bit in the appropriate instance of sonetPathCurrentStatus (defined in IETF RFC 3592) and the etherWisPathPLM bit in the corresponding instance of etherWisPathCurrentStatus.

stack: A scalable system in which modularity is achieved by interconnecting a number of different systems.

STS-Path Remote Defect Indication: IEEE Std 802.3 mandates the use of ERDI-P (Enhanced Remote Defect Indication-Path) defined in ANSI T1.231-1997 to signal remote server defects (triggered by path AIS or path LOP) and remote payload defects (triggered by Path Label Mismatch or Loss of Code-group Delineation). IETF RFC 3592 defines the one-bit RDI-P (Remote Defect Indication-Path), which signals remote server defects (i.e., path AIS and path LOP) only. An implementation of the MIB module defined in Clause 12 of IEEE Std 802.3.1-2024 sets the sonetPathSTSRDI bit in the appropriate instance of sonetPathCurrentStatus when it receives an ERDI-P server defect indication from the remote end. Both

¹⁶ The *IEEE Standards Dictionary Online* is available at <https://dictionary.ieee.org/>. An IEEE Account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

ERDI-P payload defects and ERDI-P server defects are reported in the object `etherWisFarEndPathCurrentStatus`.

system: An entity compliant with one or more Management Information Base (MIB) modules of this standard.

system interconnect segment: An internal segment allowing interconnection of ports belonging to different physical entities into the same logical managed repeater, bridge, or other system. Examples of implementation might be backplane buses in modular hubs, or chaining cables in stacks of bridges/switches. It is not uncommon for such segments to be a proprietary implementation.

trivial repeater-unit: An isolated port that can gather statistics.

4. Abbreviations

AIS	Alarm Indication Signal
ARP	address resolution protocol
ASCII	American Standard Code for Information Interchange
BER	bit error ratio
BIP	bit interleaved parity
BW	bandwidth
CID	Company ID
CO	central office
CPE	customer premises equipment
CRC	cyclic redundancy check
DTE	data terminal equipment
EFM	Ethernet in the First Mile
EFMCu	Ethernet in the First Mile copper
ELTE	Ethernet line termination equipment
EPON	Ethernet passive optical network
ERDI-P	enhanced remote defect indication—path
FCS	frame check sequence
FEC	forward error correction
GMII	gigabit media independent interface
IANA	Internet Assigned Numbers Authority
IETF	Internet Engineering Task Force
IFG	inter-frame gap
ITU	International Telecommunication Union
LAN	local area network
LCD	Loss of Code-group Delineation - Path
LLC	logical link control
LLDP	logical link discovery protocol
LLDPDU	logical link discovery protocol data unit
LLID	logical link identifier
LOP	Loss of Pointer
LTE	line termination equipment
MAC	media access control
MAU	medium attachment unit
MDI	medium dependent interface
MDIO	management data input/output
MIB	Management Information Base
MII	media independent interface
MPCP	multipoint control protocol
MPCPDU	multipoint control protocol data unit
MTU	maximum transmission unit
OAM	operations, administration, and maintenance
OAMPDU	operations, administration, and maintenance protocol data unit
OID	object identifier
OLT	optical line terminal

OMP	optical multipoint
ONU	optical network unit
OSI	Open Systems Interconnection
OUI	Organizationally Unique Identifier
P2MP	point-to-multipoint
P2PE	point-to-point emulation
PAF	PME aggregation function
PBO	power back-off
PCS	physical coding sublayer
PD	powered device
PDU	protocol data unit
PHY	Physical Layer entity
PLM	Path Label Mismatch
PMA	physical medium attachment
PMD	physical medium dependent
PME	physical medium entity
PON	passive optical network
PSD	power spectral density
PSE	power sourcing equipment
RFC	Request for Comments
ROM	read-only-memory
RS	reconciliation sublayer
RTT	round-trip time
SDH	Synchronous Digital Hierarchy
SLA	service level agreement
SLD	start of logical link identifier delimiter
SMlv2	structure of management information version 2
SNMP	simple network management protocol
SNR	signal-to-noise ratio
SONET	Synchronous Optical Network
TCPAM	trellis coded pulse amplitude modulation
TDM	time division multiplexing
TDMA	time division multiple access
TLV	type/length/value
TQ	time quanta
WAN	wide area network
WDM	wavelength division multiplexing
WIS	WAN interface sublayer

5. Ethernet logical link discovery protocol (LLDP) extension MIB module

The logical link discovery protocol (LLDP) is defined in IEEE Std 802.1AB, Station and Media Access Control Discovery. Extensions to this protocol for Ethernet are defined in Clause 79 of IEEE Std 802.3.

5.1 Structure of the IEEE 802.3 LLDP extension MIB

Table 5-1 summarizes the particular object groups that are required for each operating mode. The implemented MIB shall comply with the MIB conformance section for the particular operating mode being supported.

Table 5-1—IEEE 802.3 LLDP extension MIB object group conformance requirements

MIB group	Rx mode	Tx mode	Tx/Rx mode
lldpV2Xdot3ConfigGroup	M ^a	M	M
lldpV2Xdot1LocSysGroup	M	—	M
lldpV2Xdot1RemSysGroup	—	M	M
ifGeneralInformationGroup	M	M	M

^a M = Mandatory.

Table 5-2 shows the structure of the MIB and the relationship of the MIB objects to the LLDP operational status/control variables, LLDP statistics variables, and TLV variables.

Table 5-2—IEEE 802.3/LLDP extension MIB cross reference

MIB table	MIB object	LLDP reference
<i>Configuration group</i>		
	lldpV2Xdot3PortConfigTable	Augments lldpV2PortConfigEntry
	lldpV2Xdot3PortConfigTLVsTxEnable	Normal LLDPDUs
<i>Local devices information group</i>		
	lldpV2Xdot3LocPortTable	
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot3LocPortAutoNegSupported	Auto-Negotiation support/status
	lldpV2Xdot3LocPortAutoNegEnabled	Auto-Negotiation support/status
	lldpV2Xdot3LocPortAutoNegAdvertisedCap	Auto-Negotiation advertised
	lldpV2Xdot3LocPortOperMauType	Operational MAU type

Table 5-2—IEEE 802.3/LLDP extension MIB cross reference (*continued*)

MIB table	MIB object	LLDP reference
lldpV2Xdot3LocPowerTable		
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot3LocPowerPortClass	MDI power support
	lldpV2Xdot3LocPowerMDISupported	MDI power support
	lldpV2Xdot3LocPowerMDIEnabled	MDI power support
	lldpV2Xdot3LocPowerPairControlable	MDI power support
	lldpV2Xdot3LocPowerPairs	PSE power pair
	lldpV2Xdot3LocPowerClass	Power class
lldpV2Xdot3LocMaxFrameSizeTable		
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot3LocMaxFrameSize	Maximum frame size
<i>Remote devices information group</i>		
lldpV2Xdot3RemPortTable		
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot3RemPortAutoNegSupported	Auto-Negotiation support/status
	lldpV2Xdot3RemPortAutoNegEnabled	Auto-Negotiation support/status
	lldpV2Xdot3RemPortAutoNegAdvertisedCap	Auto-Negotiation advertised
	lldpV2Xdot3RemPortOperMauType	Operational MAU type
lldpV2Xdot3RemPowerTable		
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot3RemPowerPortClass	MDI power support
	lldpV2Xdot3RemPowerMDISupported	MDI power support
	lldpV2Xdot3RemPowerMDIEnabled	MDI power support
	lldpV2Xdot3RemPowerPairControlable	MDI power support
	lldpV2Xdot3RemPowerPairs	PSE power pair
	lldpV2Xdot3RemPowerClass	Power class

Table 5-2—IEEE 802.3/LLDP extension MIB cross reference (*continued*)

MIB table	MIB object	LLDP reference
lldpV2Xdot3RemMaxFrameSizeTable		
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot3RemMaxFrameSize	Maximum frame size

5.2 Relationship to other MIBs

Version 1 of the IEEE 802.3 LLDP extension MIB module is deprecated. LLDP-V2_MIB and LLDP-V2-TC-MIB are used instead.

5.3 Security considerations for IEEE 802.3 LLDP extension MIB module

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write.¹⁷ Such objects can be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Setting the object lldpXdot3PortConfigTLVsTxEnable to incorrect values can result in improper operation of LLDP.

The following readable objects in this MIB module can be considered sensitive or vulnerable in some network environments:

- a) Objects that are associated with the transmit mode are as follows:
 - 1) lldpV2Xdot3LocPortAutoNegSupported
 - 2) lldpV2Xdot3LocPortAutoNegEnabled
 - 3) lldpV2Xdot3LocPortAutoNegAdvertisedCap
 - 4) lldpV2Xdot3LocPortOperMauType
 - 5) lldpV2Xdot3LocPowerPortClass
 - 6) lldpV2Xdot3LocPowerMDISupported
 - 7) lldpV2Xdot3LocPowerMDIEnabled
 - 8) lldpV2Xdot3LocPowerPairControlable
 - 9) lldpV2Xdot3LocPowerPairs
 - 10) lldpV2Xdot3LocPowerClass
 - 11) lldpV2Xdot3LocMaxFrameSize
- b) Objects that are associated with the receive mode are as follows:
 - 1) lldpV2Xdot3RemPortAutoNegSupported
 - 2) lldpV2Xdot3RemPortAutoNegEnabled
 - 3) lldpV2Xdot3RemPortAutoNegAdvertisedCap
 - 4) lldpV2Xdot3RemPortOperMauType

¹⁷ In IETF MIB definitions, the MAX-ACCESS clause defines the type of access that is allowed for particular data elements in the MIB. An explanation of the MAX-ACCESS mapping is given in section 7.3 of IETF RFC 2578.

- 5) lldpV2Xdot3RemPowerPortClass
- 6) lldpV2Xdot3RemPowerMDISupported
- 7) lldpV2Xdot3RemPowerMDIEnabled
- 8) lldpV2Xdot3RemPowerPairControlable
- 9) lldpV2Xdot3RemPowerPairs
- 10) lldpV2Xdot3RemPowerClass
- 11) lldpV2Xdot3RemMaxFrameSize

This concern applies both objects that describe the configuration of the local host, as well as objects that describe information from the remote hosts, acquired via LLDP and displayed by the objects in this MIB module. It is thus also important to control GET and/or NOTIFY access to these objects and possibly to encrypt the values of these objects when sending them over the network via SNMP.

5.4 MIB module definition

In the following MIB definition, should any discrepancy between the DESCRIPTION text and the corresponding definition in 5.2 through 5.3 of this clause occur, the definitions in 5.2 through 5.3 prevail.

An ASCII text version of the MIB definition can be found at the following URL:¹⁸

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C5mib.txt

Two additional modules have to be imported when compiling the IEEE 802.3 LLDP extension MIB module, and they can be found at the following URLs:

<https://www.ieee802.org/1/files/public/MIBs/LLDP-V2-MIB-201603110000Z.mib>

<https://www.ieee802.org/1/files/public/MIBs/LLDP-V2-TC-MIB-201603110000Z.mib>

¹⁸ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

5.4.1 Tree hierarchy

```

----- ieee8023lldpV2Xdot3MTIB(1.3.111.2.802.3.1.5)
+----- lldpV2Xdot3Objects(1)
| +----- lldpV2Xdot3Config(1)
| | +----- lldpV2Xdot3PortConfigTable(1)
| | | +----- lldpV2Xdot3PortConfigEntry(1) [lldpV2PortConfigIfIndex, lldpV2PortConfigDestAddressIndex]
| | | | +---rw lldpV2Xdot3PortConfigTLVsTxEnable(1) Bits
| | | +----- lldpV2Xdot3LocalData(2)
| | | +----- lldpV2Xdot3LocPortTable(1)
| | | | +----- lldpV2Xdot3LocPortEntry(1) [lldpV2LocPortIfIndex]
| | | | | +---r- lldpV2Xdot3LocPortAutoNegSupported(1) TruthValue
| | | | | +---r- lldpV2Xdot3LocPortAutoNegEnabled(2) TruthValue
| | | | | +---r- lldpV2Xdot3LocPortAutoNegAdvertisedCap(3) OctetString
| | | | | +---r- lldpV2Xdot3LocPortOperMauType(4) Unsigned32
| | | +----- lldpV2Xdot3LocPowerTable(2)
| | | | +----- lldpV2Xdot3LocPowerEntry(1) [lldpV2LocPortIfIndex]
| | | | | +---r- lldpV2Xdot3LocPowerPortClass(1) LldpV2PowerPortClass
| | | | | +---r- lldpV2Xdot3LocPowerMDISupported(2) TruthValue
| | | | | +---r- lldpV2Xdot3LocPowerMDIEnabled(3) TruthValue
| | | | | +---r- lldpV2Xdot3LocPowerPairControlable(4) TruthValue
| | | | | +---r- lldpV2Xdot3LocPowerPairs(5) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerClass(6) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerType(7) Bits
| | | | | +---r- lldpV2Xdot3LocPowerSource(8) Enumeration
| | | | | +---rw lldpV2Xdot3LocPowerPriority(9) Enumeration
| | | | | +---r- lldpV2Xdot3LocPDRequestedPowerValue(10) Integer32
| | | | | +---r- lldpV2Xdot3LocPDRequestedPowerValueA(11) Integer32
| | | | | +---r- lldpV2Xdot3LocPDRequestedPowerValueB(12) Integer32
| | | | | +---r- lldpV2Xdot3LocPSEAllocatedPowerValue(13) Integer32
| | | | | +---r- lldpV2Xdot3LocPSEAllocatedPowerValueA(14) Integer32
| | | | | +---r- lldpV2Xdot3LocPSEAllocatedPowerValueB(15) Integer32
| | | | | +---r- lldpV2Xdot3LocPSEPoweringStatus(16) Enumeration
| | | | | +---r- lldpV2Xdot3LocPDPoweredStatus(17) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerPairsExt(18) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerClassExtA(19) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerClassExtB(20) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerClassExt(21) Enumeration
| | | | | +---r- lldpV2Xdot3LocPowerTypeExt(22) Enumeration

```

		+--r-	lldpV2Xdot3LocPDLoad (23)	TruthValue
		+--r-	lldpV2Xdot3LocPD4PID (24)	TruthValue
		+--r-	lldpV2Xdot3LocPSEMaxAvailPower (25)	Integer32
		+--r-	lldpV2Xdot3LocPSEAutoclassSupport (26)	TruthValue
		+--r-	lldpV2Xdot3LocPSEAutoclassCompleted (27)	TruthValue
		+--r-	lldpV2Xdot3LocPSEAutoclassRequest (28)	TruthValue
		+--rw	lldpV2Xdot3LocPowerDownRequest (29)	Integer32
		+--rw	lldpV2Xdot3LocPowerDownTime (30)	Integer32
		+--r-	lldpV2Xdot3LocMeasVoltageSupport (31)	TruthValue
		+--r-	lldpV2Xdot3LocMeasCurrentSupport (32)	TruthValue
		+--r-	lldpV2Xdot3LocMeasPowerSupport (33)	TruthValue
		+--r-	lldpV2Xdot3LocMeasEnergySupport (34)	TruthValue
		+--rw	lldpV2Xdot3LocMeasurementSource (35)	TruthValue
		+--r-	lldpV2Xdot3LocMeasVoltageRequest (36)	TruthValue
		+--r-	lldpV2Xdot3LocMeasCurrentRequest (37)	TruthValue
		+--r-	lldpV2Xdot3LocMeasPowerRequest (38)	TruthValue
		+--r-	lldpV2Xdot3LocMeasEnergyRequest (39)	TruthValue
		+--r-	lldpV2Xdot3LocMeasVoltageValid (40)	TruthValue
		+--r-	lldpV2Xdot3LocMeasCurrentValid (41)	TruthValue
		+--r-	lldpV2Xdot3LocMeasPowerValid (42)	TruthValue
		+--r-	lldpV2Xdot3LocMeasEnergyValid (43)	TruthValue
		+--r-	lldpV2Xdot3LocMeasVoltageUncertainty (44)	Integer32
		+--r-	lldpV2Xdot3LocMeasCurrentUncertainty (45)	Integer32
		+--r-	lldpV2Xdot3LocMeasPowerUncertainty (46)	Integer32
		+--r-	lldpV2Xdot3LocMeasEnergyUncertainty (47)	Integer32
		+--r-	lldpV2Xdot3LocVoltageMeasurement (48)	Integer32
		+--r-	lldpV2Xdot3LocCurrentMeasurement (49)	Integer32
		+--r-	lldpV2Xdot3LocPowerMeasurement (50)	Integer32
		+--r-	lldpV2Xdot3LocEnergyMeasurement (51)	Integer32
		+--r-	lldpV2Xdot3LocPSEPowerPriceIndex (52)	Integer32
		+--r-	lldpV2Xdot3LocResponseTime (53)	Integer32
		+--r-	lldpV2Xdot3LocReady (54)	TruthValue
		+-----	lldpV2Xdot3LocMaxFrameSizeTable (3)	
			+----- lldpV2Xdot3LocMaxFrameSizeEntry (1) [lldpV2LocPortIfIndex]	
			+----- lldpV2Xdot3LocMaxFrameSize (1) Unsigned32	
			+----- lldpV2Xdot3LocEEETable (4)	
			+----- lldpV2Xdot3LocEEETableEntry (1) [lldpV2LocPortIfIndex]	
			+--r- lldpV2Xdot3LocTxTwSys (1)	Integer32
			+--r- lldpV2Xdot3LocTxTwSysEcho (2)	Integer32

		+--r-	lldpV2Xdot3LocRxTwSys (3)	Integer32
		+--r-	lldpV2Xdot3LocRxTwSysEcho (4)	Integer32
		+--r-	lldpV2Xdot3LocFbTwSys (5)	Integer32
		+--r-	lldpV2Xdot3TxDllReady (6)	TruthValue
		+--r-	lldpV2Xdot3RxDllReady (7)	TruthValue
		+--r-	lldpV2Xdot3LocDllEnabled (8)	TruthValue
		+--r-	lldpV2Xdot3LocTxFw (9)	TruthValue
		+--r-	lldpV2Xdot3LocTxFwEcho (10)	TruthValue
		+--r-	lldpV2Xdot3LocRxFw (11)	TruthValue
		+--r-	lldpV2Xdot3LocRxFwEcho (12)	TruthValue
		+--r-	lldpV2Xdot3LocPreemptSupported (13)	TruthValue
		+--r-	lldpV2Xdot3LocPreemptEnabled (14)	TruthValue
		+--r-	lldpV2Xdot3LocPreemptActive (15)	TruthValue
		+--r-	lldpV2Xdot3LocAddFragSize (16)	Integer32
		+-----	lldpV2Xdot3RemoteData (3)	
		+-----	lldpV2Xdot3RemPortTable (1)	
			+----- lldpV2Xdot3RemPortEntry (1) [lldpV2RemLocalIfIndex, lldpV2RemLocalDestMACAddress, lldpV2RemIndex]	
		+--r-	lldpV2Xdot3RemPortAutoNegSupported (1)	TruthValue
		+--r-	lldpV2Xdot3RemPortAutoNegEnabled (2)	TruthValue
		+--r-	lldpV2Xdot3RemPortAutoNegAdvertisedCap (3)	OctetString
		+--r-	lldpV2Xdot3RemPortOperMauType (4)	Unsigned32
		+-----	lldpV2Xdot3RemPowerTable (2)	
			+----- lldpV2Xdot3RemPowerEntry (1) [lldpV2RemTimeMark, lldpV2RemLocalIfIndex, lldpV2RemLocalDestMACAddress, lldpV2RemIndex]	
		+--r-	lldpV2Xdot3RemPowerPortClass (1)	LldpV2PowerPortClass
		+--r-	lldpV2Xdot3RemPowerMDISupported (2)	TruthValue
		+--r-	lldpV2Xdot3RemPowerMDIEnabled (3)	TruthValue
		+--r-	lldpV2Xdot3RemPowerPairControllable (4)	TruthValue
		+--r-	lldpV2Xdot3RemPowerPairs (5)	Bits
		+--r-	lldpV2Xdot3RemPowerClass (6)	Enumeration
		+--r-	lldpV2Xdot3RemPowerType (7)	Bits
		+--r-	lldpV2Xdot3RemPowerSource (8)	Enumeration
		+--rw	lldpV2Xdot3RemPowerPriority (9)	Enumeration
		+--r-	lldpV2Xdot3RemPDRequestedPowerValue (10)	Integer32
		+--r-	lldpV2Xdot3RemPDRequestedPowerValueA (11)	Integer32
		+--r-	lldpV2Xdot3RemPDRequestedPowerValueB (12)	Integer32
		+--r-	lldpV2Xdot3RemPSEAllocatedPowerValue (13)	Integer32
		+--r-	lldpV2Xdot3RemPSEAllocatedPowerValueA (14)	Integer32

		+--r-	lldpV2Xdot3RemPSEAlllocatedPowerValueB (15)	Integer32
		+--r-	lldpV2Xdot3RemPSEPoweredStatus (16)	Enumeration
		+--r-	lldpV2Xdot3RemPDPoweredStatus (17)	Enumeration
		+--r-	lldpV2Xdot3RemPowerPairsExt (18)	Enumeration
		+--r-	lldpV2Xdot3RemPowerClassExtA (19)	Enumeration
		+--r-	lldpV2Xdot3RemPowerClassExtB (20)	Enumeration
		+--r-	lldpV2Xdot3RemPowerClassExt (21)	Enumeration
		+--r-	lldpV2Xdot3RemPowerTypeExt (22)	TruthValue
		+--r-	lldpV2Xdot3RemPDLload (23)	TruthValue
		+--r-	lldpV2Xdot3RemPD4PID (24)	Integer32
		+--r-	lldpV2Xdot3RemPSEMaxAvailPower (25)	TruthValue
		+--r-	lldpV2Xdot3RemPSEAutoclassSupport (26)	TruthValue
		+--r-	lldpV2Xdot3RemPSEAutoclassCompleted (27)	TruthValue
		+--r-	lldpV2Xdot3RemPSEAutoclassRequest (28)	Integer32
		+--rw	lldpV2Xdot3RemPowerDownRequest (29)	Integer32
		+--rw	lldpV2Xdot3RemPowerDownTime (30)	TruthValue
		+--r-	lldpV2Xdot3RemMeasVoltageSupport (31)	TruthValue
		+--r-	lldpV2Xdot3RemMeasCurrentSupport (32)	TruthValue
		+--r-	lldpV2Xdot3RemMeasPowerSupport (33)	TruthValue
		+--r-	lldpV2Xdot3RemMeasEnergySupport (34)	TruthValue
		+--rw	lldpV2Xdot3RemMeasurementSource (35)	TruthValue
		+--r-	lldpV2Xdot3RemMeasVoltageRequest (36)	TruthValue
		+--r-	lldpV2Xdot3RemMeasCurrentRequest (37)	TruthValue
		+--r-	lldpV2Xdot3RemMeasPowerRequest (38)	TruthValue
		+--r-	lldpV2Xdot3RemMeasEnergyRequest (39)	TruthValue
		+--r-	lldpV2Xdot3RemMeasVoltageValid (40)	TruthValue
		+--r-	lldpV2Xdot3RemMeasCurrentValid (41)	TruthValue
		+--r-	lldpV2Xdot3RemMeasPowerValid (42)	TruthValue
		+--r-	lldpV2Xdot3RemMeasEnergyValid (43)	TruthValue
		+--r-	lldpV2Xdot3RemMeasVoltageUncertainty (44)	Integer32
		+--r-	lldpV2Xdot3RemMeasCurrentUncertainty (45)	Integer32
		+--r-	lldpV2Xdot3RemMeasPowerUncertainty (46)	Integer32
		+--r-	lldpV2Xdot3RemMeasEnergyUncertainty (47)	Integer32
		+--r-	lldpV2Xdot3RemVoltageMeasurement (48)	Integer32
		+--r-	lldpV2Xdot3RemCurrentMeasurement (49)	Integer32
		+--r-	lldpV2Xdot3RemPowerMeasurement (50)	Integer32
		+--r-	lldpV2Xdot3RemEnergyMeasurement (51)	Integer32
		+--r-	lldpV2Xdot3RemPSEPowerPriceIndex (52)	Integer32
		+-----	lldpV2Xdot3RemMaxFrameSizeTable (3)	

```
| | +---- lldpV2Xdot3RemMaxFrameSizeEntry(1) [lldpV2RemLocalIfIndex, lldpV2RemLocalDestMACAd-  
dress, lldpV2RemIndex]  
| | +---r- lldpV2Xdot3RemMaxFrameSize(1) Unsigned32  
| +---- lldpV2Xdot3RemEEEETable(4)  
| +---- lldpV2Xdot3RemEEEEEntry(1) [lldpV2RemLocalIfIndex]  
| +---r- lldpV2Xdot3RemTxTwSys(1) Integer32  
| +---r- lldpV2Xdot3RemTxTwSysEcho(2) Integer32  
| +---r- lldpV2Xdot3RemRxTwSys(3) Integer32  
| +---r- lldpV2Xdot3RemRxTwSysEcho(4) Integer32  
| +---r- lldpV2Xdot3RemFbTwSys(5) Integer32  
| +---r- lldpV2Xdot3RemTxFw(6) TruthValue  
| +---r- lldpV2Xdot3RemTxFwEcho(7) TruthValue  
| +---r- lldpV2Xdot3RemRxFw(8) TruthValue  
| +---r- lldpV2Xdot3RemRxFwEcho(9) TruthValue  
| +---r- lldpV2Xdot3RemPreemptSupported(10) TruthValue  
| +---r- lldpV2Xdot3RemPreemptEnabled(11) TruthValue  
| +---r- lldpV2Xdot3RemPreemptActive(12) TruthValue  
| +---r- lldpV2Xdot3RemAddFragSize(13) Integer32  
+---- lldpV2Xdot3Conformance(2)  
+---- lldpV2Xdot3Compliances(1)  
| +---- lldpV2Xdot3TxRxCompliance(1)  
| +---- lldpV2Xdot3TxCompliance(2)  
| +---- lldpV2Xdot3RxCompliance(3)  
+---- lldpV2Xdot3Groups(2)  
+---- lldpV2Xdot3ConfigGroup(1)  
+---- lldpV2Xdot3LocSysGroup(2)  
+---- lldpV2Xdot3RemSysGroup(3)
```

5.4.2 MIB module definition

```
IEEE8023-DOT3-LLDP-EXT-V2-MIB DEFINITIONS ::= BEGIN

IMPORTS
    ifGeneralInformationGroup
        FROM IF-MIB
    lldpV2LocPortIfIndex, lldpV2PortConfigEntry, lldpV2RemIndex,
    lldpV2RemLocalDestMACAddress, lldpV2RemLocalIfIndex,
    lldpV2RemTimeMark
        FROM LLDP-V2-MIB
    LldpV2PowerPortClass
        FROM LLDP-V2-TC-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
        FROM SNMPv2-CONF
    Integer32, MODULE-IDENTITY, OBJECT-TYPE, Unsigned32, org
        FROM SNMPv2-SMI
    TruthValue
        FROM SNMPv2-TC;

ieee8023lldpV2Xdot3MIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "The LLDP Management Information Base extension module for
        IEEE 802.3 organizationally defined discovery information."
    REVISION      "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION      "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION      "201102020000Z"
    DESCRIPTION
        "This revision incorporated changes to the MIB module to
        add objects to support management of Energy Efficient
        Ethernet (EEE) and Enhanced DTE Power via the MDI (PoE+)."
    ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
    man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 5 }

lldpV2Xdot3Objects OBJECT IDENTIFIER
    ::= { ieee8023lldpV2Xdot3MIB 1 }

lldpV2Xdot3Config OBJECT IDENTIFIER
    ::= { lldpV2Xdot3Objects 1 }
```

```
lldpV2Xdot3PortConfigTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF LldpV2Xdot3PortConfigEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table that controls selection of LLDP TLVs to be transmitted
         on individual ports."
    ::= { lldpV2Xdot3Config 1 }

lldpV2Xdot3PortConfigEntry OBJECT-TYPE
    SYNTAX      LldpV2Xdot3PortConfigEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "LLDP configuration information that controls the
         transmission of IEEE 802.3 organizationally defined TLVs on
         LLDP transmission capable ports.

        This configuration object augments the lldpV2PortConfigEntry of
        the LLDP-MIB, therefore it is only present along with the port
        configuration defined by the associated lldpV2PortConfigEntry
        entry.

        Each active lldpV2Xdot3PortConfigEntry is restored from non-volatile
        storage (along with the corresponding lldpV2PortConfigEntry)
        after a re-initialization of the management system."
    AUGMENTS    { lldpV2PortConfigEntry }
    ::= { lldpV2Xdot3PortConfigTable 1 }

LldpV2Xdot3PortConfigEntry ::= SEQUENCE {
    lldpV2Xdot3PortConfigTLVsTxEnable    BITS
}

lldpV2Xdot3PortConfigTLVsTxEnable OBJECT-TYPE
    SYNTAX      BITS { macPhyConfigStatus(0), powerViaMDI(1), unused(2),
                       maxFrameSize(3), eeeEnabled(4),
                       eeeFastWakeEnabled(5), addEthernetCapabilities(6) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The lldpV2Xdot3PortConfigTLVsTxEnable, defined as a bitmap,
         includes the IEEE 802.3 organizationally defined set of LLDP
         TLVs whose transmission is allowed by the local LLDP agent by
         the network management. Each bit in the bitmap corresponds
         to an IEEE 802.3 subtype associated with a specific IEEE
         802.3 optional TLV.

        The bit 'macPhyConfigStatus(0)' indicates that the LLDP agent
         should transmit 'MAC/PHY configuration/status TLV'.

        The bit 'powerViaMDI(1)' indicates that the LLDP agent should
         transmit 'Power via MDI TLV'.

        The bit 'unused(2)' is no longer used; this was used for
         the 'Link Aggregation TLV' in the previous version.

        The bit 'maxFrameSize(3)' indicates that the LLDP agent should
         transmit 'Maximum-frame-size TLV'.
```

The bit 'eeeEnabled(4)' indicates that the LLDP agent should transmit EEE TLV.

The bit 'eeeFastWakeEnabled(5)' indicates that the LLDP agent should transmit EEE Fast Wake TLV.

The bit 'addEthernetCapabilities(6)' indicates that the LLDP agent should transmit Additional Ethernet Capabilities TLV.

The default value for lldpV2Xdot3PortConfigTLVsTxEnable object is an empty set, which means no enumerated values are set.

The value of this object is restored from non-volatile storage after a re-initialization of the management system."

REFERENCE

"IEEE Std 802.3, 30.12.1.1.1"

DEFVAL { {} }

::= { lldpV2Xdot3PortConfigEntry 1 }

lldpV2Xdot3LocalData OBJECT IDENTIFIER

::= { lldpV2Xdot3Objects 2 }

lldpV2Xdot3LocPortTable OBJECT-TYPE

SYNTAX SEQUENCE OF LldpV2Xdot3LocPortEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table contains one row per port of Ethernet port information (as a part of the LLDP 802.3 organizational extension) on the local system known to this agent."

::= { lldpV2Xdot3LocalData 1 }

lldpV2Xdot3LocPortEntry OBJECT-TYPE

SYNTAX LldpV2Xdot3LocPortEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Information about a particular port component."

INDEX { lldpV2LocPortIfIndex }

::= { lldpV2Xdot3LocPortTable 1 }

LldpV2Xdot3LocPortEntry ::= SEQUENCE {

lldpV2Xdot3LocPortAutoNegSupported TruthValue,

lldpV2Xdot3LocPortAutoNegEnabled TruthValue,

lldpV2Xdot3LocPortAutoNegAdvertisedCap OCTET STRING,

lldpV2Xdot3LocPortOperMauType Unsigned32

}

lldpV2Xdot3LocPortAutoNegSupported OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The truth value used to indicate whether the given port (associated with the local system) supports Auto-negotiation."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.1"

::= { lldpV2Xdot3LocPortEntry 1 }

lldpV2Xdot3LocPortAutoNegEnabled OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The truth value used to indicate whether port
 Auto-negotiation is enabled on the given port associated
 with the local system."
REFERENCE
 "IEEE Std 802.3, 30.12.2.1.2"
::= { lldpV2Xdot3LocPortEntry 2 }

lldpV2Xdot3LocPortAutoNegAdvertisedCap OBJECT-TYPE

SYNTAX OCTET STRING (SIZE(2))
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This object contains the value (bitmap) of the
 ifMauAutoNegCapAdvertisedBits object (defined in IETF RFC
 3636) which is associated with the given port on the
 local system."
REFERENCE
 "IEEE Std 802.3, 30.12.2.1.3"
::= { lldpV2Xdot3LocPortEntry 3 }

lldpV2Xdot3LocPortOperMauType OBJECT-TYPE

SYNTAX Unsigned32 (0..2147483647)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "An integer value that indicates the operational MAU type
 of the given port on the local system.

 This object contains the integer value derived from the
 list position of the corresponding dot3MauType as listed
 in Clause 13 and is equal to the last number in the
 respective dot3MauType OID.

 For example, if the ifMauType object is dot3MauType1000BaseTHD
 which corresponds to {dot3MauType 29}, the numerical value of
 this field is 29. For MAU types not listed in Clause 13,
 the value of this field shall be set to zero."
REFERENCE
 "IEEE Std 802.3, 30.12.2.1.4"
::= { lldpV2Xdot3LocPortEntry 4 }

lldpV2Xdot3LocPowerTable OBJECT-TYPE

SYNTAX SEQUENCE OF LldpV2Xdot3LocPowerEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
 "This table contains one row per port of Power over Ethernet
 information (as a part of the LLDP IEEE 802.3 organizational
 extension) on the local system known to this agent."
::= { lldpV2Xdot3LocalData 2 }

lldpV2Xdot3LocPowerEntry OBJECT-TYPE

SYNTAX LldpV2Xdot3LocPowerEntry

```

MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "Information about a particular port component."
INDEX          { lldpV2LocPortIfIndex }
 ::= { lldpV2Xdot3LocPowerTable 1 }

```

```

LldpV2Xdot3LocPowerEntry ::= SEQUENCE {
    lldpV2Xdot3LocPowerPortClass      LldpV2PowerPortClass,
    lldpV2Xdot3LocPowerMDISupported   TruthValue,
    lldpV2Xdot3LocPowerMDIEnabled     TruthValue,
    lldpV2Xdot3LocPowerPairControlable TruthValue,
    lldpV2Xdot3LocPowerPairs          INTEGER,
    lldpV2Xdot3LocPowerClass          INTEGER,
    lldpV2Xdot3LocPowerType           BITS,
    lldpV2Xdot3LocPowerSource         INTEGER,
    lldpV2Xdot3LocPowerPriority        INTEGER,
    lldpV2Xdot3LocPDRrequestedPowerValue Integer32,
    lldpV2Xdot3LocPDRrequestedPowerValueA Integer32,
    lldpV2Xdot3LocPDRrequestedPowerValueB Integer32,
    lldpV2Xdot3LocPSEAllocatedPowerValue Integer32,
    lldpV2Xdot3LocPSEAllocatedPowerValueA Integer32,
    lldpV2Xdot3LocPSEAllocatedPowerValueB Integer32,
    lldpV2Xdot3LocPSEPoweringStatus   INTEGER,
    lldpV2Xdot3LocPDPoweredStatus     INTEGER,
    lldpV2Xdot3LocPowerPairsExt       INTEGER,
    lldpV2Xdot3LocPowerClassExtA      INTEGER,
    lldpV2Xdot3LocPowerClassExtB     INTEGER,
    lldpV2Xdot3LocPowerClassExt      INTEGER,
    lldpV2Xdot3LocPowerTypeExt       INTEGER,
    lldpV2Xdot3LocPDLload             TruthValue,
    lldpV2Xdot3LocPDL4PID             TruthValue,
    lldpV2Xdot3LocPSEMaxAvailPower    Integer32,
    lldpV2Xdot3LocPSEAutoclassSupport TruthValue,
    lldpV2Xdot3LocPSEAutoclassCompleted TruthValue,
    lldpV2Xdot3LocPSEAutoclassRequest TruthValue,
    lldpV2Xdot3LocPowerDownRequest    Integer32,
    lldpV2Xdot3LocPowerDownTime       Integer32,
    lldpV2Xdot3LocMeasVoltageSupport   TruthValue,
    lldpV2Xdot3LocMeasCurrentSupport   TruthValue,
    lldpV2Xdot3LocMeasPowerSupport     TruthValue,
    lldpV2Xdot3LocMeasEnergySupport    TruthValue,
    lldpV2Xdot3LocMeasurementSource    TruthValue,
    lldpV2Xdot3LocMeasVoltageRequest   TruthValue,
    lldpV2Xdot3LocMeasCurrentRequest   TruthValue,
    lldpV2Xdot3LocMeasPowerRequest     TruthValue,
    lldpV2Xdot3LocMeasEnergyRequest    TruthValue,
    lldpV2Xdot3LocMeasVoltageValid     TruthValue,
    lldpV2Xdot3LocMeasCurrentValid     TruthValue,
    lldpV2Xdot3LocMeasPowerValid       TruthValue,
    lldpV2Xdot3LocMeasEnergyValid      TruthValue,
    lldpV2Xdot3LocMeasVoltageUncertainty Integer32,
    lldpV2Xdot3LocMeasCurrentUncertainty Integer32,
    lldpV2Xdot3LocMeasPowerUncertainty Integer32,
    lldpV2Xdot3LocMeasEnergyUncertainty Integer32,
    lldpV2Xdot3LocVoltageMeasurement   Integer32,
    lldpV2Xdot3LocCurrentMeasurement   Integer32,
    lldpV2Xdot3LocPowerMeasurement     Integer32,
    lldpV2Xdot3LocEnergyMeasurement    Integer32,

```

```

    lldpV2Xdot3LocPSEPowerPriceIndex      Integer32,
    lldpV2Xdot3LocResponseTime             Integer32,
    lldpV2Xdot3LocReady                    TruthValue
}

lldpV2Xdot3LocPowerPortClass OBJECT-TYPE
    SYNTAX      LldpV2PowerPortClass
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value that identifies the port Class of the given port
        associated with the local system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.5"
    ::= { lldpV2Xdot3LocPowerEntry 1 }

lldpV2Xdot3LocPowerMDISupported OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For a PSE, the truth value used to indicate whether the MDI
        power is supported on the given port associated with the
        local system. For a PD, this attribute is undefined."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.6"
    ::= { lldpV2Xdot3LocPowerEntry 2 }

lldpV2Xdot3LocPowerMDIEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For a PSE, the truth value used to identify whether MDI
        power is enabled on the given port associated with the
        local system. For a PD, this attribute is undefined."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.7"
    ::= { lldpV2Xdot3LocPowerEntry 3 }

lldpV2Xdot3LocPowerPairControlable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A truth value used to indicate the ability to control which
        PSE Pinout Alternative (see IEEE Std 802.3, 33.2.3 and 145.2.4)
        is used for PD detection and power. For a PSE, this attribute
        contains the value of the aPSEPowerPairsControlAbility attribute
        (see IEEE Std 802.3, 30.9.1.1.3). For a PD, the contents of this
        this attribute are undefined."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.8"
    ::= { lldpV2Xdot3LocPowerEntry 4 }

lldpV2Xdot3LocPowerPairs OBJECT-TYPE
    SYNTAX      INTEGER { signal(0), spare(1) }
    MAX-ACCESS  read-only
    STATUS      current
```


DESCRIPTION

"This attribute identifies the PSE Pinout Alternative (see IEEE Std 802.3, 33.2.3 and 145.2.4) in use for detecting and supplying power to the PD.
A Type 3 or Type 4 PSE detecting or supplying power on both PSE Pinout Alternatives may return either PSE Pinout Alternative as this configuration is communicated through the aLldpXdot3LocPowerPairsExt attribute.
A Type 3 or Type 4 PSE supplying power on only one PSE Pinout Alternative returns that PSE Pinout Alternative.
For a PD, the contents of this attribute are undefined."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.9"
::= { lldpV2Xdot3LocPowerEntry 5 }

lldpV2Xdot3LocPowerClass OBJECT-TYPE

SYNTAX INTEGER { class0(0), class1(1), class2(2), class3(3),
class4(4) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute identifies the requested Class of the PD as specified in IEEE Std 802.3, 33.2.6 and 145.2.8.
This attribute returns an enumeration of 'class4' for a PD of Class 4 or higher as such PD Classes are identified through the aLldpXdot3LocPowerClassExt attribute."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.10"
::= { lldpV2Xdot3LocPowerEntry 6 }

lldpV2Xdot3LocPowerType OBJECT-TYPE

SYNTAX BITS { typelp(0), pdpse(1) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute that returns a bit string indicating whether the local system is a PSE or a PD and whether it is Type 1 or greater than Type 1. The first bit ('typelp') indicates Type 1 or greater than Type 1.
The second bit ('pdpse') indicates PSE or PD."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.14"
::= { lldpV2Xdot3LocPowerEntry 7 }

lldpV2Xdot3LocPowerSource OBJECT-TYPE

SYNTAX INTEGER { pseprimary(0), psebackup(1), pseunknown(2),
pdpseandlocal(3), pdpseonly(4), pdunknown(5) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A GET returns an integer indicating the power sources of the local system. A PSE indicates whether it is being powered by a primary power source; a backup power source; or unknown. A PD indicates whether it is being powered by a PSE and locally; by a PSE only; or unknown."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.15"
::= { lldpV2Xdot3LocPowerEntry 8 }

```
lldpV2Xdot3LocPowerPriority OBJECT-TYPE
    SYNTAX      INTEGER { low(0), high(1), critical(2), unknown(3) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "A GET returns the priority of a PD system. For a PSE, this
         is the priority that the PSE assigns to the PD. For a PD, this
         is the priority that the PD requests from the PSE. A SET
         operation changes the priority of the PD system to the indicated
         value."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.16"
    ::= { lldpV2Xdot3LocPowerEntry 9 }

lldpV2Xdot3LocPDRequestedPowerValue OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 Watts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the PD requested power value.
         For a PD, it is the power value that the PD has currently
         requested from the remote system. PD requested power value
         is the maximum input average power the PD ever draws under
         this power allocation if accepted. For a PSE, it is the power
         value that the PSE echoes back to the remote system. This is
         the PD requested power value that was used by the PSE to compute
         the power it has currently allocated to the remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.17"
    ::= { lldpV2Xdot3LocPowerEntry 10 }

lldpV2Xdot3LocPDRequestedPowerValueA OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 Watts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the PD requested power value for the Mode A
         pairset.
         For a PD, it is the power value that the PD has currently
         requested from the remote system for the Mode A pairset.
         For a PSE, it is the power value for the Alternative A
         pairset that the PSE echoes back to the remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.18"
    ::= { lldpV2Xdot3LocPowerEntry 11 }

lldpV2Xdot3LocPDRequestedPowerValueB OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 Watts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the PD requested power value for the Mode B
         pairset.
         For a PD, it is the power value that the PD has currently
         requested from the remote system for the Mode B pairset.
         For a PSE, it is the power value for the Alternative B
```

pairset that the PSE echoes back to the remote system."
REFERENCE

"IEEE Std 802.3, 30.12.2.1.19"
::= { lldpV2Xdot3LocPowerEntry 12 }

lldpV2Xdot3LocPSEAllocatedPowerValue OBJECT-TYPE

SYNTAX Integer32
UNITS "0.1 Watts"
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"A GET returns the PSE allocated power value.
For a PSE, it is the power value that the PSE has currently allocated to the remote system. The PSE allocated power value is the maximum input average power that the PSE wants the PD to ever draw under this allocation if it is accepted. For a PD, it is the power value that the PD echoes back to the remote system. This is the PSE allocated power value that was used by the PD to compute the power that it has currently requested from the remote system."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.20"
::= { lldpV2Xdot3LocPowerEntry 13 }

lldpV2Xdot3LocPSEAllocatedPowerValueA OBJECT-TYPE

SYNTAX Integer32
UNITS "0.1 Watts"
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"A GET returns the PSE allocated power value for the Alternative A pairset.
For a PSE, it is the power value for the Alternative A pairset that the PSE has currently allocated to the remote system.
For a PD, it is the power value for the Mode A pairset that the PD echoes back to the remote system."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.21"
::= { lldpV2Xdot3LocPowerEntry 14 }

lldpV2Xdot3LocPSEAllocatedPowerValueB OBJECT-TYPE

SYNTAX Integer32
UNITS "0.1 Watts"
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"A GET returns the PSE allocated power value for the Alternative B pairset.
For a PSE, it is the power value for the Alternative B pairset that the PSE has currently allocated to the remote system.
For a PD, it is the power value for the Mode B pairset that the PD echoes back to the remote system."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.22"
::= { lldpV2Xdot3LocPowerEntry 15 }

lldpV2Xdot3LocPSEPoweringStatus OBJECT-TYPE

SYNTAX INTEGER { fourPairDualSigPD(0), fourPairSingleSigPD(1),
twoPair(2) }

```
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "This attribute indicates the powering status of the PSE.
    For a PD, the contents of this attribute are undefined."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.23"
::= { lldpV2Xdot3LocPowerEntry 16 }

lldpV2Xdot3LocPDPoweredStatus OBJECT-TYPE
SYNTAX        INTEGER { fourPairDualSigPD(0), twoPairDualSigPD(1),
                        singleSigPD(2) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "This attribute indicates the powering status of the PD.
    For a PSE, the contents of this attribute are undefined."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.24"
::= { lldpV2Xdot3LocPowerEntry 17 }

lldpV2Xdot3LocPowerPairsExt OBJECT-TYPE
SYNTAX        INTEGER { altA(0), altB(1), both(2) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "This attribute identifies the supported PSE Pinout Alternatives
    for a PSE.
    For a PD, the contents of this attribute are undefined."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.25"
::= { lldpV2Xdot3LocPowerEntry 18 }

lldpV2Xdot3LocPowerClassExtA OBJECT-TYPE
SYNTAX        INTEGER { singlesig(0), class1(1), class2(2), class3(3),
                        class4(4), class5(5) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "For a dual-signature PD, this attribute indicates the
    requested Class for Mode A during Physical Layer Classification
    (see IEEE Std 802.3, 145.3.6).
    For a single-signature PD, this attribute is set to 'singlesig'.
    For a PSE connected to a dual-signature PD, this attribute
    indicates the currently assigned Class for Mode A
    (see IEEE Std 802.3, 145.2.8).
    For a PSE connected to a single-signature PD or a PSE that operates
    only in 2-pair mode, this attribute is set to 'singlesig'."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.26"
::= { lldpV2Xdot3LocPowerEntry 19 }

lldpV2Xdot3LocPowerClassExtB OBJECT-TYPE
SYNTAX        INTEGER { singlesig(0), class1(1), class2(2), class3(3),
                        class4(4), class5(5) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "For a dual-signature PD, this attribute indicates the
```

requested Class for Mode B during Physical Layer Classification
(see IEEE Std 802.3, 145.3.6).
For a single-signature PD, this attribute is set to 'singlesig'.
For a PSE connected to a dual-signature PD, this attribute
indicates the currently assigned Class for Mode B
(see IEEE Std 802.3, 145.2.8).
For a PSE connected to a single-signature PD or a PSE that operates
only in 2-pair mode, this attribute is set to 'singlesig'."

REFERENCE
"IEEE Std 802.3, 30.12.2.1.27"
::= { lldpV2Xdot3LocPowerEntry 20 }

lldpV2Xdot3LocPowerClassExt OBJECT-TYPE

SYNTAX INTEGER { dualsig(0), class1(1), class2(2), class3(3),
class4(4), class5(5), class6(6), class7(7),
class8(8) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For a single-signature PD, this attribute indicates the
requested Class during Physical Layer Classification
(see IEEE Std 802.3, 145.3.6).
For a dual-signature PD, this attribute is set to 'dualsig'.
For a PSE connected to a single-signature PD or a PSE that
operates only in 2-pair mode, this attribute indicates the
currently assigned Class (see IEEE Std 802.3, 145.2.8).
For a PSE connected to a dual-signature PD, this attribute is
set to 'dualsig'."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.28"
::= { lldpV2Xdot3LocPowerEntry 21 }

lldpV2Xdot3LocPowerTypeExt OBJECT-TYPE

SYNTAX INTEGER { type4dualSigPD(0), type4singleSigPD(1),
type3dualSigPD(2), type3singleSigPD(3), type4PSE(4),
type3PSE(5) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute indicates if the local system is a Type 3 or Type 4
PSE or PD and, in the case of a Type 3 or Type 4 PD, if it is a
single-signature PD or a dual-signature PD."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.29"
::= { lldpV2Xdot3LocPowerEntry 22 }

lldpV2Xdot3LocPDLoad OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For a dual-signature PD, this attribute indicates whether the
load of a dual-signature PD is electrically isolated, as defined
in IEEE Std 802.3, 79.3.2.10.2.
For a single-signature PD or a PSE, the value of this attribute
is FALSE."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.30"
::= { lldpV2Xdot3LocPowerEntry 23 }

```
lldpV2Xdot3LocPD4PID OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the local PD supports
        powering of both PD Modes."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.31"
    ::= { lldpV2Xdot3LocPowerEntry 24 }

lldpV2Xdot3LocPSEMaxAvailPower OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 Watts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute reports the local PSE maximum available power."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.32"
    ::= { lldpV2Xdot3LocPowerEntry 25 }

lldpV2Xdot3LocPSEAutoclassSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the local PSE supports
        Autoclass."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.33"
    ::= { lldpV2Xdot3LocPowerEntry 26 }

lldpV2Xdot3LocPSEAutoclassCompleted OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the local PSE has
        completed the Autoclass measurement."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.34"
    ::= { lldpV2Xdot3LocPowerEntry 27 }

lldpV2Xdot3LocPSEAutoclassRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the local PSE is
        requesting an Autoclass measurement and power budget
        adjustment."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.35"
    ::= { lldpV2Xdot3LocPowerEntry 28 }

lldpV2Xdot3LocPowerDownRequest OBJECT-TYPE
    SYNTAX      Integer32
```

```
MAX-ACCESS    read-write
STATUS        current
DESCRIPTION
    "This attribute indicates the local PD is requesting
    a power down when the value is 0x1D."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.36"
::= { lldpV2Xdot3LocPowerEntry 29 }

lldpV2Xdot3LocPowerDownTime OBJECT-TYPE
    SYNTAX      Integer32
    UNITS        "Seconds"
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "This attribute indicates the time the PD requests
        to stay powered off. A value of zero indicates an
        indefinite amount of time."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.37"
    ::= { lldpV2Xdot3LocPowerEntry 30 }

lldpV2Xdot3LocMeasVoltageSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is capable of
        providing a voltage measurement. "
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.38"
    ::= { lldpV2Xdot3LocPowerEntry 31 }

lldpV2Xdot3LocMeasCurrentSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is capable of
        providing a current measurement. "
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.39"
    ::= { lldpV2Xdot3LocPowerEntry 32 }

lldpV2Xdot3LocMeasPowerSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is capable of
        providing a power measurement."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.40"
    ::= { lldpV2Xdot3LocPowerEntry 33 }

lldpV2Xdot3LocMeasEnergySupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
```

```
DESCRIPTION
    "This attribute indicates the local device is capable of
    providing an energy measurement."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.41"
::= { lldpV2Xdot3LocPowerEntry 34 }

lldpV2Xdot3LocMeasurementSource OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "This attribute indicates to local device on which Alternative
        or Mode the measurement is to be taken."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.42"
    ::= { lldpV2Xdot3LocPowerEntry 35 }

lldpV2Xdot3LocMeasVoltageRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is requesting
        a voltage measurement from the remote device."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.43"
    ::= { lldpV2Xdot3LocPowerEntry 36 }

lldpV2Xdot3LocMeasCurrentRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is requesting
        a current measurement from the remote device."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.44"
    ::= { lldpV2Xdot3LocPowerEntry 37 }

lldpV2Xdot3LocMeasPowerRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is requesting
        a power measurement from the remote device."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.45"
    ::= { lldpV2Xdot3LocPowerEntry 38 }

lldpV2Xdot3LocMeasEnergyRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the local device is requesting
        energy measurement from the remote device."
    REFERENCE
```



```
"IEEE Std 802.3, 30.12.2.1.46"
 ::= { lldpV2Xdot3LocPowerEntry 39 }

lldpV2Xdot3LocMeasVoltageValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the local device's voltage measurement
         is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.47"
    ::= { lldpV2Xdot3LocPowerEntry 40 }

lldpV2Xdot3LocMeasCurrentValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the local device's current measurement
         is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.48"
    ::= { lldpV2Xdot3LocPowerEntry 41 }

lldpV2Xdot3LocMeasPowerValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the local device's power measurement
         is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.49"
    ::= { lldpV2Xdot3LocPowerEntry 42 }

lldpV2Xdot3LocMeasEnergyValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the local device's energy measurement
         is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.50"
    ::= { lldpV2Xdot3LocPowerEntry 43 }

lldpV2Xdot3LocMeasVoltageUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the device's voltage measurement.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.51"
    ::= { lldpV2Xdot3LocPowerEntry 44 }
```

```
lldpV2Xdot3LocMeasCurrentUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the device's current measurement.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.52"
    ::= { lldpV2Xdot3LocPowerEntry 45 }

lldpV2Xdot3LocMeasPowerUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the device's power measurement.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.53"
    ::= { lldpV2Xdot3LocPowerEntry 46 }

lldpV2Xdot3LocMeasEnergyUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the device's energy measurement.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.54"
    ::= { lldpV2Xdot3LocPowerEntry 47 }

lldpV2Xdot3LocVoltageMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured device voltage.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.55"
    ::= { lldpV2Xdot3LocPowerEntry 48 }

lldpV2Xdot3LocCurrentMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured device current.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.56"
    ::= { lldpV2Xdot3LocPowerEntry 49 }

lldpV2Xdot3LocPowerMeasurement OBJECT-TYPE
```

```
SYNTAX      Integer32
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This attribute indicates the measured device power.
    See IEEE Std 802.3, Table 79-21."
REFERENCE
    "IEEE Std 802.3, 30.12.2.1.57"
::= { lldpV2Xdot3LocPowerEntry 50 }

lldpV2Xdot3LocEnergyMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured device energy.
        See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.58"
    ::= { lldpV2Xdot3LocPowerEntry 51 }

lldpV2Xdot3LocPSEPowerPriceIndex OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates an index of the price of power being
        sourced by the PSE. For a PD, this value is undefined."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.59"
    ::= { lldpV2Xdot3LocPowerEntry 52 }

lldpV2Xdot3LocResponseTime OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the response time in seconds of the local system.
        For a PD, it is the maximum time required to update the value of
        lldpV2Xdot3LocPDRequestedPowerValue when the remote system
        requests the PD to change its max power draw. For a PSE, it is
        the maximum time required to update the value of
        lldpV2Xdot3LocPDRequestedPowerValue when the remote system
        requests of the PSE a new power value."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.60"
    ::= { lldpV2Xdot3LocPowerEntry 53 }

lldpV2Xdot3LocReady OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to identify whether the local Data Link Layer
        classification engine has completed initialization and is ready to
        receive and transmit LLDPDUs."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.61"
    ::= { lldpV2Xdot3LocPowerEntry 54 }
```

```
lldpV2Xdot3LocMaxFrameSizeTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF LldpV2Xdot3LocMaxFrameSizeEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table contains one row per port of maximum frame
         size information (as a part of the LLDP IEEE 802.3 organizational
         extension) on the local system known to this agent."
    ::= { lldpV2Xdot3LocalData 3 }

lldpV2Xdot3LocMaxFrameSizeEntry OBJECT-TYPE
    SYNTAX      LldpV2Xdot3LocMaxFrameSizeEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Maximum Frame Size information about a particular port
         component."
    INDEX       { lldpV2LocPortIfIndex }
    ::= { lldpV2Xdot3LocMaxFrameSizeTable 1 }

LldpV2Xdot3LocMaxFrameSizeEntry ::= SEQUENCE {
    lldpV2Xdot3LocMaxFrameSize  Unsigned32
}

lldpV2Xdot3LocMaxFrameSize OBJECT-TYPE
    SYNTAX      Unsigned32 (0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "An integer value indicating the maximum supported frame
         size in octets on the given port of the local system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.13"
    ::= { lldpV2Xdot3LocMaxFrameSizeEntry 1 }

lldpV2Xdot3LocEEETable OBJECT-TYPE
    SYNTAX      SEQUENCE OF LldpV2Xdot3LocEEEEEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table contains one row per port of Energy Efficient Ethernet
         information (as a part of the LLDP IEEE 802.3 organizational
         extension) on the local system known to this agent."
    ::= { lldpV2Xdot3LocalData 4 }

lldpV2Xdot3LocEEEEEntry OBJECT-TYPE
    SYNTAX      LldpV2Xdot3LocEEEEEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Information about a particular port component."
    INDEX       { lldpV2LocPortIfIndex }
    ::= { lldpV2Xdot3LocEEETable 1 }

LldpV2Xdot3LocEEEEEntry ::= SEQUENCE {
    lldpV2Xdot3LocTxTwSys      Integer32,
    lldpV2Xdot3LocTxTwSysEcho  Integer32,
    lldpV2Xdot3LocRxTwSys      Integer32,
```

```
lldpV2Xdot3LocRxTwSysEcho      Integer32,
lldpV2Xdot3LocFbTwSys           Integer32,
lldpV2Xdot3TxDllReady           TruthValue,
lldpV2Xdot3RxDllReady           TruthValue,
lldpV2Xdot3LocDllEnabled        TruthValue,
lldpV2Xdot3LocTxFw              TruthValue,
lldpV2Xdot3LocTxFwEcho          TruthValue,
lldpV2Xdot3LocRxFw              TruthValue,
lldpV2Xdot3LocRxFwEcho          TruthValue,
lldpV2Xdot3LocPreemptSupported  TruthValue,
lldpV2Xdot3LocPreemptEnabled    TruthValue,
lldpV2Xdot3LocPreemptActive     TruthValue,
lldpV2Xdot3LocAddFragSize       Integer32
}

lldpV2Xdot3LocTxTwSys OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that the local system
        can support in the transmit direction.
        This object maps to the variable LocTxSystemValue as defined
        in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.62"
    ::= { lldpV2Xdot3LocEEEEEntry 1 }

lldpV2Xdot3LocTxTwSysEcho OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that the remote system is
        advertising that it can support in the transmit direction and is
        echoed by the local system under the control of the EEE DLL receiver
        state diagram. This object maps to the variable
        LocTxSystemValueEcho as defined in IEEE Std 802.3, 78.4.2.3"
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.63"
    ::= { lldpV2Xdot3LocEEEEEntry 2 }

lldpV2Xdot3LocRxTwSys OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that
        the local system is requesting in the receive direction.
        This object maps to the variable LocRxSystemValue as
        defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.64"
    ::= { lldpV2Xdot3LocEEEEEntry 3 }

lldpV2Xdot3LocRxTwSysEcho OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
```

DESCRIPTION

"A GET returns the value of Tw_sys_tx that the remote system is advertising that it is requesting in the receive direction and is echoed by the local system under the control of the EEE DLL transmitter state diagram. This object maps to the variable LocRxSystemValueEcho as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.65"
::= { lldpV2Xdot3LocEEEEEntry 4 }

lldpV2Xdot3LocFbTwSys OBJECT-TYPE

SYNTAX Integer32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"A GET returns the value of the fallback Tw_sys_tx that the local system is advertising to the remote system. This object maps to the variable LocFbSystemValue as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.66"
::= { lldpV2Xdot3LocEEEEEntry 5 }

lldpV2Xdot3TxDllReady OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"The truth value used to identify whether the local Data Link Layer EEE layer management function has completed initialization and is ready to receive and transmit LLDPDUs."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.67"
::= { lldpV2Xdot3LocEEEEEntry 6 }

lldpV2Xdot3RxDllReady OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"The truth value used to identify whether the local Data Link Layer EEE layer management function has completed initialization and is ready to receive and transmit LLDPDUs."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.68"
::= { lldpV2Xdot3LocEEEEEntry 7 }

lldpV2Xdot3LocDllEnabled OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"The truth value used to identify whether the local system has completed auto-negotiation with a link partner that has indicated at least one EEE capability."

REFERENCE

"IEEE Std 802.3, 30.12.2.1.69"
::= { lldpV2Xdot3LocEEEEEntry 8 }

```
lldpV2Xdot3LocTxFw OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value identifies the LPI_FW value that the local system can
        support in the transmit direction. This attribute maps to
        variable LocTxSystemFW as defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.70"
    ::= { lldpV2Xdot3LocEEEEEntry 9 }

lldpV2Xdot3LocTxFwEcho OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value identifies the LPI_FW value advertised by the remote
        system and echoed by the local system. This attribute maps to
        variable LocTxSystemFWEcho as defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.71"
    ::= { lldpV2Xdot3LocEEEEEntry 10 }

lldpV2Xdot3LocRxFw OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value identifies the LPI_FW value that the local system is
        requesting in the receive direction. This attribute maps to
        variable LocRxSystemFW as defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.72"
    ::= { lldpV2Xdot3LocEEEEEntry 11 }

lldpV2Xdot3LocRxFwEcho OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value identifies the LPI_FW value requested by the remote
        system and echoed by the local system. This attribute maps to
        variable LocRxSystemFWEcho as defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.73"
    ::= { lldpV2Xdot3LocEEEEEntry 12 }

lldpV2Xdot3LocPreemptSupported OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to identify whether the local system
        supports the preemption capability."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.74"
    ::= { lldpV2Xdot3LocEEEEEntry 13 }
```

```
lldpV2Xdot3LocPreemptEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to identify whether the preemption
         capability is enabled on the local system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.75"
    ::= { lldpV2Xdot3LocEEEEEntry 14 }

lldpV2Xdot3LocPreemptActive OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to identify whether the preemption
         capability is active on the local system."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.76"
    ::= { lldpV2Xdot3LocEEEEEntry 15 }

lldpV2Xdot3LocAddFragSize OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value indicates the minimum size of non-final
         fragments supported by the local system. This value
         is expressed in units of 64 octets of additional
         fragment length."
    REFERENCE
        "IEEE Std 802.3, 30.12.2.1.77"
    ::= { lldpV2Xdot3LocEEEEEntry 16 }

lldpV2Xdot3RemoteData OBJECT IDENTIFIER
    ::= { lldpV2Xdot3Objects 3 }

lldpV2Xdot3RemPortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF LldpV2Xdot3RemPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table contains Ethernet port information (as a part
         of the LLDP IEEE 802.3 organizational extension) of the remote
         system."
    ::= { lldpV2Xdot3RemoteData 1 }

lldpV2Xdot3RemPortEntry OBJECT-TYPE
    SYNTAX      LldpV2Xdot3RemPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Information about a particular physical network connection."
    INDEX
        { lldpV2RemTimeMark, lldpV2RemLocalIfIndex,
          lldpV2RemLocalDestMACAddress, lldpV2RemIndex }
    ::= { lldpV2Xdot3RemPortTable 1 }
```



```
LldpV2Xdot3RemPortEntry ::= SEQUENCE {
    lldpV2Xdot3RemPortAutoNegSupported      TruthValue,
    lldpV2Xdot3RemPortAutoNegEnabled        TruthValue,
    lldpV2Xdot3RemPortAutoNegAdvertisedCap  OCTET STRING,
    lldpV2Xdot3RemPortOperMauType           Unsigned32
}

lldpV2Xdot3RemPortAutoNegSupported OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to indicate whether the given port
         (associated with remote system) supports Auto-negotiation."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.1"
    ::= { lldpV2Xdot3RemPortEntry 1 }

lldpV2Xdot3RemPortAutoNegEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to indicate whether port
         Auto-negotiation is enabled on the given port associated
         with the remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.2"
    ::= { lldpV2Xdot3RemPortEntry 2 }

lldpV2Xdot3RemPortAutoNegAdvertisedCap OBJECT-TYPE
    SYNTAX      OCTET STRING (SIZE(2))
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This object contains the value (bitmap) of the
         ifMauAutoNegCapAdvertisedBits object (defined in IETF RFC
         3636) which is associated with the given port on the
         remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.3"
    ::= { lldpV2Xdot3RemPortEntry 3 }

lldpV2Xdot3RemPortOperMauType OBJECT-TYPE
    SYNTAX      Unsigned32 (0..2147483647)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "An integer value that indicates the operational MAU type
         of the sending device.

        This object contains the integer value derived from the
        list position of the corresponding dot3MauType as listed in
        in Clause 13 and is equal to the last number in
        the respective dot3MauType OID.

        For example, if the ifMauType object is dot3MauType1000BaseTHD
        which corresponds to {dot3MauType 29}, the numerical value of
        this field is 29. For MAU types not listed in Clause 13,
```

the value of this field shall be set to zero."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.4"

::= { lldpV2Xdot3RemPortEntry 4 }

lldpV2Xdot3RemPowerTable OBJECT-TYPE

SYNTAX SEQUENCE OF LldpV2Xdot3RemPowerEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table contains Ethernet power information (as a part of the LLDP IEEE 802.3 organizational extension) of the remote system."

::= { lldpV2Xdot3RemoteData 2 }

lldpV2Xdot3RemPowerEntry OBJECT-TYPE

SYNTAX LldpV2Xdot3RemPowerEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Information about a particular physical network connection."

INDEX { lldpV2RemTimeMark, lldpV2RemLocalIfIndex,
lldpV2RemLocalDestMACAddress, lldpV2RemIndex }

::= { lldpV2Xdot3RemPowerTable 1 }

LldpV2Xdot3RemPowerEntry ::= SEQUENCE {

lldpV2Xdot3RemPowerPortClass	LldpV2PowerPortClass,
lldpV2Xdot3RemPowerMDISupported	TruthValue,
lldpV2Xdot3RemPowerMDIEnabled	TruthValue,
lldpV2Xdot3RemPowerPairControlable	TruthValue,
lldpV2Xdot3RemPowerPairs	BITS,
lldpV2Xdot3RemPowerClass	INTEGER,
lldpV2Xdot3RemPowerType	BITS,
lldpV2Xdot3RemPowerSource	INTEGER,
lldpV2Xdot3RemPowerPriority	INTEGER,
lldpV2Xdot3RemPDRequestedPowerValue	Integer32,
lldpV2Xdot3RemPDRequestedPowerValueA	Integer32,
lldpV2Xdot3RemPDRequestedPowerValueB	Integer32,
lldpV2Xdot3RemPSEAllocatedPowerValue	Integer32,
lldpV2Xdot3RemPSEAllocatedPowerValueA	Integer32,
lldpV2Xdot3RemPSEAllocatedPowerValueB	Integer32,
lldpV2Xdot3RemPSEPoweringStatus	INTEGER,
lldpV2Xdot3RemPDPoweredStatus	INTEGER,
lldpV2Xdot3RemPowerPairsExt	INTEGER,
lldpV2Xdot3RemPowerClassExtA	INTEGER,
lldpV2Xdot3RemPowerClassExtB	INTEGER,
lldpV2Xdot3RemPowerClassExt	INTEGER,
lldpV2Xdot3RemPowerTypeExt	INTEGER,
lldpV2Xdot3RemPDLload	TruthValue,
lldpV2Xdot3RemPD4PID	TruthValue,
lldpV2Xdot3RemPSEMaxAvailPower	Integer32,
lldpV2Xdot3RemPSEAutoclassSupport	TruthValue,
lldpV2Xdot3RemPSEAutoclassCompleted	TruthValue,
lldpV2Xdot3RemPSEAutoclassRequest	TruthValue,
lldpV2Xdot3RemPowerDownRequest	Integer32,
lldpV2Xdot3RemPowerDownTime	Integer32,
lldpV2Xdot3RemMeasVoltageSupport	TruthValue,
lldpV2Xdot3RemMeasCurrentSupport	TruthValue,
lldpV2Xdot3RemMeasPowerSupport	TruthValue,

```

lldpV2Xdot3RemMeasEnergySupport      TruthValue,
lldpV2Xdot3RemMeasurementSource      TruthValue,
lldpV2Xdot3RemMeasVoltageRequest     TruthValue,
lldpV2Xdot3RemMeasCurrentRequest     TruthValue,
lldpV2Xdot3RemMeasPowerRequest       TruthValue,
lldpV2Xdot3RemMeasEnergyRequest      TruthValue,
lldpV2Xdot3RemMeasVoltageValid       TruthValue,
lldpV2Xdot3RemMeasCurrentValid       TruthValue,
lldpV2Xdot3RemMeasPowerValid         TruthValue,
lldpV2Xdot3RemMeasEnergyValid        TruthValue,
lldpV2Xdot3RemMeasVoltageUncertainty Integer32,
lldpV2Xdot3RemMeasCurrentUncertainty Integer32,
lldpV2Xdot3RemMeasPowerUncertainty   Integer32,
lldpV2Xdot3RemMeasEnergyUncertainty  Integer32,
lldpV2Xdot3RemVoltageMeasurement     Integer32,
lldpV2Xdot3RemCurrentMeasurement     Integer32,
lldpV2Xdot3RemPowerMeasurement       Integer32,
lldpV2Xdot3RemEnergyMeasurement      Integer32,
lldpV2Xdot3RemPSEPowerPriceIndex     Integer32
}

lldpV2Xdot3RemPowerPortClass OBJECT-TYPE
    SYNTAX      LldpV2PowerPortClass
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value that identifies the port Class of the given port
         associated with the remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.5"
    ::= { lldpV2Xdot3RemPowerEntry 1 }

lldpV2Xdot3RemPowerMDISupported OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to indicate whether the MDI power
         is supported on the given port associated with the remote
         system."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.6"
    ::= { lldpV2Xdot3RemPowerEntry 2 }

lldpV2Xdot3RemPowerMDIEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The truth value used to identify whether MDI power is
         enabled on the given port associated with the remote system."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.7"
    ::= { lldpV2Xdot3RemPowerEntry 3 }

lldpV2Xdot3RemPowerPairControlable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current

```

DESCRIPTION

"This attribute indicates the ability to control which PSE Pinout Alternative (see IEEE Std 802.3, 33.2.3 and 145.2.4) is used for PD detection and power on the given port on the remote system.
For a PD, this attribute contains the value of the aPSEPowerPairsControlAbility attribute (see IEEE Std 802.3, 30.9.1.1.3) on the given port on the remote system.
For a PSE, the contents of this attribute are undefined."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.8"
::= { lldpV2Xdot3RemPowerEntry 4 }

lldpV2Xdot3RemPowerPairs OBJECT-TYPE

SYNTAX BITS { signal(0), spare(1) }
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This attribute identifies the supported PSE Pinout Alternative (see IEEE Std 802.3, 33.2.3 and 145.2.4) in use for supplying power to the PD on the given port on the remote system. For a PD, this attribute contains a value derived from the aPSEPowerPairs attribute (see IEEE Std 802.3, 30.9.1.1.4) on the given port on the remote system.
For a PSE, the contents of this attribute are undefined. When the remote system is a Type 3 or Type 4 PSE supplying power on both PSE Pinout Alternatives, the value of this attribute can indicate either pinout. If the aLldpXdot3RemPowerPairsExt attribute is available, it reports this configuration."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.9"
::= { lldpV2Xdot3RemPowerEntry 5 }

lldpV2Xdot3RemPowerClass OBJECT-TYPE

SYNTAX INTEGER { class0(0), class1(1), class2(2), class3(3),
class4(4) }
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This attribute identifies the requested Class of the PD as specified in IEEE Std 802.3, 33.2.6 and 145.2.8 on the given port on the remote system. This attribute returns an enumeration of 'class4' for a PD of Class 4 or higher as such PD Classes are identified through the aLldpXdot3RemPowerClassExt attribute."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.10"
::= { lldpV2Xdot3RemPowerEntry 6 }

lldpV2Xdot3RemPowerType OBJECT-TYPE

SYNTAX BITS { typelp(0), pdpse(1) }
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This attribute that returns a bit string indicating whether the remote system is a PSE or a PD and whether it is Type 1 or greater than Type 1. The first bit ('typelp') indicates Type 1 or greater than Type 1.
The second bit ('pdpse') indicates PSE or PD."

REFERENCE

```
"IEEE Std 802.3, 30.12.3.1.14"
 ::= { lldpV2Xdot3RemPowerEntry 7 }

lldpV2Xdot3RemPowerSource OBJECT-TYPE
    SYNTAX      INTEGER { pseprimary(0), psebackup(1), pseunknown(2),
                        pdpseandlocal(3), pdlocalonly(4), pdpseonly(5),
                        pdunknown(6) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A GET returns an integer indicating the power sources of the
        remote system. When the remote system is a PSE, it indicates
        whether it is being powered by a primary power source; a backup
        power source; or unknown. When the remote system is a PD, it
        indicates whether it is being powered by a PSE and locally;
        locally only; by a PSE only; or unknown."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.15"
    ::= { lldpV2Xdot3RemPowerEntry 8 }

lldpV2Xdot3RemPowerPriority OBJECT-TYPE
    SYNTAX      INTEGER { low(0), high(1), critical(2), unknown(3) }
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "A GET returns the priority of a PD system. For a PSE, this
        is the priority that the remote system requests. For a PD, this
        is the priority that the remote system has assigned."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.16"
    ::= { lldpV2Xdot3RemPowerEntry 9 }

lldpV2Xdot3RemPDRequestedPowerValue OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A GET returns the PD requested power value that was used
        by the remote system to compute the power value that is has
        currently allocated to the PD. For a PSE, it is the PD requested
        power value received from the remote system. The definition and
        encoding of PD requested power value is the same as described in
        lldpV2Xdot3LocPDRequestedPowerValue."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.17"
    ::= { lldpV2Xdot3RemPowerEntry 10 }

lldpV2Xdot3RemPDRequestedPowerValueA OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute identifies the PD requested power value for
        the Mode A pairset that was used by the remote system to
        compute the power value that it has currently allocated to the
        PD. For a PSE, it is the PD requested power value for the
        Alternative A pairset received from the remote system. For a PD,
        it is the PD requested power value for the Alternative A pairset
        that the PSE echoes back to the remote system. The definition and
```

encoding of PD requested power value for the Mode A pairset is the same as described in aLldpXdot3LocPDRequestedPowerValueA (see IEEE Std 802.3, 30.12.2.1.18)."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.18"

::= { lldpV2Xdot3RemPowerEntry 11 }

lldpV2Xdot3RemPDRequestedPowerValueB OBJECT-TYPE

SYNTAX Integer32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute identifies the PD requested power value for the Mode B pairset that was used by the remote system to compute the power value that it has currently allocated to the PD. For a PSE, it is the PD requested power value for the Alternative B pairset received from the remote system. For a PD, it is the PD requested power value for the Alternative B pairset that the PSE echoes back to the remote system. The definition and encoding of PD requested power value for the Mode B pairset is the same as described in aLldpXdot3LocPDRequestedPowerValueB (see IEEE Std 802.3, 30.12.2.1.19)."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.19"

::= { lldpV2Xdot3RemPowerEntry 12 }

lldpV2Xdot3RemPSEAllocatedPowerValue OBJECT-TYPE

SYNTAX Integer32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute identifies the PSE allocated power value received from the remote system. For a PSE, it is the PSE allocated power value that was echoed back by the remote PD. For a PD, it is the PSE allocated power value received from the remote system. The definition and encoding of PSE allocated power value is the same as described in aLldpXdot3LocPSEAllocatedPowerValue (see IEEE Std 802.3, 30.12.2.1.20)."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.20"

::= { lldpV2Xdot3RemPowerEntry 13 }

lldpV2Xdot3RemPSEAllocatedPowerValueA OBJECT-TYPE

SYNTAX Integer32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This attribute identifies the PSE allocated power value for the Alternative A pairset received from the remote system. For a PSE, it is the PSE allocated power value for the Alternative A pairset that was echoed back by the remote PD. For a PD, it is the PSE allocated power value for the Mode A pairset received from the remote system. The definition and encoding of PSE allocated power value for the Alternative A pairset is the same as described in aLldpXdot3LocPSEAllocatedPowerValueA (see IEEE Std 802.3, 30.12.2.1.21)."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.21"

::= { lldpV2Xdot3RemPowerEntry 14 }

```
lldpV2Xdot3RemPSEAllocatedPowerValueB OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute identifies the PSE allocated power value for the
        Alternative B pairset received from the remote system. For a PSE,
        it is the PSE allocated power value for the Alternative B pairset
        that was echoed back by the remote PD. For a PD, it is the PSE
        allocated power value for the Mode B pairset received from the
        remote system. The definition and encoding of PSE allocated power
        value for the Alternative B pairset is the same as described in
        aLldpXdot3LocPSEAllocatedPowerValueB
        (see IEEE Std 802.3, 30.12.2.1.22)."
```

```
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.22"
 ::= { lldpV2Xdot3RemPowerEntry 15 }
```

```
lldpV2Xdot3RemPSEPoweringStatus OBJECT-TYPE
    SYNTAX      INTEGER { fourPairDualSigPD(0), fourPairSingleSigPD(1),
                          twoPair(2) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the powering status of the remote PSE.
        For a PSE, the contents of this attribute are undefined."
```

```
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.23"
 ::= { lldpV2Xdot3RemPowerEntry 16 }
```

```
lldpV2Xdot3RemPDPoweredStatus OBJECT-TYPE
    SYNTAX      INTEGER { fourPairDualSigPD(0), twoPairDualSigPD(1),
                          singleSigPD(2) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the powering status of the remote PD.
        For a PD, the contents of this attribute are undefined."
```

```
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.24"
 ::= { lldpV2Xdot3RemPowerEntry 17 }
```

```
lldpV2Xdot3RemPowerPairsExt OBJECT-TYPE
    SYNTAX      INTEGER { altA(0), altB(1), both(2) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute identifies the supported PSE Pinout Alternative
        specified in IEEE Std 802.3, 145.2.4.
        For a PD, this attribute contains the value of the aPSEPowerPairs
        attribute (see IEEE Std 802.3, 30.9.1.1.4) as sent by the remote PSE.
        For a PSE, the contents of this attribute are undefined."
```

```
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.25"
 ::= { lldpV2Xdot3RemPowerEntry 18 }
```

```
lldpV2Xdot3RemPowerClassExtA OBJECT-TYPE
    SYNTAX      INTEGER { singlesig(0), class1(1), class2(2), class3(3),
```

```

        class4(4), class5(5) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "For a dual-signature PD, this attribute indicates the currently
    assigned Class for Mode A by the remote 4-pair PSE.
    For a single-signature PD or a dual-signature PD connected to a
    2-pair only PSE, this attribute is set to 'singlesig' by the remote
    PSE.
    For a PSE connected to a dual-signature PD, this attribute indicates
    the requested Class for Mode A during Physical Layer classification
    (see IEEE Std 802.3, 145.2.8) by the remote PD.
    For a PSE connected to a single-signature PD, this attribute is set to
    'singlesig' by the remote PD."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.26"
::= { lldpV2Xdot3RemPowerEntry 19 }

lldpV2Xdot3RemPowerClassExtB OBJECT-TYPE
SYNTAX        INTEGER { singlesig(0), class1(1), class2(2), class3(3),
                        class4(4), class5(5) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "For a dual-signature PD, this attribute indicates the currently
    assigned Class for Mode B by the remote 4-pair PSE.
    For a single-signature PD or a dual-signature PD connected to a
    2-pair only PSE, this attribute is set to 'singlesig' by the remote
    PSE.
    For a PSE connected to a dual-signature PD, this attribute indicates
    the requested Class for Mode B during Physical Layer classification
    (see IEEE Std 802.3, 145.2.8) by the remote PD.
    For a PSE connected to a single-signature PD, this attribute is set to
    'singlesig' by the remote PD."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.27"
::= { lldpV2Xdot3RemPowerEntry 20 }

lldpV2Xdot3RemPowerClassExt OBJECT-TYPE
SYNTAX        INTEGER { dualsig(0), class1(1), class2(2), class3(3),
                        class4(4), class5(5), class6(6), class7(7),
                        class8(8) }
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "For a single-signature PD or a dual-signature PD connected to
    a 2-pair only PSE, this attribute indicates the currently
    assigned Class by the remote PSE.
    For a dual-signature PD connected to a 4-pair capable PSE, this
    attribute is set to 'dualsig' by the remote PSE.
    For a PSE connected to a single-signature PD, this attribute
    indicates the requested Class during Physical Layer classification
    (see IEEE Std 802.3, 145.2.8) by the remote PD.
    For a PSE connected to a dual-signature PD, this attribute is set to
    'dualsig' by the remote PD."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.28"
::= { lldpV2Xdot3RemPowerEntry 21 }
```



```
lldpV2Xdot3RemPowerTypeExt OBJECT-TYPE
    SYNTAX      INTEGER { type4dualSigPD(0), type4singleSigPD(1),
                          type3dualSigPD(2), type3singleSigPD(3), type4PSE(4),
                          type3PSE(5) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicate if the remote system is a Type 3 or Type 4
        PSE or PD and, in the case of a Type 3 or Type 4 PD, if it is a
        single-signature PD or a dual-signature PD."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.29"
    ::= { lldpV2Xdot3RemPowerEntry 22 }

lldpV2Xdot3RemPDLoad OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For a dual-signature PD, this attribute indicates whether the
        load of a dual-signature PD is electrically isolated, as defined
        in IEEE Std 802.3, 79.3.2.10.2.
        For a PD, the value of this attribute is FALSE."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.30"
    ::= { lldpV2Xdot3RemPowerEntry 23 }

lldpV2Xdot3RemPD4PID OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the remote PD system supports
        powering of both PD Modes."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.31"
    ::= { lldpV2Xdot3RemPowerEntry 24 }

lldpV2Xdot3RemPSEMaxAvailPower OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute reports the remote PSE maximum available power
        value in units of 0.1 W."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.32"
    ::= { lldpV2Xdot3RemPowerEntry 25 }

lldpV2Xdot3RemPSEAutoclassSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates whether the remote PSE system supports
        Autoclass."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.33"
    ::= { lldpV2Xdot3RemPowerEntry 26 }
```

```
lldpV2Xdot3RemPSEAutoclassCompleted OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates whether the remote PSE system has
        completed the Autoclass measurement."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.34"
    ::= { lldpV2Xdot3RemPowerEntry 27 }

lldpV2Xdot3RemPSEAutoclassRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates whether the remote PSE system is
        requesting an Autoclass measurement and power budget
        adjustment."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.35"
    ::= { lldpV2Xdot3RemPowerEntry 28 }

lldpV2Xdot3RemPowerDownRequest OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "This attribute indicates the remote PD system is requesting
        a power down when the value is 0x1D."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.36"
    ::= { lldpV2Xdot3RemPowerEntry 29 }

lldpV2Xdot3RemPowerDownTime OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "This attribute indicates the number of seconds the remote PD
        requests to stay powered off. A value of zero indicates an
        indefinite amount of time."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.37"
    ::= { lldpV2Xdot3RemPowerEntry 30 }

lldpV2Xdot3RemMeasVoltageSupport OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the remote device is capable of
        providing a voltage measurement. "
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.38"
    ::= { lldpV2Xdot3RemPowerEntry 31 }

lldpV2Xdot3RemMeasCurrentSupport OBJECT-TYPE
```

```
SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This attribute indicates the remote device is capable of
    providing a current measurement. "
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.39"
::= { lldpV2Xdot3RemPowerEntry 32 }

lldpV2Xdot3RemMeasPowerSupport OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This attribute indicates the remote device is capable of
    providing a power measurement."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.40"
::= { lldpV2Xdot3RemPowerEntry 33 }

lldpV2Xdot3RemMeasEnergySupport OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This attribute indicates the remote device is capable of
    providing an energy measurement."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.41"
::= { lldpV2Xdot3RemPowerEntry 34 }

lldpV2Xdot3RemMeasurementSource OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION
    "This attribute indicates to remote device on which Alternative
    or Mode the measurement is to be taken."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.42"
::= { lldpV2Xdot3RemPowerEntry 35 }

lldpV2Xdot3RemMeasVoltageRequest OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This attribute indicates the remote device is requesting
    a voltage measurement from the local device."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.43"
::= { lldpV2Xdot3RemPowerEntry 36 }

lldpV2Xdot3RemMeasCurrentRequest OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
```

```
"This attribute indicates the remote device is requesting
a current measurement from the local device."
REFERENCE
  "IEEE Std 802.3, 30.12.3.1.44"
::= { lldpV2Xdot3RemPowerEntry 37 }

lldpV2Xdot3RemMeasPowerRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the remote device is requesting
        a power measurement from the local device."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.45"
    ::= { lldpV2Xdot3RemPowerEntry 38 }

lldpV2Xdot3RemMeasEnergyRequest OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the remote device is requesting
        energy measurement from the local device."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.46"
    ::= { lldpV2Xdot3RemPowerEntry 39 }

lldpV2Xdot3RemMeasVoltageValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the remote device's voltage measurement
        is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.47"
    ::= { lldpV2Xdot3RemPowerEntry 40 }

lldpV2Xdot3RemMeasCurrentValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the remote device's current measurement
        is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.48"
    ::= { lldpV2Xdot3RemPowerEntry 41 }

lldpV2Xdot3RemMeasPowerValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the remote device's power measurement
        is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.49"
```

```
::= { lldpV2Xdot3RemPowerEntry 42 }

lldpV2Xdot3RemMeasEnergyValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the remote device's energy measurement
         is valid."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.50"
    ::= { lldpV2Xdot3RemPowerEntry 43 }

lldpV2Xdot3RemMeasVoltageUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the remote device's voltage
         measurement. See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.51"
    ::= { lldpV2Xdot3RemPowerEntry 44 }

lldpV2Xdot3RemMeasCurrentUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the remote device's current
         measurement. See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.52"
    ::= { lldpV2Xdot3RemPowerEntry 45 }

lldpV2Xdot3RemMeasPowerUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the remote device's power
         measurement. See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.53"
    ::= { lldpV2Xdot3RemPowerEntry 46 }

lldpV2Xdot3RemMeasEnergyUncertainty OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This attribute indicates the expanded uncertainty
         (coverage factor k = 2) for the remote device's energy
         measurement. See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.54"
```

```
::= { lldpV2Xdot3RemPowerEntry 47 }

lldpV2Xdot3RemVoltageMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured remote device voltage.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.55"
    ::= { lldpV2Xdot3RemPowerEntry 48 }

lldpV2Xdot3RemCurrentMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured remote device current.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.56"
    ::= { lldpV2Xdot3RemPowerEntry 49 }

lldpV2Xdot3RemPowerMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured remote device power.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.57"
    ::= { lldpV2Xdot3RemPowerEntry 50 }

lldpV2Xdot3RemEnergyMeasurement OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates the measured remote device energy.
         See IEEE Std 802.3, Table 79-21."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.58"
    ::= { lldpV2Xdot3RemPowerEntry 51 }

lldpV2Xdot3RemPSEPowerPriceIndex OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This attribute indicates an index of the price of power being
         sourced by the remote PSE. For a PSE, this value is undefined."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.59"
    ::= { lldpV2Xdot3RemPowerEntry 52 }

lldpV2Xdot3RemMaxFrameSizeTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF LldpV2Xdot3RemMaxFrameSizeEntry
```

```
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "This table contains one row per port/destination
    address pair of maximum frame
    size information (as a part of the LLDP IEEE 802.3
    organizational extension) of the remote system."
 ::= { lldpV2Xdot3RemoteData 3 }

lldpV2Xdot3RemMaxFrameSizeEntry OBJECT-TYPE
SYNTAX        LldpV2Xdot3RemMaxFrameSizeEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "Maximum Frame Size information about a particular port
    component."
INDEX         { lldpV2RemTimeMark, lldpV2RemLocalIfIndex,
                lldpV2RemLocalDestMACAddress, lldpV2RemIndex }
 ::= { lldpV2Xdot3RemMaxFrameSizeTable 1 }

lldpV2Xdot3RemMaxFrameSizeEntry ::= SEQUENCE {
    lldpV2Xdot3RemMaxFrameSize  Unsigned32
}

lldpV2Xdot3RemMaxFrameSize OBJECT-TYPE
SYNTAX        Unsigned32 (0..65535)
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "An integer value indicating the maximum supported frame
    size in octets on the port component associated with the
    remote system."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.13"
 ::= { lldpV2Xdot3RemMaxFrameSizeEntry 1 }

lldpV2Xdot3RemEEETable OBJECT-TYPE
SYNTAX        SEQUENCE OF LldpV2Xdot3RemEEEEEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "This table contains one row per port of Energy Efficient Ethernet
    information (as a part of the LLDP IEEE 802.3 organizational
    extension) on the local system known to this agent."
 ::= { lldpV2Xdot3RemoteData 4 }

lldpV2Xdot3RemEEEEEntry OBJECT-TYPE
SYNTAX        LldpV2Xdot3RemEEEEEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "Information about a particular port component."
INDEX         { lldpV2RemLocalIfIndex }
 ::= { lldpV2Xdot3RemEEETable 1 }

lldpV2Xdot3RemEEEEEntry ::= SEQUENCE {
    lldpV2Xdot3RemTxTwSys          Integer32,
    lldpV2Xdot3RemTxTwSysEcho      Integer32,
    lldpV2Xdot3RemRxTwSys          Integer32,
```

```
lldpV2Xdot3RemRxTwSysEcho      Integer32,
lldpV2Xdot3RemFbTwSys          Integer32,
lldpV2Xdot3RemTxFw             TruthValue,
lldpV2Xdot3RemTxFwEcho         TruthValue,
lldpV2Xdot3RemRxFw             TruthValue,
lldpV2Xdot3RemRxFwEcho         TruthValue,
lldpV2Xdot3RemPreemptSupported TruthValue,
lldpV2Xdot3RemPreemptEnabled   TruthValue,
lldpV2Xdot3RemPreemptActive     TruthValue,
lldpV2Xdot3RemAddFragSize       Integer32
}

lldpV2Xdot3RemTxTwSys OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that the remote system
        can support in the transmit direction.
        This object maps to the variable RemTxSystemValue as defined
        in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.60"
    ::= { lldpV2Xdot3RemEEEEEntry 1 }

lldpV2Xdot3RemTxTwSysEcho OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that the local system is
        advertising that it can support in the transmit direction as
        echoed by the remote system under the control of the EEE DLL receiver
        state diagram. This object maps to the variable
        RemTxSystemValueEcho as defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.61"
    ::= { lldpV2Xdot3RemEEEEEntry 2 }

lldpV2Xdot3RemRxTwSys OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that
        the remote system is requesting in the receive direction.
        This object maps to the variable RemRxSystemValue as
        defined in IEEE Std 802.3, 78.4.2.3."
    REFERENCE
        "IEEE Std 802.3, 30.12.3.1.62"
    ::= { lldpV2Xdot3RemEEEEEntry 3 }

lldpV2Xdot3RemRxTwSysEcho OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A GET returns the value of Tw_sys_tx that
        the local system is advertising that it is requesting in the
```


receive direction and is echoed by the remote system under the control of the EEE DLL transmitter state diagram. This object maps to the variable RemRxSystemValueEcho as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.63"

::= { lldpV2Xdot3RemEEEEEntry 4 }

lldpV2Xdot3RemFbTwSys OBJECT-TYPE

SYNTAX Integer32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A GET returns the value of the fallback Tw_sys_tx that the remote system is advertising.

This object maps to the variable RemFbSystemValue as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.64"

::= { lldpV2Xdot3RemEEEEEntry 5 }

lldpV2Xdot3RemTxFw OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This value identifies the LPI_FW value that the remote system can support in the transmit direction. This attribute maps to variable RemTxSystemFW as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.65"

::= { lldpV2Xdot3RemEEEEEntry 6 }

lldpV2Xdot3RemTxFwEcho OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This value identifies the LPI_FW value advertised by the local system and echoed by the remote system. This attribute maps to variable RemTxSystemFWEcho as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.66"

::= { lldpV2Xdot3RemEEEEEntry 7 }

lldpV2Xdot3RemRxFw OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This value identifies the LPI_FW value that the remote system is requesting in the receive direction. This attribute maps to variable RemRxSystemFW as defined in IEEE Std 802.3, 78.4.2.3."

REFERENCE

"IEEE Std 802.3, 30.12.3.1.67"

::= { lldpV2Xdot3RemEEEEEntry 8 }

lldpV2Xdot3RemRxFwEcho OBJECT-TYPE

SYNTAX TruthValue

```
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "This value identifies the LPI_FW value requested by the local
    system and echoed by the remote system. This attribute maps to
    variable RemRxSystemFWEcho as defined in IEEE Std 802.3, 78.4.2.3."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.68"
::= { lldpV2Xdot3RemEEEEEntry 9 }

lldpV2Xdot3RemPreemptSupported OBJECT-TYPE
SYNTAX        TruthValue
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The truth value used to identify whether the remote system
    supports the preemption capability."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.69"
::= { lldpV2Xdot3RemEEEEEntry 10 }

lldpV2Xdot3RemPreemptEnabled OBJECT-TYPE
SYNTAX        TruthValue
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The truth value used to identify whether the preemption
    capability is enabled on the remote system."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.70"
::= { lldpV2Xdot3RemEEEEEntry 11 }

lldpV2Xdot3RemPreemptActive OBJECT-TYPE
SYNTAX        TruthValue
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "The truth value used to identify whether the preemption
    capability is active on the remote system."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.71"
::= { lldpV2Xdot3RemEEEEEntry 12 }

lldpV2Xdot3RemAddFragSize OBJECT-TYPE
SYNTAX        Integer32
MAX-ACCESS    read-only
STATUS        current
DESCRIPTION
    "This value indicates the minimum size of non-final
    fragments supported by the remote system. This value
    is expressed in units of 64 octets of additional
    fragment length."
REFERENCE
    "IEEE Std 802.3, 30.12.3.1.72"
::= { lldpV2Xdot3RemEEEEEntry 13 }

lldpV2Xdot3Conformance OBJECT IDENTIFIER
::= { ieee8023lldpV2Xdot3MIB 2 }
```

```
lldpV2Xdot3Compliances OBJECT IDENTIFIER
    ::= { lldpV2Xdot3Conformance 1 }

lldpV2Xdot3Groups OBJECT IDENTIFIER
    ::= { lldpV2Xdot3Conformance 2 }

lldpV2Xdot3ConfigGroup OBJECT-GROUP
    OBJECTS      { lldpV2Xdot3PortConfigTLVsTxEnable }
    STATUS       current
    DESCRIPTION   "The collection of objects that are used to configure the
        LLDP IEEE 802.3 organizational extension implementation behavior."
    ::= { lldpV2Xdot3Groups 1 }

lldpV2Xdot3LocSysGroup OBJECT-GROUP
    OBJECTS      { lldpV2Xdot3LocPortAutoNegSupported,
        lldpV2Xdot3LocPortAutoNegEnabled,
        lldpV2Xdot3LocPortAutoNegAdvertisedCap,
        lldpV2Xdot3LocPortOperMauType,
        lldpV2Xdot3LocPowerPortClass,
        lldpV2Xdot3LocPowerMDISupported,
        lldpV2Xdot3LocPowerMDIEnabled,
        lldpV2Xdot3LocPowerPairControlable,
        lldpV2Xdot3LocPowerPairs, lldpV2Xdot3LocPowerClass,
        lldpV2Xdot3LocMaxFrameSize, lldpV2Xdot3LocPowerType,
        lldpV2Xdot3LocPowerSource, lldpV2Xdot3LocPowerPriority,
        lldpV2Xdot3LocPDRequestedPowerValue,
        lldpV2Xdot3LocPSEAllocatedPowerValue,
        lldpV2Xdot3LocResponseTime, lldpV2Xdot3LocReady,
        lldpV2Xdot3LocTxTwSys, lldpV2Xdot3LocTxTwSysEcho,
        lldpV2Xdot3LocRxTwSys, lldpV2Xdot3LocRxTwSysEcho,
        lldpV2Xdot3LocFbTwSys, lldpV2Xdot3TxD11Ready,
        lldpV2Xdot3RxD11Ready, lldpV2Xdot3LocD11Enabled,
        lldpV2Xdot3LocPDRequestedPowerValueA,
        lldpV2Xdot3LocPDRequestedPowerValueB,
        lldpV2Xdot3LocPSEAllocatedPowerValueA,
        lldpV2Xdot3LocPSEAllocatedPowerValueB,
        lldpV2Xdot3LocPSEPoweringStatus,
        lldpV2Xdot3LocPDPoweredStatus,
        lldpV2Xdot3LocPowerPairsExt,
        lldpV2Xdot3LocPowerClassExtA,
        lldpV2Xdot3LocPowerClassExtB,
        lldpV2Xdot3LocPowerClassExt,
        lldpV2Xdot3LocPowerTypeExt, lldpV2Xdot3LocPDLload,
        lldpV2Xdot3LocPD4PID, lldpV2Xdot3LocPSEMaxAvailPower,
        lldpV2Xdot3LocPSEAutoclassSupport,
        lldpV2Xdot3LocPSEAutoclassCompleted,
        lldpV2Xdot3LocPSEAutoclassRequest,
        lldpV2Xdot3LocPowerDownRequest,
        lldpV2Xdot3LocPowerDownTime,
        lldpV2Xdot3LocMeasVoltageSupport,
        lldpV2Xdot3LocMeasCurrentSupport,
        lldpV2Xdot3LocMeasPowerSupport,
        lldpV2Xdot3LocMeasEnergySupport,
        lldpV2Xdot3LocMeasurementSource,
        lldpV2Xdot3LocMeasVoltageRequest,
        lldpV2Xdot3LocMeasCurrentRequest,
        lldpV2Xdot3LocMeasPowerRequest,
        lldpV2Xdot3LocMeasEnergyRequest,
```

```
        lldpV2Xdot3LocMeasVoltageValid,
        lldpV2Xdot3LocMeasCurrentValid,
        lldpV2Xdot3LocMeasPowerValid,
        lldpV2Xdot3LocMeasEnergyValid,
        lldpV2Xdot3LocMeasVoltageUncertainty,
        lldpV2Xdot3LocMeasCurrentUncertainty,
        lldpV2Xdot3LocMeasPowerUncertainty,
        lldpV2Xdot3LocMeasEnergyUncertainty,
        lldpV2Xdot3LocVoltageMeasurement,
        lldpV2Xdot3LocCurrentMeasurement,
        lldpV2Xdot3LocPowerMeasurement,
        lldpV2Xdot3LocEnergyMeasurement,
        lldpV2Xdot3LocPSEPowerPriceIndex, lldpV2Xdot3LocTxFw,
        lldpV2Xdot3LocTxFwEcho, lldpV2Xdot3LocRxFw,
        lldpV2Xdot3LocRxFwEcho, lldpV2Xdot3LocPreemptSupported,
        lldpV2Xdot3LocPreemptEnabled,
        lldpV2Xdot3LocPreemptActive, lldpV2Xdot3LocAddFragSize }
STATUS      current
DESCRIPTION
    "The collection of objects that are used to represent LLDP
    IEEE 802.3 organizational extension Local Device Information."
 ::= { lldpV2Xdot3Groups 2 }

lldpV2Xdot3RemSysGroup OBJECT-GROUP
OBJECTS      { lldpV2Xdot3RemPortAutoNegSupported,
                lldpV2Xdot3RemPortAutoNegEnabled,
                lldpV2Xdot3RemPortAutoNegAdvertisedCap,
                lldpV2Xdot3RemPortOperMauType,
                lldpV2Xdot3RemPowerPortClass,
                lldpV2Xdot3RemPowerMDISupported,
                lldpV2Xdot3RemPowerMDIEnabled,
                lldpV2Xdot3RemPowerPairControlable,
                lldpV2Xdot3RemPowerPairs, lldpV2Xdot3RemPowerClass,
                lldpV2Xdot3RemMaxFrameSize, lldpV2Xdot3RemPowerType,
                lldpV2Xdot3RemPowerSource, lldpV2Xdot3RemPowerPriority,
                lldpV2Xdot3RemPDRequestedPowerValue,
                lldpV2Xdot3RemPSEAllocatedPowerValue,
                lldpV2Xdot3RemTxTwSys, lldpV2Xdot3RemTxTwSysEcho,
                lldpV2Xdot3RemRxTwSys, lldpV2Xdot3RemRxTwSysEcho,
                lldpV2Xdot3RemFbTwSys,
                lldpV2Xdot3RemPDRequestedPowerValueA,
                lldpV2Xdot3RemPDRequestedPowerValueB,
                lldpV2Xdot3RemPSEAllocatedPowerValueA,
                lldpV2Xdot3RemPSEAllocatedPowerValueB,
                lldpV2Xdot3RemPSEPoweringStatus,
                lldpV2Xdot3RemPDPoweredStatus,
                lldpV2Xdot3RemPowerPairsExt,
                lldpV2Xdot3RemPowerClassExtA,
                lldpV2Xdot3RemPowerClassExtB,
                lldpV2Xdot3RemPowerClassExt,
                lldpV2Xdot3RemPowerTypeExt, lldpV2Xdot3RemPDLoad,
                lldpV2Xdot3RemPD4PID, lldpV2Xdot3RemPSEMaxAvailPower,
                lldpV2Xdot3RemPSEAutoclassSupport,
                lldpV2Xdot3RemPSEAutoclassCompleted,
                lldpV2Xdot3RemPSEAutoclassRequest,
                lldpV2Xdot3RemPowerDownRequest,
                lldpV2Xdot3RemPowerDownTime,
                lldpV2Xdot3RemMeasVoltageSupport,
                lldpV2Xdot3RemMeasCurrentSupport,
```

```
        lldpV2Xdot3RemMeasPowerSupport,
        lldpV2Xdot3RemMeasEnergySupport,
        lldpV2Xdot3RemMeasurementSource,
        lldpV2Xdot3RemMeasVoltageRequest,
        lldpV2Xdot3RemMeasCurrentRequest,
        lldpV2Xdot3RemMeasPowerRequest,
        lldpV2Xdot3RemMeasEnergyRequest,
        lldpV2Xdot3RemMeasVoltageValid,
        lldpV2Xdot3RemMeasCurrentValid,
        lldpV2Xdot3RemMeasPowerValid,
        lldpV2Xdot3RemMeasEnergyValid,
        lldpV2Xdot3RemMeasVoltageUncertainty,
        lldpV2Xdot3RemMeasCurrentUncertainty,
        lldpV2Xdot3RemMeasPowerUncertainty,
        lldpV2Xdot3RemMeasEnergyUncertainty,
        lldpV2Xdot3RemVoltageMeasurement,
        lldpV2Xdot3RemCurrentMeasurement,
        lldpV2Xdot3RemPowerMeasurement,
        lldpV2Xdot3RemEnergyMeasurement,
        lldpV2Xdot3RemPSEPowerPriceIndex, lldpV2Xdot3RemTxFw,
        lldpV2Xdot3RemTxFwEcho, lldpV2Xdot3RemRxFw,
        lldpV2Xdot3RemRxFwEcho, lldpV2Xdot3RemPreemptSupported,
        lldpV2Xdot3RemPreemptEnabled,
        lldpV2Xdot3RemPreemptActive, lldpV2Xdot3RemAddFragSize }
STATUS      current
DESCRIPTION
    "The collection of objects that are used to represent LLDP
    IEEE 802.3 organizational extension Local Device Information."
 ::= { lldpV2Xdot3Groups 3 }

lldpV2Xdot3TxRxCompliance MODULE-COMPLIANCE
STATUS      current
DESCRIPTION
    "A compliance statement for SNMP entities that implement
    the LLDP IEEE 802.3 organizational extension MIB.

    This group is mandatory for all agents that implement the
    LLDP IEEE 802.3 organizational extension in TX and/or RX mode.

    This version defines compliance requirements for
    V2 of the LLDP MIB."

MODULE      -- this module

    MANDATORY-GROUPS      { lldpV2Xdot3ConfigGroup }

MODULE      IF-MIB

    MANDATORY-GROUPS      { ifGeneralInformationGroup }

 ::= { lldpV2Xdot3Compliances 1 }

lldpV2Xdot3TxCompliance MODULE-COMPLIANCE
STATUS      current
DESCRIPTION
    "The compliance statement for SNMP entities that implement
    the LLDP IEEE 802.3 organizational extension MIB.

    This group is mandatory for agents that implement the
```

```
LLDP IEEE 802.3 organizational extension in the TX mode.

This version defines compliance requirements for
V2 of the LLDP MIB."

MODULE      -- this module

    MANDATORY-GROUPS      { lldpV2Xdot3LocSysGroup }

    ::= { lldpV2Xdot3Compliances 2 }

lldpV2Xdot3RxCompliance MODULE-COMPLIANCE
    STATUS      current
    DESCRIPTION
        "The compliance statement for SNMP entities that implement
        the LLDP IEEE 802.3 organizational extension MIB.

        This group is mandatory for agents that implement the
        LLDP IEEE 802.3 organizational extension in the RX mode.

        This version defines compliance requirements for
        V2 of the LLDP MIB."

MODULE      -- this module

    MANDATORY-GROUPS      { lldpV2Xdot3RemSysGroup }

    ::= { lldpV2Xdot3Compliances 3 }

END -- end of module IEEE8023-DOT3-LLDP-EXT-V2-MIB.
```

6. Ethernet operations, administration, and maintenance (OAM) MIB module

6.1 Introduction

The IEEE 802.3ah Ethernet in the First Mile (EFM) Task Force added management capabilities to Ethernet-like interfaces to provide some basic operations, administration, and maintenance (OAM) functions. The defined functionality includes discovery, error signaling, loopback, and link monitoring.

This clause defines a MIB module for use with SNMP to manage these Ethernet-like interface capabilities.

6.2 Overview

Ethernet OAM is composed of a core set of functions and a set of optional functional groups as described in Clause 57 of IEEE Std 802.3. The core functions include discovery operations (determining if the other end of the link is OAM capable and what OAM functions it supports), state machine implementation, and some critical event flows. The optional functional groups are for (1) link events, (2) remote loopback, and (3) variable retrieval and response. Each optional functional group is controlled by a separate MIB table(s).

Ethernet OAM is complementary with SNMP management in that it provides some basic management functions at layer 2, rather than using layer 3 and above as required by SNMP over an IP infrastructure. Ethernet OAM provides single-hop functionality in that it works only between two directly connected Ethernet stations. SNMP can be used to manage the Ethernet OAM interactions of one Ethernet station with another.

Ethernet OAM has three functional objectives, which are detailed in 6.2.1 through 6.2.3. The definition of a basic Ethernet OAM protocol data unit is given in 6.2.4.

6.2.1 Remote fault indication

Remote fault indication provides a mechanism for one end of an Ethernet link to signal the other end that the receive path is non-operational. Some Ethernet Physical Layers offer mechanisms to signal this condition at the Physical Layer. Ethernet OAM added a mechanism so that some Ethernet Physical Layers can operate in unidirectional mode, allowing frames to be transmitted in one direction even when the other direction is non-operational. Traditionally, Ethernet PHYs do not allow frame transmission in one direction if the other direction is not operational. Using this mode, Ethernet OAM allows frame-based signaling of remote fault conditions while still not allowing higher layer applications to be aware of the unidirectional capability. This clause includes mechanisms for capturing that fault information and reflecting such information in objects and notifications within the SNMP management framework.

6.2.2 Link monitoring

Ethernet OAM includes event signaling capability so that one end of an Ethernet link can indicate the occurrence of certain important events to the other end of the link. This happens via layer 2 protocols. This clause defines methods for incorporating the occurrence of these layer 2 events, at both the local end and the far end of the link, into the SNMP management framework.

Ethernet OAM also includes mechanisms for one Ethernet station to query another directly connected Ethernet station about the status of its Ethernet interface variables and status. This clause does not include mechanisms for controlling how one Ethernet endpoint uses this functionality to query the status or statistics of a peer Ethernet entity.

6.2.3 Remote loopback

Remote loopback is a link state where the peer Ethernet entity echoes every received packet (without modifications) back onto the link. Remote loopback is intrusive in that the other end of the link is not forwarding traffic from higher layers out over the link. This clause defines objects controlling loopback operation and reading the status of the loopback state.

6.2.4 Ethernet OAM protocol data units

An Ethernet OAM protocol data unit is a valid Ethernet frame with a destination Media Access Control (MAC) address equal to the reserved MAC address for Slow Protocols (see Annex 57A of IEEE Std 802.3), a lengthOrType field equal to the reserved type for Slow Protocols, and a Slow Protocols subtype equal to that of the subtype reserved for Ethernet OAM.

OAMPDU is used throughout this clause as an abbreviation for Ethernet OAM protocol data unit. OAMPDUs are the mechanism by which two directly connected Ethernet interfaces exchange OAM information.

6.3 Relation to other MIB modules

The definitions presented here are based on Clause 30 and Clause 57 of IEEE Std 802.3. Note that these clauses describe many of these variables and their effects on the MAC sublayer. In some cases, there is a one-to-one relationship between an object in this clause and an object in the Clause 30 MIB. In other cases, the objects of this clause reflect a more complex entity and are reflected by more than one object in the Clause 30 MIB.

The objects defined in this clause manage OAM functionality introduced as part of the IEEE 802.3ah project. These objects do not overlap with the Interfaces Group MIB module defined in IETF RFC 2863, the Ethernet-like interface MIB module defined in Clause 10, or any other MIB module currently used to manage various aspects of an Ethernet interface. The objects defined here are defined for Ethernet-like interfaces only and use the same ifIndex as the associated Ethernet interface. Ethernet OAM can be implemented on any Ethernet-like interface.

6.3.1 Relation to other EFM MIB modules

The Ethernet OAM functionality and MIB module is independent of the other functionality and MIB modules derived from IEEE Std 802.3 for copper and EPON. Ethernet OAM may be implemented (or not) on the new EFM interface types, just as it can on any other Ethernet interface.

6.3.2 Mapping of IEEE 802.3 managed objects

Table 6-1 contains the mapping between managed objects defined in Clause 30 of IEEE Std 802.3 and managed objects defined in this clause.

All IEEE 802.3 OAM managed objects are reflected in this MIB module.

Table 6-1—Mapping between IEEE 802.3 managed objects and SNMP objects

IEEE 802.3 managed object	Corresponding SNMP object
oOAM	
.aOAMID	IF-MIB ifIndex
.aOAMAdminState	dot3OamAdminState
.aOAMMode	dot3OamMode
.aOAMDiscoveryState	dot3OamOperStatus
.aOAMRemoteMACAddress	dot3OamPeerMacAddress
.aOAMLocalConfiguration	dot3OamFunctionsSupported
.aOAMRemoteConfiguration	dot3OamPeerFunctionsSupported, dot3Oam-PeerMode
.aOAMLocalPDUConfiguration	dot3OamMaxOamPduSize
.aOAMRemotePDUConfiguration	dot3OamPeerMaxOamPduSize
.aOAMLocalFlagsField	dot3OamOperStatus, dot3OamEventLogEntry
.aOAMRemoteFlagsField	dot3OamOperStatus, dot3OamEventLogEntry
.aOAMLocalRevision	dot3OamConfigRevision
.aOAMRemoteRevision	dot3OamPeerConfigRevision
.aOAMLocalState	dot3OamLoopbackStatus
.aOAMRemoteState	dot3OamLoopbackStatus
.aOAMRemoteVendorOUI	dot3OamPeerVendorOui
.aOAMRemoteVendorSpecificInfo	dot3OamPeerVendorInfo
.aOAMUnsupportedCodesTx	dot3OamUnsupportedCodesTx
.aOAMUnsupportedCodesRx	dot3OamUnsupportedCodesRx
.aOAMInformationTx	dot3OamInformationTx
.aOAMInformationRx	dot3OamInformationRx
.aOAMUniqueEventNotificationTx	dot3OamUniqueEventNotificationTx
.aOAMUniqueEventNotificationRx	dot3OamUniqueEventNotificationRx
.aOAMDuplicateEventNotificationTx	dot3OamDuplicateEventNotificationTx
.aOAMDuplicateEventNotificationRx	dot3OamDuplicateEventNotificationRx
.aOAMLoopbackControlTx	dot3OamLoopbackControlTx

Table 6-1—Mapping between IEEE 802.3 managed objects and SNMP objects (*continued*)

IEEE 802.3 managed object	Corresponding SNMP object
.aOAMLoopbackControlRx	dot3OamLoopbackControlRx
.aOAMVariableRequestTx	dot3OamVariableRequestTx
.aOAMVariableRequestRx	dot3OamVariableRequestRx
.aOAMVariableResponseTx	dot3OamVariableResponseTx
.aOAMVariableResponseRx	dot3OamVariableResponseRx
.aOAMOrganizationSpecificTx	dot3OamOrgSpecificTx
.aOAMOrganizationSpecificRx	dot3OamOrgSpecificTx
.aOAMLocalErrSymPeriodConfig	dot3OamErrSymPeriodWindow, dot3OamErrSymPeriodThreshold
.aOAMLocalErrSymPeriodEvent	dot3OamEventLogEntry
.aOAMLocalErrFrameConfig	dot3OamErrFrameWindow, dot3OamErr- FrameThreshold
.aOAMLocalErrFrameEvent	dot3OamEventLogEntry
.aOAMLocalErrFramePeriodConfig	dot3OamErrFramePeriodWindow, dot3OamEr- rFramePeriodThreshold
.aOAMLocalErrFramePeriodEvent	dot3OamEventLogEntry
.aOAMLocalErrFrameSecsSummaryConfig	dot3OamErrFrameSecsSummaryWindow, dot3OamErrFrameSecsummaryThreshold
.aOAMLocalErrFrameSecsSummaryEvent	dot3OamEventLogEntry
.aOAMRemoteErrSymPeriodEvent	dot3OamEventLogEntry
.aOAMRemoteErrFrameEvent	dot3OamEventLogEntry
.aOAMRemoteErrFramePeriodEvent	dot3OamEventLogEntry
.aOAMRemoteErrFrameSecsSummaryEvent	dot3OamEventLogEntry
.aFramesLostDueToOAMError	dot3OamFramesLostDueToOam
.acOAMAdminControl	dot3OamAdminState

6.4 MIB structure

The Ethernet OAM MIB objects of this clause focus on the OAM capabilities introduced in IEEE Std 802.3. The MIB objects are partitioned into six different MIB groups.

The dot3OamTable group manages the primary OAM objects of the Ethernet interface. This group controls the state and status of OAM as well as the mode in which it operates. The dot3OamPeerTable maintains the current information on the status and configuration of the peer OAM entity on the Ethernet interface. Managed information includes the capabilities and function available on the peer OAM entity.

The dot3OamLoopbackTable manages the loopback function introduced in IEEE Std 802.3. This table controls enabling and disabling loopback, as well as indicating the loopback status of Ethernet OAM on this interface.

The dot3OamStatsTable maintains statistics on the number and type of Ethernet OAM frames being transmitted and received on the Ethernet interface.

The dot3OamEventConfigTable defines the objects for managing the event notification capability available in Ethernet OAM. With Ethernet OAM, one device may send notifications to its peer devices whenever an important event happens on the local device. This table provides management of which events result in notifications via Ethernet OAM notifications and/or via SNMP notifications.

The dot3OamEventLogTable manages the current status of local and remote events detected via Ethernet OAM. This table is updated whenever local events are detected by Ethernet OAM or whenever Ethernet OAM Event Notifications are received from the peer OAM entity.

There are two notifications defined to report Ethernet OAM events (one for threshold crossing events and one for non-threshold crossing events). Both notifications are contained within the same conformance group.

6.5 Security considerations for Ethernet operations, administration, and maintenance (OAM) MIB module

The readable objects in this module can provide information about network traffic, and therefore, they can be considered sensitive. In particular, OAM provides mechanisms for reading the Clause 30 IEEE 802.3 MIB attributes from a link partner via a specialized layer 2 protocol. Unlike SNMP, IEEE 802.3 OAM does not include encryption or authentication mechanisms. It should not be used in environments where this interface information is considered sensitive, and where the facility terminations are unprotected. By default, OAM is disabled on Ethernet-like interfaces and is therefore not a risk.

IEEE 802.3 OAM is designed to support deployment in access and enterprise networks. In access networks, one end of a link is the CO-side, and the other is the CPE-side, and the facilities are often protected in wiring cages or closets. In such deployments, it is often the case that the CO-side is protected from access from the CPE-side. Within IEEE 802.3 OAM, this protection from remote access is accomplished by configuring the CPE-side in passive mode using the dot3OamMode attribute. This prevents the CPE from accessing functions and information at the CO-side of the connection. In enterprise networks, read-only interface information is often considered non-sensitive.

The frequency of OAM PDUs on an Ethernet interface does not adversely affect data traffic, as OAM is a slow protocol with very limited bandwidth potential, and it is not required for normal link operation. Although there are a number of objects in this module with read-write or read-create MAX-ACCESS, they have limited effects on user data.

The loopback capability of OAM can have potentially disruptive effects; when remote loopback is enabled, the remote station automatically transmits all received traffic back to the local station except for OAM traffic. This completely disrupts all higher layer protocols such as bridging, IP, and SNMP. Therefore, an attribute (`dot3OamLoopbackIgnoreRx`) was introduced to control whether the local station processes or ignores received loopback commands.

The administrative state and mode are also read-write objects. Disabling OAM can interrupt management activities between peer devices, potentially causing serious problems. Setting the `dot3OamMode` to an undesired value can allow access to Ethernet monitoring, events, and functions that might not be acceptable in a particular deployment scenario. In addition to loopback functionality, Ethernet interface statistics and events can be accessed via the OAM protocol, which might not be desired in some circumstances.

OAM event configuration also contains read-write objects. These objects control whether events are sent, and at what thresholds. Note that the frequency of event communication is limited by the frequency limits of Slow Protocols on Ethernet interfaces. Also, the information available via OAM events is also available via OAM Variable Requests. Access to this information via either OAM events or Variable Requests is controlled by the `dot3OamAdminState` and `dot3OamMode` objects. As mentioned previously, inadequate protection of these variables can result in access to link information and functions.

6.6 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:¹⁹

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C6mib.txt

¹⁹ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

6.6.1 Tree hierarchy

```
----- ieee8023Dot3OamMIB(1.3.111.2.802.3.1.6)
+----- dot3OamNotifications(0)
| +----- dot3OamThresholdEvent(1)
| {dot3OamEventLogTimestamp, dot3OamEventLogOui, dot3OamEventLogType, dot3OamEventLogLocation, dot3OamEventLogWindowHi, dot3Oam
EventLogWindowLo, dot3OamEventLogThresholdHi, dot3OamEventLogThresholdLo, dot3OamEventLogValue, dot3OamEventLogRunningTotal,
dot3OamEventLogEventTotal}
| +----- dot3OamNonThresholdEvent(2) {dot3OamEventLogTimestamp, dot3OamEventLogOui, dot3OamEventLogType, dot3OamEventLo-
gLocation, dot3OamEventLogEventTotal}
+----- dot3OamObjects(1)
| +----- dot3OamTable(1)
| | +----- dot3OamEntry(1) [ifIndex]
| | +---rw dot3OamAdminState(1)
| | +---r- dot3OamOperStatus(2)
| | +---rw dot3OamMode(3)
| | +---r- dot3OamMaxOamPduSize(4)
| | +---r- dot3OamConfigRevision(5)
| | +---r- dot3OamFunctionsSupported(6) Bits
| | +----- dot3OamPeerTable(2)
| | | +----- dot3OamPeerEntry(1) [ifIndex]
| | | +---r- dot3OamPeerMacAddress(1)
| | | +---r- dot3OamPeerVendorOui(2)
| | | +---r- dot3OamPeerVendorInfo(3)
| | | +---r- dot3OamPeerMode(4)
| | | +---r- dot3OamPeerMaxOamPduSize(5)
| | | +---r- dot3OamPeerConfigRevision(6)
| | | +---r- dot3OamPeerFunctionsSupported(7) Bits
| | +----- dot3OamLoopbackTable(3)
| | | +----- dot3OamLoopbackEntry(1) [ifIndex]
| | | +---rw dot3OamLoopbackStatus(1)
| | | +---rw dot3OamLoopbackIgnoreRx(2)
| | +----- dot3OamStatsTable(4)
| | | +----- dot3OamStatsEntry(1) [ifIndex]
| | | +---r- dot3OamInformationTx(1)
| | | +---r- dot3OamInformationRx(2)
| | | +---r- dot3OamUniqueEventNotificationTx(3)
| | | +---r- dot3OamUniqueEventNotificationRx(4)
| | +---r- dot3OamDuplicateEventNotificationTx(5)
| | Counter32
| | Counter32
| | Counter32
| | Counter32
| | Counter32
```

		+	--r-	dot3OamDuplicateEventNotificationRx (6)	Counter32
		+	--r-	dot3OamLoopbackControlTx (7)	Counter32
		+	--r-	dot3OamLoopbackControlRx (8)	Counter32
		+	--r-	dot3OamVariableRequestTx (9)	Counter32
		+	--r-	dot3OamVariableRequestRx (10)	Counter32
		+	--r-	dot3OamVariableResponseTx (11)	Counter32
		+	--r-	dot3OamVariableResponseRx (12)	Counter32
		+	--r-	dot3OamOrgSpecificTx (13)	Counter32
		+	--r-	dot3OamOrgSpecificRx (14)	Counter32
		+	--r-	dot3OamUnsupportedCodesTx (15)	Counter32
		+	--r-	dot3OamUnsupportedCodesRx (16)	Counter32
		+	--r-	dot3OamFramesLostDueToOam (17)	Counter32
		+	-----	dot3OamEventConfigTable (5)	
			+	-----	dot3OamEventConfigEntry (1) [ifIndex]
			+	--rw	dot3OamErrSymPeriodWindowHi (1)
			+	--rw	dot3OamErrSymPeriodWindowLo (2)
			+	--rw	dot3OamErrSymPeriodThresholdHi (3)
			+	--rw	dot3OamErrSymPeriodThresholdLo (4)
			+	--rw	dot3OamErrSymPeriodEvNotifEnable (5)
			+	--rw	dot3OamErrFramePeriodWindow (6)
			+	--rw	dot3OamErrFramePeriodThreshold (7)
			+	--rw	dot3OamErrFramePeriodEvNotifEnable (8)
			+	--rw	dot3OamErrFrameWindow (9)
			+	--rw	dot3OamErrFrameThreshold (10)
			+	--rw	dot3OamErrFrameEvNotifEnable (11)
			+	--rw	dot3OamErrFrameSecsSummaryWindow (12)
			+	--rw	dot3OamErrFrameSecsSummaryThreshold (13)
			+	--rw	dot3OamErrFrameSecsEvNotifEnable (14)
			+	--rw	dot3OamDyingGaspEnable (15)
			+	--rw	dot3OamCriticalEventEnable (16)
		+	-----	dot3OamEventLogTable (6)	
			+	-----	dot3OamEventLogEntry (1) [ifIndex, dot3OamEventLogIndex]
			+	----	dot3OamEventLogIndex (1)
			+	--r-	dot3OamEventLogTimestamp (2)
			+	--r-	dot3OamEventLogOui (3)
			+	--r-	dot3OamEventLogType (4)
			+	--r-	dot3OamEventLogLocation (5)
			+	--r-	dot3OamEventLogWindowHi (6)
			+	--r-	dot3OamEventLogWindowLo (7)

```

| | +---r- dot3OamEventLogThresholdHi (8) Unsigned32
| | +---r- dot3OamEventLogThresholdLo (9) Unsigned32
| | +---r- dot3OamEventLogValue (10) CounterBasedGauge64
| | +---r- dot3OamEventLogRunningTotal (11) CounterBasedGauge64
| | +---r- dot3OamEventLogEventTotal (12) Unsigned32
+----- dot3OamConformance (2)
+----- dot3OamGroups (1)
| +----- dot3OamControlGroup (1)
| +----- dot3OamPeerGroup (2)
| +----- dot3OamStatsBaseGroup (3)
| +----- dot3OamLoopbackGroup (4)
| +----- dot3OamErrSymbolPeriodEventGroup (5)
| +----- dot3OamErrFramePeriodEventGroup (6)
| +----- dot3OamErrFrameEventGroup (7)
| +----- dot3OamErrFrameSecsSummaryEventGroup (8)
| +----- dot3OamFlagEventGroup (9)
| +----- dot3OamEventLogGroup (10)
| +----- dot3OamNotificationGroup (11)
+----- dot3OamCompliances (2)
+----- dot3OamCompliance (1)

```

6.6.2 MIB module definition

```
IEEE8023-DOT3-OAM-MIB DEFINITIONS ::= BEGIN

IMPORTS
    CounterBasedGauge64
        FROM HCNUM-TC
    ifIndex
        FROM IF-MIB
    MODULE-COMPLIANCE, NOTIFICATION-GROUP, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Integer32, MODULE-IDENTITY, NOTIFICATION-TYPE,
    OBJECT-TYPE, Unsigned32, org
        FROM SNMPv2-SMI
    MacAddress, TEXTUAL-CONVENTION, TimeStamp, TruthValue
        FROM SNMPv2-TC;

ieee8023Dot3OamMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "The MIB module for managing the new Ethernet OAM features
        introduced by the Ethernet in the First Mile Task Force (IEEE
        802.3ah). The functionality presented here is based on IEEE
        Std 802.3ah, released in October, 2004, which was prepared as
        an amendment to IEEE Std 802.3. Since then, IEEE Std 802.3ah
        has been merged into the base IEEE 802.3 standard.

        In particular, this MIB focuses on the new OAM functions
        introduced in Clause 57 of IEEE Std 802.3. The OAM functionality
        of Clause 57 is controlled by new management attributes
        introduced in Clause 30 of IEEE Std 802.3. The OAM functions are
        not specific to any particular Ethernet Physical Layer, and
        can be generically applied to any Ethernet interface.

        An Ethernet OAM protocol data unit is a valid Ethernet frame
        with a destination MAC address equal to the reserved MAC
        address for Slow Protocols (See Annex 57A of IEEE Std 802.3), a
        lengthOrType field equal to the reserved type for Slow
        Protocols, and a Slow Protocols subtype equal to that of the
        subtype reserved for Ethernet OAM. OAMPDU is used throughout
        this document as an abbreviation for Ethernet OAM protocol
        data unit."
    REVISION      "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
```



```
REVISION      "201304110000Z"
DESCRIPTION
    "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
REVISION      "201102020000Z"
DESCRIPTION
    "Initial version, based on an earlier version in RFC 4878."
 ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 6 }

EightOTwoOui ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "3x:"
    STATUS      current
    DESCRIPTION
        "24-bit Organizationally Unique Identifier (OUI) or Company ID (CID).
        Information on OUIs and CIDs can be found in [B3]."
    SYNTAX      OCTET STRING (SIZE(3))

dot3OamNotifications OBJECT IDENTIFIER
 ::= { ieee8023Dot3OamMIB 0 }

dot3OamObjects OBJECT IDENTIFIER
 ::= { ieee8023Dot3OamMIB 1 }

dot3OamTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3OamEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table contains the primary controls and status for the
        OAM capabilities of an Ethernet-like interface. There will be
        one row in this table for each Ethernet-like interface in the
        system that supports the OAM functions defined in IEEE Std 802.3."
 ::= { dot3OamObjects 1 }

dot3OamEntry OBJECT-TYPE
    SYNTAX      Dot3OamEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table that contains information on the
        Ethernet OAM function for a single Ethernet-like interface.
        Entries in the table are created automatically for each
        interface supporting Ethernet OAM. The status of the row
        entry can be determined from dot3OamOperStatus.

        A dot3OamEntry is indexed in the dot3OamTable by the ifIndex
        object of the Interfaces Group MIB."
    INDEX      { ifIndex }
 ::= { dot3OamTable 1 }

Dot3OamEntry ::= SEQUENCE {
    dot3OamAdminState      INTEGER,
    dot3OamOperStatus      INTEGER,
    dot3OamMode             INTEGER,
    dot3OamMaxOamPduSize    Unsigned32,
    dot3OamConfigRevision   Unsigned32,
    dot3OamFunctionsSupported BITS
}
```

dot3OamAdminState OBJECT-TYPE

SYNTAX INTEGER { enabled(1), disabled(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object is used to provision the default administrative OAM mode for this interface. This object represents the desired state of OAM for this interface.

The dot3OamAdminState starts in the disabled(2) state until an explicit management action or configuration information retained by the system causes a transition to the enabled(1) state. When enabled(1), Ethernet OAM will attempt to operate over this interface."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.2"

::= { dot3OamEntry 1 }

dot3OamOperStatus OBJECT-TYPE

SYNTAX INTEGER { disabled(1), linkFault(2), passiveWait(3),
activeSendLocal(4), sendLocalAndRemote(5),
sendLocalAndRemoteOk(6),
oamPeeringLocallyRejected(7),
oamPeeringRemotelyRejected(8), operational(9),
nonOperHalfDuplex(10) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"At initialization and failure conditions, two OAM entities on the same full-duplex Ethernet link begin a discovery phase to determine what OAM capabilities are available on that link. The progress of this initialization is controlled by the OAM sublayer.

This value is disabled(1) if OAM is disabled on this interface via the dot3OamAdminState.

If the link has detected a fault and is transmitting OAMPDUs with a link fault indication, the value is linkFault(2). Also, if the interface is not operational (ifOperStatus is not up(1)), linkFault(2) is returned. Note that the object ifOperStatus might be other than up(1) as a result of link failure or administrative action (ifAdminState being down(2) or testing(3)).

The passiveWait(3) state is returned only by OAM entities in passive mode (dot3OamMode) and reflects the state in which the OAM entity is waiting to see if the peer device is OAM capable. The activeSendLocal(4) value is used by active mode devices (dot3OamMode) and reflects the OAM entity actively trying to discover whether the peer has OAM capability but has not yet made that determination.

The state sendLocalAndRemote(5) reflects that the local OAM entity has discovered the peer but has not yet accepted or rejected the configuration of the peer. The local device can, for whatever reason, decide that the peer device is unacceptable and decline OAM peering. If the local OAM entity

rejects the peer OAM entity, the state becomes `oamPeeringLocallyRejected(7)`. If the OAM peering is allowed by the local device, the state moves to `sendLocalAndRemoteOk(6)`. Note that both the `sendLocalAndRemote(5)` and `oamPeeringLocallyRejected(7)` states fall within the state `SEND_LOCAL_REMOTE` of the OAM Discovery state diagram (see IEEE Std 802.3, Figure 57-5), with the difference being whether the local OAM client has actively rejected the peering or has just not indicated any decision yet. Whether a peering decision has been made is indicated via the local flags field in the OAMPDU (reflected in the `aOAMLocalFlagsField` of IEEE Std 802.3, 30.3.6.1.10).

If the remote OAM entity rejects the peering, the state becomes `oamPeeringRemotelyRejected(8)`. Note that both the `sendLocalAndRemoteOk(6)` and `oamPeeringRemotelyRejected(8)` states fall within the state `SEND_LOCAL_REMOTE_OK` of the OAM Discovery state diagram (see IEEE Std 802.3, Figure 57-5), with the difference being whether the remote OAM client has rejected the peering or has just not yet decided. This is indicated via the remote flags field in the OAMPDU (reflected in the `aOAMRemoteFlagsField` of IEEE Std 802.3, 30.3.6.1.11).

When the local OAM entity learns that both it and the remote OAM entity have accepted the peering, the state moves to `operational(9)` corresponding to the `SEND_ANY` state of the OAM Discovery state diagram (see IEEE Std 802.3, Figure 57-5).

Since Ethernet OAM functions are not designed to work completely over half-duplex interfaces, the value `nonOperHalfDuplex(10)` is returned whenever Ethernet OAM is enabled (`dot3OamAdminState` is enabled(1)), but the interface is in half-duplex operation."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.4, 30.3.6.1.10, 30.3.6.1.11"
::= { dot3OamEntry 2 }

dot3OamMode OBJECT-TYPE

SYNTAX INTEGER { passive(1), active(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object configures the mode of OAM operation for this Ethernet-like interface. OAM on Ethernet interfaces might be in 'active' mode or 'passive' mode. These two modes differ in that active mode provides additional capabilities to initiate monitoring activities with the remote OAM peer entity, while passive mode generally waits for the peer to initiate OAM actions with it. As an example, an active OAM entity can put the remote OAM entity in a loopback state, where a passive OAM entity cannot.

The default value of `dot3OamMode` is dependent on the type of system on which this Ethernet-like interface resides. The default value should be 'active(2)' unless it is known that this system should take on a subservient role to the other device connected over this interface.

Changing this value results in incrementing the configuration

revision field of locally generated OAMPDUs (IEEE Std 802.3, 30.3.6.1.12) and potentially rerunning the OAM discovery process if the dot3OamOperStatus was already operational(9)."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.3"

::= { dot3OamEntry 3 }

dot3OamMaxOamPduSize OBJECT-TYPE

SYNTAX Unsigned32 (64..1518)

UNITS "octets"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The largest OAMPDU that the OAM entity supports. OAM entities exchange maximum OAMPDU sizes and negotiate to use the smaller of the two maximum OAMPDU sizes between the peers. This value is determined by the local implementation."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.8"

::= { dot3OamEntry 4 }

dot3OamConfigRevision OBJECT-TYPE

SYNTAX Unsigned32 (0..65535)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The configuration revision (see IEEE Std 802.3, 57.5.2.1) of the OAM entity as reflected in the latest OAMPDU sent by the OAM entity. The config revision is used by OAM entities to indicate that configuration changes have occurred, which might require the peer OAM entity to re-evaluate whether OAM peering is allowed."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.12"

::= { dot3OamEntry 5 }

dot3OamFunctionsSupported OBJECT-TYPE

SYNTAX BITS { unidirectionalSupport(0), loopbackSupport(1),
eventSupport(2), variableSupport(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The OAM functions supported on this Ethernet-like interface. OAM consists of separate functional sets beyond the basic discovery process that is required. These functional groups can be supported independently by any implementation. These values are communicated to the peer via the local configuration field of Information OAMPDUs.

Setting 'unidirectionalSupport(0)' indicates that the OAM entity supports the transmission of OAMPDUs on links that are operating in unidirectional mode (traffic flowing in one direction only). Setting 'loopbackSupport(1)' indicates that the OAM entity can initiate and respond to loopback commands. Setting 'eventSupport(2)' indicates that the OAM entity can send and receive Event Notification OAMPDUs. Setting 'variableSupport(3)' indicates that the OAM entity can send and receive Variable Request and Response OAMPDUs."

REFERENCE

```
"IEEE Std 802.3, 30.3.6.1.6"
 ::= { dot3OamEntry 6 }

dot3OamPeerTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3OamPeerEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table contains information about the OAM peer for a
        particular Ethernet-like interface. OAM entities communicate
        with a single OAM peer entity on Ethernet links on which OAM
        is enabled and operating properly. There is one entry in this
        table for each entry in the dot3OamTable for which information
        on the peer OAM entity is available."
    ::= { dot3OamObjects 2 }

dot3OamPeerEntry OBJECT-TYPE
    SYNTAX      Dot3OamPeerEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table containing information on the peer OAM
        entity for a single Ethernet-like interface.

        Note that there is at most one OAM peer for each Ethernet-like
        interface. Entries are automatically created when information
        about the OAM peer entity becomes available, and automatically
        deleted when the OAM peer entity is no longer in
        communication. Peer information is not available when
        dot3OamOperStatus is disabled(1), linkFault(2),
        passiveWait(3), activeSendLocal(4), or nonOperHalfDuplex(10)."
    INDEX      { ifIndex }
    ::= { dot3OamPeerTable 1 }

Dot3OamPeerEntry ::= SEQUENCE {
    dot3OamPeerMacAddress      MacAddress,
    dot3OamPeerVendorOui      EightOTwoOui,
    dot3OamPeerVendorInfo      Unsigned32,
    dot3OamPeerMode            INTEGER,
    dot3OamPeerMaxOamPduSize    Unsigned32,
    dot3OamPeerConfigRevision   Unsigned32,
    dot3OamPeerFunctionsSupported BITS
}

dot3OamPeerMacAddress OBJECT-TYPE
    SYNTAX      MacAddress
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The MAC address of the peer OAM entity. The MAC address is
        derived from the most recently received OAMPDU.
        This value is updated on reception of a valid frame with
        (1) a destinationField equal to the reserved multicast address
        for Slow Protocols specified in IEEE Std 802.3, Table 57A-1,
        (2) lengthOrType field value equal to the reserved Type for
        Slow Protocols as specified in IEEE Std 802.3, Table 57A-2,
        (3) a Slow Protocols subtype value equal to the subtype reserved
        for OAM as specified in IEEE Std 802.3, Table 57A-3.;"
    REFERENCE
```

```
"IEEE Std 802.3, 30.3.6.1.5."
 ::= { dot3OamPeerEntry 1 }

dot3OamPeerVendorOui OBJECT-TYPE
    SYNTAX      EightOTwoOui
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The OUI/CID of the OAM peer as reflected in the latest
        Information OAMPDU received with a Local Information TLV. The
        OUI or CID can be used to identify the vendor of the remote OAM
        entity. This value is initialized to three octets of zero
        before any Local Information TLV is received
        (see IEEE Std 802.3, 57.5.2.1)."
```

```
REFERENCE
    "IEEE Std 802.3, 30.3.6.1.16."
 ::= { dot3OamPeerEntry 2 }

dot3OamPeerVendorInfo OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The Vendor Info of the OAM peer as reflected in the latest
        Information OAMPDU received with a Local Information TLV.
        The semantics of the Vendor Information field is proprietary
        and specific to the vendor (identified by the
        dot3OamPeerVendorOui). This information could, for example,
        be used to identify a specific product or product family.
        This value is initialized to zero before any Local
        Information TLV is received
        (see IEEE Std 802.3, 57.5.2.1)."
```

```
REFERENCE
    "IEEE Std 802.3, 30.3.6.1.17."
 ::= { dot3OamPeerEntry 3 }

dot3OamPeerMode OBJECT-TYPE
    SYNTAX      INTEGER { passive(1), active(2), unknown(3) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The mode of the OAM peer as reflected in the latest
        Information OAMPDU received with a Local Information TLV. The
        mode of the peer can be determined from the Configuration
        field in the Local Information TLV of the last Information
        OAMPDU received from the peer. The value is unknown(3)
        whenever no Local Information TLV has been received. The
        values of active(2) and passive(1) are returned when a Local
        Information TLV has been received indicating that the peer is
        in active or passive mode, respectively."
```

```
REFERENCE
    "IEEE Std 802.3, 30.3.6.1.7."
 ::= { dot3OamPeerEntry 4 }

dot3OamPeerMaxOamPduSize OBJECT-TYPE
    SYNTAX      Unsigned32 (0 | 64..1518)
    UNITS       "octets"
    MAX-ACCESS  read-only
    STATUS      current
```

DESCRIPTION

"The maximum size of OAMPDU supported by the peer as reflected in the latest Information OAMPDU received with a Local Information TLV. Ethernet OAM on this interface shall not use OAMPDUs that exceed this size. The maximum OAMPDU size can be determined from the PDU Configuration field of the Local Information TLV of the last Information OAMPDU received from the peer. A value of zero is returned if no Local Information TLV has been received. Otherwise, the value of the OAM peer's maximum OAMPDU size is returned in this value."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.9."

::= { dot3OamPeerEntry 5 }

dot3OamPeerConfigRevision OBJECT-TYPE

SYNTAX Unsigned32 (0..65535)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The configuration revision (see IEEE Std 802.3, 57.5.2.1) of the OAM peer as reflected in the latest OAMPDU. This attribute is changed by the peer whenever it has a local configuration change for Ethernet OAM on this interface. The configuration revision can be determined from the Revision field of the Local Information TLV of the most recently received Information OAMPDU with a Local Information TLV. A value of zero is returned if no Local Information TLV has been received."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.13."

::= { dot3OamPeerEntry 6 }

dot3OamPeerFunctionsSupported OBJECT-TYPE

SYNTAX BITS { unidirectionalSupport(0), loopbackSupport(1),
eventSupport(2), variableSupport(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The OAM functions supported on this Ethernet-like interface. OAM consists of separate functionality sets above the basic discovery process. This value indicates the capabilities of the peer OAM entity with respect to these functions. This value is initialized so all bits are clear.

If unidirectionalSupport(0) is set, then the peer OAM entity supports sending OAM frames on Ethernet interfaces when the receive path is known to be inoperable. If loopbackSupport(1) is set, then the peer OAM entity can send and receive OAM loopback commands. If eventSupport(2) is set, then the peer OAM entity can send and receive event OAMPDUs to signal various error conditions. If variableSupport(3) is set, then the peer OAM entity can send and receive variable requests to monitor the attribute value as described in IEEE Std 802.3, Clause 57.

The capabilities of the OAM peer can be determined from the configuration field of the Local Information TLV of the most recently received Information OAMPDU with a Local Information TLV. All zeros are returned if no Local Information TLV has yet been received."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.7."
 ::= { dot3OamPeerEntry 7 }

dot3OamLoopbackTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3OamLoopbackEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table contains controls for the loopback state of the local link as well as indicates the status of the loopback function. There is one entry in this table for each entry in dot3OamTable that supports loopback functionality (where dot3OamFunctionsSupported includes the loopbackSupport bit set).

Loopback can be used to place the remote OAM entity in a state where every received frame (except OAMPDUs) is echoed back over the same interface on which it was received. In this state, at the remote entity, 'normal' traffic is disabled as only the looped back frames are transmitted on the interface. Loopback is thus an intrusive operation that prohibits normal data flow and should be used accordingly."

::= { dot3OamObjects 3 }

dot3OamLoopbackEntry OBJECT-TYPE

SYNTAX Dot3OamLoopbackEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the table, containing information on the loopback status for a single Ethernet-like interface. Entries in the table are automatically created whenever the local OAM entity supports loopback capabilities. The loopback status on the interface can be determined from the dot3OamLoopbackStatus object."

INDEX { ifIndex }

::= { dot3OamLoopbackTable 1 }

Dot3OamLoopbackEntry ::= SEQUENCE {

dot3OamLoopbackStatus INTEGER,

dot3OamLoopbackIgnoreRx INTEGER

}

dot3OamLoopbackStatus OBJECT-TYPE

SYNTAX INTEGER { noLoopback(1), initiatingLoopback(2),
remoteLoopback(3), terminatingLoopback(4),
localLoopback(5), unknown(6) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The loopback status of the OAM entity. This status is determined by a combination of the local parser and multiplexer states, the remote parser and multiplexer states, as well as by the actions of the local OAM client. When operating in normal mode with no loopback in progress, the status reads noLoopback(1).

The values initiatingLoopback(2) and terminatingLoopback(4)

can be read or written. The other values can only be read - they can never be written. Writing `initiatingLoopback` causes the local OAM entity to start the loopback process with its peer. This value can only be written when the status is `noLoopback(1)`. Writing the value `initiatingLoopback(2)` in any other state has no effect. When in `remoteLoopback(3)`, writing `terminatingLoopback(4)` causes the local OAM entity to initiate the termination of the loopback state. Writing `terminatingLoopback(4)` in any other state has no effect.

If the OAM client initiates a loopback and has sent a Loopback OAMPDU and is waiting for a response, where the local parser and multiplexer states are DISCARD (see IEEE Std 802.3, 57.2.11.1), the status is 'initiatingLoopback'. In this case, the local OAM entity has yet to receive any acknowledgment that the remote OAM entity has received its loopback command request.

If the local OAM client knows that the remote OAM entity is in loopback mode (via the remote state information as described in IEEE Std 802.3, 57.2.11.1, 30.3.6.1.15), the status is `remoteLoopback(3)`. If the local OAM client is in the process of terminating the remote loopback (see IEEE Std 802.3, 57.2.11.3, 30.3.6.1.14) with its local multiplexer and parser states in DISCARD, the status is `terminatingLoopback(4)`. If the remote OAM client has put the local OAM entity in loopback mode as indicated by its local parser state, the status is `localLoopback(5)`.

The `unknown(6)` status indicates that the parser and multiplexer combination is unexpected. This status can be returned if the OAM loopback is in a transition state but should not persist.

The values of this attribute correspond to the following values of the local and remote parser and multiplexer states.

value	LclPrsr	LclMux	RmtPrsr	RmtMux
<code>noLoopback</code>	FWD	FWD	FWD	FWD
<code>initLoopback</code>	DISCARD	DISCARD	FWD	FWD
<code>rmtLoopback</code>	DISCARD	FWD	LPBK	DISCARD
<code>tmtngLoopback</code>	DISCARD	DISCARD	LPBK	DISCARD
<code>lclLoopback</code>	LPBK	DISCARD	DISCARD	FWD
<code>unknown</code>	***	any other combination	***	***

REFERENCE

"IEEE Std 802.3, 57.2.11, 30.3.6.1.14, 30.3.6.1.15"

```
::= { dot3OamLoopbackEntry 1 }
```

dot3OamLoopbackIgnoreRx OBJECT-TYPE

SYNTAX INTEGER { ignore(1), process(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Since OAM loopback is a disruptive operation (user traffic does not pass), this attribute provides a mechanism to provide controls over whether received OAM loopback commands are processed or ignored. When the value is `ignore(1)`, received loopback commands are ignored. When the value is `process(2)`, OAM loopback commands are processed. The default value is to ignore loopback commands (`ignore(1)`)."

REFERENCE

"IEEE Std 802.3, 57.2.11, 30.3.6.1.14, 30.3.6.1.15"
::= { dot3OamLoopbackEntry 2 }

dot3OamStatsTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3OamStatsEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table contains statistics for the OAM function on a particular Ethernet-like interface. There is an entry in the table for every entry in the dot3OamTable.

The counters in this table are defined as 32-bit entries to match the counter size as defined in IEEE Std 802.3. Given that the OAM protocol is a slow protocol, the counters increment at a slow rate."

::= { dot3OamObjects 4 }

dot3OamStatsEntry OBJECT-TYPE

SYNTAX Dot3OamStatsEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the table containing statistics information on the Ethernet OAM function for a single Ethernet-like interface. Entries are automatically created for every entry in the dot3OamTable. Counters are maintained across transitions in dot3OamOperStatus."

INDEX { ifIndex }

::= { dot3OamStatsTable 1 }

Dot3OamStatsEntry ::= SEQUENCE {

dot3OamInformationTx	Counter32,
dot3OamInformationRx	Counter32,
dot3OamUniqueEventNotificationTx	Counter32,
dot3OamUniqueEventNotificationRx	Counter32,
dot3OamDuplicateEventNotificationTx	Counter32,
dot3OamDuplicateEventNotificationRx	Counter32,
dot3OamLoopbackControlTx	Counter32,
dot3OamLoopbackControlRx	Counter32,
dot3OamVariableRequestTx	Counter32,
dot3OamVariableRequestRx	Counter32,
dot3OamVariableResponseTx	Counter32,
dot3OamVariableResponseRx	Counter32,
dot3OamOrgSpecificTx	Counter32,
dot3OamOrgSpecificRx	Counter32,
dot3OamUnsupportedCodesTx	Counter32,
dot3OamUnsupportedCodesRx	Counter32,
dot3OamFramesLostDueToOam	Counter32

}

dot3OamInformationTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Information OAMPDUs transmitted on

this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.20."

::= { dot3OamStatsEntry 1 }

dot3OamInformationRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Information OAMPDUs received on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.21."

::= { dot3OamStatsEntry 2 }

dot3OamUniqueEventNotificationTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of unique Event OAMPDUs transmitted on this interface. Event Notifications might be sent in duplicate to increase the probability of successfully being received, given the possibility that a frame might be lost in transit. Duplicate Event Notification transmissions are counted by dot3OamDuplicateEventNotificationTx.

A unique Event Notification OAMPDU is indicated as an Event Notification OAMPDU with a Sequence Number field that is distinct from the previously transmitted Event Notification OAMPDU Sequence Number.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.22."

::= { dot3OamStatsEntry 3 }

dot3OamUniqueEventNotificationRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of unique Event OAMPDUs received on this interface. Event Notification OAMPDUs may be sent in duplicate to increase the probability of successfully being

received, given the possibility that a frame might be lost in transit. Duplicate Event Notification receptions are counted by dot3OamDuplicateEventNotificationRx.

A unique Event Notification OAMPDU is indicated as an Event Notification OAMPDU with a Sequence Number field that is distinct from the previously received Event Notification OAMPDU Sequence Number.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.24."

::= { dot3OamStatsEntry 4 }

dot3OamDuplicateEventNotificationTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of duplicate Event OAMPDUs transmitted on this interface. Event Notification OAMPDUs might be sent in duplicate to increase the probability of successfully being received, given the possibility that a frame might be lost in transit.

A duplicate Event Notification OAMPDU is indicated as an Event Notification OAMPDU with a Sequence Number field that is identical to the previously transmitted Event Notification OAMPDU Sequence Number.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.23."

::= { dot3OamStatsEntry 5 }

dot3OamDuplicateEventNotificationRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of duplicate Event OAMPDUs received on this interface. Event Notification OAMPDUs might be sent in duplicate to increase the probability of successfully being received, given the possibility that a frame might be lost in transit.

A duplicate Event Notification OAMPDU is indicated as an Event Notification OAMPDU with a Sequence Number field that is identical to the previously received Event Notification OAMPDU Sequence Number.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by

the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.25."

::= { dot3OamStatsEntry 6 }

dot3OamLoopbackControlTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Loopback Control OAMPDUs transmitted on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.26."

::= { dot3OamStatsEntry 7 }

dot3OamLoopbackControlRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Loopback Control OAMPDUs received on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.27."

::= { dot3OamStatsEntry 8 }

dot3OamVariableRequestTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Variable Request OAMPDUs transmitted on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.28."

::= { dot3OamStatsEntry 9 }

dot3OamVariableRequestRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Variable Request OAMPDUs received on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.29."

::= { dot3OamStatsEntry 10 }

dot3OamVariableResponseTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Variable Response OAMPDUs transmitted on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.30."

::= { dot3OamStatsEntry 11 }

dot3OamVariableResponseRx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Variable Response OAMPDUs received on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.31."

::= { dot3OamStatsEntry 12 }

dot3OamOrgSpecificTx OBJECT-TYPE

SYNTAX Counter32

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of Organization Specific OAMPDUs transmitted on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.32."

::= { dot3OamStatsEntry 13 }

dot3OamOrgSpecificRx OBJECT-TYPE

SYNTAX Counter32
UNITS "frames"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A count of the number of Organization Specific OAMPDUs received on this interface.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."
REFERENCE
"IEEE Std 802.3, 30.3.6.1.33."
::= { dot3OamStatsEntry 14 }

dot3OamUnsupportedCodesTx OBJECT-TYPE

SYNTAX Counter32
UNITS "frames"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A count of the number of OAMPDUs transmitted on this interface with an unsupported op-code.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."
REFERENCE
"IEEE Std 802.3, 30.3.6.1.18."
::= { dot3OamStatsEntry 15 }

dot3OamUnsupportedCodesRx OBJECT-TYPE

SYNTAX Counter32
UNITS "frames"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A count of the number of OAMPDUs received on this interface with an unsupported op-code.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."
REFERENCE
"IEEE Std 802.3, 30.3.6.1.19."
::= { dot3OamStatsEntry 16 }

dot3OamFramesLostDueToOam OBJECT-TYPE

SYNTAX Counter32
UNITS "frames"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A count of the number of frames that were dropped by the OAM multiplexer. Since the OAM multiplexer has multiple inputs and a single output, there might be cases where frames are dropped due to transmit resource contention. This counter is incremented whenever a frame is dropped by the OAM layer. Note that any Ethernet frame, not just OAMPDUs, might be dropped

by the OAM layer. This can occur when an OAMPDU takes precedence over a 'normal' frame resulting in the 'normal' frame being dropped.

When this counter is incremented, no other counters in this MIB are incremented.

Discontinuities of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of the ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.46."

::= { dot3OamStatsEntry 17 }

dot3OamEventConfigTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3OamEventConfigEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Ethernet OAM includes the ability to generate and receive Event Notification OAMPDUs to indicate various link problems. This table contains the mechanisms to enable Event Notifications and configure the thresholds to generate the standard Ethernet OAM events. There is one entry in the table for every entry in dot3OamTable that supports OAM events (where dot3OamFunctionsSupported includes the eventSupport bit set). The values in the table are maintained across changes to dot3OamOperStatus.

The standard threshold crossing events are:

- Errored Symbol Period Event. Generated when the number of symbol errors exceeds a threshold within a given window defined by a number of symbols (for example, 1,000 symbols out of 1,000,000 had errors).
- Errored Frame Period Event. Generated when the number of frame errors exceeds a threshold within a given window defined by a number of frames (for example, 10 frames out of 1000 had errors).
- Errored Frame Event. Generated when the number of frame errors exceeds a threshold within a given window defined by a period of time (for example, 10 frames in 1 second had errors).
- Errored Frame Seconds Summary Event. Generated when the number of errored frame seconds exceeds a threshold within a given time period (for example, 10 errored frame seconds within the last 100 seconds). An errored frame second is defined as a 1 second interval which had >0 frame errors.

There are other events (dying gasp, critical events) that are not threshold crossing events but that can be enabled/disabled via this table."

::= { dot3OamObjects 5 }

dot3OamEventConfigEntry OBJECT-TYPE

SYNTAX Dot3OamEventConfigEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Entries are automatically created and deleted from this table, and exist whenever the OAM entity supports Ethernet OAM

events (as indicated by the eventSupport bit in dot3OamFunctionsSupported). Values in the table are maintained across changes to the value of dot3OamOperStatus.

Event configuration controls when the local management entity sends Event Notification OAMPDUs to its OAM peer, and when certain event flags are set or cleared in OAMPDUs."

```
INDEX      { ifIndex }
 ::= { dot3OamEventConfigTable 1 }
```

```
Dot3OamEventConfigEntry ::= SEQUENCE {
    dot3OamErrSymPeriodWindowHi      Unsigned32,
    dot3OamErrSymPeriodWindowLo      Unsigned32,
    dot3OamErrSymPeriodThresholdHi    Unsigned32,
    dot3OamErrSymPeriodThresholdLo    Unsigned32,
    dot3OamErrSymPeriodEvNotifEnable  TruthValue,
    dot3OamErrFramePeriodWindow      Unsigned32,
    dot3OamErrFramePeriodThreshold    Unsigned32,
    dot3OamErrFramePeriodEvNotifEnable TruthValue,
    dot3OamErrFrameWindow             Unsigned32,
    dot3OamErrFrameThreshold          Unsigned32,
    dot3OamErrFrameEvNotifEnable      TruthValue,
    dot3OamErrFrameSecsSummaryWindow  Integer32,
    dot3OamErrFrameSecsSummaryThreshold Integer32,
    dot3OamErrFrameSecsEvNotifEnable  TruthValue,
    dot3OamDyingGaspEnable            TruthValue,
    dot3OamCriticalEventEnable        TruthValue
}
```

dot3OamErrSymPeriodWindowHi OBJECT-TYPE

```
SYNTAX      Unsigned32
UNITS       "2^32 symbols"
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION

    "The two objects dot3OamErrSymPeriodWindowHi and
    dot3OamErrSymPeriodLo together form an unsigned 64-bit
    integer representing the number of symbols over which this
    threshold event is defined. This is defined as
    dot3OamErrSymPeriodWindow = ((2^32)*dot3OamErrSymPeriodWindowHi)
                                + dot3OamErrSymPeriodWindowLo
```

If dot3OamErrSymPeriodThreshold symbol errors occur within a window of dot3OamErrSymPeriodWindow symbols, an Event Notification OAMPDU should be generated with an Errored Symbol Period Event TLV indicating that the threshold has been crossed in this window.

The default value for dot3OamErrSymPeriodWindow is the number of symbols in one second for the underlying Physical Layer."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.34"

```
::= { dot3OamEventConfigEntry 1 }
```

dot3OamErrSymPeriodWindowLo OBJECT-TYPE

```
SYNTAX      Unsigned32
UNITS       "symbols"
MAX-ACCESS  read-write
STATUS      current
```

DESCRIPTION

"The two objects dot3OamErrSymPeriodWindowHi and dot3OamErrSymPeriodWindowLo together form an unsigned 64-bit integer representing the number of symbols over which this threshold event is defined. This is defined as

$$\text{dot3OamErrSymPeriodWindow} = ((2^{32}) * \text{dot3OamErrSymPeriodWindowHi}) + \text{dot3OamErrSymPeriodWindowLo}$$

If dot3OamErrSymPeriodThreshold symbol errors occur within a window of dot3OamErrSymPeriodWindow symbols, an Event Notification OAMPDU should be generated with an Errored Symbol Period Event TLV indicating that the threshold has been crossed in this window.

The default value for dot3OamErrSymPeriodWindow is the number of symbols in one second for the underlying Physical Layer."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.34"
::= { dot3OamEventConfigEntry 2 }

dot3OamErrSymPeriodThresholdHi OBJECT-TYPE

SYNTAX Unsigned32
UNITS "2^32 symbols"
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"The two objects dot3OamErrSymPeriodThresholdHi and dot3OamErrSymPeriodThresholdLo together form an unsigned 64-bit integer representing the minimum number of symbol errors occurring within a given window to cause an Errored Symbol Period Event.

This is defined as

$$\text{dot3OamErrSymPeriodThreshold} = ((2^{32}) * \text{dot3OamErrSymPeriodThresholdHi}) + \text{dot3OamErrSymPeriodThresholdLo}$$

If dot3OamErrSymPeriodThreshold symbol errors occur within a window of dot3OamErrSymPeriodWindow symbols, an Event Notification OAMPDU is generated with an Errored Symbol Period Event TLV indicating that the threshold has been crossed in this window.

The default value for dot3OamErrSymPeriodThreshold is one symbol errors. If the threshold value is zero, then an Event Notification OAMPDU is sent periodically (at the end of every window). This can be used as an asynchronous notification to the peer OAM entity of the statistics related to this threshold crossing alarm."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.34"
::= { dot3OamEventConfigEntry 3 }

dot3OamErrSymPeriodThresholdLo OBJECT-TYPE

SYNTAX Unsigned32
UNITS "symbols"
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"The two objects dot3OamErrSymPeriodThresholdHi and dot3OamErrSymPeriodThresholdLo together form an unsigned 64-bit integer representing the minimum number of symbol errors occurring within a given window to cause an Errored Symbol Period Event.

This is defined as

$$\text{dot3OamErrSymPeriodThreshold} = ((2^{32}) * \text{dot3OamErrSymPeriodThresholdHi}) + \text{dot3OamErrSymPeriodThresholdLo}$$

If dot3OamErrSymPeriodThreshold symbol errors occur within a window of dot3OamErrSymPeriodWindow symbols, an Event Notification OAMPDU is generated with an Errored Symbol Period Event TLV indicating that the threshold has been crossed in this window.

The default value for dot3OamErrSymPeriodThreshold is one symbol error. If the threshold value is zero, then an Event Notification OAMPDU is sent periodically (at the end of every window). This can be used as an asynchronous notification to the peer OAM entity of the statistics related to this threshold crossing alarm."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.34"
::= { dot3OamEventConfigEntry 4 }

dot3OamErrSymPeriodEvNotifEnable OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"If true, the OAM entity sends an Event Notification OAMPDU when an Errored Symbol Period Event occurs.

The default value for this object is true for Ethernet-like interfaces that support OAM. If the OAM layer does not support Event Notifications (as indicated via the dot3OamFunctionsSupported attribute), this value is ignored."

::= { dot3OamEventConfigEntry 5 }

dot3OamErrFramePeriodWindow OBJECT-TYPE

SYNTAX Unsigned32
UNITS "frames"
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"The number of frames over which the threshold is defined. The default value of the window is the number of minimum size Ethernet frames that can be received over the Physical Layer in one second.

If dot3OamErrFramePeriodThreshold frame errors occur within a window of dot3OamErrFramePeriodWindow frames, an Event Notification OAMPDU should be generated with an Errored Frame Period Event TLV indicating that the threshold has been crossed in this window."

REFERENCE

```
"IEEE Std 802.3, 30.3.6.1.38"
 ::= { dot3OamEventConfigEntry 6 }

dot3OamErrFramePeriodThreshold OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "frames"
    MAX-ACCESS   read-write
    STATUS      current
    DESCRIPTION
        "The minimum number of frame errors that cause an Errored Frame
        Period Event. The default value is one frame error. If the
        threshold value is zero, then an Event Notification OAMPDU is
        sent periodically (at the end of every window). This can be
        used as an asynchronous notification to the peer OAM entity of
        the statistics related to this threshold crossing alarm.

        If dot3OamErrFramePeriodThreshold frame errors occur within a
        window of dot3OamErrFramePeriodWindow frames, an Event
        Notification OAMPDU is generated with an Errored Frame
        Period Event TLV indicating that the threshold has been
        crossed in this window."
    REFERENCE
        "IEEE Std 802.3, 30.3.6.1.38"
    ::= { dot3OamEventConfigEntry 7 }

dot3OamErrFramePeriodEvNotifEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-write
    STATUS      current
    DESCRIPTION
        "If true, the OAM entity should send an Event Notification
        OAMPDU when an Errored Frame Period Event occurs.

        By default, this object should have the value true for
        Ethernet-like interfaces that support OAM. If the OAM layer
        does not support Event Notifications (as indicated via the
        dot3OamFunctionsSupported attribute), this value is ignored."
    ::= { dot3OamEventConfigEntry 8 }

dot3OamErrFrameWindow OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "tenths of a second"
    MAX-ACCESS   read-write
    STATUS      current
    DESCRIPTION
        "The amount of time (in 100 ms increments) over which the
        threshold is defined. The default value is 10 (1 second).

        If dot3OamErrFrameThreshold frame errors occur within a window
        of dot3OamErrFrameWindow seconds (measured in tenths of
        seconds), an Event Notification OAMPDU should be generated
        with an Errored Frame Event TLV indicating that the threshold
        has been crossed in this window."
    REFERENCE
        "IEEE Std 802.3, 30.3.6.1.36"
    DEFVAL      { 10 }
    ::= { dot3OamEventConfigEntry 9 }

dot3OamErrFrameThreshold OBJECT-TYPE
```

```
SYNTAX      Unsigned32
UNITS       "frames"
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION
    "The minimum number of frame errors that cause an Errored Frame
    Event. The default value is one frame error. If the
    threshold value is zero, then an Event Notification OAMPDU is
    sent periodically (at the end of every window). This can be
    used as an asynchronous notification to the peer OAM entity of
    the statistics related to this threshold crossing alarm.

    If dot3OamErrFrameThreshold frame errors occur within a window
    of dot3OamErrFrameWindow (in tenths of seconds), an Event
    Notification OAMPDU is generated with an Errored Frame
    Event TLV indicating the threshold has been crossed in this
    window."
REFERENCE
    "IEEE Std 802.3, 30.3.6.1.36"
DEFVAL      { 1 }
::= { dot3OamEventConfigEntry 10 }

dot3OamErrFrameEvNotifEnable OBJECT-TYPE
SYNTAX      TruthValue
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION
    "If true, the OAM entity should send an Event Notification
    OAMPDU when an Errored Frame Event occurs.

    By default, this object should have the value true for
    Ethernet-like interfaces that support OAM. If the OAM layer
    does not support Event Notifications (as indicated via the
    dot3OamFunctionsSupported attribute), this value is ignored."
DEFVAL      { true }
::= { dot3OamEventConfigEntry 11 }

dot3OamErrFrameSecsSummaryWindow OBJECT-TYPE
SYNTAX      Integer32 (100..9000)
UNITS       "tenths of a second"
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION
    "The amount of time (in 100 ms intervals) over which the
    threshold is defined. The default value is 100 (10 seconds).

    If dot3OamErrFrameSecsSummaryThreshold frame errors occur
    within a window of dot3OamErrFrameSecsSummaryWindow (in tenths
    of seconds), an Event Notification OAMPDU should be generated
    with an Errored Frame Seconds Summary Event TLV indicating
    that the threshold has been crossed in this window."
REFERENCE
    "IEEE Std 802.3, 30.3.6.1.40"
DEFVAL      { 100 }
::= { dot3OamEventConfigEntry 12 }

dot3OamErrFrameSecsSummaryThreshold OBJECT-TYPE
SYNTAX      Integer32 (1..900)
UNITS       "errored frame seconds"
```

MAX-ACCESS read-write
STATUS current
DESCRIPTION

"The minimum number of errored frame seconds that cause an Errored Frame Seconds Summary Event. The default value is one errored frame second. If the threshold value is zero, then an Event Notification OAMPDU is sent periodically (at the end of every window). This can be used as an asynchronous notification to the peer OAM entity of the statistics related to this threshold crossing alarm.

If dot3OamErrFrameSecsSummaryThreshold frame errors occur within a window of dot3OamErrFrameSecsSummaryWindow (in tenths of seconds), an Event Notification OAMPDU is generated with an Errored Frame Seconds Summary Event TLV indicating that the threshold has been crossed in this window."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.40"

DEFVAL { 1 }
::= { dot3OamEventConfigEntry 13 }

dot3OamErrFrameSecsEvNotifEnable OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION

"If true, the local OAM entity sends an Event Notification OAMPDU when an Errored Frame Seconds Event occurs.

The default value for this object is true for Ethernet-like interfaces that support OAM. If the OAM layer does not support Event Notifications (as indicated via the dot3OamFunctionsSupported attribute), this value is ignored."

DEFVAL { true }
::= { dot3OamEventConfigEntry 14 }

dot3OamDyingGaspEnable OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION

"If true, the local OAM entity should attempt to indicate a dying gasp via the OAMPDU flags field to its peer OAM entity when a dying gasp event occurs. The exact definition of a dying gasp event is implementation dependent. If the system does not support dying gasp capability, setting this object has no effect, and reading the object returns 'false'.

The default value for this object is true for Ethernet-like interfaces that support OAM. If the OAM layer does not support Event Notifications (as indicated via the dot3OamFunctionsSupported attribute), this value is ignored."

DEFVAL { true }
::= { dot3OamEventConfigEntry 15 }

dot3OamCriticalEventEnable OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"If true, the local OAM entity should attempt to indicate a critical event via the OAMPDU flags to its peer OAM entity when a critical event occurs. The exact definition of a critical event is implementation dependent. If the system does not support critical event capability, setting this object has no effect, and reading the object should result in 'false'.

By default, this object should have the value true for Ethernet-like interfaces that support OAM. If the OAM layer does not support Event Notifications (as indicated via the dot3OamFunctionsSupported attribute), this value is ignored."

DEFVAL { true }
::= { dot3OamEventConfigEntry 16 }

dot3OamEventLogTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3OamEventLogEntry
MAX-ACCESS not-accessible
STATUS current

DESCRIPTION

"This table records a history of the events that have occurred at the Ethernet OAM level. These events can include locally detected events, potentially resulting in locally generated OAMPDUs, and remotely detected events, which are detected by the OAM peer entity and signaled to the local entity via Ethernet OAM. Ethernet OAM events can be signaled by Event Notification OAMPDUs or by the flags field in any OAMPDU.

This table contains both threshold crossing events and non-threshold crossing events. The parameters for the threshold window, threshold value, and actual value (dot3OamEventLogWindowXX, dot3OamEventLogThresholdXX, dot3OamEventLogValue) are only applicable to threshold crossing events, and are returned as all F's ($2^{32} - 1$) for non-threshold crossing events.

Entries in the table are automatically created when such events are detected. The size of the table is implementation dependent. When the table reaches its maximum size, older entries are automatically deleted to make room for newer entries."

::= { dot3OamObjects 6 }

dot3OamEventLogEntry OBJECT-TYPE

SYNTAX Dot3OamEventLogEntry
MAX-ACCESS not-accessible
STATUS current

DESCRIPTION

"An entry in the dot3OamEventLogTable. Entries are automatically created whenever Ethernet OAM events occur at the local OAM entity, and when Event Notification OAMPDUs are received at the local OAM entity (indicating that events have occurred at the peer OAM entity). The size of the table is implementation dependent, but when the table becomes full, older events are automatically deleted to make room for newer events. The table index dot3OamEventLogIndex increments for each new entry, and when the maximum value is reached, the value restarts at zero."

```
INDEX      { ifIndex, dot3OamEventLogIndex }
::= { dot3OamEventLogTable 1 }

Dot3OamEventLogEntry ::= SEQUENCE {
    dot3OamEventLogIndex      Unsigned32,
    dot3OamEventLogTimestamp  TimeStamp,
    dot3OamEventLogOui        EightOTwoOui,
    dot3OamEventLogType       Unsigned32,
    dot3OamEventLogLocation   INTEGER,
    dot3OamEventLogWindowHi   Unsigned32,
    dot3OamEventLogWindowLo   Unsigned32,
    dot3OamEventLogThresholdHi Unsigned32,
    dot3OamEventLogThresholdLo Unsigned32,
    dot3OamEventLogValue      CounterBasedGauge64,
    dot3OamEventLogRunningTotal CounterBasedGauge64,
    dot3OamEventLogEventTotal Unsigned32
}

dot3OamEventLogIndex OBJECT-TYPE
    SYNTAX      Unsigned32 (1..4294967295)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An arbitrary integer for identifying individual events
        within the event log."
    ::= { dot3OamEventLogEntry 1 }

dot3OamEventLogTimestamp OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value of sysUpTime at the time of the logged event. For
        locally generated events, the time of the event can be
        accurately retrieved from sysUpTime. For remotely generated
        events, the time of the event is indicated by the reception of
        the Event Notification OAMPDU indicating that the event
        occurred on the peer. A system can attempt to adjust the
        timestamp value to more accurately reflect the time of the
        event at the peer OAM entity by using other information, such
        as that found in the timestamp found of the Event Notification
        TLVs, which provides an indication of the relative time
        between events at the peer entity."
    ::= { dot3OamEventLogEntry 2 }

dot3OamEventLogOui OBJECT-TYPE
    SYNTAX      EightOTwoOui
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A value indicating the entity defining the object type.
        All IEEE 802.3 defined events (as appearing in IEEE Std 802.3
        except for the Organizationally Unique Event TLVs) use the
        value 71874 (0x0180C2 in hex). Organizations defining their
        own Event Notification TLVs include their OUI in the Event
        Notification TLV that gets reflected here."
    ::= { dot3OamEventLogEntry 3 }

dot3OamEventLogType OBJECT-TYPE
```


SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The type of event that generated this entry in the event log.
 When the value of the dot3OamEventLogOui object is 71874
 (0x0180C2 in hex), the following event types are defined:
 erroredSymbolEvent(1),
 erroredFramePeriodEvent(2),
 erroredFrameEvent(3),
 erroredFrameSecondsEvent(4),
 linkFault(256),
 dyingGaspEvent(257),
 criticalLinkEvent(258)
 The first four are considered threshold crossing events, as
 they are generated when a metric exceeds a given value within
 a specified window. The other three are not threshold
 crossing events.

 When the value of the dot3OamEventLogOui object is not 71874
 (0x0180C2 in hex), the OUI value of the dot3OamEventLogOui
 object identifies the organization that has defined the
 event space."
REFERENCE
 "IEEE Std 802.3, 30.3.6.1.10 and 57.5.3."
::= { dot3OamEventLogEntry 4 }

dot3OamEventLogLocation OBJECT-TYPE
SYNTAX INTEGER { local(1), remote(2) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "Whether this event occurred locally (local(1)), or was
 received from the OAM peer via Ethernet OAM (remote(2))."
::= { dot3OamEventLogEntry 5 }

dot3OamEventLogWindowHi OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "If the event represents a threshold crossing event, the two
 objects dot3OamEventWindowHi and dot3OamEventWindowLo, form
 an unsigned 64-bit integer yielding the window over which the
 value was measured for the threshold crossing event (for
 example, 5, when 11 occurrences happened in 5 seconds while
 the threshold was 10). The two objects are combined as:
 dot3OamEventLogWindow = ((2³²) * dot3OamEventLogWindowHi)
 + dot3OamEventLogWindowLo

 Otherwise, this value is returned as all F's (2³² - 1) and
 adds no useful information."
REFERENCE
 "IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."
::= { dot3OamEventLogEntry 6 }

dot3OamEventLogWindowLo OBJECT-TYPE
SYNTAX Unsigned32
MAX-ACCESS read-only

STATUS current

DESCRIPTION

"If the event represents a threshold crossing event, the two objects dot3OamEventWindowHi and dot3OamEventWindowLo form an unsigned 64-bit integer yielding the window over which the value was measured for the threshold crossing event (for example, 5, when 11 occurrences happened in 5 seconds while the threshold was 10). The two objects are combined as:

$$\text{dot3OamEventLogWindow} = ((2^{32}) * \text{dot3OamEventLogWindowHi}) + \text{dot3OamEventLogWindowLo}$$

Otherwise, this value is returned as all F's ($2^{32} - 1$) and adds no useful information."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."

::= { dot3OamEventLogEntry 7 }

dot3OamEventLogThresholdHi OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"If the event represents a threshold crossing event, the two objects dot3OamEventThresholdHi and dot3OamEventThresholdLo form an unsigned 64-bit integer yielding the value that was crossed for the threshold crossing event (for example, 10, when 11 occurrences happened in 5 seconds while the threshold was 10). The two objects are combined as:

$$\text{dot3OamEventLogThreshold} = ((2^{32}) * \text{dot3OamEventLogThresholdHi}) + \text{dot3OamEventLogThresholdLo}$$

Otherwise, this value is returned as all F's ($2^{32} - 1$) and adds no useful information."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."

::= { dot3OamEventLogEntry 8 }

dot3OamEventLogThresholdLo OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"If the event represents a threshold crossing event, the two objects dot3OamEventThresholdHi and dot3OamEventThresholdLo form an unsigned 64-bit integer yielding the value that was crossed for the threshold crossing event (for example, 10, when 11 occurrences happened in 5 seconds while the threshold was 10). The two objects are combined as:

$$\text{dot3OamEventLogThreshold} = ((2^{32}) * \text{dot3OamEventLogThresholdHi}) + \text{dot3OamEventLogThresholdLo}$$

Otherwise, this value is returned as all F's ($2^{32} - 1$) and adds no useful information."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."

::= { dot3OamEventLogEntry 9 }

dot3OamEventLogValue OBJECT-TYPE

SYNTAX CounterBasedGauge64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"If the event represents a threshold crossing event, this value indicates the value of the parameter within the given window that generated this event (for example, 11, when 11 occurrences happened in 5 seconds while the threshold was 10).

Otherwise, this value is returned as all F's ($2^{64} - 1$) and adds no useful information."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."

::= { dot3OamEventLogEntry 10 }

dot3OamEventLogRunningTotal OBJECT-TYPE

SYNTAX CounterBasedGauge64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Each Event Notification TLV contains a running total of the number of times an event has occurred, as well as the number of times an Event Notification for the event has been transmitted. For non-threshold crossing events, the number of events (dot3OamLogRunningTotal) and the number of resultant Event Notifications (dot3OamLogEventTotal) should be identical.

For threshold crossing events when multiple occurrences cross the threshold, the number of events and the number of resulting Event Notifications are likely to be different. This value represents the total number of times this event has happened since the last reset (for example, 3253, when 3253 symbol errors have occurred since the last reset, which has resulted in 51 symbol error threshold crossing events since the last reset)."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."

::= { dot3OamEventLogEntry 11 }

dot3OamEventLogEventTotal OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Each Event Notification TLV contains a running total of the number of times an event has occurred, as well as the number of times an Event Notification for the event has been transmitted. For non-threshold crossing events, the number of events (dot3OamLogRunningTotal) and the number of resultant Event Notifications (dot3OamLogEventTotal) should be identical.

For threshold crossing events when multiple occurrences cross the threshold, the number of events and the number of resulting Event Notifications are likely to be different. This value represents the total number of times one or more of these occurrences have resulted in an Event Notification

(for example, 51 when 3253 symbol errors have occurred since the last reset, which has resulted in 51 symbol error threshold crossing events since the last reset)."

REFERENCE

"IEEE Std 802.3, 30.3.6.1.37 and 57.5.3.2."
 ::= { dot3OamEventLogEntry 12 }

dot3OamConformance OBJECT IDENTIFIER
 ::= { ieee8023Dot3OamMIB 2 }

dot3OamGroups OBJECT IDENTIFIER
 ::= { dot3OamConformance 1 }

dot3OamCompliances OBJECT IDENTIFIER
 ::= { dot3OamConformance 2 }

dot3OamThresholdEvent NOTIFICATION-TYPE

OBJECTS { dot3OamEventLogTimestamp, dot3OamEventLogOui,
 dot3OamEventLogType, dot3OamEventLogLocation,
 dot3OamEventLogWindowHi, dot3OamEventLogWindowLo,
 dot3OamEventLogThresholdHi, dot3OamEventLogThresholdLo,
 dot3OamEventLogValue, dot3OamEventLogRunningTotal,
 dot3OamEventLogEventTotal }

STATUS current

DESCRIPTION

"A dot3OamThresholdEvent notification is sent when a local or remote threshold crossing event is detected. A local threshold crossing event is detected by the local entity, while a remote threshold crossing event is detected by the reception of an Ethernet OAM Event Notification OAMPDU that indicates a threshold event.

This notification should not be sent more than once per second.

The OAM entity can be derived from extracting the ifIndex from the variable bindings. The objects in the notification correspond to the values in a row instance in the dot3OamEventLogTable.

The management entity should periodically check dot3OamEventLogTable to detect any missed events."

::= { dot3OamNotifications 1 }

dot3OamNonThresholdEvent NOTIFICATION-TYPE

OBJECTS { dot3OamEventLogTimestamp, dot3OamEventLogOui,
 dot3OamEventLogType, dot3OamEventLogLocation,
 dot3OamEventLogEventTotal }

STATUS current

DESCRIPTION

"A dot3OamNonThresholdEvent notification is sent when a local or remote non-threshold crossing event is detected. A local event is detected by the local entity, while a remote event is detected by the reception of an Ethernet OAM Event Notification OAMPDU that indicates a non-threshold crossing event.

This notification should not be sent more than once per second.

The OAM entity can be derived from extracting the ifIndex from the variable bindings. The objects in the notification correspond to the values in a row instance of the dot3OamEventLogTable.

The management entity should periodically check dot3OamEventLogTable to detect any missed events."

```
::= { dot3OamNotifications 2 }
```

dot3OamControlGroup OBJECT-GROUP

```
OBJECTS      { dot3OamAdminState, dot3OamOperStatus, dot3OamMode,
                dot3OamMaxOamPduSize, dot3OamConfigRevision,
                dot3OamFunctionsSupported }
```

```
STATUS       current
```

```
DESCRIPTION
```

```
"A collection of objects providing the abilities,
configuration, and status of an Ethernet OAM entity."
```

```
::= { dot3OamGroups 1 }
```

dot3OamPeerGroup OBJECT-GROUP

```
OBJECTS      { dot3OamPeerMacAddress, dot3OamPeerVendorOui,
                dot3OamPeerVendorInfo, dot3OamPeerMode,
                dot3OamPeerFunctionsSupported,
                dot3OamPeerMaxOamPduSize, dot3OamPeerConfigRevision }
```

```
STATUS       current
```

```
DESCRIPTION
```

```
"A collection of objects providing the abilities,
configuration, and status of a peer Ethernet OAM entity."
```

```
::= { dot3OamGroups 2 }
```

dot3OamStatsBaseGroup OBJECT-GROUP

```
OBJECTS      { dot3OamInformationTx, dot3OamInformationRx,
                dot3OamUniqueEventNotificationTx,
                dot3OamUniqueEventNotificationRx,
                dot3OamDuplicateEventNotificationTx,
                dot3OamDuplicateEventNotificationRx,
                dot3OamLoopbackControlTx, dot3OamLoopbackControlRx,
                dot3OamVariableRequestTx, dot3OamVariableRequestRx,
                dot3OamVariableResponseTx, dot3OamVariableResponseRx,
                dot3OamOrgSpecificTx, dot3OamOrgSpecificRx,
                dot3OamUnsupportedCodesTx, dot3OamUnsupportedCodesRx,
                dot3OamFramesLostDueToOam }
```

```
STATUS       current
```

```
DESCRIPTION
```

```
"A collection of objects providing the statistics for the
number of various transmit and receive events for OAM on an
Ethernet-like interface. Note that all of these counters shall
be supported even if the related function (as described in
dot3OamFunctionsSupported) is not supported."
```

```
::= { dot3OamGroups 3 }
```

dot3OamLoopbackGroup OBJECT-GROUP

```
OBJECTS      { dot3OamLoopbackStatus, dot3OamLoopbackIgnoreRx }
```

```
STATUS       current
```

```
DESCRIPTION
```

```
"A collection of objects for controlling the OAM remote
loopback function."
```

```
::= { dot3OamGroups 4 }
```

```
dot3OamErrSymbolPeriodEventGroup OBJECT-GROUP
  OBJECTS      { dot3OamErrSymPeriodWindowHi,
                  dot3OamErrSymPeriodWindowLo,
                  dot3OamErrSymPeriodThresholdHi,
                  dot3OamErrSymPeriodThresholdLo,
                  dot3OamErrSymPeriodEvNotifEnable }
  STATUS       current
  DESCRIPTION
    "A collection of objects for configuring the thresholds for an
    Errored Symbol Period Event.

    Each IEEE Std 802.3 defined Event Notification TLV has its own
    conformance group because each event can be implemented
    independently of any other."
    ::= { dot3OamGroups 5 }

dot3OamErrFramePeriodEventGroup OBJECT-GROUP
  OBJECTS      { dot3OamErrFramePeriodWindow,
                  dot3OamErrFramePeriodThreshold,
                  dot3OamErrFramePeriodEvNotifEnable }
  STATUS       current
  DESCRIPTION
    "A collection of objects for configuring the thresholds for an
    Errored Frame Period Event.

    Each IEEE Std 802.3 defined Event Notification TLV has its own
    conformance group because each event can be implemented
    independently of any other."
    ::= { dot3OamGroups 6 }

dot3OamErrFrameEventGroup OBJECT-GROUP
  OBJECTS      { dot3OamErrFrameWindow, dot3OamErrFrameThreshold,
                  dot3OamErrFrameEvNotifEnable }
  STATUS       current
  DESCRIPTION
    "A collection of objects for configuring the thresholds for an
    Errored Frame Event.

    Each IEEE Std 802.3 defined Event Notification TLV has its own
    conformance group because each event can be implemented
    independently of any other."
    ::= { dot3OamGroups 7 }

dot3OamErrFrameSecsSummaryEventGroup OBJECT-GROUP
  OBJECTS      { dot3OamErrFrameSecsSummaryWindow,
                  dot3OamErrFrameSecsSummaryThreshold,
                  dot3OamErrFrameSecsEvNotifEnable }
  STATUS       current
  DESCRIPTION
    "A collection of objects for configuring the thresholds for an
    Errored Frame Seconds Summary Event.

    Each IEEE Std 802.3 defined Event Notification TLV has its own
    conformance group because each event can be implemented
    independently of any other."
    ::= { dot3OamGroups 8 }

dot3OamFlagEventGroup OBJECT-GROUP
```

```
OBJECTS      { dot3OamDyingGaspEnable, dot3OamCriticalEventEnable }
STATUS       current
DESCRIPTION
    "A collection of objects for configuring the sending OAMPDUs
    with the critical event flag or dying gasp flag enabled."
::= { dot3OamGroups 9 }

dot3OamEventLogGroup OBJECT-GROUP
OBJECTS      { dot3OamEventLogTimestamp, dot3OamEventLogOui,
                dot3OamEventLogType, dot3OamEventLogLocation,
                dot3OamEventLogWindowHi, dot3OamEventLogWindowLo,
                dot3OamEventLogThresholdHi, dot3OamEventLogThresholdLo,
                dot3OamEventLogValue, dot3OamEventLogRunningTotal,
                dot3OamEventLogEventTotal }
STATUS       current
DESCRIPTION
    "A collection of objects for configuring the thresholds for an
    Errored Frame Seconds Summary Event and maintaining the event
    information."
::= { dot3OamGroups 10 }

dot3OamNotificationGroup NOTIFICATION-GROUP
NOTIFICATIONS { dot3OamThresholdEvent, dot3OamNonThresholdEvent }
STATUS       current
DESCRIPTION
    "A collection of notifications used by Ethernet OAM to signal
    to a management entity that local or remote events have
    occurred on a specified Ethernet link."
::= { dot3OamGroups 11 }

dot3OamCompliance MODULE-COMPLIANCE
STATUS       current
DESCRIPTION
    "The compliance statement for managed entities
    supporting OAM on Ethernet-like interfaces."

MODULE       -- this module

MANDATORY-GROUPS { dot3OamControlGroup, dot3OamPeerGroup,
                    dot3OamStatsBaseGroup }

GROUP dot3OamLoopbackGroup
DESCRIPTION
    "This group is mandatory for all IEEE 802.3 OAM
    implementations that support loopback functionality."

GROUP dot3OamErrSymbolPeriodEventGroup
DESCRIPTION
    "This group is mandatory for all IEEE 802.3 OAM
    implementations that support event functionality."

GROUP dot3OamErrFramePeriodEventGroup
DESCRIPTION
    "This group is mandatory for all IEEE 802.3 OAM
    implementations that support event functionality."

GROUP dot3OamErrFrameEventGroup
DESCRIPTION
    "This group is mandatory for all IEEE 802.3 OAM
```

```
        implementations that support event functionality."

GROUP    dot3OamErrFrameSecsSummaryEventGroup
DESCRIPTION
    "This group is mandatory for all IEEE 802.3 OAM
    implementations that support event functionality."

GROUP    dot3OamFlagEventGroup
DESCRIPTION
    "This group is optional for all IEEE 802.3 OAM
    implementations. The ability to send critical events or dying
    gasp events is not required in any system."

GROUP    dot3OamEventLogGroup
DESCRIPTION
    "This group is optional for all IEEE 802.3 OAM
    implementations. Entries in this table are dependent on what
    event functionality is supported in the local OAM
    implementation. At least one type of event shall be supported
    for entries to appear in this table."

GROUP    dot3OamNotificationGroup
DESCRIPTION
    "This group is optional for all IEEE 802.3 OAM
    implementations. Since the information in the notifications
    is dependent on the dot3OamEventLogTable, that table shall be
    implemented for notifications."

 ::= { dot3OamCompliances 1 }

END -- end of module IEEE8023-DOT3-OAM-MIB.
```


7. Ethernet repeater device MIB module

7.1 Overview

This clause defines a portion of the MIB for use with SNMP. In particular, it defines objects for managing IEEE 802.3 repeaters.

A group is a recommended, but optional, entity defined in Clause 30 of IEEE Std 802.3, in order to support a modular numbering scheme. The classic example allows an implementer to represent field-replaceable units as groups of ports, with the port numbering matching the modular hardware implementation.

7.1.1 Repeater management

Instances of the object types defined in this clause represent attributes of an IEEE 802.3 (Ethernet-like) repeater, as defined by Clause 9 and Clause 27 of IEEE Std 802.3. Implementers of these MIB objects should note that IEEE Std 802.3 explicitly describes when, where, and how various repeater attributes are measured. IEEE Std 802.3 also describes the effects of repeater actions that may be invoked by manipulating instances of the MIB objects defined here. The definitions presented here are based on 30.4 of IEEE Std 802.3. The counters in this clause are defined to be the same as the counters defined in IEEE Std 802.3, with the intention that the same instrumentation can be used to implement both standards.

These repeater MIB module objects might be used to manage non-standard repeater-like devices; however, defining objects to describe implementation-specific properties of non-standard repeater-like devices is outside the scope of this standard.

7.1.2 Structure of the MIB

Objects in this MIB module are arranged into packages, each of which contains a set of related objects within a broad functional category. Objects within a package are generally defined under the same OID subtree. These packages are intended for organizational convenience only and have no relation to the conformance groups defined later in the document.

7.1.2.1 Basic definitions

The basic definitions include objects that are applicable to all repeaters: status, parameter, and control objects for each repeater within the managed system, for the port groups within the system, and for the individual ports themselves.

7.1.2.2 Monitor definitions

The monitor definitions include monitoring statistics for each repeater within the system and for individual ports.

7.1.2.3 Address tracking definitions

This collection includes objects for tracking the MAC addresses of the DTEs attached to the ports within the system and for mapping the topology of a network.

7.1.2.4 Top N definitions

These objects may be used for tracking the ports with the most activity within the system or within particular repeaters.

7.1.3 Relationship to MIB-II

It is assumed that a repeater implementing this MIB will also implement (at least) the “system” group defined in IETF RFC 1213 (MIB-II).

7.1.3.1 Relationship to the “system” group

In MIB-II, the “system” group is defined as being mandatory for all systems such that each managed entity contains one instance of each object in the “system” group. Thus, those objects apply to the entity even if the entity’s sole functionality is management of repeaters.

7.1.3.2 Relationship to the “interfaces” group

In MIB-II, the “interfaces” group is defined as being mandatory for all systems and contains information on an entity’s interfaces, where each interface is thought of as being attached to a “subnetwork”. (Note that this term is not to be confused with “subnet”, which refers to an addressing partitioning scheme used in the Internet suite of protocols.)

This repeater MIB module uses the notion of ports on a repeater. The concept of a MIB-II interface has no specific relationship to a repeater’s port. Therefore, the “interfaces” group applies only to the one (or more) network interfaces on which the entity managing the repeater sends and receives management protocol operations, and does not apply to the repeater’s ports. This is consistent with the physical-layer nature of a repeater. A repeater-unit is a bitwise store-and-forward device. A repeater port has no MAC address, no MAC implementation, and does not pass packets up to higher level protocol entities for processing.

NOTE—When a network management entity is observing a repeater, and observes the repeater passing packets to a higher level protocol entity, this is only a means of implementing management, and this passing of management information is not part of the repeater functionality.²⁰

7.2 Topology mapping

Network topology mapping is described in section 4 of IETF RFC 2108 [B6].

7.3 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²¹

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C7mib.txt

²⁰ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement this standard.

²¹ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

7.3.1 Tree hierarchy

```

----- ieee8023snmpRptrMIB(1.3.111.2.802.3.1.7)
+----- ieee8023snmpDot3RptrMgt(1)
| +----- ieee8023snmpDot3RptrNotifications(0)
| | +----- rpTrInfoHealth(4) {rpTrInfoOperStatus}
| | +----- rpTrInfoResetEvent(5) {rpTrInfoOperStatus}
| +----- rpTrBasicPackage(1)
| | +----- rpTrGroupInfo(1)
| | | +----- rpTrGroupTable(1)
| | | | +----- rpTrGroupEntry(1) [rpTrGroupIndex]
| | | | +----- rpTrGroupIndex(1) Integer32
| | | | +--r- rpTrGroupObjectId(2) ObjectIdentifier
| | | | +--r- rpTrGroupOperStatus(3) Enumeration
| | | | +--r- rpTrGroupPortCapacity(4) Integer32
| | +----- rpTrPortInfo(2)
| | | +----- rpTrPortTable(1)
| | | | +----- rpTrPortEntry(1) [rpTrPortGroupIndex, rpTrPortIndex]
| | | | +----- rpTrPortGroupIndex(1) Integer32
| | | | +----- rpTrPortIndex(2) Integer32
| | | | +--rw rpTrPortAdminStatus(3) Enumeration
| | | | +--r- rpTrPortAutoPartitionState(4) Enumeration
| | | | +--r- rpTrPortOperStatus(5) Enumeration
| | | | +--r- rpTrPortRptrId(6) Integer32
| | +----- rpTrAllRptrInfo(3)
| | | +----- rpTrInfoTable(1)
| | | | +----- rpTrInfoEntry(1) [rpTrInfoId]
| | | | | +----- rpTrInfoId(1) Integer32
| | | | | +--r- rpTrInfoRptrType(2) Enumeration
| | | | | +--r- rpTrInfoOperStatus(3) Enumeration
| | | | | +--rw rpTrInfoReset(4) Enumeration
| | | | | +--r- rpTrInfoPartitionedPorts(5) Gauge32
| | | | | +--r- rpTrInfoLastChange(6) TimeStamp
| +----- rpTrMonitorPackage(2)
| | +----- rpTrMonitorRptrInfo(1)
| | | +----- rpTrMonitorGroupInfo(2)
| | | | +----- rpTrMonitorPortInfo(3)
| | | | | +----- rpTrMonitorPortTable(1)
| | | | | | +----- rpTrMonitorPortEntry(1) [rpTrMonitorPortGroupIndex, rpTrMonitorPortIndex]

```

					+----	rpPtrMonitorPortGroupIndex(1)	Integer32
					+----	rpPtrMonitorPortIndex(2)	Integer32
					+--r-	rpPtrMonitorPortReadableFrames(3)	Counter32
					+--r-	rpPtrMonitorPortReadableOctets(4)	Counter32
					+--r-	rpPtrMonitorPortFCSErrors(5)	Counter32
					+--r-	rpPtrMonitorPortAlignmentErrors(6)	Counter32
					+--r-	rpPtrMonitorPortFrameTooLongs(7)	Counter32
					+--r-	rpPtrMonitorPortShortEvents(8)	Counter32
					+--r-	rpPtrMonitorPortRuntts(9)	Counter32
					+--r-	rpPtrMonitorPortCollisions(10)	Counter32
					+--r-	rpPtrMonitorPortLateEvents(11)	Counter32
					+--r-	rpPtrMonitorPortVeryLongEvents(12)	Counter32
					+--r-	rpPtrMonitorPortDataRateMismatches(13)	Counter32
					+--r-	rpPtrMonitorPortAutoPartitions(14)	Counter32
					+--r-	rpPtrMonitorPortTotalErrors(15)	Counter32
					+--r-	rpPtrMonitorPortLastChange(16)	TimeStamp
					+-----	rpPtrMonitor100PortTable(2)	
					+-----	rpPtrMonitor100PortEntry(1) [rpPtrMonitorPortGroupIndex, rpPtrMonitorPortIndex]	
					+--r-	rpPtrMonitorPortIsolates(1)	Counter32
					+--r-	rpPtrMonitorPortSymbolErrors(2)	Counter32
					+--r-	rpPtrMonitorPortUpper32Octets(3)	Counter32
					+--r-	rpPtrMonitorPortHCRreadableOctets(4)	Counter64
					+-----	rpPtrMonitorAllRpPtrInfo(4)	
					+-----	rpPtrMonTable(1)	
					+-----	rpPtrMonEntry(1) [rpPtrInfoId]	
					+--r-	rpPtrMonTxCollisions(1) Counter32	
					+--r-	rpPtrMonTotalFrames(3) Counter32	
					+--r-	rpPtrMonTotalErrors(4) Counter32	
					+--r-	rpPtrMonTotalOctets(5) Counter32	
					+-----	rpPtrMon100Table(2)	
					+-----	rpPtrMon100Entry(1) [rpPtrInfoId]	
					+--r-	rpPtrMonUpper32TotalOctets(1) Counter32	
					+--r-	rpPtrMonHCtotalOctets(2) Counter64	
					+-----	rpPtrAddrTrackPackage(3)	
					+-----	rpPtrAddrTrackRpPtrInfo(1)	
					+-----	rpPtrAddrSearchTable(1)	
					+-----	rpPtrAddrSearchEntry(1) [rpPtrInfoId]	
					+--rw	rpPtrAddrSearchLock(1) TestAndIncr	
					+--rw	rpPtrAddrSearchStatus(2) Enumeration	

```

+--rw rptrAddrSearchAddress (3) MacAddress
+--r- rptrAddrSearchState (4) Enumeration
+--r- rptrAddrSearchGroup (5) Integer32
+--r- rptrAddrSearchPort (6) Integer32
+--rw rptrAddrSearchOwner (7) OwnerString
+----- rptrAddrTrackGroupInfo (2)
+----- rptrAddrTrackPortInfo (3)
+----- rptrAddrTrackTable (1)
+----- rptrAddrTrackEntry (1) [rptrAddrTrackGroupIndex, rptrAddrTrackPortIndex]
+----- rptrAddrTrackGroupIndex (1) Integer32
+----- rptrAddrTrackPortIndex (2) Integer32
+--r- rptrAddrTrackSourceAddrChanges (3) Counter32
+--r- rptrAddrTrackNewLastSrcAddress (4) OptMacAddr
+--r- rptrAddrTrackCapacity (5) Integer32
+----- rptrExtAddrTrackTable (2)
+----- rptrExtAddrTrackEntry (1) [rptrAddrTrackGroupIndex, rptrAddrTrackPortIndex, rptrExtAddrTrackMacIndex]
+----- rptrExtAddrTrackMacIndex (1) Integer32
+--r- rptrExtAddrTrackSourceAddress (2) MacAddress
+----- rptrTopNPackage (4)
+----- rptrTopNRptrInfo (1)
+----- rptrTopNGroupInfo (2)
+----- rptrTopNPortInfo (3)
+----- rptrTopNPortControlTable (1)
+----- rptrTopNPortControlEntry (1) [rptrTopNPortControlIndex]
+----- rptrTopNPortControlIndex (1) Integer32
+--rc rptrTopNPortRepeaterId (2) Integer32
+--rc rptrTopNPortRateBase (3) Enumeration
+--rc rptrTopNPortTimeRemaining (4) Integer32
+--r- rptrTopNPortDuration (5) Integer32
+--rc rptrTopNPortRequestedSize (6) Integer32
+--r- rptrTopNPortGrantedSize (7) Integer32
+--r- rptrTopNPortStartTime (8) TimeStamp
+--rc rptrTopNPortOwner (9) OwnerString
+--rc rptrTopNPortRowStatus (10) RowStatus
+----- rptrTopNPortTable (2)
+----- rptrTopNPortEntry (1) [rptrTopNPortControlIndex, rptrTopNPortIndex]
+----- rptrTopNPortIndex (1) Integer32
+--r- rptrTopNPortGroupIndex (2) Integer32
+--r- rptrTopNPortPortIndex (3) Integer32

```

```
|
+----- snmpRptrModConf (2)                               Gauge32
|
+----- snmpRptrModCompls (1)
| +----- snmpRptrModCompl (1)
+----- snmpRptrModObjGrps (2)
| +----- snmpRptrGrpBasic (1)
| +----- snmpRptrGrpMonitor (2)
| +----- snmpRptrGrpMonitor100 (3)
| +----- snmpRptrGrpMonitor100w64 (4)
| +----- snmpRptrGrpAddrTrack (5)
| +----- snmpRptrGrpExtAddrTrack (6)
| +----- snmpRptrGrpRptrAddrSearch (7)
| +----- snmpRptrGrpTopNPort (8)
+----- snmpRptrModNotGrps (3)
+----- ieee8023snmpDot3RptrNotGroup (1)
```

7.3.2 MIB module definition

```
IEEE8023-SNMP-REPEATER-MIB DEFINITIONS ::= BEGIN

IMPORTS
    OwnerString
        FROM RFC1271-MIB
    MODULE-COMPLIANCE, NOTIFICATION-GROUP, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Counter64, Gauge32, Integer32, MODULE-IDENTITY,
    NOTIFICATION-TYPE, OBJECT-TYPE, org
        FROM SNMPv2-SMI
    MacAddress, RowStatus, TEXTUAL-CONVENTION, TestAndIncr, TimeStamp
        FROM SNMPv2-TC;

ieee8023snmpRptrMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-E-Mail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "Management information for IEEE 802.3 repeaters."
    REVISION    "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION    "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION    "201102020000Z"
    DESCRIPTION
        "Initial revision, based on an earlier version in RFC 2108"
    ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
    man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 7 }

OptMacAddr ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "1x:"
    STATUS      current
    DESCRIPTION
        "Either a 6-octet address in the 'canonical'
        order defined by IEEE Std 802.1a, i.e., as if it
        were transmitted least significant bit first
        if a value is available or a zero length string."
    REFERENCE
        "See MacAddress in SNMPv2-TC. The only difference
        is that a zero length string is allowed as a value
        for OptMacAddr and not for MacAddress."
    SYNTAX      OCTET STRING (SIZE(0 | 6))
```

```
ieee8023snmpDot3RptrMgt OBJECT IDENTIFIER
    ::= { ieee8023snmpRptrMIB 1 }

ieee8023snmpDot3RptrNotifications OBJECT IDENTIFIER
    ::= { ieee8023snmpDot3RptrMgt 0 }

rptrBasicPackage OBJECT IDENTIFIER
    ::= { ieee8023snmpDot3RptrMgt 1 }

rptrGroupInfo OBJECT IDENTIFIER
    ::= { rptrBasicPackage 1 }

rptrGroupTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RptrGroupEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of descriptive and status information about
         the groups of ports."
    ::= { rptrGroupInfo 1 }

rptrGroupEntry OBJECT-TYPE
    SYNTAX      RptrGroupEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
         about a single group of ports."
    INDEX       { rptrGroupIndex }
    ::= { rptrGroupTable 1 }

RptrGroupEntry ::= SEQUENCE {
    rptrGroupIndex      Integer32,
    rptrGroupObjectID   OBJECT IDENTIFIER,
    rptrGroupOperStatus INTEGER,
    rptrGroupPortCapacity Integer32
}

rptrGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the group within the
         repeater system for which this entry contains
         information."
    REFERENCE
        "IEEE Std 802.3, 30.4.2.1.1"
    ::= { rptrGroupEntry 1 }

rptrGroupObjectID OBJECT-TYPE
    SYNTAX      OBJECT IDENTIFIER
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The vendor's authoritative identification of the
         group. This value might be allocated within the SMI
         enterprises subtree (1.3.6.1.4.1) and provides a
```


straightforward and unambiguous means for determining what kind of group is being managed.

For example, this object could take the value 1.3.6.1.4.1.4242.1.2.14 if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.4242, and had assigned the identifier 1.3.6.1.4.1.4242.1.2.14 to its 'Wilma Flintstone 6-Port FOIRL Plug-in module.'

::= { rptrGroupEntry 2 }

rpPtrGroupOperStatus OBJECT-TYPE

SYNTAX INTEGER { other(1), operational(2), malfunctioning(3),
notPresent(4), underTest(5), resetInProgress(6) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that indicates the operational status of the group.

A status of notPresent(4) indicates that the group is temporarily or permanently physically and/or logically not a part of the repeater. It is an implementation-specific matter as to whether the agent effectively removes notPresent entries from the table.

A status of operational(2) indicates that the group is functioning, and a status of malfunctioning(3) indicates that the group is malfunctioning in some way."

::= { rpPtrGroupEntry 3 }

rpPtrGroupPortCapacity OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The rpPtrGroupPortCapacity is the number of ports that can be contained within the group. Valid range is 1-2147483647. Within each group, the ports are uniquely numbered in the range from 1 to rpPtrGroupPortCapacity.

Some ports might not be present in the repeater system, in which case the actual number of ports present will be less than the value of rpPtrGroupPortCapacity. The number of ports present in the group will never be greater than the value of rpPtrGroupPortCapacity.

Note: In practice, this will generally be the number of ports on a module, card, or board, and the port numbers will correspond to numbers marked on the physical embodiment."

REFERENCE

"IEEE Std 802.3, 30.4.2.1.2"

::= { rpPtrGroupEntry 4 }

rpPtrPortInfo OBJECT IDENTIFIER

```
::= { rpPtrBasicPackage 2 }

rpPtrPortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RpPtrPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of descriptive and status information about
         the repeater ports in the repeater system. The number of
         entries is independent of the number of repeaters
         in the managed repeater system."
    ::= { rpPtrPortInfo 1 }

rpPtrPortEntry OBJECT-TYPE
    SYNTAX      RpPtrPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
         about a single port."
    INDEX       { rpPtrPortGroupIndex, rpPtrPortIndex }
    ::= { rpPtrPortTable 1 }

RpPtrPortEntry ::= SEQUENCE {
    rpPtrPortGroupIndex      Integer32,
    rpPtrPortIndex           Integer32,
    rpPtrPortAdminStatus     INTEGER,
    rpPtrPortAutoPartitionState  INTEGER,
    rpPtrPortOperStatus      INTEGER,
    rpPtrPortRpPtrId         Integer32
}

rpPtrPortGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the group containing the
         port for which this entry contains information."
    ::= { rpPtrPortEntry 1 }

rpPtrPortIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the port within the group
         for which this entry contains information. This
         identifies the port independently from the repeater
         to which it is attached. The numbering scheme for
         ports is implementation specific; however, this
         value can never be greater than
         rpPtrGroupPortCapacity for the associated group."
    REFERENCE
        "IEEE Std 802.3, 30.4.3.1.1"
    ::= { rpPtrPortEntry 2 }

rpPtrPortAdminStatus OBJECT-TYPE
    SYNTAX      INTEGER { enabled(1), disabled(2) }
```

MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Setting this object to disabled(2) disables the port. A disabled port neither transmits nor receives. Once disabled, a port shall be explicitly enabled to restore operation. A port that is disabled when power is lost or when a reset is exerted shall remain disabled when normal operation resumes.

The admin status takes precedence over auto-partition and functionally operates between the auto-partition mechanism and the AUI/PMA, PCS/PMA or GMII/PCS as applicable.

Setting this object to enabled(1) enables the port and exerts a BEGIN on the port's auto-partition state machine.

(In effect, when a port is disabled, the value of rpPtrPortAutoPartitionState for that port is frozen until the port is next enabled. When the port becomes enabled, the rpPtrPortAutoPartitionState becomes notAutoPartitioned(1), regardless of its pre-disabling state.)"

REFERENCE

"IEEE Std 802.3, 30.4.3.1.2 and 30.4.3.2.1"

::= { rpPtrPortEntry 3 }

rpPtrPortAutoPartitionState OBJECT-TYPE

SYNTAX INTEGER { notAutoPartitioned(1), autoPartitioned(2) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The autoPartitionState flag indicates whether the port is currently partitioned by the repeater's auto-partition protection.

The conditions that cause port partitioning are specified in partition state machine in IEEE Std 802.3 Clause 9 and Clause 27. They are not differentiated here. An IEEE Std 802.3, Clause 27 and Clause 41 repeater port partitions on entry to the PARTITION WAIT state of the partition state diagram (see IEEE Std 802.3, Figure 27-8 and Figure 41-4)."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.3"

::= { rpPtrPortEntry 4 }

rpPtrPortOperStatus OBJECT-TYPE

SYNTAX INTEGER { operational(1), notOperational(2),
notPresent(3) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"This object indicates the port's operational status. The notPresent(3) status indicates the port is physically removed (Note that for some port types

```
it is not possible to remove the port.)
The operational(1) status indicates that the port
is enabled (see rpPtrPortAdminStatus) and working,
even though it might be auto-partitioned (see
rpPtrPortAutoPartitionState).

If this object has the value operational(1) and
rpPtrPortAdminStatus is set to disabled(2), it is
expected that this object's value will soon change
to notOperational(2)."
```

```
::= { rpPtrPortEntry 5 }
```

```
rpPtrPortRpPtrId OBJECT-TYPE
    SYNTAX      Integer32 (0..2147483647)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This object identifies the repeater to
        which this port belongs. The repeater
        identified by a particular value of this object
        is the same as that identified by the same
        value of rpPtrInfoId. A value of zero
        indicates that this port currently is not
        a member of any repeater."
    ::= { rpPtrPortEntry 6 }
```

```
rpPtrAllRpPtrInfo OBJECT IDENTIFIER
    ::= { rpPtrBasicPackage 3 }
```

```
rpPtrInfoTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RpPtrInfoEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of information about each
        non-trivial repeater. The number of entries
        depends on the physical configuration of the
        managed repeater system."
    ::= { rpPtrAllRpPtrInfo 1 }
```

```
rpPtrInfoEntry OBJECT-TYPE
    SYNTAX      RpPtrInfoEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
        about a single non-trivial repeater."
    INDEX       { rpPtrInfoId }
    ::= { rpPtrInfoTable 1 }
```

```
RpPtrInfoEntry ::= SEQUENCE {
    rpPtrInfoId          Integer32,
    rpPtrInfoRpPtrType   INTEGER,
    rpPtrInfoOperStatus  INTEGER,
    rpPtrInfoReset       INTEGER,
    rpPtrInfoPartitionedPorts Gauge32,
    rpPtrInfoLastChange  TimeStamp
}
```

rpPtrInfoId OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
 "This object identifies the repeater for which
 this entry contains information."
 ::= { rpPtrInfoEntry 1 }

rpPtrInfoRpPtrType OBJECT-TYPE

SYNTAX INTEGER { other(1), tenMb(2), onehundredMbClassI(3),
 onehundredMbClassII(4), onethousandMb(5) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The rpPtrInfoRpPtrType returns a value that identifies
 the CSMA/CD repeater type."
REFERENCE
 "IEEE Std 802.3, 30.4.1.1.2"
 ::= { rpPtrInfoEntry 2 }

rpPtrInfoOperStatus OBJECT-TYPE

SYNTAX INTEGER { other(1), ok(2), failure(3), failureGroup(4),
 failurePort(5), failureGeneral(6) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The rpPtrInfoOperStatus object indicates the
 operational state of the repeater."
REFERENCE
 "IEEE Std 802.3, 30.4.1.1.5"
 ::= { rpPtrInfoEntry 3 }

rpPtrInfoReset OBJECT-TYPE

SYNTAX INTEGER { noReset(1), reset(2) }
MAX-ACCESS read-write
STATUS current
DESCRIPTION
 "Setting this object to reset(2) causes a
 transition to the START state of IEEE Std 802.3,
 Figure 9-2 for a 10 Mb/s repeater, and to the START
 state of IEEE Std 802.3, Figure 27-2 for a
 100 Mb/s repeater.

 Setting this object to noReset(1) has no effect.
 The agent will return the value noReset(1)
 when this object is read.

 After receiving a request to set this variable to
 reset(2), the agent is allowed to delay the reset
 for a short period. For example, it is possible
 to delay the reset to allow the SNMP response
 to be transmitted. In any event, SNMP requires
 that a response be transmitted.

 This action does not reset the management counters
 defined in this document nor does it affect the
 portAdminStatus parameters. Included in this
 action is the execution of a disruptive Self-Test

with the following characteristics: a) The nature of the tests is not specified. b) The test resets the repeater but without affecting management information about the repeater. c) The test does not inject packets onto any segment. d) Packets received during the test can be transferred. e) The test does not interfere with management functions.

After performing this self-test, the agent will update the repeater health information (including rpPtrInfoOperStatus), and send a rpPtrInfoResetEvent notification."

REFERENCE

"IEEE Std 802.3, 30.4.1.2.1"
::= { rpPtrInfoEntry 4 }

rpPtrInfoPartitionedPorts OBJECT-TYPE

SYNTAX Gauge32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This object returns the total number of ports in the repeater whose current state meets all three of the following criteria: rpPtrPortOperStatus does not have the value notPresent(3), rpPtrPortAdminStatus is enabled(1), and rpPtrPortAutoPartitionState is autoPartitioned(2)."
::= { rpPtrInfoEntry 5 }

rpPtrInfoLastChange OBJECT-TYPE

SYNTAX TimeStamp
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"The value of sysUpTime when any of the following conditions occurred:
1) agent cold- or warm-started;
2) this instance of repeater was created (such as when a device or module was added to the repeater system);
3) a change in the value of rpPtrInfoOperStatus;
4) ports were added or removed as members of the repeater; or
5) any of the counters associated with this repeater had a discontinuity."
::= { rpPtrInfoEntry 6 }

rpPtrMonitorPackage OBJECT IDENTIFIER

::= { ieee8023snmpDot3RpPtrMgt 2 }

rpPtrMonitorRpPtrInfo OBJECT IDENTIFIER

::= { rpPtrMonitorPackage 1 }

rpPtrMonitorGroupInfo OBJECT IDENTIFIER

::= { rpPtrMonitorPackage 2 }

rpPtrMonitorPortInfo OBJECT IDENTIFIER

::= { rpPtrMonitorPackage 3 }

```
rpPtrMonitorPortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RpPtrMonitorPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of performance and error statistics for the
        ports. The number of entries is the same as that
        in the rpPtrPortTable.

        The columnar object rpPtrMonitorPortLastChange
        is used to indicate possible discontinuities
        of counter type columnar objects in the table."
    ::= { rpPtrMonitorPortInfo 1 }

rpPtrMonitorPortEntry OBJECT-TYPE
    SYNTAX      RpPtrMonitorPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing performance and
        error statistics for a single port."
    INDEX       { rpPtrMonitorPortGroupIndex, rpPtrMonitorPortIndex }
    ::= { rpPtrMonitorPortTable 1 }

RpPtrMonitorPortEntry ::= SEQUENCE {
    rpPtrMonitorPortGroupIndex      Integer32,
    rpPtrMonitorPortIndex           Integer32,
    rpPtrMonitorPortReadableFrames  Counter32,
    rpPtrMonitorPortReadableOctets  Counter32,
    rpPtrMonitorPortFCSErrors       Counter32,
    rpPtrMonitorPortAlignmentErrors Counter32,
    rpPtrMonitorPortFrameTooLongs   Counter32,
    rpPtrMonitorPortShortEvents     Counter32,
    rpPtrMonitorPortRunts           Counter32,
    rpPtrMonitorPortCollisions      Counter32,
    rpPtrMonitorPortLateEvents      Counter32,
    rpPtrMonitorPortVeryLongEvents  Counter32,
    rpPtrMonitorPortDataRateMismatchCounter32,
    rpPtrMonitorPortAutoPartitions  Counter32,
    rpPtrMonitorPortTotalErrors     Counter32,
    rpPtrMonitorPortLastChange      TimeStamp
}

rpPtrMonitorPortGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the group containing the
        port for which this entry contains information."
    ::= { rpPtrMonitorPortEntry 1 }

rpPtrMonitorPortIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the port within the group
        for which this entry contains information."
```

REFERENCE

"IEEE Std 802.3, 30.4.3.1.1"
::= { rpPtrMonitorPortEntry 2 }

rpPtrMonitorPortReadableFrames OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object is the number of frames of valid frame length that have been received on this port. This counter is incremented by one for each frame received on this port whose OctetCount is greater than or equal to minFrameSize and less than or equal to maxFrameSizeLimit (see IEEE Std 802.3, 4.2.7.1) and for which the FCSError and CollisionEvent signals are not asserted.

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes.

This statistic provides one of the parameters necessary for obtaining the packet error ratio. The approximate minimum time for rollover of this counter is 80 hours at 10 Mb/s."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.4"
::= { rpPtrMonitorPortEntry 3 }

rpPtrMonitorPortReadableOctets OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object is the number of octets contained in valid frames that have been received on this port. This counter is incremented by OctetCount for each frame received on this port that has been determined to be a readable frame (i.e., including FCS octets but excluding framing bits and dribble bits).

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes.

This statistic provides an indicator of the total data transferred. The approximate minimum time for rollover of this counter in a 10 Mb/s repeater is 58 minutes.

For ports receiving traffic at a maximum rate in a 100 Mb/s repeater, this counter can roll over in less than 6 minutes. Since that amount of time could be less than a management station's poll cycle time, in order to avoid a loss of information a management station is advised to also poll the rpPtrMonitorPortUpper32Octets object, or to use the

64-bit counter defined by
rpPtrMonitorPortHCReadableOctets instead of the
two 32-bit counters."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.5"

::= { rpPtrMonitorPortEntry 4 }

rpPtrMonitorPortFCSErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each frame received on this port with the FCSError signal asserted and the FramingError and CollisionEvent signals deasserted and whose OctetCount is greater than or equal to minFrameSize and less than or equal to maxFrameSizeLimit (See IEEE Std 802.3, 4.2.7.1)."

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 80 hours at 10 Mb/s."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.6"

::= { rpPtrMonitorPortEntry 5 }

rpPtrMonitorPortAlignmentErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each frame received on this port with the FCSError and FramingError signals asserted and CollisionEvent signal deasserted and whose OctetCount is greater than or equal to minFrameSize and less than or equal to maxFrameSizeLimit (See IEEE Std 802.3, 4.2.7.1). If rpPtrMonitorPortAlignmentErrors is incremented then the rpPtrMonitorPortFCSErrors Counter shall not be incremented for the same frame."

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 80 hours at 10 Mb/s."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.7"

::= { rpPtrMonitorPortEntry 6 }

rpPtrMonitorPortFrameTooLongs OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each frame received on this port whose OctetCount is greater than maxFrameSizeLimit (See IEEE Std 802.3, 4.2.7.1). If rpPtrMonitorPortFrameTooLongs is incremented then neither the rpPtrMonitorPortAlignmentErrors nor the rpPtrMonitorPortFCSErrors counter shall be incremented for the frame.

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 61 days in a 10 Mb/s repeater."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.8"
::= { rpPtrMonitorPortEntry 7 }

rpPtrMonitorPortShortEvents OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This counter is incremented by one for each CarrierEvent on this port with ActivityDuration less than ShortEventMaxTime. ShortEventMaxTime is greater than 74 bit times and less than 82 bit times. ShortEventMaxTime has tolerances included to provide for circuit losses between a conformance test point at the AUI and the measurement point within the state machine.

Notes:

ShortEvents might indicate externally generated noise hits that will cause the repeater to transmit RuntS to its other ports, or propagate a collision (which might be late) back to the transmitting DTE and damaged frames to the rest of the network.

It may be desirable to select the ShortEventMaxTime towards the lower end of the allowed tolerance range to accommodate bit losses suffered through physical channel devices not budgeted for within this standard.

The significance of this attribute is different in 10 Mb/s and 100 Mb/s collision domains. IEEE Std 802.3, Clause 9 repeaters perform fragment extension of short events which would be counted as runts on the interconnect ports of other repeaters. IEEE Std 802.3, Clause 27 repeaters do not perform fragment extension.

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this

counter is 16 hours in a 10 Mb/s repeater."
REFERENCE

"IEEE Std 802.3, 30.4.3.1.9"
::= { rpPtrMonitorPortEntry 8 }

rpPtrMonitorPortRuns OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each CarrierEvent on this port that meets one of the following two conditions. Only one test need be made. a) The ActivityDuration is greater than ShortEventMaxTime and less than ValidPacketMinTime and the CollisionEvent signal is deasserted. b) The OctetCount is less than 64, the ActivityDuration is greater than ShortEventMaxTime and the CollisionEvent signal is deasserted. ValidPacketMinTime is greater than or equal to 552 bit times and less than 565 bit times.

An event whose length is greater than 74 bit times but less than 82 bit times shall increment either the shortEvents counter or the runs counter but not both. A CarrierEvent greater than or equal to 552 bit times but less than 565 bit times may be counted as a runt.

ValidPacketMinTime has tolerances included to provide for circuit losses between a conformance test point at the AUI and the measurement point within the state machine.

Runs usually indicate collision fragments, a normal network event. In certain situations associated with large diameter networks a percentage of collision fragments can exceed ValidPacketMinTime.

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 16 hours in a 10 Mb/s repeater."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.10"
::= { rpPtrMonitorPortEntry 9 }

rpPtrMonitorPortCollisions OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For an IEEE Std 802.3, Clause 9 repeater, this counter is incremented by one for any CarrierEvent signal on any port for which the CollisionEvent signal on this port is asserted. For an IEEE Std 802.3, Clause 27 repeater port the counter increments on

entering the Collision Count Increment state of the partition state diagram (see IEEE Std 802.3, Figure 27-8).

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 16 hours in a 10 Mb/s repeater."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.11"

::= { rpPtrMonitorPortEntry 10 }

rpPtrMonitorPortLateEvents OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For an IEEE Std 802.3, Clause 9 repeater port, this counter is incremented by one for each CarrierEvent on this port in which the CollIn(X) variable transitions to the value SQE (see IEEE Std 802.3, 9.6.6.2) while the ActivityDuration is greater than the LateEventThreshold. For an IEEE Std 802.3, Clause 27 repeater port, this counter is incremented by one on entering the Collision Count Increment state of the partition state diagram (see IEEE Std 802.3, Figure 27-8) while the ActivityDuration is greater than the LateEventThreshold. Such a CarrierEvent is counted twice, as both a collision and as a lateEvent.

The LateEventThreshold is greater than 480 bit times and less than 565 bit times. LateEventThreshold has tolerances included to permit an implementation to build a single threshold to serve as both the LateEventThreshold and ValidPacketMinTime threshold.

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 81 hours in a 10 Mb/s repeater."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.12"

::= { rpPtrMonitorPortEntry 11 }

rpPtrMonitorPortVeryLongEvents OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For an IEEE Std 802.3, Clause 9 repeater port, this counter is incremented by one for each CarrierEvent whose ActivityDuration is greater than

the MAU Jabber Lockup Protection timer TW3
(see IEEE Std 802.3 9.6.1 and 9.6.5).

For an IEEE Std 802.3, Clause 27 repeater port,
this counter is incremented by one on entry to the
Rx Jabber state of the receiver timer state
diagram (see IEEE Std 802.3, Figure 27-7).
Other counters might be incremented as appropriate.

A discontinuity can occur in the value
when the value of object
rpPtrMonitorPortLastChange changes."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.13"
::= { rpPtrMonitorPortEntry 12 }

rpPtrMonitorPortDataRateMismatches OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each
frame received by this port that meets all
of the conditions required by only one of the
following two measurement methods:

Measurement method A: 1) The CollisionEvent
signal is not asserted (10 Mb/s operation) or
the Collision Count Increment state of the
partition state diagram (see IEEE Std 802.3,
Figure 27-8) has not been entered
(100 Mb/s operation). 2) The ActivityDuration
is greater than ValidPacketMinTime. 3) The
frequency (data rate) is detectably mismatched
from the local transmit frequency.

Measurement method B: 1) The CollisionEvent
signal is not asserted (10 Mb/s operation)
or the Collision Count Increment state of the
partition state diagram (see IEEE Std 802.3,
Figure 27-8) has not been entered
(100 Mb/s operation). 2) The OctetCount is
greater than 63. 3) The frequency (data
rate) is detectably mismatched from the local
transmit frequency. The exact degree of
mismatch is vendor specific and is to be
defined by the vendor for conformance testing.

When this event occurs, other counters whose
increment conditions were satisfied might
also be incremented, at the implementor's
discretion. Whether or not the repeater was able
to maintain data integrity is beyond the scope of
this standard.

A discontinuity can occur in the value
when the value of object
rpPtrMonitorPortLastChange changes."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.14"
::= { rpPtrMonitorPortEntry 13 }

rpPtrMonitorPortAutoPartitions OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each time the repeater has automatically partitioned this port.

The conditions that cause an IEEE Std 802.3, Clause 9 repeater port to partition are specified in the partition state diagram in IEEE Std 802.3, Clause 9. They are not differentiated here. An IEEE Std 802.3, Clause 27 repeater port partitions on entry to the Partition Wait state of the partition state diagram (see IEEE Std 802.3, Figure 27-8).

A discontinuity can occur in the value when the value of object rpPtrMonitorPortLastChange changes."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.15"
::= { rpPtrMonitorPortEntry 14 }

rpPtrMonitorPortTotalErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The total number of errors which have occurred on this port. This counter is the summation of the values of other error counters (for the same port), namely:

rpPtrMonitorPortFCSErrors,
rpPtrMonitorPortAlignmentErrors,
rpPtrMonitorPortFrameTooLongs,
rpPtrMonitorPortShortEvents,
rpPtrMonitorPortLateEvents,
rpPtrMonitorPortVeryLongEvents,
rpPtrMonitorPortDataRateMismatches, and
rpPtrMonitorPortSymbolErrors.

This counter is redundant in the sense that it is the summation of information already available through other objects. However, it is included specifically because the regular retrieval of this object as a means of tracking the health of a port provides a considerable optimization of network management traffic over the otherwise necessary retrieval of the summed counters.

Note that rpPtrMonitorPortRunts is not included in this total; this is because runts usually indicate collision fragments, a normal network

```
event.

A discontinuity can occur in the value
when the value of object
rpPtrMonitorPortLastChange changes."
::= { rpPtrMonitorPortEntry 15 }

rpPtrMonitorPortLastChange OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value of sysUpTime when the last of
        the following occurred:
        1) the agent cold- or warm-started;
        2) the row for the port was created
           (such as when a device or module was added
           to the repeater system); or
        3) any condition that would cause one of
           the counters for the row to experience
           a discontinuity."
    ::= { rpPtrMonitorPortEntry 16 }

rpPtrMonitor100PortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RpPtrMonitor100PortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of additional performance and error
        statistics for 100 Mb/s ports, above and
        beyond those parameters that apply to both
        10 Mb/s and 100 Mb/s ports. Entries exist only for
        ports attached to 100 Mb/s repeaters.

        The columnar object rpPtrMonitorPortLastChange
        is used to indicate possible discontinuities
        of counter type columnar objects in this table."
    ::= { rpPtrMonitorPortInfo 2 }

rpPtrMonitor100PortEntry OBJECT-TYPE
    SYNTAX      RpPtrMonitor100PortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing performance
        and error statistics for a single 100 Mb/s port."
    INDEX      { rpPtrMonitorPortGroupIndex, rpPtrMonitorPortIndex }
    ::= { rpPtrMonitor100PortTable 1 }

RpPtrMonitor100PortEntry ::= SEQUENCE {
    rpPtrMonitorPortIsolates      Counter32,
    rpPtrMonitorPortSymbolErrors Counter32,
    rpPtrMonitorPortUpper32Octets Counter32,
    rpPtrMonitorPortHCReadableOctets Counter64
}

rpPtrMonitorPortIsolates OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
```

STATUS current

DESCRIPTION

"This counter is incremented by one each time that the repeater port automatically isolates as a consequence of false carrier events. The conditions which cause a port to automatically isolate are defined by the transition from the False Carrier state to the Link Unstable state of the carrier integrity state diagram (see IEEE Std 802.3, Figure 27-9).

Note: Isolates do not affect the value of the PortOperStatus object.

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.16"

::= { rpPtrMonitor100PortEntry 1 }

rpPtrMonitorPortSymbolErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one each time when a valid length packet was received at the port and there was at least one occurrence of an invalid data symbol. This can increment only once per valid carrier event. A collision presence at any port of the repeater containing port N, will not cause this attribute to increment.

A discontinuity can occur in the value when the value of object
rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this counter is 7.4 hours at 100 Mb/s."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.17"

::= { rpPtrMonitor100PortEntry 2 }

rpPtrMonitorPortUpper32Octets OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object is the number of octets contained in valid frames that have been received on this port, modulo 2^{32} . That is, it contains the upper 32 bits of a 64-bit octets counter, of which the lower 32 bits are contained in the
rpPtrMonitorPortReadableOctets object.

This two-counter mechanism is provided for those network management protocols that do not support 64-bit counters (e.g. SNMP V1) and are used to

manage a repeater type of 100 Mb/s.

Conformance clauses for this MIB are defined such that implementation of this object is not required in a repeater system which does not support 100 Mb/s. However, repeater systems with mixed 10 Mb/s and 100 Mb/s ports might implement this object across all ports, including 10 Mb/s. If this object is implemented, the value shall be a valid count as defined in the first paragraph of this description.

A discontinuity can occur in the value when the value of object

rpPtrMonitorPortLastChange changes."

::= { rpPtrMonitor100PortEntry 3 }

rpPtrMonitorPortHCReadableOctets OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object is the number of octets contained in valid frames that have been received on this port. This counter is incremented by OctetCount for each frame received on this port which has been determined to be a readable frame (i.e., including FCS octets but excluding framing bits and dribble bits).

This statistic provides an indicator of the total data transferred.

This counter is a 64-bit version of rpPtrMonitorPortReadableOctets. It should be used by network management protocols which support 64-bit counters (e.g., SNMPv2).

Conformance clauses for this MIB are defined such that implementation of this object is not required in a repeater system which does not support 100 Mb/s. However, repeater systems with mixed 10 Mb/s and 100 Mb/s ports might implement this object across all ports, including 10 Mb/s. If this object is implemented, the value shall be a valid count as defined in the first paragraph of this description.

A discontinuity can occur in the value when the value of object

rpPtrMonitorPortLastChange changes."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.5"

::= { rpPtrMonitor100PortEntry 4 }

rpPtrMonitorAllRpPtrInfo OBJECT IDENTIFIER

::= { rpPtrMonitorPackage 4 }

rpPtrMonTable OBJECT-TYPE

SYNTAX SEQUENCE OF RpPtrMonEntry

MAX-ACCESS not-accessible

```
STATUS      current
DESCRIPTION
    "A table of information about each
    non-trivial repeater. The number of entries
    in this table is the same as the number of
    entries in the rptrInfoTable.

    The columnar object rptrInfoLastChange is
    used to indicate possible discontinuities of
    counter type columnar objects in this table."
::= { rptrMonitorAllRptrInfo 1 }

rptrMonEntry OBJECT-TYPE
    SYNTAX      RptrMonEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
        about a single non-trivial repeater."
    INDEX       { rptrInfoId }
    ::= { rptrMonTable 1 }

RptrMonEntry ::= SEQUENCE {
    rptrMonTxCollisions      Counter32,
    rptrMonTotalFrames      Counter32,
    rptrMonTotalErrors      Counter32,
    rptrMonTotalOctets      Counter32
}

rptrMonTxCollisions OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For an IEEE Std 802.3, Clause 9 (10 Mb/s) repeater,
        this counter is incremented every time the repeater
        state machine enters the TRANSMIT COLLISION state
        from any state other than ONE PORT LEFT
        (see IEEE Std 802.3, Figure 9-2).

        For an IEEE Std 802.3, Clause 27 repeater, this counter
        is incremented every time the repeater core state
        diagram enters the Jam state as a result of
        activity(ALL) > 1 (see IEEE Std 802.3, Figure 27-2).

        The approximate minimum time for rollover of this
        counter is 16 hours in a 10 Mb/s repeater and 1.6
        hours in a 100 Mb/s repeater."
    REFERENCE
        "IEEE Std 802.3, 30.4.1.1.8"
    ::= { rptrMonEntry 1 }

rptrMonTotalFrames OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The number of frames of valid frame length
        that have been received on the ports in this repeater
```

and for which the FCSError and CollisionEvent signals were not asserted. If an implementation can not obtain a count of frames as seen by the repeater itself, this counter might be implemented as the summation of the values of the rpPtrMonitorPortReadableFrames counters for all of the ports in the repeater.

This statistic provides one of the parameters necessary for obtaining the packet error ratio. The approximate minimum time for rollover of this counter is 80 hours in a 10 Mb/s repeater."

::= { rpPtrMonEntry 3 }

rpPtrMonTotalErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The total number of errors which have occurred on all of the ports in this repeater. The errors included in this count are the same as those listed for the rpPtrMonitorPortTotalErrors counter. If an implementation can not obtain a count of these errors as seen by the repeater itself, this counter might be implemented as the summation of the values of the rpPtrMonitorPortTotalErrors counters for all of the ports in the repeater."

::= { rpPtrMonEntry 4 }

rpPtrMonTotalOctets OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The total number of octets contained in the valid frames that have been received on the ports in this group. If an implementation can not obtain a count of octets as seen by the repeater itself, this counter might be the summation of the values of the rpPtrMonitorPortReadableOctets counters for all of the ports in the group.

This statistic provides an indicator of the total data transferred. The approximate minimum time for rollover of this counter in a 10 Mb/s repeater is 58 minutes divided by the number of ports in the repeater.

For 100 Mb/s repeaters processing traffic at a maximum rate, this counter can roll over in less than 6 minutes divided by the number of ports in the repeater. Since that amount of time could be less than a management station's poll cycle time, in order to avoid a loss of information a management station is advised to also poll the rpPtrMonUpper32TotalOctets object, or to use the 64-bit counter defined by rpPtrMonHCTotalOctets instead of the two 32-bit counters."

```
::= { rpPtrMonEntry 5 }
```

rpPtrMon100Table OBJECT-TYPE
SYNTAX SEQUENCE OF RptrMon100Entry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"A table of additional information about each
100 Mb/s repeater, augmenting the entries in
the rpPtrMonTable. Entries exist in this table
only for 100 Mb/s repeaters.

The columnar object rpPtrInfoLastChange is
used to indicate possible discontinuities of
counter type columnar objects in this table."
::= { rpPtrMonitorAllRptrInfo 2 }

rpPtrMon100Entry OBJECT-TYPE
SYNTAX RptrMon100Entry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"An entry in the table, containing information
about a single 100 Mb/s repeater."
INDEX { rpPtrInfoId }
::= { rpPtrMon100Table 1 }

RptrMon100Entry ::= SEQUENCE {
rpPtrMonUpper32TotalOctets Counter32,
rpPtrMonHCTotalOctets Counter64
}

rpPtrMonUpper32TotalOctets OBJECT-TYPE
SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The total number of octets contained in the valid
frames that have been received on the ports in
this repeater, modulo 2^{32} . That is, it contains
the upper 32 bits of a 64-bit counter, of which
the lower 32 bits are contained in the
rpPtrMonTotalOctets object. If an implementation
can not obtain a count of octets as seen
by the repeater itself, the 64-bit value
might be the summation of the values of the
rpPtrMonitorPortReadableOctets counters combined
with the corresponding rpPtrMonitorPortUpper32Octets
counters for all of the ports in the repeater.

This statistic provides an indicator of the total
data transferred within the repeater.

This two-counter mechanism is provided for those
network management protocols that do not support
64-bit counters (e.g., SNMP v1) and are used to
manage a repeater type of 100 Mb/s.

Conformance clauses for this MIB are defined such

that implementation of this object is not required in a repeater system which does not support 100 Mb/s. However, repeater systems with mixed 10 Mb/s and 100 Mb/s ports might implement this object across all ports, including 10 Mb/s. If this object is implemented, the value shall be a valid count as defined in the first paragraph of this description."

::= { rpPtrMon100Entry 1 }

rpPtrMonHCTotalOctets OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The total number of octets contained in the valid frames that have been received on the ports in this group. If a implementation can not obtain a count of octets as seen by the repeater itself, this counter might be the summation of the values of the rpPtrMonitorPortReadableOctets counters for all of the ports in the group.

This statistic provides an indicator of the total data transferred.

This counter is a 64-bit (high-capacity) version of rpPtrMonUpper32TotalOctets and rpPtrMonTotalOctets. It should be used by network management protocols which support 64-bit counters (e.g. SNMPv2).

Conformance clauses for this MIB are defined such that implementation of this object is not required in a repeater system which does not support 100 Mb/s. However, repeater systems with mixed 10 Mb/s and 100 Mb/s ports might implement this object across all ports, including 10 Mb/s. If this object is implemented, the value shall be a valid count as defined in the first paragraph of this description."

::= { rpPtrMon100Entry 2 }

rpPtrAddrTrackPackage OBJECT IDENTIFIER

::= { ieee8023snmpDot3RPtrMgt 3 }

rpPtrAddrTrackRPtrInfo OBJECT IDENTIFIER

::= { rpPtrAddrTrackPackage 1 }

rpPtrAddrSearchTable OBJECT-TYPE

SYNTAX SEQUENCE OF RPtrAddrSearchEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table contains one entry per repeater in the repeater system. It defines objects that allow a network management application to instruct an agent to watch for a given MAC address and report which port it was seen on. Only one address search can be in progress on each repeater at any one time. Before starting an address search, a management application should obtain 'ownership' of the entry in

rptrAddrSearchTable for the repeater that is to perform the search. This is accomplished with the rptrAddrSearchLock and rptrAddrSearchStatus as follows:

```
try_again:
    get(rptrAddrSearchLock, rptrAddrSearchStatus)
    while (rptrAddrSearchStatus != notInUse)
    {
        /* Loop waiting for objects to be available*/
        short delay
        get(rptrAddrSearchLock, rptrAddrSearchStatus)
    }

    /* Try to claim map objects */
    lock_value = rptrAddrSearchLock
    if ( set(rptrAddrSearchLock = lock_value,
            rptrAddrSearchStatus = inUse,
            rptrAddrSearchOwner = 'my-IP-address')
        == FAILURE)
        /* Another manager got the lock */
        goto try_again

    /* I have the lock */
    set (rptrAddrSearchAddress = <search target>)

    wait for rptrAddrSearchState to change from none

    if (rptrAddrSearchState == single)
        get (rptrAddrSearchGroup, rptrAddrSearchPort)
    /* release the lock, making sure not to overwrite
       anyone else's lock */
    set (rptrAddrSearchLock = lock_value+1,
        rptrAddrSearchStatus = notInUse,
        rptrAddrSearchOwner = '')
```

A management station first retrieves the values of the appropriate instances of the rptrAddrSearchLock and rptrAddrSearchStatus objects, periodically repeating the retrieval if necessary, until the value of rptrAddrSearchStatus is 'notInUse'. The management station then tries to set the same instance of the rptrAddrSearchLock object to the value it just retrieved, the same instance of the rptrAddrSearchStatus object to 'inUse', and the corresponding instance of rptrAddrSearchOwner to a value indicating itself. If the set operation succeeds, then the management station has obtained ownership of the rptrAddrSearchEntry, and the value of rptrAddrSearchLock is incremented by the agent (as per the semantics of TestAndIncr). Failure of the set operation indicates that some other manager has obtained ownership of the rptrAddrSearchEntry.

Once ownership is obtained, the management station can proceed with the search operation. Note that the agent will reset rptrAddrSearchStatus to 'notInUse' if it has been in the 'inUse' state for an abnormally long period of time, to prevent a misbehaving manager

from permanently locking the entry. It is suggested that this timeout period be between one and five minutes.

When the management station has completed its search operation, it should free the entry by setting the instance of the `rpPtrAddrSearchLock` object to the previous value + 1, the instance of the `rpPtrAddrSearchStatus` to 'notInUse', and the instance of `rpPtrAddrSearchOwner` to a zero length string. This is done to prevent overwriting another station's lock."

```
::= { rpPtrAddrTrackRpPtrInfo 1 }
```

`rpPtrAddrSearchEntry` OBJECT-TYPE

SYNTAX `RpPtrAddrSearchEntry`

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry containing objects for invoking an address search on a repeater."

INDEX { `rpPtrInfoId` }

```
::= { rpPtrAddrSearchTable 1 }
```

`RpPtrAddrSearchEntry` ::= SEQUENCE {

`rpPtrAddrSearchLock` TestAndIncr,

`rpPtrAddrSearchStatus` INTEGER,

`rpPtrAddrSearchAddress` MacAddress,

`rpPtrAddrSearchState` INTEGER,

`rpPtrAddrSearchGroup` Integer32,

`rpPtrAddrSearchPort` Integer32,

`rpPtrAddrSearchOwner` OwnerString

}

`rpPtrAddrSearchLock` OBJECT-TYPE

SYNTAX TestAndIncr

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object is used by a management station as an advisory lock for this `rpPtrAddrSearchEntry`."

```
::= { rpPtrAddrSearchEntry 1 }
```

`rpPtrAddrSearchStatus` OBJECT-TYPE

SYNTAX INTEGER { notInUse(1), inUse(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object is used to indicate that some management station is currently using this `rpPtrAddrSearchEntry`. Cooperating managers should set this object to 'notInUse' when they are finished using this entry. The agent will automatically set the value of this object to 'notInUse' if it has been set to 'inUse' for an unusually long period of time."

```
::= { rpPtrAddrSearchEntry 2 }
```

`rpPtrAddrSearchAddress` OBJECT-TYPE

SYNTAX MacAddress

```
MAX-ACCESS    read-write
STATUS        current
DESCRIPTION
    "This object is used to search for a specified MAC
    address. When this object is set, an address search
    begins. This automatically sets the corresponding
    instance of the rptrAddrSearchState object to 'none'
    and the corresponding instances of the
    rptrAddrSearchGroup and rptrAddrSearchPort objects to
    0.

    When a valid frame is received by this repeater with
    a source MAC address that matches the current value
    of rptrAddrSearchAddress, the agent will update the
    corresponding instances of rptrAddrSearchState,
    rptrAddrSearchGroup and rptrAddrSearchPort to reflect
    the current status of the search, and the group and
    port on which the frame was seen."
::= { rptrAddrSearchEntry 3 }
```

```
rptrAddrSearchState OBJECT-TYPE
    SYNTAX      INTEGER { none(1), single(2), multiple(3) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The current state of the MAC address search on this
        repeater. This object is initialized to 'none' when
        the corresponding instance of rptrAddrSearchAddress
        is set. If the agent detects the address on exactly
        one port, it will set this object to 'single', and
        set the corresponding instances of
        rptrAddrSearchGroup and rptrAddrSearchPort to reflect
        the group and port on which the address was heard.
        If the agent detects the address on more than one
        port, it will set this object to 'multiple'."
    ::= { rptrAddrSearchEntry 4 }
```

```
rptrAddrSearchGroup OBJECT-TYPE
    SYNTAX      Integer32 (0..2147483647)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The group from which an error-free frame whose
        source address is equal to the corresponding instance
        of rptrAddrSearchAddress has been received. The
        value of this object is undefined when the
        corresponding instance of rptrAddrSearchState is
        equal to 'none' or 'multiple'."
    ::= { rptrAddrSearchEntry 5 }
```

```
rptrAddrSearchPort OBJECT-TYPE
    SYNTAX      Integer32 (0..2147483647)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The port from which an error-free frame whose
        source address is equal to the corresponding instance
        of rptrAddrSearchAddress has been received. The
        value of this object is undefined when the
```



```
corresponding instance of rptrAddrSearchState is
equal to 'none' or 'multiple'."
::= { rptrAddrSearchEntry 6 }

rptrAddrSearchOwner OBJECT-TYPE
    SYNTAX      OwnerString
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "The entity that currently has 'ownership' of this
        rptrAddrSearchEntry."
    ::= { rptrAddrSearchEntry 7 }

rptrAddrTrackGroupInfo OBJECT IDENTIFIER
    ::= { rptrAddrTrackPackage 2 }

rptrAddrTrackPortInfo OBJECT IDENTIFIER
    ::= { rptrAddrTrackPackage 3 }

rptrAddrTrackTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RptrAddrTrackEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "Table of address mapping information about the
        ports."
    ::= { rptrAddrTrackPortInfo 1 }

rptrAddrTrackEntry OBJECT-TYPE
    SYNTAX      RptrAddrTrackEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "An entry in the table, containing address mapping
        information about a single port."
    INDEX       { rptrAddrTrackGroupIndex, rptrAddrTrackPortIndex }
    ::= { rptrAddrTrackTable 1 }

RptrAddrTrackEntry ::= SEQUENCE {
    rptrAddrTrackGroupIndex      Integer32,
    rptrAddrTrackPortIndex       Integer32,
    rptrAddrTrackSourceAddrChanges Counter32,
    rptrAddrTrackNewLastSrcAddress OptMacAddr,
    rptrAddrTrackCapacity        Integer32
}

rptrAddrTrackGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "This object identifies the group containing the
        port for which this entry contains information."
    ::= { rptrAddrTrackEntry 1 }

rptrAddrTrackPortIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS   not-accessible
    STATUS       current
```

DESCRIPTION

"This object identifies the port within the group
for which this entry contains information."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.1"

::= { rptrAddrTrackEntry 2 }

rpPtrAddrTrackSourceAddrChanges OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented by one for each time
that the rpPtrAddrTrackNewLastSrcAddress attribute
for this port has changed.

This might indicate whether a link is connected to a
single DTE or another multi-user segment.

A discontinuity can occur in the value when the
value of object rpPtrMonitorPortLastChange changes.

The approximate minimum time for rollover of this
counter is 81 hours in a 10 Mb/s repeater."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.19"

::= { rpPtrAddrTrackEntry 3 }

rpPtrAddrTrackNewLastSrcAddress OBJECT-TYPE

SYNTAX OptMacAddr

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object is the SourceAddress of the last
readable frame (i.e., counted by
rpPtrMonitorPortReadableFrames) received by this
port. If no frames have been received by this
port since the agent began monitoring the port
activity, the agent shall return a string of
length zero."

REFERENCE

"IEEE Std 802.3, 30.4.3.1.18"

::= { rpPtrAddrTrackEntry 4 }

rpPtrAddrTrackCapacity OBJECT-TYPE

SYNTAX Integer32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The maximum number of addresses that can be
detected on this port. This value indicates
to the maximum number of entries in the
rpPtrExtAddrTrackTable relative to this port.

If this object has the value of 1, the agent
implements only the LastSourceAddress mechanism
described by IETF RFC 1368 or IETF RFC 1516."

::= { rpPtrAddrTrackEntry 5 }

rpPtrExtAddrTrackTable OBJECT-TYPE

SYNTAX SEQUENCE OF RPtrExtAddrTrackEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"A table to extend the address tracking table (i.e., rpPtrAddrTrackTable) with a list of source MAC addresses that were recently received on each port. The number of ports is the same as the number of entries in table rpPtrPortTable. The number of entries in this table depends on the agent/repeater implementation and the number of different addresses received on each port.

The first entry for each port contains the same MAC address that is given by the rpPtrAddrTrackNewLastSrcAddress for that port.

Entries in this table for a particular port are retained when that port is switched from one repeater to another.

The ordering of MAC addresses listed for a particular port is implementation dependent."

::= { rpPtrAddrTrackPortInfo 2 }

rpPtrExtAddrTrackEntry OBJECT-TYPE

SYNTAX RPtrExtAddrTrackEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"A row in the table of extended address tracking information for ports. Entries cannot be directly created or deleted via SNMP operations."

INDEX { rpPtrAddrTrackGroupIndex, rpPtrAddrTrackPortIndex,
rpPtrExtAddrTrackMacIndex }

::= { rpPtrExtAddrTrackTable 1 }

RPtrExtAddrTrackEntry ::= SEQUENCE {
rpPtrExtAddrTrackMacIndex Integer32,
rpPtrExtAddrTrackSourceAddress MacAddress
}

rpPtrExtAddrTrackMacIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"The index of a source MAC address seen on the port.

The ordering of MAC addresses listed for a particular port is implementation dependent.

There is no implied relationship between a particular index and a particular MAC address. The index for a particular MAC address can change without notice."

::= { rpPtrExtAddrTrackEntry 1 }

```
rpPtrExtAddrTrackSourceAddress OBJECT-TYPE
    SYNTAX      MacAddress
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The source MAC address from a readable frame
         (i.e., counted by rpPtrMonitorPortReadableFrames)
         recently received by the port."
    REFERENCE
        "IEEE Std 802.3, 30.4.3.1.18"
    ::= { rpPtrExtAddrTrackEntry 2 }

rpPtrTopNPackage OBJECT IDENTIFIER
    ::= { ieee8023snmpDot3RPtrMgt 4 }

rpPtrTopNRPtrInfo OBJECT IDENTIFIER
    ::= { rpPtrTopNPackage 1 }

rpPtrTopNGroupInfo OBJECT IDENTIFIER
    ::= { rpPtrTopNPackage 2 }

rpPtrTopNPortInfo OBJECT IDENTIFIER
    ::= { rpPtrTopNPackage 3 }

rpPtrTopNPortControlTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RPtrTopNPortControlEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of control records for reports on the top 'N'
         ports for the rate of a selected counter. The number
         of entries depends on the configuration of the agent.
         The maximum number of entries is implementation
         dependent."
    ::= { rpPtrTopNPortInfo 1 }

rpPtrTopNPortControlEntry OBJECT-TYPE
    SYNTAX      RPtrTopNPortControlEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A set of parameters that control the creation of a
         report of the top N ports according to several metrics."
    INDEX      { rpPtrTopNPortControlIndex }
    ::= { rpPtrTopNPortControlTable 1 }

RPtrTopNPortControlEntry ::= SEQUENCE {
    rpPtrTopNPortControlIndex      Integer32,
    rpPtrTopNPortRepeaterId        Integer32,
    rpPtrTopNPortRateBase          INTEGER,
    rpPtrTopNPortTimeRemaining     Integer32,
    rpPtrTopNPortDuration           Integer32,
    rpPtrTopNPortRequestedSize     Integer32,
    rpPtrTopNPortGrantedSize       Integer32,
    rpPtrTopNPortStartTime         TimeStamp,
    rpPtrTopNPortOwner              OwnerString,
    rpPtrTopNPortRowStatus          RowStatus
}
```

```
rpPtrTopNPortControlIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..65535)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index that uniquely identifies an entry in the
         rpPtrTopNPortControl table. Each such entry defines
         one top N report prepared for a repeater or repeater system."
    ::= { rpPtrTopNPortControlEntry 1 }

rpPtrTopNPortRepeaterId OBJECT-TYPE
    SYNTAX      Integer32 (0..2147483647)
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "Identifies the repeater for which a top N report will
         be prepared (see rpPtrInfoId). If the value of this
         object is positive, only ports assigned to this repeater
         will be used to form the list in which to order the
         Top N table. If this value is zero, all ports will be
         eligible for inclusion on the list.

        The value of this object might be modified if the
         associated rpPtrTopNPortRowStatus object is equal to
         active(1).
        If, for a particular row in this table, the repeater
         specified by the value of this object goes away (is
         removed from the rpPtrInfoTable) while the associated
         rpPtrTopNPortRowStatus object is equal to active(1),
         the row in this table is preserved by the agent but
         the value of rpPtrTopNPortRowStatus is changed to
         notInService(2), and the agent might time out the row
         if appropriate. If the specified repeater comes
         back (reappears in the rpPtrInfoTable) before the row
         has been timed out, the management station sets
         the value of the rpPtrTopNPortRowStatus object back
         to active(1) if desired (the agent doesn't do this
         automatically)."
```

```
    ::= { rpPtrTopNPortControlEntry 2 }

rpPtrTopNPortRateBase OBJECT-TYPE
    SYNTAX      INTEGER { readableFrames(1), readableOctets(2),
                           fcsErrors(3), alignmentErrors(4), frameTooLongs(5),
                           shortEvents(6), runs(7), collisions(8),
                           lateEvents(9), veryLongEvents(10),
                           dataRateMismatches(11), autoPartitions(12),
                           totalErrors(13), isolates(14), symbolErrors(15) }
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "The monitored variable, which the rpPtrTopNPortRate
         variable is based upon.

        The value of this object may not be modified if
         the associated rpPtrTopNPortRowStatus object has
         a value of active(1)."
```

```
    ::= { rpPtrTopNPortControlEntry 3 }
```

rpPtrTopNPortTimeRemaining OBJECT-TYPE
SYNTAX Integer32 (0..2147483647)
MAX-ACCESS read-create
STATUS current
DESCRIPTION
 "The number of seconds left in the report currently being collected. When this object is modified by the management station, a new collection is started, possibly aborting a currently running report. The new value is used as the requested duration of this report, which is loaded into the associated rpPtrTopNPortDuration object.

 When this object is set to a non-zero value, any associated rpPtrTopNPortEntries shall be made inaccessible by the agent. While the value of this object is non-zero, it decrements by one per second until it reaches zero. During this time, all associated rpPtrTopNPortEntries shall remain inaccessible. At the time that this object decrements to zero, the report is made accessible in the rpPtrTopNPortTable. Thus, the rpPtrTopNPort table needs to be created only at the end of the collection interval.

 If the value of this object is set to zero while the associated report is running, the running report is aborted and no associated rpPtrTopNPortEntries are created."
DEFVAL { 0 }
 ::= { rpPtrTopNPortControlEntry 4 }

rpPtrTopNPortDuration OBJECT-TYPE
SYNTAX Integer32 (0..2147483647)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "The number of seconds that this report has collected during the last sampling interval, or if this report is currently being collected, the number of seconds that this report is being collected during this sampling interval.

 When the associated rpPtrTopNPortTimeRemaining object is set, this object shall be set by the agent to the same value and shall not be modified until the next time the rpPtrTopNPortTimeRemaining is set.

 This value shall be zero if no reports have been requested for this rpPtrTopNPortControlEntry."
 ::= { rpPtrTopNPortControlEntry 5 }

rpPtrTopNPortRequestedSize OBJECT-TYPE
SYNTAX Integer32
MAX-ACCESS read-create
STATUS current
DESCRIPTION

"The maximum number of repeater ports requested for the Top N Table.

When this object is created or modified, the agent should set `rpPtrTopNPortGrantedSize` as close to this object as is possible for the particular implementation and available resources."

DEFVAL { 10 }
::= { `rpPtrTopNPortControlEntry` 6 }

`rpPtrTopNPortGrantedSize` OBJECT-TYPE
SYNTAX Integer32 (0..65535)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The maximum number of repeater ports in the top N table.

When the associated `rpPtrTopNPortRequestedSize` object is created or modified, the agent should set this object as closely to the requested value as is possible for the particular implementation and available resources. The agent shall not lower this value except as a result of a set to the associated `rpPtrTopNPortRequestedSize` object."

::= { `rpPtrTopNPortControlEntry` 7 }

`rpPtrTopNPortStartTime` OBJECT-TYPE
SYNTAX TimeStamp
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The value of `sysUpTime` when this top N report was last started. In other words, this is the time that the associated `rpPtrTopNPortTimeRemaining` object was modified to start the requested report.

If the report has not yet been started, the value of this object is zero."

::= { `rpPtrTopNPortControlEntry` 8 }

`rpPtrTopNPortOwner` OBJECT-TYPE
SYNTAX OwnerString
MAX-ACCESS read-create
STATUS current
DESCRIPTION
"The entity that configured this entry and is using the resources assigned to it."

::= { `rpPtrTopNPortControlEntry` 9 }

`rpPtrTopNPortRowStatus` OBJECT-TYPE
SYNTAX RowStatus
MAX-ACCESS read-create
STATUS current
DESCRIPTION
"The status of this row.

If the value of this object is not equal to `active(1)`, all associated entries in the `rpPtrTopNPortTable` shall be deleted by the

```
agent."
 ::= { rpPtrTopNPortControlEntry 10 }

rpPtrTopNPortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF RptrTopNPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of reports for the top 'N' ports based on
        setting of associated control table entries. The
        maximum number of entries depends on the number
        of entries in table rpPtrTopNPortControlTable and
        the value of object rpPtrTopNPortGrantedSize for
        each entry.

        For each entry in the rpPtrTopNPortControlTable,
        repeater ports with the highest value of
        rpPtrTopNPortRate shall be placed in this table
        in decreasing order of that rate until there is
        no more room or until there are no more ports."
    ::= { rpPtrTopNPortInfo 2 }

rpPtrTopNPortEntry OBJECT-TYPE
    SYNTAX      RptrTopNPortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A set of statistics for a repeater port that is
        part of a top N report."
    INDEX       { rpPtrTopNPortControlIndex, rpPtrTopNPortIndex }
    ::= { rpPtrTopNPortTable 1 }

RptrTopNPortEntry ::= SEQUENCE {
    rpPtrTopNPortIndex      Integer32,
    rpPtrTopNPortGroupIndex Integer32,
    rpPtrTopNPortPortIndex  Integer32,
    rpPtrTopNPortRate       Gauge32
}

rpPtrTopNPortIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..65535)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index that uniquely identifies an entry in
        the rpPtrTopNPort table among those in the same
        report. This index is between 1 and N, where N
        is the number of entries in this report. Increasing
        values of rpPtrTopNPortIndex shall be assigned to
        entries with decreasing values of rpPtrTopNPortRate
        until index N is assigned to the entry with the
        lowest value of rpPtrTopNPortRate or there are no
        more rpPtrTopNPortEntries.

        No ports are included in a report where their
        value of rpPtrTopNPortRate would be zero."
    ::= { rpPtrTopNPortEntry 1 }

rpPtrTopNPortGroupIndex OBJECT-TYPE
```



```
SYNTAX      Integer32 (1..2147483647)
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This object identifies the group containing
    the port for this entry. (See also object
    type rpPtrGroupIndex.)"
 ::= { rpPtrTopNPortEntry 2 }

rpPtrTopNPortPortIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The index of the repeater port.
        (See object type rpPtrPortIndex.)"
    ::= { rpPtrTopNPortEntry 3 }

rpPtrTopNPortRate OBJECT-TYPE
    SYNTAX      Gauge32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The amount of change in the selected variable
        during this sampling interval for the identified
        port. The selected variable is that port's
        instance of the object selected by
        rpPtrTopNPortRateBase."
    ::= { rpPtrTopNPortEntry 4 }

snmpRptrModConf OBJECT IDENTIFIER
    ::= { ieee8023snmpRptrMIB 2 }

snmpRptrModCompls OBJECT IDENTIFIER
    ::= { snmpRptrModConf 1 }

snmpRptrModObjGrps OBJECT IDENTIFIER
    ::= { snmpRptrModConf 2 }

snmpRptrModNotGrps OBJECT IDENTIFIER
    ::= { snmpRptrModConf 3 }

rpPtrInfoHealth NOTIFICATION-TYPE
    OBJECTS      { rpPtrInfoOperStatus }
    STATUS      current
    DESCRIPTION
        "In a repeater system containing multiple managed repeaters,
        the rpPtrInfoHealth notification conveys information
        related to the operational status of a repeater.
        It is sent either when the value of rpPtrInfoOperStatus
        changes, or upon completion of a non-disruptive test.

        The agent shall limit the generation of
        consecutive rpPtrInfoHealth notifications for
        the same repeater so that there is at least
        a five-second gap between notifications of this type.
        When notifications are throttled, they are dropped,
        not queued for sending at a future time. (Note
        that 'generating' a notification means sending
```

```
        to all configured recipients.)"
REFERENCE
    "IEEE Std 802.3, 30.4.1.3.1
    notification."
::= { ieee8023snmpDot3RptrNotifications 4 }

rptrInfoResetEvent NOTIFICATION-TYPE
OBJECTS      { rptrInfoOperStatus }
STATUS       current
DESCRIPTION
    "In a repeater system containing multiple managed
    repeaters, the rptrInfoResetEvent notification
    conveys information related to the operational
    status of a repeater. This notification is sent
    on completion of a repeater reset action. A
    repeater reset action is defined as a transition
    to the START state of IEEE Std 802.3, Figure 9-2,
    when triggered by a management command
    (e.g., an SNMP Set on the rptrInfoReset object).

    The agent shall limit the generation of
    consecutive rptrInfoResetEvent notifications for
    a single repeater so that there is at least
    a five-second gap between notifications of
    this type. When notifications are throttled,
    they are dropped, not queued for sending at
    a future time. (Note that 'generating' a
    notification means sending to all configured
    recipients.)

    The rptrInfoResetEvent is not sent when the
    agent restarts and sends an SNMP coldStart or
    warmStart trap. However, it is recommended that
    a repeater agent send the rptrInfoOperStatus
    object as an optional object with its coldStart
    and warmStart trap PDUs."
REFERENCE
    "IEEE Std 802.3, 30.4.1.3.2"
::= { ieee8023snmpDot3RptrNotifications 5 }

snmpRptrGrpBasic OBJECT-GROUP
OBJECTS      { rptrGroupObjectID, rptrGroupOperStatus,
                rptrGroupPortCapacity, rptrPortAdminStatus,
                rptrPortAutoPartitionState, rptrPortOperStatus,
                rptrPortRpPtrId, rptrInfoRpPtrType, rptrInfoOperStatus,
                rptrInfoReset, rptrInfoPartitionedPorts,
                rptrInfoLastChange }
STATUS       current
DESCRIPTION
    "Basic group for a repeater system with one or more
    repeater-units in multisegment (post-RFC 1516)
    version of the MIB module."
::= { snmpRpPtrModObjGrps 1 }

snmpRpPtrGrpMonitor OBJECT-GROUP
OBJECTS      { rptrMonitorPortReadableFrames,
                rptrMonitorPortReadableOctets,
                rptrMonitorPortFCSErrors,
                rptrMonitorPortAlignmentErrors,
```

```
        rpPtrMonitorPortFrameTooLongs,
        rpPtrMonitorPortShortEvents, rpPtrMonitorPortRuns,
        rpPtrMonitorPortCollisions, rpPtrMonitorPortLateEvents,
        rpPtrMonitorPortVeryLongEvents,
        rpPtrMonitorPortDataRateMismatches,
        rpPtrMonitorPortAutoPartitions,
        rpPtrMonitorPortTotalErrors, rpPtrMonitorPortLastChange,
        rpPtrMonTxCollisions, rpPtrMonTotalFrames,
        rpPtrMonTotalErrors, rpPtrMonTotalOctets }
STATUS      current
DESCRIPTION
    "Monitor group for a repeater system with one or more
    repeater-units in multisegment (post-RFC 1516)
    version of the MIB module."
::= { snmpRptrModObjGrps 2 }

snmpRptrGrpMonitor100 OBJECT-GROUP
OBJECTS      { rpPtrMonitorPortIsolates, rpPtrMonitorPortSymbolErrors,
               rpPtrMonitorPortUpper32Octets,
               rpPtrMonUpper32TotalOctets }
STATUS      current
DESCRIPTION
    "Monitor group for 100 Mb/s ports and repeaters
    in a repeater system with one or more repeater-units in
    multisegment (post-RFC 1516) version of the MIB
    module. Repeater systems which support Counter64 should
    also implement snmpRptrGrpMonitor100w64."
::= { snmpRptrModObjGrps 3 }

snmpRptrGrpMonitor100w64 OBJECT-GROUP
OBJECTS      { rpPtrMonitorPortHCReadableOctets, rpPtrMonHCTotalOctets }
STATUS      current
DESCRIPTION
    "Monitor group for 100 Mb/s ports and repeaters in a
    repeater system with one or more repeater-units and support
    for Counter64."
::= { snmpRptrModObjGrps 4 }

snmpRptrGrpAddrTrack OBJECT-GROUP
OBJECTS      { rpPtrAddrTrackSourceAddrChanges,
               rpPtrAddrTrackNewLastSrcAddress, rpPtrAddrTrackCapacity }
STATUS      current
DESCRIPTION
    "Passive address tracking group for post-RFC 1516
    version of the MIB module."
::= { snmpRptrModObjGrps 5 }

snmpRptrGrpExtAddrTrack OBJECT-GROUP
OBJECTS      { rpPtrExtAddrTrackSourceAddress }
STATUS      current
DESCRIPTION
    "Extended passive address tracking group for
    a repeater system with one or more repeater-units in
    post-RFC 1516 version of the MIB module."
::= { snmpRptrModObjGrps 6 }

snmpRptrGrpRptrAddrSearch OBJECT-GROUP
OBJECTS      { rpPtrAddrSearchLock, rpPtrAddrSearchStatus,
               rpPtrAddrSearchAddress, rpPtrAddrSearchState,
```

```

        rptrAddrSearchGroup, rptrAddrSearchPort,
        rptrAddrSearchOwner }
STATUS      current
DESCRIPTION
    "Active MAC address search group and topology
    mapping support for repeaters."
::= { snmpRptrModObjGrps 7 }

snmpRptrGrpTopNPort OBJECT-GROUP
OBJECTS     { rptrTopNPortRepeaterId, rptrTopNPortRateBase,
              rptrTopNPortTimeRemaining, rptrTopNPortDuration,
              rptrTopNPortRequestedSize, rptrTopNPortGrantedSize,
              rptrTopNPortStartTime, rptrTopNPortOwner,
              rptrTopNPortRowStatus, rptrTopNPortGroupIndex,
              rptrTopNPortPortIndex, rptrTopNPortRate }
STATUS      current
DESCRIPTION
    "Top 'N' group for repeater ports."
::= { snmpRptrModObjGrps 8 }

ieee8023snmpDot3RptrNotGroup NOTIFICATION-GROUP
NOTIFICATIONS { rptrInfoHealth, rptrInfoResetEvent }
STATUS        current
DESCRIPTION
    "Conformance Group for repeater notifications.
    Formerly an empty group."
::= { snmpRptrModNotGrps 1 }

snmpRptrModCompl MODULE-COMPLIANCE
STATUS      current
DESCRIPTION
    "Compliance for the multisegment version of the
    MIB module for a repeater system with one or more
    repeater-units."

MODULE      -- this module

MANDATORY-GROUPS { snmpRptrGrpBasic, snmpRptrGrpMonitor,
                  snmpRptrGrpAddrTrack }

GROUP      snmpRptrGrpMonitor100
DESCRIPTION
    "Implementation of this group is
    mandatory for managed repeater systems that
    contain 100 Mb/s repeaters."

GROUP      snmpRptrGrpMonitor100w64
DESCRIPTION
    "Implementation of this group is
    mandatory for managed repeater systems that
    contain 100 Mb/s repeaters and that
    can support Counter64."

GROUP      snmpRptrGrpExtAddrTrack
DESCRIPTION
    "Implementation of this group is
    recommended for repeater systems that have
    the necessary instrumentation to track
    MAC addresses of multiple DTEs attached
```

to a single repeater port."

```
GROUP    snmpRptrGrpRptrAddrSearch
DESCRIPTION
    "Implementation of this group is
    recommended for repeater systems that allow
    read-write access and that have
    the necessary instrumentation to
    search all incoming data streams
    for a particular MAC address."
```

```
GROUP    snmpRptrGrpTopNPort
DESCRIPTION
    "Implementation of this group is
    recommended for repeater systems that have
    the necessary resources to support
    TopN statistics reporting."
```

```
GROUP    ieee8023snmpDot3RptrNotGroup
DESCRIPTION
    "Implementation of this group is
    recommended for repeaters that
    support notifications."
```

```
::= { snmpRptrModCompls 1 }
```

```
END -- end of module IEEE8023-SNMP-REPEATER-MIB.
```

8. Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) MIB module

8.1 Introduction

This clause defines a portion of the MIB for use with SNMP. In particular, it defines a set of MIB objects to manage Power via MDI Power Sourcing Equipment (PSE).

8.2 Overview

IEEE Std 802.3 defines the hardware registers that will allow for management interfaces to be built for a DTE Power via MDI device. The MIB module defined in this clause extends the Ethernet-like interface MIB defined in Clause 10 with the management objects required for the management of the DTE Power via MDI devices and ports.

8.3 MIB structure

These MIB objects are categorized into three MIB groups.

The `pethPsePortTable` defines the objects used for configuring and describing the status of ports on a PSE device. Examples of PSE devices are Ethernet switches that support Power over Ethernet and mid-span devices.

The `pethMainPseObjects` MIB group defines the management objects for a managed main power source in a PSE device. Ethernet switches are one example of devices that would support these objects.

The `pethNotificationControlTable` includes objects that control the transmission of notifications from the agent to a management application.

8.4 Security considerations for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) MIB module

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write. Such objects can be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Setting the following objects to incorrect values can result in improper operation of the PSE, including the possibility that the PD does not receive power from the PSE port:

- `pethPsePortAdminEnable`
- `pethPsePortPowerPairs`
- `pethPsePortPowerPriority`
- `pethPsePortType`

Setting the following objects to incorrect values can result in an excessive number of traps being sent to network management stations:

- `pethMainPseUsageThreshold`
- `pethNotificationControlEnable`

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not-accessible) can be considered sensitive or vulnerable in some network environments. These are as follows:

- pethPsePortPowerPairsControlAbility
- pethPsePortPowerPriority
- pethPsePortPowerClassifications

It is thus important to control GET and/or NOTIFY access to these objects and possibly to encrypt their values when sending them over the network via SNMP.

8.5 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²²

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C8mib.txt

²² Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

8.5.1 Tree hierarchy

```

----- ieee8023powerEthernetMIB(1.3.111.2.802.3.1.8)
+----- pethNotifications(0)
| +----- pethPsePortOnOffNotification(1) {pethPsePortDetectionStatus}
| +----- pethMainPowerUsageOnNotification(2) {pethMainPseConsumptionPower}
| +----- pethMainPowerUsageOffNotification(3) {pethMainPseConsumptionPower}
+----- pethObjects(1)
| +----- pethPsePortTable(1)
| | +----- pethPsePortEntry(1) [pethPsePortGroupIndex,pethPsePortIndex]
| | | Integer32
| | | Integer32
| | | TruthValue
| | | TruthValue
| | | Enumeration
| | | Enumeration
| | | Enumeration
| | | Counter32
| | | SnmpAdminString
| | | Enumeration
| | | Counter32
| | | Counter32
| | | Counter32
| | | Counter32
| | | Integer32
| | | Integer32
| | | Counter32
| | | TruthValue
| +----- pethMainPseObjects(3)
| | +----- pethMainPseTable(1)
| | | +----- pethMainPseEntry(1) [pethMainPseGroupIndex]
| | | | +----- pethMainPseGroupIndex(1) Integer32
| | | | +---r- pethMainPsePower(2) Gauge32
| | | | +---r- pethMainPseOperStatus(3) Enumeration
| | | | +---r- pethMainPseConsumptionPower(4) Gauge32
| | | | +---rw pethMainPseUsageThreshold(5) Integer32

```



```

| +----- pethNotificationControl(4)
| | +----- pethNotificationControlTable(1)
| | | +----- pethNotificationControlEntry(1) [pethNotificationControlGroupIndex]
| | | +----- pethNotificationControlGroupIndex(1) Integer32
| | | +---rw pethNotificationControlEnable(2) TruthValue
+----- pethConformance(2)
| +----- pethCompliances(1)
| | +----- pethCompliance(1)
| | +----- pethGroups(2)
| | | +----- pethPsePortGroup(1)
| | | +----- pethMainPseGroup(2)
| | | +----- pethNotificationControlGroup(3)
| | | +----- pethPsePortNotificationGroup(4)
| | | +----- pethMainPowerNotificationGroup(5)

```

8.5.2 MIB module definition

```
IEEE8023-POWER-ETHERNET-MIB DEFINITIONS ::= BEGIN

IMPORTS
    SnmpAdminString
        FROM SNMP-FRAMEWORK-MIB
    MODULE-COMPLIANCE, NOTIFICATION-GROUP, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Gauge32, Integer32, MODULE-IDENTITY, NOTIFICATION-TYPE,
    OBJECT-TYPE, org
        FROM SNMPv2-SMI
    TruthValue
        FROM SNMPv2-TC;

ieee8023powerEthernetMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "The MIB module for managing Power Source Equipment
        (PSE) specified in IEEE Std 802.3 Clause 33."
    REVISION    "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION    "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION    "201102020000Z"
    DESCRIPTION
        "Initial version, based on an earlier version published
        as RFC 3621."
    ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
    man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 8 }

pethNotifications OBJECT IDENTIFIER
    ::= { ieee8023powerEthernetMIB 0 }

pethObjects OBJECT IDENTIFIER
    ::= { ieee8023powerEthernetMIB 1 }
```

```
pethPsePortTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF PethPsePortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of objects that display and control the power
        characteristics of Power over Ethernet ports on a Power Source
        Equipment (PSE) device. This group will be implemented in
        managed Power over Ethernet switches and mid-span devices.
        Values of all read-write objects in this table are
        persistent at restart/reboot."
    ::= { pethObjects 1 }
```

```
pethPsePortEntry OBJECT-TYPE
    SYNTAX      PethPsePortEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A set of objects that display and control the power
        characteristics of a Power over Ethernet PSE port."
    INDEX       { pethPsePortGroupIndex, pethPsePortIndex }
    ::= { pethPsePortTable 1 }
```

```
PethPsePortEntry ::= SEQUENCE {
    pethPsePortGroupIndex      Integer32,
    pethPsePortIndex           Integer32,
    pethPsePortAdminEnable     TruthValue,
    pethPsePortPowerPairsControlAbility TruthValue,
    pethPsePortPowerPairs      INTEGER,
    pethPsePortDetectionStatus INTEGER,
    pethPsePortPowerPriority    INTEGER,
    pethPsePortMPSAbsentCounter Counter32,
    pethPsePortType            SnmpAdminString,
    pethPsePortPowerClassifications INTEGER,
    pethPsePortInvalidSignatureCounter Counter32,
    pethPsePortPowerDeniedCounter Counter32,
    pethPsePortOverLoadCounter Counter32,
    pethPsePortShortCounter    Counter32,
    pethPsePortActualPower     Integer32,
    pethPsePortPowerAccuracy    Integer32,
    pethPsePortCumulativeEnergy Counter32,
    pethPsePortAdminState      TruthValue
}
```

```
pethPsePortGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This variable uniquely identifies the group
        containing the port to which a Power over Ethernet PSE is
        connected. Group means box in the stack, module in a
        rack and the value 1 shall be used for non-modular devices.
        Furthermore, the same value shall be used in this variable,
```

pethMainPseGroupIndex, and pethNotificationControlGroupIndex
to refer to a given box in a stack or module in a rack."
::= { pethPsePortEntry 1 }

pethPsePortIndex OBJECT-TYPE
SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"This variable uniquely identifies the Power over Ethernet PSE
port within group pethPsePortGroupIndex to which the
Power over Ethernet PSE entry is connected."
::= { pethPsePortEntry 2 }

pethPsePortAdminEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"true (1) Enable PSE functions.
false (2) Disable PSE functions."
REFERENCE
"IEEE Std 802.3, 30.9.1.2.1"
::= { pethPsePortEntry 3 }

pethPsePortPowerPairsControlAbility OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"Describes the capability of controlling the power pairs
functionality to switch pins for sourcing power.
The value true indicates that the device has the capability
to control the power pairs. When false the PSE Pinout
Alternative used cannot be controlled through the
PethPsePortAdminEnable attribute."
REFERENCE
"IEEE Std 802.3, 30.9.1.1.3"
::= { pethPsePortEntry 4 }

pethPsePortPowerPairs OBJECT-TYPE
SYNTAX INTEGER { signal(1), spare(2), both(3) }
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Describes or controls the pairs in use. If the value of
pethPsePortPowerPairsControl is true, this object is
writeable.
A value of signal(1) means that the signal pairs
only are in use.
A value of spare(2) means that the spare pairs
only are in use.
A value of both(3) means that both signal and spare
pairs are in use"

REFERENCE

"IEEE Std 802.3, 30.9.1.1.4"
::= { pethPsePortEntry 5 }

pethPsePortDetectionStatus OBJECT-TYPE

SYNTAX INTEGER { disabled(1), searching(2), deliveringPower(3),
fault(4), test(5), otherFault(6) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Describes the operational status of the port PD detection.
A value of disabled(1)- indicates that the PSE State diagram
is in the state DISABLED.
A value of searching(2)- indicates the PSE State diagram is
in a state other than one of the states listed here.
A value of deliveringPower(3) - indicates that the PSE State
diagram is in the state POWER_ON for a duration greater than
tlim max (see IEEE Std 802.3, Table 33-11).
A value of fault(4) - indicates that the PSE State diagram is
in the state TEST_ERROR.
A value of test(5) - indicates that the PSE State diagram is
in the state TEST_MODE.
A value of otherFault(6) - indicates that the PSE State
diagram is in the state IDLE due to the variable
error_conditions."

REFERENCE

"IEEE Std 802.3, 30.9.1.1.5"
::= { pethPsePortEntry 6 }

pethPsePortPowerPriority OBJECT-TYPE

SYNTAX INTEGER { critical(1), high(2), low(3) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object controls the priority of the port from the point
of view of a power management algorithm. The priority that
is set by this variable could be used by a control mechanism
that prevents overcurrent situations by disconnecting first
ports with lower power priority. Ports that connect devices
critical to the operation of the network - like the E911
telephones ports - should be set to higher priority."

REFERENCE

"IEEE Std 802.3, 79.3.2.4"
::= { pethPsePortEntry 7 }

pethPsePortMPSAbsentCounter OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This counter is incremented when the PSE doesn't detect a PD when
powering up."

REFERENCE

"IEEE Std 802.3, 30.9.1.1.20"

```
::= { pethPsePortEntry 8 }

pethPsePortType OBJECT-TYPE
    SYNTAX      SnmpAdminString
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "A manager will set the value of this variable to indicate
        the type of powered device that is connected to the port.
        The default value supplied by the agent if no value has
        ever been set should be a zero-length octet string."
    REFERENCE
        "IEEE Std 802.3, 79.3.2.4"
    ::= { pethPsePortEntry 9 }

pethPsePortPowerClassifications OBJECT-TYPE
    SYNTAX INTEGER { class0(1), class1(2), class2(3), class3(4),
                    class4(5), class5(6), class6(7), class7(8), class8(9) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "Classification is a way to tag different terminals on the
        Power over Ethernet network according to their power consumption.

        The meaning of the classification labels is defined in
        IEEE Std 802.3 Clause 145.

        This variable is valid only while a PD is being powered,
        that is, while the attribute pethPsePortDetectionStatus
        is reporting the enumeration deliveringPower."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.8"
    ::= { pethPsePortEntry 10 }

pethPsePortInvalidSignatureCounter OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This counter is incremented when a Type 1 or Type 2 PSE
        detects an invalid signature from the port."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.11"
    ::= { pethPsePortEntry 11 }

pethPsePortPowerDeniedCounter OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This counter is incremented when the PSE denies power to the PD."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.14"
    ::= { pethPsePortEntry 12 }
```

```
pethPsePortOverLoadCounter OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This counter is incremented when the PSE ouput current
        has gone into overload."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.17"
    ::= { pethPsePortEntry 13 }
```

```
pethPsePortShortCounter OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      obsolete
    DESCRIPTION
        "Obsolete."
    ::= { pethPsePortEntry 14 }
```

```
pethPsePortActualPower OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "Milliwatts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The power being supplied by the PSE to the PD
        measured at the MDI."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.23"
    ::= { pethPsePortEntry 15 }
```

```
pethPsePortPowerAccuracy OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "Milliwatts"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The accuracy of the power measurement at MDI."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.24"
    ::= { pethPsePortEntry 16 }
```

```
pethPsePortCumulativeEnergy OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Millijoules"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The cumulative energy supplied by the PSE ."
    REFERENCE
        "IEEE Std 802.3, 30.9.1.1.25"
    ::= { pethPsePortEntry 17 }
```

pethPsePortAdminState OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"true (1) PSE functions are enabled.
false(2) PSE functions are disabled."
REFERENCE
"IEEE Std 802.3, 30.9.1.1.2"
::= { pethPsePortEntry 18 }

pethMainPseObjects OBJECT IDENTIFIER

::= { pethObjects 3 }

pethMainPseTable OBJECT-TYPE

SYNTAX SEQUENCE OF PethMainPseEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"A table of objects that display and control attributes
of the main power source in a PSE device. Ethernet
switches are one example of devices that would support
these objects.
Values of all read-write objects in this table are
persistent at restart/reboot."
::= { pethMainPseObjects 1 }

pethMainPseEntry OBJECT-TYPE

SYNTAX PethMainPseEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"A set of objects that display and control the Main
power of a PSE."
INDEX { pethMainPseGroupIndex }
::= { pethMainPseTable 1 }

PethMainPseEntry ::= SEQUENCE {
pethMainPseGroupIndex Integer32,
pethMainPsePower Gauge32,
pethMainPseOperStatus INTEGER,
pethMainPseConsumptionPower Gauge32,
pethMainPseUsageThreshold Integer32
}

pethMainPseGroupIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"This variable uniquely identifies the group to which the
Power over Ethernet PSE is connected. Group means (box in
the stack, module in a rack) and the value 1 shall be
used for non-modular devices. Furthermore, the same

value shall be used in this variable, pethPsePortGroupIndex,
and pethNotificationControlGroupIndex to refer to a
given box in a stack or module in a rack."
::= { pethMainPseEntry 1 }

pethMainPsePower OBJECT-TYPE
SYNTAX Gauge32 (1..65535)
UNITS "Watts"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The nominal power of the PSE expressed in Watts."
::= { pethMainPseEntry 2 }

pethMainPseOperStatus OBJECT-TYPE
SYNTAX INTEGER { on(1), off(2), faulty(3) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"The operational status of the main PSE."
::= { pethMainPseEntry 3 }

pethMainPseConsumptionPower OBJECT-TYPE
SYNTAX Gauge32
UNITS "Watts"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"Measured usage power expressed in Watts."
::= { pethMainPseEntry 4 }

pethMainPseUsageThreshold OBJECT-TYPE
SYNTAX Integer32 (1..99)
UNITS "%"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"The usage threshold expressed in percents for
comparing the measured power and initiating
an alarm if the threshold is exceeded."
::= { pethMainPseEntry 5 }

pethNotificationControl OBJECT IDENTIFIER
::= { pethObjects 4 }

pethNotificationControlTable OBJECT-TYPE
SYNTAX SEQUENCE OF PethNotificationControlEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"A table of objects that display and control the
Notification on a PSE device.
Values of all read-write objects in this table are
persistent at restart/reboot."

```
::= { pethNotificationControl 1 }

pethNotificationControlEntry OBJECT-TYPE
    SYNTAX      PethNotificationControlEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A set of objects that control the Notification events."
    INDEX       { pethNotificationControlGroupIndex }
    ::= { pethNotificationControlTable 1 }

PethNotificationControlEntry ::= SEQUENCE {
    pethNotificationControlGroupIndex  Integer32,
    pethNotificationControlEnable      TruthValue
}

pethNotificationControlGroupIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This variable uniquely identifies the group. Group
        means box in the stack, module in a rack and the value
        1 shall be used for non-modular devices. Furthermore,
        the same value shall be used in this variable,
        pethPsePortGroupIndex, and
        pethMainPseGroupIndex to refer to a given box in a
        stack or module in a rack."
    ::= { pethNotificationControlEntry 1 }

pethNotificationControlEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "This object controls, on a per-group basis, whether
        or not notifications from the agent are enabled. The
        value true(1) means that notifications are enabled; the
        value false(2) means that they are not."
    ::= { pethNotificationControlEntry 2 }

pethConformance OBJECT IDENTIFIER
    ::= { ieee8023powerEthernetMIB 2 }

pethCompliances OBJECT IDENTIFIER
    ::= { pethConformance 1 }

pethGroups OBJECT IDENTIFIER
    ::= { pethConformance 2 }

pethPsePortOnOffNotification NOTIFICATION-TYPE
    OBJECTS      { pethPsePortDetectionStatus }
    STATUS      current
    DESCRIPTION
```

```
"This Notification indicates if Pse Port is delivering power
to the PD or not. This Notification should be sent on
every status change except in the searching mode.
At least 500 msec shall elapse between notifications
being emitted by the same object instance."
::= { pethNotifications 1 }

pethMainPowerUsageOnNotification NOTIFICATION-TYPE
OBJECTS      { pethMainPseConsumptionPower }
STATUS       current
DESCRIPTION
    "This Notification indicates PSE Threshold usage
    indication is on, the usage power is above the
    threshold. At least 500 msec shall elapse between
    notifications being emitted by the same object
    instance."
::= { pethNotifications 2 }

pethMainPowerUsageOffNotification NOTIFICATION-TYPE
OBJECTS      { pethMainPseConsumptionPower }
STATUS       current
DESCRIPTION
    "This Notification indicates PSE Threshold usage indication
    off, the usage power is below the threshold.
    At least 500 msec shall elapse between notifications being
    emitted by the same object instance."
::= { pethNotifications 3 }

pethPsePortGroup OBJECT-GROUP
OBJECTS      { pethPsePortAdminEnable,
                pethPsePortPowerPairsControlAbility,
                pethPsePortPowerPairs, pethPsePortDetectionStatus,
                pethPsePortPowerPriority, pethPsePortMPSAbsentCounter,
                pethPsePortInvalidSignatureCounter,
                pethPsePortPowerDeniedCounter,
                pethPsePortOverLoadCounter, pethPsePortType,
                pethPsePortPowerClassifications,
                pethPsePortActualPower, pethPsePortPowerAccuracy,
                pethPsePortCumulativeEnergy, pethPsePortAdminState }
STATUS       current
DESCRIPTION
    "PSE Port objects."
::= { pethGroups 1 }

pethMainPseGroup OBJECT-GROUP
OBJECTS      { pethMainPsePower, pethMainPseOperStatus,
                pethMainPseConsumptionPower, pethMainPseUsageThreshold }
STATUS       current
DESCRIPTION
    "Main PSE Objects."
::= { pethGroups 2 }

pethNotificationControlGroup OBJECT-GROUP
OBJECTS      { pethNotificationControlEnable }
```

```
STATUS      current
DESCRIPTION
    "Notification Control Objects."
::= { pethGroups 3 }

pethPsePortNotificationGroup NOTIFICATION-GROUP
    NOTIFICATIONS { pethPsePortOnOffNotification }
    STATUS      current
    DESCRIPTION
        "Pse Port Notifications."
    ::= { pethGroups 4 }

pethMainPowerNotificationGroup NOTIFICATION-GROUP
    NOTIFICATIONS { pethMainPowerUsageOnNotification,
                    pethMainPowerUsageOffNotification }
    STATUS      current
    DESCRIPTION
        "Main PSE Notifications."
    ::= { pethGroups 5 }

pethCompliance MODULE-COMPLIANCE
    STATUS      current
    DESCRIPTION
        "Describes the requirements for conformance to the
        Power Ethernet MIB."

    MODULE      -- this module

        MANDATORY-GROUPS      { pethPsePortGroup,
                                pethPsePortNotificationGroup,
                                pethNotificationControlGroup }

        GROUP      pethMainPseGroup
        DESCRIPTION
            "The pethMainPseGroup is mandatory for PSE systems
            that implement a main power supply."

        GROUP      pethMainPowerNotificationGroup
        DESCRIPTION
            "The pethMainPowerNotificationGroup is mandatory for
            PSE systems that implement a main power supply."

    ::= { pethCompliances 1 }

END -- end of module IEEE8023-POWER-ETHERNET-MIB.
```

9. Ethernet passive optical networks (EPON) MIB module

9.1 Overview

This clause defines a MIB module for use with SNMP to manage 1G-EPON and 10G-EPON (collectively referred to as EPON) interfaces for Ethernet Passive Optical Networks. The clause contains a list of management objects based on the attributes defined in the relevant parts of Clause 30 of IEEE Std 802.3, referring to EPON.

9.1.1 EPON architecture highlights

9.1.1.1 Introduction

The EPON standard, now part of IEEE Std 802.3, defines the Physical Layer and Media Access Control sublayer of EPON interfaces. EPON is a variant of Gigabit Ethernet used in optical access. The passive optical network (PON) comprises sections of single-mode fiber connected with passive optical splitter/coupler devices, forming a passive optical tree, as shown in Figure 9-1. Individual branches of the PON are terminated with the optical line terminal (OLT) in the central office and optical network units (ONUs) near the subscribers. ONUs can be located either in some remote location (e.g., basement in a multidwelling unit) or directly at the subscriber premises. Various types of customer premises equipment (CPE) can be connected to ONUs or even integrated with such devices. Figure 9-1 presents an example PON topology.

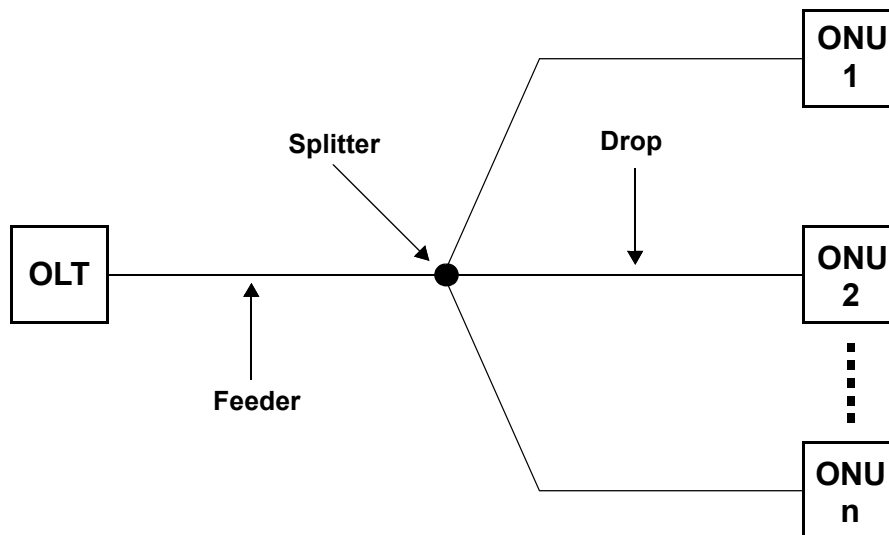


Figure 9-1—PON topology example

The IEEE layering architecture of an EPON interface is defined in the diagram of Figure 56-3 in IEEE Std 802.3. The following clauses in IEEE Std 802.3 define the corresponding layers of an EPON interface:

- Clause 30: Management.
- Clause 60: PMD for 1G-EPON media (burst-mode PMD), and Clause 75: PMD for 10G-EPON media (burst-mode PMD).
- Clause 64: MPCP (Multipoint Control Protocol), which defines the Multipoint architecture, and control protocol for the media access of 1G-EPON, and Clause 77: MPCP for 10G-EPON.

- Clause 65: Reconciliation Sublayer and Physical Coding Sublayer, which defines a number of extensions to the standard Gigabit Ethernet PCS:
 - a) Definition of Point-to-Point emulation function for 1G-EPON.
 - b) Definition of the optional (frame-based) FEC for 1G-EPON.
 - c) PMA for 1G-EPON.
- Clause 76: Reconciliation Sublayer and Physical Coding Sublayer for 10G-EPON.

9.1.1.2 Principles of operation

The 1G-EPON interface specification extends the specification of Gigabit Ethernet as described in Clause 35 and Clause 36 of IEEE Std 802.3. The Ethernet MAC operates at the data rate of 1 Gb/s, and it is connected to a media-dependent interface through the GMII interface, as described in Clause 35. The EPON PCS layer extends the Gigabit Ethernet PCS as described in Clause 36. New EPON-specific layers are added to Gigabit Ethernet layers in the following locations:

- MPCP is placed in the MAC control layer, providing EPON media access, station discovery, and registration protocol.
- Functionality of the reconciliation sublayer (RS) of Gigabit Ethernet was extended, creating logical links over shared passive optical medium, providing private transmission channels to each of the connected ONUs.
- (Optional) FEC functionality located between the PCS and PMA layers was added, extending the Gigabit Ethernet PCS layer, enhancing reach and split performance of the EPON optical link.

Similarly, the 10G-EPON interface specification extends the specification of 10 Gigabit Ethernet. The Ethernet MAC operates at the data rate of 10 Gb/s, and it is connected to a media-dependent interface through the XGMII interface. The 10G-EPON PCS layer extends the 10 Gigabit Ethernet PCS.

Figure 9-2 presents the EPON layering model.

9.1.1.3 Physical media

The physical link in EPON comprises single-mode fiber. The OLT and ONUs are connected through a passive optical network comprising sections of single-mode fiber interconnected with passive splitter/coupler devices.

The term *downstream* denotes transmission from the OLT to all connected ONUs, while the term *upstream* denotes transmission from the connected ONUs (one at the time) to the OLT. Upstream and downstream transmissions are wavelength division multiplexed (WDM) into a single strand of single-mode fiber, sharing the same physical link.

The downstream transmission channel is continuously available to the OLT; thus, Time Division Multiplexing (TDM) is used. Transmissions from the OLT arrive at all of the connected ONUs and the individual ONUs filter data from the OLT's transmission based on the logical link identifiers (LLIDs) assigned to them during the registration and discovery process.

The upstream transmission channel is shared among a number of connected and registered ONUs using Time Division Multiple Access (TDMA). Access to the upstream channel is controlled via the Multipoint Control Protocol (MPCP), where the OLT plays the role of the master and ONUs play the role of slave devices. An ONU upon registration remains silent until registered, and once registered, it transmits data toward the OLT only when granted a transmission opportunity (slot).

NOTE—IEEE Std 802.3, Annex K defines optional alternative terminology for “master” and “slave”.

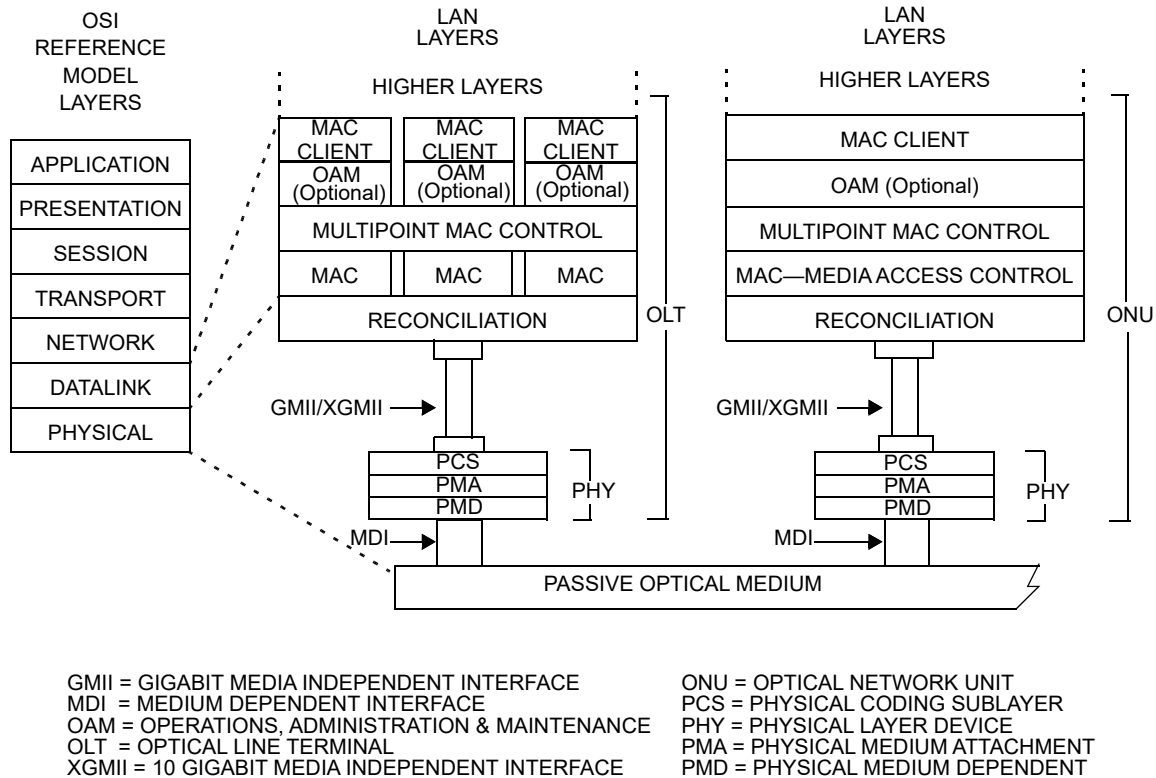


Figure 9-2—Relationship of Multipoint MAC control and the OSI protocol stack

9.1.1.4 PMD specifications

The EPON PMD specifications are based on a wavelength plan similar to that used by ITU-T G.983.1. The OLT and ONU optical parameters were derived in part from earlier Ethernet PMD specifications, with the addition of WDM capabilities, and burst mode operation for ONU transmitters and the OLT receiver.

The upstream burst mode operation capability corresponds directly to the TDMA operation in the upstream direction, where queued data is burst from individual ONUs at full data rate for the duration of the allocated transmission period. Once completed, the ONU goes silent and another ONU starts transmitting its data.

9.1.1.5 Point-to-point emulation

The downstream link is a broadcast medium, which means that all data transmitted by the OLT is received by all connected ONUs. In order to facilitate compliance of EPON with Ethernet architecture, the P2PE function was included in the RS, creating a series of logical links between the OLT and connected ONUs. An additional broadcast link is also provided for delivery of any broadcast content. In this way, EPON becomes a collection of logical P2P connections established between the OLT and the ONUs. Therefore, the OLT can be seen as an Ethernet device with $N + 1$ logical ports (N P2P logical interfaces and 1 broadcast interface, where N designates the number of connected ONUs).

Logical links also provide a solution for privacy of data, which otherwise would be shared by all subscribers connected to a single OLT port. In this way, each subscriber is isolated and restricted to accessing data streams addressed only to that particular subscriber.

This concept is illustrated in Figure 9-3, which shows an example of an EPON with a single OLT and three connected ONUs.

The single copy broadcast channel (addressed with a special, reserved LLID, see 65.1.3.1 of IEEE Std 802.3 for 1G-EPON, and see 76.2.6.1.3.2 of IEEE Std 802.3 for 10G-EPON) was added to take advantage of the broadcast transmission capability of the underlying physical medium. In this way, it is very simple and very bandwidth efficient to deliver broadcast content to all ONUs at the same time, avoiding the need to replicate data into a series of P2P links.

The ONUs filter all downstream data and drop all frames addressed to other devices. Only broadcast frames and frames with correct unicast logical link ID (LLID) are admitted and processed. The LLID replaces two octets of the Ethernet frame preamble, identifying a logical link established between the OLT and the given ONU during the discovery and registration process. The LLID indicates the destination port in the downstream and the source port in the upstream. The logical links are used effectively to prevent EPON from violating the IEEE 802.1Q bridging rules.

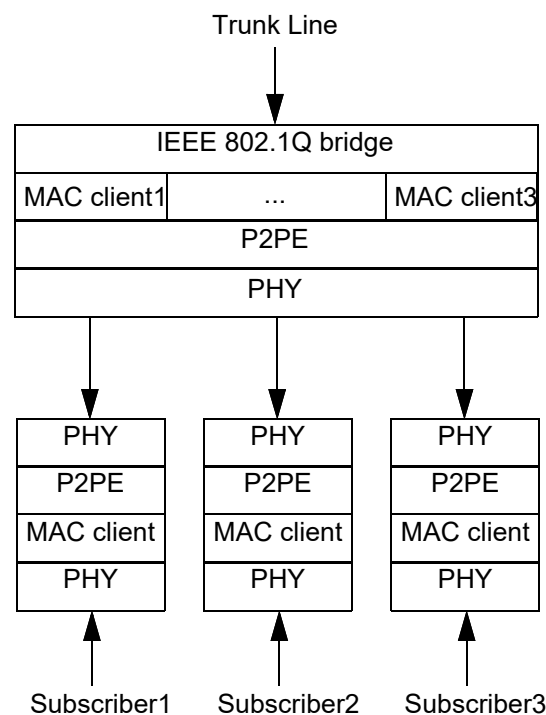


Figure 9-3—Example of point-to-point emulation used between an OLT and three ONUs

9.1.1.6 Principles of the MPCP

The EPON standard comprises a mechanism for media access control, referred to as the Multipoint Control Protocol (MPCP). An access network architecture is different from a typical LAN environment, primarily in terms of network provisioning. An access network is an administrated environment, with an operator providing services and subscribers consuming them depending on service provisioning contracts. The operator controls the network, manages traffic and medium access, and enforces the service level agreements (SLAs). For instance, the available bandwidth is controlled and subscribers may be billed for services. In this sense, the access network (and EPON specifically) requires a media access control protocol

that provides a mechanism for station discovery and registration as well as bandwidth provisioning capabilities.

In the MPCP, the OLT is considered to be the master, controlling a series of connected ONUs (slave devices). The OLT manages the network and controls access to network resources from individual slave devices. The MPCP is also used for provisioning upstream channel access to individual slave devices via a MPCPDU pair, that is, GATE and REPORT. The MPCP is part of the MAC control layer, and MPCPDUs are considered MAC control messages, carrying a specific Ethertype of 0x8808. These messages are not forwarded outside of the EPON domain and are used to manage the EPON link only.

A concept of time exists in the MPCP in order to schedule the upstream transmission. A timestamp, which is transmitted in the MPCPDUs downstream by the OLT and received by the connected ONUs, is used to synchronize slave devices to the master device clock. This coordinates upstream transmissions from individual ONUs so that the transmissions arrive at the OLT at precisely the anticipated time, and thus, data from different ONUs does not overlap.

The MPCP plane is also used to measure the round-trip time (RTT) for each connected ONU. Each MPCPDU carries a generalized timestamp field, which is filled in by the transmitting station with the current value of its MPCP clock at the time when the given MPCPDU is transmitted. The RTT is measured first during the discovery and registration process and then updated regularly upon each exchange of MPCPDUs between the OLT and one of the ONUs. RTT is used by the OLT bandwidth scheduler to schedule upstream transmission slots for individual ONUs in a non-overlapping manner. The IEEE 802.3 EPON standard provides support for the network diameter (distance between the OLT and the farthest ONU) of nominally up to 20 km, which corresponds to a RTT of approximately 200 μ s. However, nothing in the standard precludes support for larger network diameters.

The TDMA control is performed using a pair of MPDPUs, namely GATE generated by the OLT to indicate a future transmission opportunity to an ONU and REPORT generated by the ONU with information on the current queue status (bandwidth demand). Internal structure and possible encoding of GATE and REPORT MPCPDUs are defined in Clause 64 of IEEE Std 802.3 for 1G-EPON and Clause 77 of IEEE Std 802.3 for 10G-EPON.

A scheduling algorithm at the OLT, which is not defined in IEEE Std 802.3, is responsible for dividing the bandwidth and controlling the transmission delay of each ONU according to its SLA. The MPCP defines a closed loop operation in order for this algorithm to be efficient. The MPCP allows the ONUs to report on the amount of bandwidth they require for transmission using a special REPORT message. This allows allocating bandwidth to an ONU only when requested, relying on the statistical burst property of the traffic, and allowing different peak bandwidths for different ONUs at different times, hence, allowing over-subscription of the bandwidth. The REPORT message reports the amount of data waiting in the ONU queues.

In addition, the MPCP defines a protocol of auto-discovery and registration of ONUs.

The MPCP registration process is presented in Figure 9-4, while details are described in Clause 64 of IEEE Std 802.3 for 1G-EPON and Clause 77 of IEEE Std 802.3 for 10G-EPON.

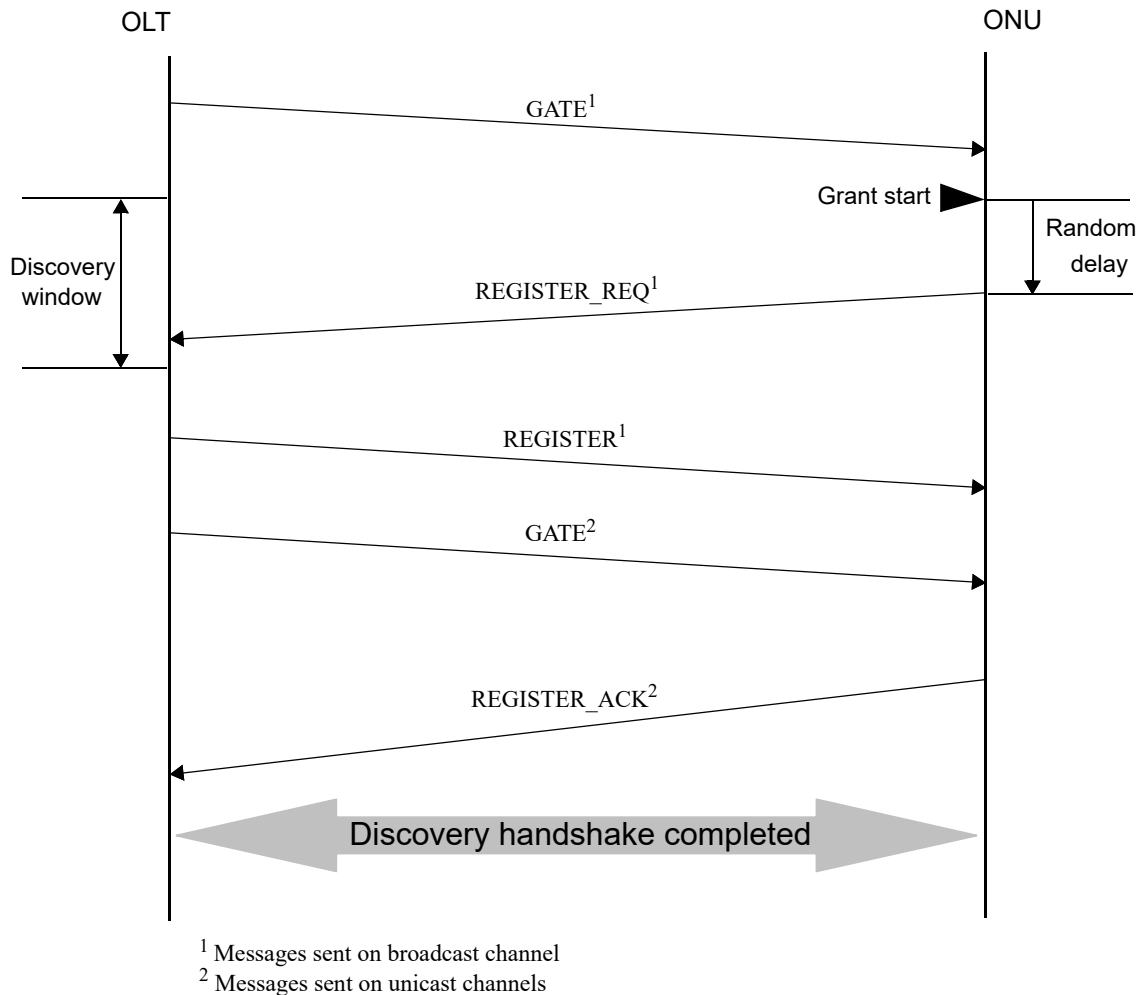


Figure 9-4—Discovery handshake message exchange

A new ONU requests to register during a special upstream window (called Discovery Window), sending the REGISTER_REQ MPCPDU. More than one ONU may attempt registration during that window, which means that their REGISTER_REQ MPCPDUs can potentially collide at the OLT receiver, since the ONU-specific RTT is not yet known and transmissions from individual ONUs cannot be scheduled in a non-overlapping manner. A random backoff mechanism was therefore developed and is used to increase the registration success probability.

When the OLT receives a REGISTER_REQ MPCPDU from an ONU, a decision on registration is taken and an LLID is assigned to that ONU. Next, the OLT sends a REGISTER MPCPDU to that ONU, informing the given slave device whether it is admitted to a network or not. The registration process is completed with the ONU sending REGISTER_ACK MPCPDU to the OLT, confirming assigned parameters and registration in the network. From that point onward, the OLT can schedule transmissions from that ONU using its LLID, using the measured RTT so that its transmissions do not collide with other ONUs.

Additional higher layer protocols may be employed to authenticate the ONU and allow it to participate in the network; however, their specification is outside the scope of IEEE Std 802.3.

9.1.1.7 Forward error correction (FEC) for 1G-EPON

The optional FEC mechanism is defined to enhance the EPON link budget. All the passive components of the fiber plant attenuate the optical signal; thus, the target distance (network diameter) and the number of supported splits are limited by the available link budget. The optional FEC mechanism increases the available link budget by improving the link BER from 10^{-4} to 10^{-12} (the target BER at the MAC), effectively increasing the target network diameter and/or split ratio. The target use of the increased power budget remains at the sole discretion of the network architects and is out of the scope of IEEE Std 802.3.

The optional FEC used in EPON is frame-based, meaning that parity information is added at the end of each Ethernet packet. Extra space between individual Ethernet packets is provided by the MAC rate adaptation function, while extra idle symbols are replaced within the FEC function.

The start and end of packet codewords also define the FEC boundaries, and they are outside the FEC protection. They are replaced by a series of symbols to reduce their vulnerability to link errors.

Figure 9-5 presents the structure of an FEC-protected EPON frame.

The optional FEC function is added to the extended Gigabit Ethernet PCS per 65.2 in IEEE Std 802.3. The added, optional FEC function introduces a fixed delay in the receive path and in the transmit path.

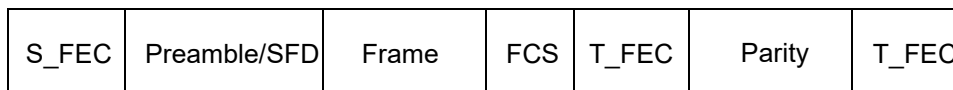


Figure 9-5—FEC-protected frame

9.1.2 Management architecture

All EPON layers are accompanied by a management interface that is controlled through mechanisms defined in Clause 30 of IEEE Std 802.3. Since IEEE Std 802.3 specifications may be used for different applications (and hence are extensible), and some of the clauses may be used separately, the management clause allocates a separate package for each independent layer. The structure of the MIB modules follows this separation.

Figure 9-6 presents the relation of the MIB module groups to the individual IEEE 802.3 layers.

The association is straightforward for the ONU interface. There is one logical and one physical interface, and a single copy of each layer can be remotely queried by the OLT.

The OLT has a single physical interface and N logical interfaces, one for each logical link connected to an ONU. There is also one logical interface for the single copy broadcast link. Per the layering diagram in Figure 9-6, the MAC sublayer is virtually replicated. Therefore, in this clause it was elected that management of logical interfaces is performed in the manner identical to management of any physical interfaces—an interface index is allocated for each one of the logical links, and an additional interface index is allocated for the OLT.

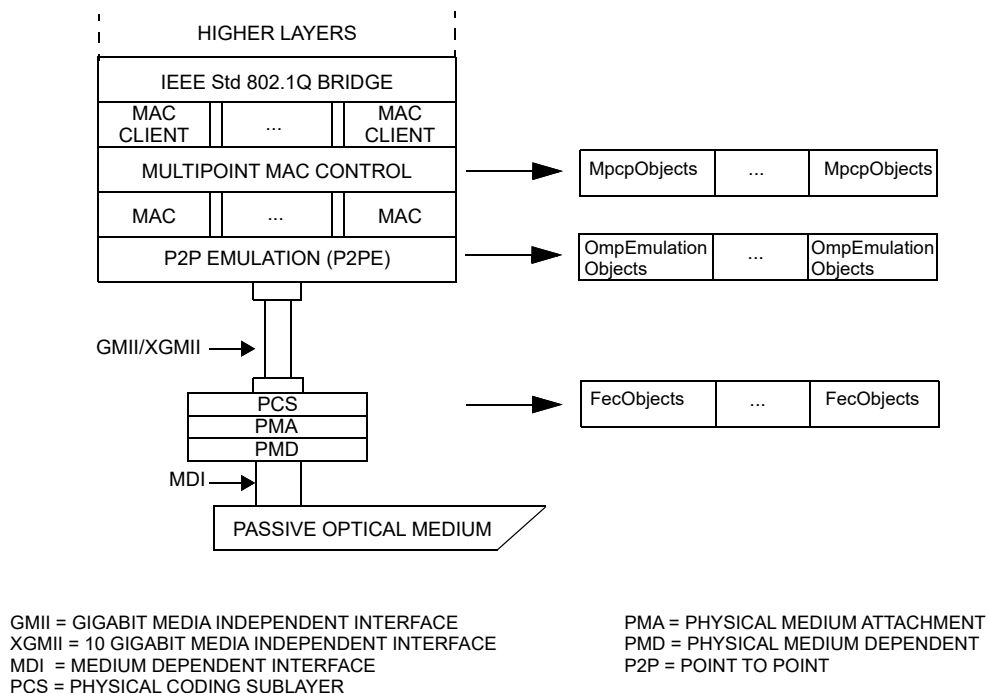


Figure 9-6—Relationship of the MIB groups to the EPON sublayers

For each physical interface, there would be an entry (ifIndex) in the tables of the interface MIB module defined in IETF RFC 2863, the MAU MIB module defined in Clause 13, and the Ethernet-like MIB module defined in Clause 10. Additionally, there would be entries (ifIndexes) for the virtual interfaces of the OLT interface. The justification for the additional allocation of indexes is that the virtual interfaces are quite well distinguished, as they connect different physical ONUs from the OLT side. For instance, there is a meaning for separate bad frames counter or bad octets counter for each virtual link, as the ONUs can be differently distanced. This is quite similar to a case of separate physical interfaces.

The same partition concept exists for the MIB module of this clause. Each row in the tables is indexed according to the ifIndex; specifically, there is a row for each virtual link. There are some control objects that are shared and are the same for the virtual interfaces (and they should have the same value for each ifIndex), but most of the objects have different values for $N + 1$ logical interfaces at the OLT. This is done for each MIB group. It is different from the EPON layering diagram, which presents the P2MP layer as a single layer, while duplicating the MAC and MAC client layers (please see Figure 9-6). However, from a management perspective, it is more convenient to partition the management of the layers for the virtual links, as the atomic managed entity is the virtual link. It is also convenient to use the interface index of the virtual link for that purpose, as it is already used to index the rows of the virtual links at the Interface, MAU, and Ethernet-like interface MIBs.

9.2 MIB structure

This subclause defines the DOT3 EPON MIB module. The DOT3 EPON MIB module defines the objects used for management of the IEEE 802.3 EPON interfaces. These MIB objects are included in the following four groups:

- a) MPCP MIB objects—MIB objects related to Clause 64 of IEEE Std 802.3 for 1G-EPON and Clause 77 of IEEE Std 802.3 for 10G-EPON, Multipoint Control Protocol attributes. The following tables are presented in this group:
 - 1) The dot3MpcpControlTable defines the objects used for the configuration and status indication, which are per logical link, of MPCP compliant interfaces.
 - 2) The dot3MpcpStatTable defines the statistics objects that are per logical link, of MPCP compliant interfaces.
 - 3) The operational mode of an OLT/ONU for the tables is defined by the dot3MpcpMode object in the dot3MpcpControlTable.
- b) The OMPEmulation MIB objects—MIB objects related to Clause 65 of IEEE Std 802.3 for 1G-EPON and Clause 76 of IEEE Std 802.3 for 10G-EPON, point-to-point emulation attributes. The following tables are presented in this group:
 - 1) The dot3OmpEmulationTable defines the objects used for the configuration and status indication, which are per logical link, of OMPEmulation compliant interfaces.
 - 2) The dot3OmpEmulationStatTable defines the statistics objects that are per logical link, of OMPEmulation compliant interfaces.
 - 3) The operational mode of an OLT/ONU for the tables is defined by the dot3OmpEmulationType object in the dot3OmpEmulationTable.
- c) The FEC MIB objects—MIB objects related to Clause 60 and Clause 65 of IEEE Std 802.3 for 1G-EPON and Clause 76 of IEEE Std 802.3 for 10G-EPON, EPON FEC attributes. The following table is presented in this group:
 - 1) The dot3EponFecTable defines the objects used for the configuration and status indication, which are per logical link, of FEC EPON compliant interfaces.
- d) The EPON extended package MIB objects—MIB objects used for configuration and status indication with extended capabilities of the EPON interfaces. The following tables are presented in this group:
 - 1) The dot3ExtPkgControlTable defines the objects, which are per logical link, used for the configuration and status indication of EPON compliant interfaces.
 - 2) The dot3ExtPkgQueueTable defines the objects, which are per logical link and per queue, used for the configuration and status indication of the ONU queues reported in the MPCP REPORT message, of EPON compliant interfaces.
 - 3) The dot3ExtPkgQueueSetsTable defines the objects, which are per logical link, per queue, and per queue_set, used for the configuration and status indication of the ONU queue_sets reported in the MPCP REPORT message, of EPON compliant interfaces.
 - 4) The dot3ExtPkgOptIfTable defines the objects, which are per logical link, used for the control and status indication of the optical interface of EPON compliant interfaces.

The interface MIB module defined in IETF RFC 2863 defines the interface index (ifIndex). Interface Index, as specified in IETF RFC 2863, is used in this MIB module as an index to the EPON MIB tables. The ifIndex is used to denote the physical interface and the virtual link interfaces at the OLT. The OLT interface and the virtual link interfaces are stacked using the ifStack table defined in IETF RFC 2863 and the ifInvStack defined in IETF RFC 2864. The OLT interface is the lower layer of all other interfaces associated with the virtual links.

As described in 9.1.2, each row in the tables is indexed according to the ifIndex; specifically, there is a row for each virtual link. There are a few control objects that are shared and have the same value for the virtual interfaces (and they should have the same value for each ifIndex), but most of the objects have different values for $N + 1$ logical interfaces at the OLT. This is done for each MIB group. It is a bit different from the

EPON layering diagram, which presents the P2MP layer as a single layer while duplicating the MAC and MAC client layers. However, from a management perspective, it is more convenient to partition the management of the layers for the virtual links, as the atomic managed entity is the virtual link. It is also convenient to use the interface index of the virtual link for that purpose, as it is already used to index the rows of the virtual links at the Interface, MAU, and Ethernet-like interface MIB modules.

The creation of the rows of the ONU interface is done at initialization. Table 9-1 presents the MPCP control table of ONU1 after initialization. A single row exists in the table.

Table 9-1—MPCP control table of ONU1 after initialization

MPCP control MIB object	Value
ifIndex	100
dot3MpcpOperStatus	true
dot3MpcpAdminState	true
dot3MpcpMode	onu
dot3MpcpSyncTime	0
dot3MpcpLinkID	0
dot3MpcpRemoteMACAddress	00:00:00:00:00:00
dot3MpcpRegistrationState	unregistered
dot3MpcpTransmitElapsed	0
dot3MpcpReceiveElapsed	0
dot3MpcpRoundTripTime	0

Table 9-2 presents the MPCP control table of ONU1 in working mode. A single row exists in the table.

Table 9-2—MPCP control table of ONU1 in working mode

MPCP control MIB object	Value
ifIndex	100
dot3MpcpOperStatus	true
dot3MpcpAdminState	true
dot3MpcpMode	onu
dot3MpcpSyncTime	25
dot3MpcpLinkID	1

Table 9-2—MPCP control table of ONU1 in working mode (*continued*)

MPCP control MIB object	Value
dot3MpcpRemoteMACAddress	OLT_MAC_Address ^a
dot3MpcpRegistrationState	registered
dot3MpcpTransmitElapsed	10
dot3MpcpReceiveElapsed	10
dot3MpcpRoundTripTime	100

^a OLT_MAC_Address is the MAC address of the OLT EPON interface.

The creation of the rows of the OLT interface and the broadcast virtual interface is done at initialization.

The creation of rows of the virtual interfaces at the OLT is done when the link is established (ONU registers) and the deletion is done when the link is deleted (ONU deregisters).

Table 9-3 presents the MPCP control table of the OLT after initialization, before the ONUs register. A single row exists in this table associated with the virtual broadcast link.

Table 9-3—MPCP control table of the OLT after initialization

MPCP control MIB object	Value
ifIndex	165535
dot3MpcpOperStatus	true
dot3MpcpAdminState	true
dot3MpcpMode	olt
dot3MpcpSyncTime	25
dot3MpcpLinkID	65535
dot3MpcpRemoteMACAddress	BRCT_MAC_Address ^a
dot3MpcpRegistrationState	registered
dot3MpcpTransmitElapsed	10
dot3MpcpReceiveElapsed	100000
dot3MpcpRoundTripTime	0

^a BRCT_MAC_Address is the MAC address of the broadcast EPON interface, which is the OLT MAC address.

Table 9-4 presents the MPCP control table of the OLT in working mode. Three rows exist in the table associated with the virtual links.

Table 9-4—MPCP control table of the OLT in working mode

MPCP control MIB object	Value	Value	Value
ifIndex	100001	100002	165535
dot3MpcpOperStatus	true	true	true
dot3MpcpAdminState	true	true	true
dot3MpcpMode	olt	olt	olt
dot3MpcpSyncTime	25	25	25
dot3MpcpLinkID	1	2	65535
dot3MpcpRemote MACAddress	ONU1_MAC_Address ^a	ONU2_MAC_Address ^b	BRCT_MAC_Address ^c
dot3MpcpRegistrationState	registered	registered	registered
dot3MpcpTransmitElapsed	10	10	10
dot3MpcpReceiveElapsed	10	10	10
dot3MpcpRoundTripTime	100	60	0

^a ONU1_MAC_Address is the MAC address of the ONU1 EPON interface.

^b ONU2_MAC_Address is the MAC address of the ONU2 EPON interface.

^c BRCT_MAC_Address is the MAC address of the broadcast EPON interface, which is the OLT MAC address.

9.3 Relationship to other MIB modules

9.3.1 Relation to the Interfaces Group MIB and Ethernet-like interface MIB

This MIB module extends the objects of the Interfaces Group MIB and the Ethernet-like interface MIB for the EPON type interface. Therefore, if this module is implemented, the Interfaces Group MIB module defined in IETF RFC 2863 and the Ethernet-like interface MIB module defined in Clause 10 shall also be implemented.

Thus, each managed EPON interface would have a corresponding entry in the mandatory tables of the Ethernet-like MIB module found in Clause 10, and likewise in the tables of the Interfaces Group MIB module found in IETF RFC 2863. Also, each managed virtual EPON interface would have a corresponding entry in the mandatory tables of the Ethernet-like MIB module found in Clause 10, and likewise in the tables of the Interfaces Group MIB module found in IETF RFC 2863 with a dedicated ifIndex for this interface.

In this clause, there is no replication of the objects from these MIBs. Therefore, for instance, the clause is defining the dot3MpcpRemoteMACAddress only while assuming that the local MAC address object is already defined in Clause 10.

This clause defines the specific EPON objects of an ONU interface and an OLT interface. Information in the tables is per LLID. The rows in the EPON MIB tables referring to the LLIDs are denoted with the corresponding ifIndexes of the virtual link interfaces.

Note that all virtual interfaces have the same physical MAC address at the OLT since the physical OLT interface used by all virtual interfaces is the same. The value of this physical MAC interface is specified in 64.1.2 of IEEE Std 802.3 for 1G-EPON and 77.1.2 of IEEE Std 802.3 for 10G-EPON. The corresponding object of the Ethernet-like interface MIB is replicated for all virtual interfaces.

For example, the values of the Interfaces Group MIB objects are presented in the following tables, for an OLT with three registered ONUs.

Table 9-5 presents the objects of the Interfaces Group MIB of an ONU in working mode.

Table 9-5—Interfaces Group MIB of an ONU in working mode, example for 1G-EPON

Interfaces Group MIB object	Value
ifIndex	1
ifDescr	“interface description”
ifType	ethernetCsmacd (6) 1000base-Px
ifMtu	MTU size (1522)
ifSpeed	1000000000
ifPhysAddress	ONU_MAC_Address ^a
ifAdminStatus	up
ifOperStatus	Up
ifLastChange	ONUup_time
ifInOctets	ONU_octets_number
ifInUcastPkts	ONU_unicast_frame_number
ifInNUcastPkts	ONU_non_unicast_frame_number
ifInDiscards	ONU_discard_frame_number
ifInErrors	ONU_error_frame_number
ifInUnknownProtos	ONU_unknown_frame_number
ifOutOctets	ONU_octets_number
ifOutUcastPkts	ONU_unicast_frame_number
ifOutNUcastPkts	ONU_non_unicast_frame_number

Table 9-5—Interfaces Group MIB of an ONU in working mode, example for 1G-EPON (*continued*)

Interfaces Group MIB object	Value
ifOutDiscards	ONU_discard_frame_number
ifOutErrors	ONU_error_frame_number
ifOutQLen	ONU_queue_frame_number

^a ONU_MAC_Address is the MAC address of the ONU EPON interface.

Table 9-6 presents the objects of the Interfaces Group MIB of the ONU interface.

Table 9-6—Interfaces Group MIB of the ONU interface, example for 1G-EPON

Interfaces Group MIB object	Value
ifIndex	100
ifDescr	“interface description”
ifType	ethernetCsmacd (6) 1000base-Px
ifMtu	MTU size (1522)
ifSpeed	1000000000
ifPhysAddress	ONU_MAC_Address ^a
ifAdminStatus	up
ifOperStatus	Up
ifLastChange	up_time
ifInOctets	ONU1_octets_number
ifInUcastPkts	ONU1_unicast_frame_number
ifInNUcastPkts	ONU1_non_unicast_frame_number
ifInDiscards	ONU1_discard_frame_number
ifInErrors	ONU1_error_frame_number
ifInUnknownProtos	ONU1_unknown_frame_number
ifOutOctets	ONU1_octets_number
ifOutUcastPkts	ONU1_unicast_frame_number

Table 9-6—Interfaces Group MIB of the ONU interface, example for 1G-EPON (*continued*)

Interfaces Group MIB object	Value
ifOutNUcastPkts	ONU1_non_unicast_frame_number
ifOutDiscards	ONU1_discard_frame_number
ifOutErrors	ONU1_error_frame_number
ifOutQLen	ONU1_queue_frame_number

^a ONU_MAC_Address is the MAC address of the ONU EPON interface.

The following values will be set in the ifStack and ifInvStack tables related to this example.

ifStackTable:

- ifStackHigherLayer = 100, ifStackLowerLayer = 1 – map between the physical interface and the ONU

ifInvStackTable:

- ifStackLowerLayer = 1, ifStackHigherLayer = 100 – map between the ONU and the physical interface

Table 9-7 presents the Interfaces Group MIB objects of an OLT interface.

Table 9-7—Interfaces Group MIB objects of an OLT interface, example for 1G-EPON

Interfaces Group MIB object	Value
ifIndex	2
ifDescr	“interface description”
ifType	ethernetCsmacd (6) 1000base-Px
ifMtu	MTU size (1522)
ifSpeed	1000000000
ifPhysAddress	OLT_MAC_Address ^a
ifAdminStatus	up
ifOperStatus	Up
ifLastChange	OLTup_time
ifInOctets	OLT_octets_number
ifInUcastPkts	OLT_unicast_frame_number

Table 9-7—Interfaces Group MIB objects of an OLT interface, example for 1G-EPON (*continued*)

Interfaces Group MIB object	Value
ifInNUcastPkts	OLT_non_unicast_frame_number
ifInDiscards	OLT_discard_frame_number
ifInErrors	OLT_error_frame_number
ifInUnknownProtos	OLT_unknown_frame_number
ifOutOctets	OLT_octets_number
ifOutUcastPkts	OLT_unicast_frame_number
ifOutNUcastPkts	OLT_non_unicast_frame_number
ifOutDiscards	OLT_discard_frame_number
ifOutErrors	OLT_error_frame_number
ifOutQLen	OLT_queue_frame_number

^a OLT_MAC_Address is the MAC address of the OLT EPON interface.

Table 9-8 presents the Interfaces Group MIB objects of an OLT interface, associated with the virtual link interfaces.

Table 9-8—Interfaces Group MIB objects of an OLT interface, associated with the virtual link interfaces, example for 1G-EPON

Interface MIB object	Value	Value	Value
ifIndex	200001	200002	265535
ifDescr	“interface description”	“interface description”	“interface description”
ifType	ethernetCsmacd (6)	ethernetCsmacd (6)	ethernetCsmacd (6)
ifMtu	MTUsize(1522)	MTUsize(1522)	MTUsize(1522)
ifSpeed	1000000000	1000000000	1000000000
ifPhysAddress	OLT_MAC_Address ^a	OLT_MAC_Address	OLT_MAC_Address
ifAdminStatus	up	up	up
ifOperStatus	Up	Up	Up
ifLastChange	ONU1_up_time	ONU2_up_time	up_time
ifInOctets	ONU1_octets_number	ONU2_octets_number	BRCT_octets_number
ifInUcastPkts	ONU1_unic_frame_num	ONU2_unic_frame_num	BRCT_unic_frame_num

Table 9-8—Interfaces Group MIB objects of an OLT interface, associated with the virtual link interfaces, example for 1G-EPON (*continued*)

Interface MIB object	Value	Value	Value
ifInNUcastPkts	ONU1_non_unic_frame_num	ONU2_non_unic_frame_num	BRCT_non_unic_frame_num
ifInDiscards	ONU1_disc_frame_num	ONU2_disc_frame_num	BRCT_disc_frame_numr
ifInErrors	ONU1_err_frame_num	ONU2_err_frame_num	BRCT_err_frame_num
ifInUnknownProtos	ONU1_unknw_frame_num	ONU2_unknw_frame_num	BRCT_unknw_frame_num
ifOutOctets	ONU1_octets_number	ONU2_octets_number	BRCT_octets_number
ifOutUcastPkts	ONU1_unic_frame_num	ONU2_unic_frame_num	BRCT_unic_frame_num
ifOutNUcastPkts	ONU1_non_unic_frame_num	ONU2_non_unic_frame_num	BRCT_non_unic_frame_num
ifOutDiscards	ONU1_disc_frame_num	ONU2_disc_frame_num	BRCT_disc_frame_num
ifOutErrors	ONU1_err_frame_num	ONU2_err_frame_num	BRCT_err_frame_num
ifOutQLen	ONU1_queue_frame_num	ONU2_queue_frame_num	BRCT_queue_frame_num

^a OLT_MAC_Address is the MAC address of the OLT EPON interface.

The following values will be set in the ifStack and ifInvStack tables related to this example:

ifStackTable:

- ifStackHigherLayer = 265535, ifStackLowerLayer = 2 – map between the OLT physical interface and its broadcast virtual interface
- ifStackHigherLayer = 200001, ifStackLowerLayer = 2 – map between the OLT physical interface and its virtual interface of the 1st ONU
- ifStackHigherLayer = 200002, ifStackLowerLayer = 2 – map between the OLT physical interface and its virtual interface of the 2nd ONU
- ifStackHigherLayer = 200003, ifStackLowerLayer = 2 – map between the OLT physical interface and its virtual interface of the 3rd ONU

ifInvStackTable:

- ifStackLowerLayer = 2, ifStackHigherLayer = 265535 – map between the broadcast interface of the OLT and the OLT physical interface
- ifStackLowerLayer = 2, ifStackHigherLayer = 200001 – map between the OLT virtual interface of the 1st ONU and the OLT physical interface
- ifStackLowerLayer = 2, ifStackHigherLayer = 200002 – map between the OLT virtual interface of the 2nd ONU and the OLT physical interface
- ifStackLowerLayer = 2, ifStackHigherLayer = 200003 – map between the OLT virtual interface of the 3rd ONU and the OLT physical interface

The rows for the ONU interface, the OLT interface, and the OLT broadcast interface are created in initialization. The creation of a row for a virtual link is done when the virtual link is established (ONU registers), and deletion is done when the virtual link is deleted (ONU deregisters).

The EPON MIB module also extends the Interfaces Group MIB module with a set of counters, which are specific for the EPON interface. The EPON MIB module implements the same handling of the counters when the operation of the interface starts or stops. The interface MIB clause describes the possible behavior of counters when an interface is re-initialized using the `ifCounterDiscontinuityTime` indicator, indicating the discontinuity of the counters. See section 3.1.5 of IETF RFC 2863 for more information. The counters of the EPON MIB should be handled in a similar manner.

9.3.2 Relation to the IEEE 802.3 MAU MIBs

The MAU types of the EPON Interface are defined in Clause 13. This clause assumes the implementation of the MAU MIB for this purpose and does not repeat the EPON MAU types. Therefore, if this module is implemented, the MAU-MIB module defined in Clause 13 shall also be implemented.

The handling of the `ifMAU` tables for the EPON case is similar to the handling described in 9.3.1 for the Interface and Ethernet-like interface MIBs. A single row exists for the ONU in the `ifMauTable`. A row for each virtual link ($N + 1$ rows) exists at the OLT, with a separate value of `ifMauIfIndex` for each virtual link.

As specified above, the rows for the ONU interface, the OLT interface, and the OLT broadcast interface are created in initialization. The creation of a row for a virtual link is done when the virtual link is established (ONU registers), and deletion is done when the virtual link is deleted (ONU deregisters).

9.3.3 Relation to the Ethernet OAM MIB

The EPON interfaces are intended for use in optical subscriber access networks and most probably will be accompanied with the implementation of the OAM protocol defined in Clause 57 of IEEE Std 802.3. Therefore, the Ethernet OAM MIB module defined in Clause 6 may be implemented when this MIB module is implemented defining managed objects for the OAM protocol that are complementary to the EPON MIB module.

9.3.4 Relation to the bridge MIB

It is very probable that an EPON OLT will implement a bridging functionality above the EPON interface layer, bridging between the EPON users and the network. Bridge functionality is specified in IEEE Std 802.1Q. In this scenario, the virtual ports of the EPON are corresponding to the virtual bridge ports. There is a direct mapping between the bridge ports and the LLIDs, which are virtual EPON channels.

Therefore, the bridge MIB modules defined in IEEE Std 802.1Q may be implemented when the EPON MIB module is implemented for an EPON OLT, defining managed objects for the bridge layer.

The values of `dot1dBasePortIfIndex` would correspond to the `ifIndex` of the virtual port (1 for LLID1, 2 for LLID2, etc.).

The broadcast virtual EPON interface of the OLT has no direct mapping to a virtual bridge port as it is not port specific but used for broadcast traffic.

9.4 Mapping of IEEE 802.3 managed objects

This subclause contains the mapping between the managed objects defined in this clause and the attributes defined in Clause 30 of IEEE Std 802.3. Table 9-9 provides the mapping between the dot3EPON MIB module MPCP objects and the MPCP attributes of Clause 30 of IEEE Std 802.3.

Table 9-9—oMPCP managed object class (30.3.5 of IEEE Std 802.3)

dot3EPON MIB module object	IEEE 802.3 attribute	Reference
ifIndex	aMPCPID	30.3.5.1.1
dot3MpcpOperStatus	aMPCPAdminState	30.3.5.1.2
dot3MpcpMode	aMPCPMode	30.3.5.1.3
dot3MpcpLinkID	aMPCPLinkID	30.3.5.1.4
dot3MpcpRemoteMACAddress	aMPCPRemoteMACAddress	30.3.5.1.5
dot3MpcpRegistrationState	aMPCPRegistrationState	30.3.5.1.6
dot3MpcpMACCtrlFramesTransmitted	aMPCPMACCtrlFramesTransmitted	30.3.5.1.7
dot3MpcpMACCtrlFramesReceived	aMPCPMACCtrlFramesReceived	30.3.5.1.8
dot3MpcpTxGate	aMPCPTxGate	30.3.5.1.9
dot3MpcpTxRegAck	aMPCPTxRegAck	30.3.5.1.10
dot3MpcpTxRegister	aMPCPTxRegister	30.3.5.1.11
dot3MpcpTxRegRequest	aMPCPTxRegRequest	30.3.5.1.12
dot3MpcpTxReport	aMPCPTxReport	30.3.5.1.13
dot3MpcpRxGate	aMPCPRxGate	30.3.5.1.14
dot3MpcpRxRegAck	aMPCPRxRegAck	30.3.5.1.15
dot3MpcpRxRegister	aMPCPRxRegister	30.3.5.1.16
dot3MpcpRxRegRequest	aMPCPRxRegRequest	30.3.5.1.17
dot3MpcpRxReport	aMPCPRxReport	30.3.5.1.18
dot3MpcpTransmitElapsed	aMPCPTransmitElapsed	30.3.5.1.19
dot3MpcpReceiveElapsed	aMPCPReceiveElapsed	30.3.5.1.20
dot3MpcpRoundTripTime	aMPCPRoundTripTime	30.3.5.1.21
dot3MpcpDiscoveryWindowsSent	aMPCPDiscoveryWindowsSent	30.3.5.1.22
dot3MpcpDiscoveryTimeout	aMPCPDiscoveryTimeout	30.3.5.1.23

Table 9-9—oMPCP managed object class (30.3.5 of IEEE Std 802.3) (continued)

dot3EPON MIB module object	IEEE 802.3 attribute	Reference
dot3MpcpMaximumPendingGrants	aMPCPMaximumPendingGrants	30.3.5.1.24
dot3MpcpAdminState	aMPCPAdminControl	30.3.5.2.1
dot3MpcpSyncTime	SyncTime	64.3.3.2 for 1G-EPON or 77.3.3.2 for 10G-EPON

Table 9-10 provides the mapping between the dot3EPON MIB module OMPEmulation objects and the OMPE attributes of Clause 30 of IEEE Std 802.3.

Table 9-10—oOMPEmulation managed object class (30.3.7 of IEEE Std 802.3)

dot3EPON MIB module object	IEEE 802.3 attribute	Reference
ifIndex	aOMPEmulationID	30.3.7.1.1
dot3OmpEmulationType	aOMPEmulationType	30.3.7.1.2
dot3OmpEmulationSLDErrors	aSLDErrors	30.3.7.1.3
dot3OmpEmulationCRC8Errors	aCRC8Errors	30.3.7.1.4
dot3OmpEmulationGoodLLID	aGoodLLID	30.3.7.1.5
dot3OmpEmulationOnuPonCastLLID	aONUPONcastLLID	30.3.7.1.6
dot3OmpEmulationOltPonCastLLID	aOLTPONcastLLID	30.3.7.1.7
dot3OmpEmulationBadLLID	aBadLLID	30.3.7.1.8
dot3OmpEmulationBroadcastBitNotOnuLLid	N/A	—
dot3OmpEmulationOnuLLIDNotBroadcast	N/A	—
dot3OmpEmulationBroadcastBitPlusOnuLlid	N/A	—
dot3OmpEmulationNotBroadcastBitNotOnuLlid	N/A	—

Table 9-11 provides the mapping between the dot3EPON MIB module FEC objects and the MAU attributes of Clause 30 of IEEE Std 802.3.

Table 9-11—oMAU managed object class (30.5.1 of IEEE Std 802.3)

dot3EPON MIB module object	IEEE 802.3 attribute	Reference
dot3EponFecPCSCodingViolation	aPCSCodingViolation	30.5.1.1.14
dot3EponFecAbility	aFECAbility	30.5.1.1.15
dot3EponFecMode	aFECmode	30.5.1.1.16
dot3EponFecCorrectedBlocks	aFECCorrectedBlocks	30.5.1.1.17
dot3EponFecUncorrectableBlocks	aFECUncorrectableBlocks	30.5.1.1.18
dot3EponFecBufferHeadCodingViolation	N/A	—

9.5 Security considerations for Ethernet passive optical network (EPON) MIB module

There are number of managed objects defined in this MIB module that have a MAX-ACCESS clause of read-write or read-create. Writing to these objects can have potentially disruptive effects on network operation, including those listed in 9.5.1 to 9.5.13.

9.5.1 dot3MpcpAdminState

Changing the dot3MpcpAdminState state can lead to disabling the Multipoint Control Protocol on the respective interface, leading to the interruption of service for the users connected to the respective EPON interface.

9.5.2 dot3EponFecMode

Changing the dot3EponFecMode state can lead to disabling the Forward Error Correction on the respective interface, which can lead to a degradation of the optical link, and therefore, it may lead to an interruption of service for the users connected to the respective EPON interface.

9.5.3 dot3ExtPkgObjectReset

Changing the dot3ExtPkgObjectReset state can lead to a reset of the respective interface leading to an interruption of service for the users connected to the respective EPON interface.

9.5.4 dot3ExtPkgObjectPowerDown

Changing the dot3ExtPkgObjectPowerDown state can lead to a power down of the respective interface, leading to an interruption of service for the users connected to the respective EPON interface.

9.5.5 dot3ExtPkgObjectFecEnabled

Changing the dot3ExtPkgObjectFecEnabled state can lead to disabling the Forward Error Correction on the respective interface, which can lead to a degradation of the optical link, and therefore, it may lead to an interruption of service for the users connected to the respective EPON interface.

9.5.6 dot3ExtPkgObjectRegisterAction

Changing the dot3ExtPkgObjectRegisterAction state can lead to a change in the registration state of the respective interface, leading to a deregistration and an interruption of service for the users connected to the respective EPON interface.

9.5.7 dot3ExtPkgObjectReportNumThreshold

Changing the dot3ExtPkgObjectReportNumThreshold can lead to a change in the reporting of the ONU interface and therefore to a change in the bandwidth allocation of the respective interface. This change may lead to a degradation or an interruption of service for the users connected to the respective EPON interface.

9.5.8 dot3ExtPkgObjectReportThreshold

Changing the dot3ExtPkgObjectReportThreshold can lead to a change in the reporting of the ONU interface and therefore to a change in the bandwidth allocation of the respective interface. This change may lead to a degradation or an interruption of service for the users connected to the respective EPON interface.

9.5.9 dot3ExtPkgOptIfLowerInputPowerThreshold

Changing the dot3ExtPkgOptIfLowerInputPowerThreshold can lead to a Threshold Crossing Alert (TCA) being sent for the respective interface. This alert may be leading to an interruption of service for the users connected to the respective EPON interface, depending on the system action on such an alert.

9.5.10 dot3ExtPkgOptIfUpperInputPowerThreshold

Changing the dot3ExtPkgOptIfUpperInputPowerThreshold can lead to a Threshold Crossing Alert (TCA) being sent for the respective interface. This alert may be leading to an interruption of service for the users connected to the respective EPON interface, depending on the system action on such an alert.

9.5.11 dot3ExtPkgOptIfLowerOutputPowerThreshold

Changing the dot3ExtPkgOptIfLowerOutputPowerThreshold can lead to a Threshold Crossing Alert (TCA) being sent for the respective interface. This alert may be leading to an interruption of service for the users connected to the respective EPON interface, depending on the system action on such an alert.

9.5.12 dot3ExtPkgOptIfUpperOutputPowerThreshold

Changing the dot3ExtPkgOptIfUpperOutputPowerThreshold can lead to a Threshold Crossing Alert (TCA) being sent for the respective interface. This alert may be leading to an interruption of service for the users connected to the respective EPON interface, depending on the system action on such an alert.

9.5.13 dot3ExtPkgOptIfTransmitEnable

Changing the dot3ExtPkgOptIfTransmitEnable state can lead to a halt in the optical transmission of the respective interface, leading to an interruption of service for the users connected to the respective EPON interface.

9.6 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²³

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C9mib.txt

²³ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

9.6.1 Tree hierarchy

```

----- ieee8023dot3EponMIB(1.3.111.2.802.3.1.9)
+----- dot3EponObjects(1)
| +----- dot3EponMpcpObjects(1)
| | +----- dot3MpcpControlTable(1)
| | | +----- dot3MpcpControlEntry(1) [ifIndex]
| | | | +--r- dot3MpcpOperStatus(1)
| | | | +--rw dot3MpcpAdminState(2)
| | | | +--r- dot3MpcpMode(3)
| | | | +--r- dot3MpcpSyncTime(4)
| | | | +--r- dot3MpcpLinkID(5)
| | | | +--r- dot3MpcpRemoteMACAddress(6)
| | | | +--r- dot3MpcpRegistrationState(7)
| | | | +--r- dot3MpcpTransmitElapsed(8)
| | | | +--r- dot3MpcpReceiveElapsed(9)
| | | | +--r- dot3MpcpRoundTripTime(10)
| | | | +--r- dot3MpcpMaximumPendingGrants(11)
| | | +----- dot3MpcpStatTable(2)
| | | +----- dot3MpcpStatEntry(1) [ifIndex]
| | | | +--r- dot3MpcpMACCtrlIFramesTransmitted(1) Counter64
| | | | +--r- dot3MpcpMACCtrlIFramesReceived(2) Counter64
| | | | +--r- dot3MpcpDiscoveryWindowsSent(3) Counter32
| | | | +--r- dot3MpcpDiscoveryTimeout(4) Counter32
| | | | +--r- dot3MpcpTxRegRequest(5) Counter64
| | | | +--r- dot3MpcpRxRegRequest(6) Counter64
| | | | +--r- dot3MpcpTxRegAck(7) Counter64
| | | | +--r- dot3MpcpRxRegAck(8) Counter64
| | | | +--r- dot3MpcpTxReport(9) Counter64
| | | | +--r- dot3MpcpRxReport(10) Counter64
| | | | +--r- dot3MpcpTxGate(11) Counter64
| | | | +--r- dot3MpcpRxGate(12) Counter64
| | | | +--r- dot3MpcpTxRegister(13) Counter64
| | | | +--r- dot3MpcpRxRegister(14) Counter64
| | +----- dot3OmpEmulationObjects(2)
| | +----- dot3OmpEmulationTable(1)
| | | +----- dot3OmpEmulationEntry(1) [ifIndex]
| | | | +--r- dot3OmpEmulationType(1) Enumeration
| | | +----- dot3OmpEmulationStatTable(2)

```

		+-----	dot3OmpEmulationStatEntry(1) [ifIndex]		
		+--r-	dot3OmpEmulationSLDErrors(1)	Counter64	
		+--r-	dot3OmpEmulationCRC8Errors(2)	Counter64	
		+--r-	dot3OmpEmulationBadLLID(3)	Counter64	
		+--r-	dot3OmpEmulationGoodLLID(4)	Counter64	
		+--r-	dot3OmpEmulationOnuPonCastLLID(5)	Counter64	
		+--r-	dot3OmpEmulationOltPonCastLLID(6)	Counter64	
		+--r-	dot3OmpEmulationBroadcastBitNotOnuLLid(7)	Counter64	
		+--r-	dot3OmpEmulationOnuLLIDNotBroadcast(8)	Counter64	
		+--r-	dot3OmpEmulationBroadcastBitPlusOnuLLid(9)	Counter64	
		+--r-	dot3OmpEmulationNotBroadcastBitNotOnuLLid(10)	Counter64	
		+-----	dot3EponFecObjects(3)		
		+-----	dot3EponFecTable(1)		
		+-----	dot3EponFecEntry(1) [ifIndex]		
		+--r-	dot3EponFecPCSCodingViolation(1)	Counter64	
		+--r-	dot3EponFecAbility(2)	Enumeration	
		+--rw	dot3EponFecMode(3)	Enumeration	
		+--r-	dot3EponFecCorrectedBlocks(4)	Counter64	
		+--r-	dot3EponFecUncorrectableBlocks(5)	Counter64	
		+--r-	dot3EponFecBufferHeadCodingViolation(6)	Counter64	
		+-----	dot3ExtPkgObjects(4)		
		+-----	dot3ExtPkgControlObjects(1)		
		+-----	dot3ExtPkgControlTable(1)		
			+-----	dot3ExtPkgControlEntry(1) [ifIndex]	
			+--rw	dot3ExtPkgObjectReset(1)	Enumeration
			+--rw	dot3ExtPkgObjectPowerDown(2)	TruthValue
			+--r-	dot3ExtPkgObjectNumberOfLLIDs(3)	Unsigned32
			+--rw	dot3ExtPkgObjectFecEnabled(4)	Enumeration
			+--r-	dot3ExtPkgObjectReportMaximumNumQueues(5)	Unsigned32
			+--rw	dot3ExtPkgObjectRegisterAction(6)	Enumeration
		+-----	dot3ExtPkgQueueTable(2)		
			+-----	dot3ExtPkgQueueEntry(1) [ifIndex, dot3QueueIndex]	
			+-----	dot3QueueIndex(1)	Unsigned32
			+--rw	dot3ExtPkgObjectReportNumThreshold(2)	Unsigned32
			+--r-	dot3ExtPkgObjectReportMaximumNumThreshold(3)	Unsigned32
			+--r-	dot3ExtPkgStatTxFramesQueue(4)	Counter64
			+--r-	dot3ExtPkgStatRxFramesQueue(5)	Counter64
			+--r-	dot3ExtPkgStatDroppedFramesQueue(6)	Counter64
		+-----	dot3ExtPkgQueueSetsTable(3)		

```

| | | +----- dot3ExtPkgQueueSetsEntry(1) [ifIndex, dot3QueueSetQueueIndex, dot3QueueSetIndex]
| | | +----- dot3QueueSetQueueIndex(1) Unsigned32
| | | +----- dot3QueueSetIndex(2) Unsigned32
| | | +--rw dot3ExtPkgObjectReportThreshold(3) Unsigned32
| | | +----- dot3ExtPkgOptIfTable(5)
| | | +----- dot3ExtPkgOptIfEntry(1) [ifIndex]
| | | +--r- dot3ExtPkgOptIfSuspectedFlag(1) TruthValue
| | | +--r- dot3ExtPkgOptIfInputPower(2) Integer32
| | | +--r- dot3ExtPkgOptIfLowInputPower(3) Integer32
| | | +--r- dot3ExtPkgOptIfHighInputPower(4) Integer32
| | | +--rw dot3ExtPkgOptIfLowerInputPowerThreshold(5) Integer32
| | | +--rw dot3ExtPkgOptIfUpperInputPowerThreshold(6) Integer32
| | | +--r- dot3ExtPkgOptIfOutputPower(7) Integer32
| | | +--r- dot3ExtPkgOptIfLowOutputPower(8) Integer32
| | | +--r- dot3ExtPkgOptIfHighOutputPower(9) Integer32
| | | +--rw dot3ExtPkgOptIfLowerOutputPowerThreshold(10) Integer32
| | | +--rw dot3ExtPkgOptIfUpperOutputPowerThreshold(11) Integer32
| | | +--r- dot3ExtPkgOptIfSignalDetect(12) TruthValue
| | | +--r- dot3ExtPkgOptIfTransmitAlarm(13) TruthValue
| | | +--rw dot3ExtPkgOptIfTransmitEnable(14) TruthValue
| | | +----- dot3RecognizedMulticastIDsTable(5)
| | | +----- dot3RecognizedMulticastIDsEntry(1) [ifIndex, dot3RecognizedMulticastIDIndex]
| | | +----- dot3RecognizedMulticastIDIndex(1) Unsigned32
| | | +--rw dot3RecognizedMulticastID(2) Unsigned32
| | | +----- dot3EponConformance(2)
| | | +----- dot3EponGroups(1)
| | | +----- dot3MpcpGroupBase(1)
| | | +----- dot3MpcpGroupStat(2)
| | | +----- dot3OmpeGroupID(3)
| | | +----- dot3OmpeGroupStat(4)
| | | +----- dot3EponFecGroupAll(5)
| | | +----- dot3ExtPkgGroupControl(6)
| | | +----- dot3ExtPkgGroupQueue(7)
| | | +----- dot3ExtPkgGroupQueueSets(8)
| | | +----- dot3ExtPkgGroupOptIf(9)
| | | +----- dot3EponGroupMulticastIDs(10)
| | | +----- dot3EponCompliances(2)
| | | +----- dot3MPCPCompliance(1)
| | | +----- dot3OmpeCompliance(2)

```

+----- dot3EponFecCompliance (3)
+----- dot3ExtPkgCompliance (4)
+----- dot3EponMulticastIDsCompliance (5)

9.6.2 MIB module definition

```
IEEE8023-DOT3-EPON-MIB DEFINITIONS ::= BEGIN

IMPORTS
    ifIndex
        FROM IF-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Counter64, Integer32, MODULE-IDENTITY, OBJECT-TYPE,
    Unsigned32, org
        FROM SNMPv2-SMI
    MacAddress, TruthValue
        FROM SNMPv2-TC;

ieee8023dot3EponMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "The objects in this MIB module are used to manage the
        Ethernet in the First Mile (EFM) Ethernet Passive Optical
        Network (EPON) Interfaces as defined in IEEE Std 802.3 Clauses 60,
        64, and 65 for 1G-EPON, and Clauses 75, 76, and 77 for 10G-EPON.

        For 1G-EPON, of particular interest are Clause 64 (MultiPoint Control
        Protocol - MPCP), Clause 65 (Point-to-Multipoint
        Reconciliation Sublayer - P2MP RS), Clause 60 (Ethernet
        Passive Optical Network Physical Medium Dependent - EPON
        PMDs), Clause 30, 'Management', and Clause 45, 'Management
        Data Input/Output (MDIO) Interface'.

        For 10G-EPON, of particular interest are Clause 77 (MultiPoint Control
        Protocol - MPCP), Clause 76 (Point-to-Multipoint
        Reconciliation Sublayer - P2MP RS), Clause 75 (Ethernet
        Passive Optical Network Physical Medium Dependent - EPON
        PMDs), Clause 30, 'Management', and Clause 45, 'Management
        Data Input/Output (MDIO) Interface'."
    REVISION      "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION      "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION      "201102020000Z"
    DESCRIPTION
        "Initial version, based on an earlier version published
```



```
as RFC 4837."
 ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 9 }

dot3EponObjects OBJECT IDENTIFIER
 ::= { ieee8023dot3EponMIB 1 }

dot3EponMpcpObjects OBJECT IDENTIFIER
 ::= { dot3EponObjects 1 }

dot3MpcpControlTable OBJECT-TYPE
 SYNTAX      SEQUENCE OF Dot3MpcpControlEntry
 MAX-ACCESS  not-accessible
 STATUS      current
 DESCRIPTION
    "A Table of dot3 MultiPoint Control Protocol (MPCP)
    MIB objects. The entries in the table are control and
    status objects of the MPCP.
    Each object has a row for every virtual link denoted by
    the corresponding ifIndex.
    The LLID field, as defined in IEEE Std 802.3, is a 2-byte
    register (15-bit field and a broadcast bit) limiting the
    number of virtual links to 32768. Typically the number
    of expected virtual links in a PON is like the number of
    ONUs, which is 32-64, plus an additional entry for
    broadcast LLID."
 ::= { dot3EponMpcpObjects 1 }

dot3MpcpControlEntry OBJECT-TYPE
 SYNTAX      Dot3MpcpControlEntry
 MAX-ACCESS  not-accessible
 STATUS      current
 DESCRIPTION
    "An entry in the dot3 MPCP Control table.
    Rows exist for an OLT interface and an ONU interface.
    A row in the table is denoted by the ifIndex of the link
    and it is created when the ifIndex is created.
    The rows in the table for an ONU interface are created
    at system initialization.
    The row in the table corresponding to the OLT ifIndex
    and the row corresponding to the broadcast virtual link
    are created at system initialization.
    A row in the table corresponding to the ifIndex of a
    virtual links is created when a virtual link is
    established (ONU registers) and deleted when the virtual
    link is deleted (ONU deregisters)."
```

```
INDEX      { ifIndex }
 ::= { dot3MpcpControlTable 1 }

Dot3MpcpControlEntry ::= SEQUENCE {
    dot3MpcpOperStatus      TruthValue,
    dot3MpcpAdminState      TruthValue,
    dot3MpcpMode             INTEGER,
    dot3MpcpSyncTime         Unsigned32,
    dot3MpcpLinkID           Unsigned32,
    dot3MpcpRemoteMACAddress MacAddress,
    dot3MpcpRegistrationState INTEGER,
    dot3MpcpTransmitElapsed  Unsigned32,
```

```
dot3MpcpReceiveElapsed      Unsigned32,
dot3MpcpRoundTripTime      Unsigned32,
dot3MpcpMaximumPendingGrants Unsigned32
}

dot3MpcpOperStatus OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This object reflects the operational state of the
        MultiPoint MAC Control sublayer as defined in
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        When the value is true(1), the interface will act as if the
        MultiPoint Control Protocol is enabled. When the value is
        false(2), the interface will act as if the MultiPoint Control
        Protocol is disabled. The operational state can be changed using
        the dot3MpcpAdminState object.
        This object is applicable for an OLT, with the same
        value for all virtual interfaces, and for an ONU."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.1.2"
    ::= { dot3MpcpControlEntry 1 }

dot3MpcpAdminState OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "This object is used to define the admin state of the
        MultiPoint MAC Control sublayer, as defined in
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON,
        and to reflect its state.
        When selecting the value as true(1), the MultiPoint
        Control Protocol of the interface is enabled.
        When selecting the value as false(2), the MultiPoint
        Control Protocol of the interface is disabled.
        This object reflects the administrative state of the
        MultiPoint Control Protocol of the interface.
        The write operation is not restricted in this document
        and can be done at any time. Changing
        dot3MpcpAdminState state can lead to disabling the
        MultiPoint Control Protocol on the respective interface,
        leading to the interruption of service for the users
        connected to the respective EPON interface.
        This object is applicable for an OLT, with the same
        value for all virtual interfaces, and for an ONU."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.2.1"
    DEFVAL      { false }
    ::= { dot3MpcpControlEntry 2 }

dot3MpcpMode OBJECT-TYPE
    SYNTAX      INTEGER { olt(1), onu(2) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This object is used to identify the operational
        state of the MultiPoint MAC Control sublayer as
```

defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. Reading olt(1) for an OLT (server) mode and onu(2) for an ONU (client) mode. This object is used to identify the operational mode for the MPCP tables.
This object is applicable for an OLT, with the same value for all virtual interfaces, and for an ONU."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.3"

DEFVAL { olt }

::= { dot3MpcpControlEntry 3 }

dot3MpcpSyncTime OBJECT-TYPE

SYNTAX Unsigned32

UNITS "TQ (16 ns)"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that reports the 'sync lock time' of the OLT receiver in increments of Time Quanta (TQ)-16ns as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. The value returned shall be (sync lock time ns)/16, rounded up to the nearest TQ. If this value exceeds (2³²-1), the value (2³²-1) shall be returned. This object is applicable for an OLT, with distinct values for all virtual interfaces, and for an ONU."

REFERENCE

"IEEE Std 802.3, 64.3.3.2 for 1G-EPON and 77.3.3.2 for 10G-EPON"

::= { dot3MpcpControlEntry 4 }

dot3MpcpLinkID OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that identifies the Logical Link Identifier (LLID) associated with the MAC of the virtual link as specified in IEEE Std 802.3, 65.1.3.2.2 for 1G-EPON or 76.2.6.1.3.2 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. The ONU and the corresponding virtual MAC of the OLT, for the same virtual link, have the same value.
Value is assigned when the ONU registers.
Value is freed when the ONU deregisters."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.4"

::= { dot3MpcpControlEntry 5 }

dot3MpcpRemoteMACAddress OBJECT-TYPE

SYNTAX MacAddress

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that identifies the source_address parameter of the last MPCPDUs passed to the MAC Control. This value is updated on reception of a valid frame with
1) a destination Field equal to the reserved multicast address for MAC Control as specified in IEEE Std 802.3, Annex 31A; 2) the lengthOrType field value equal to the reserved

Type for MAC Control as specified in IEEE Std 802.3, Annex 31A; 3) an MPCP subtype value equal to the subtype reserved for MPCP as specified in IEEE Std 802.3, Annex 31A. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. The value reflects the MAC address of the remote entity and therefore the OLT holds a value for each LLID, which is the MAC address of the ONU; the ONU has a single value that is the OLT MAC address."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.5"

::= { dot3MpcpControlEntry 6 }

dot3MpcpRegistrationState OBJECT-TYPE

SYNTAX INTEGER { unregistered(1), registering(2), registered(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that identifies the registration state of the MultiPoint MAC Control sublayer as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. When this object has the enumeration unregistered(1), the interface is unregistered and may be used for registering a link partner. When this object has the enumeration registering(2), the interface is in the process of registering a link-partner. When this object has the enumeration registered(3), the interface has an established link-partner. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.6"

::= { dot3MpcpControlEntry 7 }

dot3MpcpTransmitElapsed OBJECT-TYPE

SYNTAX Unsigned32

UNITS "TQ (16 ns)"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that reports the interval from the last MPCP frame transmission in increments of Time Quanta (TQ)-16ns. The value returned shall be (interval from last MPCP frame transmission in ns)/16. If this value exceeds (2³²-1), the value (2³²-1) shall be returned. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.19"

::= { dot3MpcpControlEntry 8 }

dot3MpcpReceiveElapsed OBJECT-TYPE

SYNTAX Unsigned32

UNITS "TQ (16 ns)"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that reports the interval from last MPCP frame

reception in increments of Time Quanta (TQ)-16ns. The value returned shall be (interval from last MPCP frame reception in ns)/16. If this value exceeds ($2^{32}-1$), the value ($2^{32}-1$) shall be returned.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.20"

::= { dot3MpcpControlEntry 9 }

dot3MpcpRoundTripTime OBJECT-TYPE

SYNTAX Unsigned32 (0..65535)

UNITS "TQ (16 ns)"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that reports the MPCP round trip time in increments of Time Quanta (TQ)-16ns. The value returned shall be (round trip time in ns)/16. If this value exceeds ($2^{16}-1$), the value ($2^{16}-1$) shall be returned.
This object is applicable for an OLT. At the OLT, it has a distinct value for each virtual interface."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.21"

::= { dot3MpcpControlEntry 10 }

dot3MpcpMaximumPendingGrants OBJECT-TYPE

SYNTAX Unsigned32 (0..255)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that reports the maximum number of grants that an ONU can store for handling. The maximum number of grants that an ONU can store for handling has a range of 0 to 255.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface.
At the OLT, the value should be zero."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.24"

::= { dot3MpcpControlEntry 11 }

dot3MpcpStatTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3MpcpStatEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table defines the list of statistics counters of an interface implementing the IEEE Std 802.3, Clause 64 or Clause 77 MPCP.
Each object has a row for every virtual link denoted by the corresponding ifIndex.
The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID."

::= { dot3EponMpcpObjects 2 }

```
dot3MpcpStatEntry OBJECT-TYPE
    SYNTAX      Dot3MpcpStatEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table of statistics counters of
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON
        MPCP interface. Rows exist for an OLT interface and an
        ONU interface. A row in the table is denoted by the ifIndex
        of the link and it is created when the ifIndex is created.
        The rows in the table for an ONU interface are created
        at system initialization.
        The row in the table corresponding to the OLT ifIndex
        and the row corresponding to the broadcast virtual link
        are created at system initialization.
        A row in the table corresponding to the ifIndex of a
        virtual link is created when a virtual link is
        established (ONU registers) and deleted when the virtual
        link is deleted (ONU deregisters)."
```

```
INDEX          { ifIndex }
 ::= { dot3MpcpStatTable 1 }
```

```
Dot3MpcpStatEntry ::= SEQUENCE {
    dot3MpcpMACCtrlFramesTransmitted    Counter64,
    dot3MpcpMACCtrlFramesReceived       Counter64,
    dot3MpcpDiscoveryWindowsSent        Counter32,
    dot3MpcpDiscoveryTimeout             Counter32,
    dot3MpcpTxRegRequest                 Counter64,
    dot3MpcpRxRegRequest                 Counter64,
    dot3MpcpTxRegAck                     Counter64,
    dot3MpcpRxRegAck                     Counter64,
    dot3MpcpTxReport                     Counter64,
    dot3MpcpRxReport                     Counter64,
    dot3MpcpTxGate                       Counter64,
    dot3MpcpRxGate                       Counter64,
    dot3MpcpTxRegister                   Counter64,
    dot3MpcpRxRegister                   Counter64
}
```

```
dot3MpcpMACCtrlFramesTransmitted OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of MPCP frames passed to the MAC sublayer for
        transmission. This counter is incremented when a
        MA_CONTROL.request service primitive is generated within
        the MAC control sublayer with an opcode indicating an
        MPCP frame.
        This object is applicable for an OLT and an ONU. At the
        OLT it has a distinct value for each virtual interface.
        Discontinuities of this counter can occur at
        re-initialization of the management system, and at other
        times as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
```

```
REFERENCE
```

```
"IEEE Std 802.3, 30.3.5.1.7"
 ::= { dot3MpcpStatEntry 1 }

dot3MpcpMACCtrlFramesReceived OBJECT-TYPE
    SYNTAX      Counter64
    UNITS        "frames"
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of MPCP frames passed by the MAC sublayer to the
        MAC Control sublayer. This counter is incremented when a
        ReceiveFrame function call returns a valid frame with
        1) a lengthOrType field value equal to the reserved
        Type for 802.3_MAC_Control as specified in IEEE Std 802.3
        31.4.1.3, and
        2) an opcode indicating an MPCP frame.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.1.8"
 ::= { dot3MpcpStatEntry 2 }

dot3MpcpDiscoveryWindowsSent OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of discovery windows generated. The counter is
        incremented by one for each generated discovery window.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        At the ONU, the value should be zero.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.1.22"
 ::= { dot3MpcpStatEntry 3 }

dot3MpcpDiscoveryTimeout OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of the number of times a discovery timeout
        occurs. Increment the counter by one for each discovery
        processing state-machine reset resulting from timeout
        waiting for message arrival.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
```

```
times, as indicated by the value of the
ifCounterDiscontinuityTime object of the Interfaces Group MIB
module."
REFERENCE
    "IEEE Std 802.3, 30.3.5.1.23"
::= { dot3MpcpStatEntry 4 }

dot3MpcpTxRegRequest OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times a REGISTER_REQ MPCP
        frame transmission occurs. Increment the counter by one
        for each REGISTER_REQ MPCP frame transmitted as defined
        in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        At the OLT, the value should be zero.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.1.12"
    ::= { dot3MpcpStatEntry 5 }

dot3MpcpRxRegRequest OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times a REGISTER_REQ MPCP
        frame reception occurs.
        Increment the counter by one for each REGISTER_REQ MPCP
        frame received as defined in IEEE Std 802.3, Clause 64 for
        1G-EPON or Clause 77 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        At the ONU, the value should be zero.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.5.1.17"
    ::= { dot3MpcpStatEntry 6 }

dot3MpcpTxRegAck OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times a REGISTER_ACK MPCP
```


frame transmission occurs. Increment the counter by one for each REGISTER_ACK MPCP frame transmitted as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the OLT, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.10"

::= { dot3MpcpStatEntry 7 }

dot3MpcpRxRegAck OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a REGISTER_ACK MPCP frame reception occurs. Increment the counter by one for each REGISTER_ACK MPCP frame received as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the ONU, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.15"

::= { dot3MpcpStatEntry 8 }

dot3MpcpTxReport OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a REPORT MPCP frame transmission occurs. Increment the counter by one for each REPORT MPCP frame transmitted as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the OLT, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.13"

::= { dot3MpcpStatEntry 9 }

dot3MpcpRxReport OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a REPORT MPCP frame reception occurs.
Increment the counter by one for each REPORT MPCP frame received as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the ONU, the value should be zero.
Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.18"

::= { dot3MpcpStatEntry 10 }

dot3MpcpTxGate OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a GATE MPCP frame transmission occurs.
Increment the counter by one for each GATE MPCP frame transmitted as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the ONU, the value should be zero.
Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.9"

::= { dot3MpcpStatEntry 11 }

dot3MpcpRxGate OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a GATE MPCP frame reception occurs.
Increment the counter by one for each GATE MPCP frame received as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the

OLT, it has a distinct value for each virtual interface.
At the OLT, the value should be zero.
Discontinuities of this counter can occur at
re-initialization of the management system and at other
times, as indicated by the value of the
ifCounterDiscontinuityTime object of the Interfaces Group MIB
module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.14"

::= { dot3MpcpStatEntry 12 }

dot3MpcpTxRegister OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a REGISTER MPCP frame
transmission occurs.
Increment the counter by one for each REGISTER MPCP
frame transmitted as defined in IEEE Std 802.3, Clause 64 for
1G-EPON or Clause 77 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface.
At the ONU, the value should be zero.
Discontinuities of this counter can occur at
re-initialization of the management system and at other
times, as indicated by the value of the
ifCounterDiscontinuityTime object of the Interfaces Group MIB
module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.11"

::= { dot3MpcpStatEntry 13 }

dot3MpcpRxRegister OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of times a REGISTER MPCP frame
reception occurs.
Increment the counter by one for each REGISTER MPCP
frame received as defined in IEEE Std 802.3, Clause 64 for
1G-EPON or Clause 77 for 10G-EPON.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface.
At the ONU, the value should be zero.
Discontinuities of this counter can occur at
re-initialization of the management system and at other
times, as indicated by the value of the
ifCounterDiscontinuityTime object of the Interfaces Group MIB
module."

REFERENCE

"IEEE Std 802.3, 30.3.5.1.16"

::= { dot3MpcpStatEntry 14 }

dot3OmpEmulationObjects OBJECT IDENTIFIER

::= { dot3EponObjects 2 }

dot3OmpEmulationTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3OmpEmulationEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table of dot3 OmpEmulation MIB objects. The table contain objects for the management of the OMPEmulation sublayer.

Each object has a row for every virtual link denoted by the corresponding ifIndex.

The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID."

::= { dot3OmpEmulationObjects 1 }

dot3OmpEmulationEntry OBJECT-TYPE

SYNTAX Dot3OmpEmulationEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the dot3 OmpEmulation table.

Rows exist for an OLT interface and an ONU interface.

A row in the table is denoted by the ifIndex of the link and it is created when the ifIndex is created.

The rows in the table for an ONU interface are created at system initialization.

The row in the table corresponding to the OLT ifIndex and the row corresponding to the broadcast virtual link are created at system initialization.

A row in the table corresponding to the ifIndex of a virtual links is created when a virtual link is established (ONU registers) and deleted when the virtual link is deleted (ONU deregisters)."

INDEX { ifIndex }

::= { dot3OmpEmulationTable 1 }

Dot3OmpEmulationEntry ::= SEQUENCE {

dot3OmpEmulationType INTEGER

}

dot3OmpEmulationType OBJECT-TYPE

SYNTAX INTEGER { unknown(1), olt(2), onu(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object that indicates the mode of operation of the Reconciliation Sublayer for Point-to-Point Emulation (see IEEE Std 802.3, 65.1 for 1G-EPON or 76.2 for 10G-EPON). unknown(1) value is assigned in initialization; true state or type is not yet known. olt(2) value is assigned when the sublayer is operating in OLT mode. onu(3) value is assigned when the sublayer is operating in ONU mode.

This object is applicable for an OLT, with the same value for all virtual interfaces, and for an ONU."

REFERENCE

```
"IEEE Std 802.3, 30.3.7.1.2"
 ::= { dot3OmpEmulationEntry 1 }

dot3OmpEmulationStatTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3OmpEmulationStatEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table defines the list of statistics counters of
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON,
        OMPEmulation sublayer. Each object has a row for every virtual
        link denoted by the corresponding ifIndex.
        The LLID field, as defined in IEEE Std 802.3, is a 2-byte
        register (15-bit field and a broadcast bit) limiting the
        number of virtual links to 32768. Typically the number
        of expected virtual links in a PON is like the number of
        ONUs, which is 32-64, plus an additional entry for
        broadcast LLID."
    ::= { dot3OmpEmulationObjects 2 }

dot3OmpEmulationStatEntry OBJECT-TYPE
    SYNTAX      Dot3OmpEmulationStatEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table of statistics counters of
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON,
        OMPEmulation sublayer. Rows exist for an OLT interface and an
        ONU interface. A row in the table is denoted by the ifIndex of
        the link and it is created when the ifIndex is created.
        The rows in the table for an ONU interface are created
        at system initialization.
        The row in the table corresponding to the OLT ifIndex
        and the row corresponding to the broadcast virtual link
        are created at system initialization.
        A row in the table corresponding to the ifIndex of a
        virtual links is created when a virtual link is
        established (ONU registers) and deleted when the virtual
        link is deleted (ONU deregisters)."
```

```
INDEX      { ifIndex }
 ::= { dot3OmpEmulationStatTable 1 }

Dot3OmpEmulationStatEntry ::= SEQUENCE {
    dot3OmpEmulationSLDErrors          Counter64,
    dot3OmpEmulationCRC8Errors         Counter64,
    dot3OmpEmulationBadLLID            Counter64,
    dot3OmpEmulationGoodLLID           Counter64,
    dot3OmpEmulationOnuPonCastLLID     Counter64,
    dot3OmpEmulationOltPonCastLLID     Counter64,
    dot3OmpEmulationBroadcastBitNotOnuLlid Counter64,
    dot3OmpEmulationOnuLLIDNotBroadcast Counter64,
    dot3OmpEmulationBroadcastBitPlusOnuLlid Counter64,
    dot3OmpEmulationNotBroadcastBitNotOnuLlid Counter64
}

dot3OmpEmulationSLDErrors OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
```

STATUS current

DESCRIPTION

"A count of frames received that do not contain a valid SLD field as defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1 for 10G-EPON, as appropriate. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.7.1.3"

::= { dot3OmpEmulationStatEntry 1 }

dot3OmpEmulationCRC8Errors OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames received that contain a valid SLD field, as defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1 for 10G-EPON, but do not pass the CRC-8 check as defined in IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.7.1.4"

::= { dot3OmpEmulationStatEntry 2 }

dot3OmpEmulationBadLLID OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames received that contain a valid SLD field in an OLT, and pass the CRC-8 check, but are discarded due to the LLID check. The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1 for 10G-EPON. The CRC-8 check is defined in IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON. The LLID check is defined in IEEE Std 802.3, 65.1.3.3.2 for 1G-EPON or 76.2.6.1.3.2 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.7.1.8"

```
::= { dot3OmpEmulationStatEntry 3 }

dot3OmpEmulationGoodLLID OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS   read-only
    STATUS      current
    DESCRIPTION
        "A count of frames received that contain a valid SLD
        field, as defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or
        76.2.6.1.3.1 for 10G-EPON, and pass the CRC-8 check as defined in
        IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.7.1.5"
    ::= { dot3OmpEmulationStatEntry 4 }

dot3OmpEmulationOnuPonCastLLID OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS   read-only
    STATUS      current
    DESCRIPTION
        "A count of frames received that: 1) contain a valid SLD field
        in an ONU, 2) meet the rules for frame acceptance, and
        3) pass the CRC-8 check. The SLD is defined in
        IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1 for 10G-EPON. The
        rules for LLID acceptance are defined in IEEE Std 802.3, 65.1.3.3.2
        for 1G-EPON or 76.2.6.1.3.2 for 10G-EPON. The CRC-8 check is defined
        in IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface.
        At the OLT, the value should be zero.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.3.7.1.6"
    ::= { dot3OmpEmulationStatEntry 5 }

dot3OmpEmulationOltPonCastLLID OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS   read-only
    STATUS      current
    DESCRIPTION
        "A count of frames received that contain a valid SLD field, as
        defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1
        for 10G-EPON, pass the CRC-8 check, as defined in
        IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON,
        and meet the rules of acceptance for an OLT defined in
```

IEEE Std 802.3, 65.1.3.3.2 for 1G-EPON or 76.2.6.1.3.2 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the ONU, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.3.7.1.7"

::= { dot3OmpEmulationStatEntry 6 }

dot3OmpEmulationBroadcastBitNotOnuLlid OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames received that contain a valid SLD field, as defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON or 76.2.6.1.3.1 for 10G-EPON, pass the CRC-8 check, as defined in IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON, and contain the broadcast bit in the LLID and not the ONU's LLID (frame accepted) as defined in IEEE Std 802.3, Clause 65 for 1G-EPON or Clause 76 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the OLT, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

::= { dot3OmpEmulationStatEntry 7 }

dot3OmpEmulationOnuLLIDNotBroadcast OBJECT-TYPE

SYNTAX Counter64

UNITS "frames"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames received that contain a valid SLD field, as defined in IEEE Std 802.3, 65.1.3.3.1 for 1G-EPON, pass the CRC-8 check, as defined in IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON, and contain the ONU's LLID as defined in IEEE Std 802.3, Clause 65 for 1G-EPON or Clause 76 for 10G-EPON. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. At the OLT, the value should be zero. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

::= { dot3OmpEmulationStatEntry 8 }

dot3OmpEmulationBroadcastBitPlusOnuLlid OBJECT-TYPE


```
SYNTAX      Counter64
UNITS       "frames"
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "A count of frames received that contain a valid SLD
    field, as defined in IEEE Std 802.3,
    65.1.3.3.1 for 1G-EPON, pass the CRC-8 check, as defined in
    IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON,
    and contain the broadcast bit in the LLID and match the ONU's LLID (frame
    reflected) as defined in IEEE Std 802.3, Clause 65 for 1G-EPON
    or Clause 76 for 10G-EPON.
    This object is applicable for an OLT and an ONU. At the
    OLT, it has a distinct value for each virtual interface.
    At the OLT, the value should be zero.
    Discontinuities of this counter can occur at
    re-initialization of the management system and at other
    times, as indicated by the value of the
    ifCounterDiscontinuityTime object of the Interfaces Group MIB
    module."
 ::= { dot3OmpEmulationStatEntry 9 }

dot3OmpEmulationNotBroadcastBitNotOnuLlid OBJECT-TYPE
SYNTAX      Counter64
UNITS       "frames"
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "A count of frames received that contain a valid SLD
    field, as defined in IEEE Std 802.3,
    65.1.3.3.1 for 1G-EPON, pass the CRC-8 check, as defined in
    IEEE Std 802.3, 65.1.3.3.3 for 1G-EPON or 76.2.6.1.3.3 for 10G-EPON,
    and do not contain the ONU's LLID as defined in IEEE Std 802.3, Clause 65
    for 1G-EPON or Clause 76 for 10G-EPON.
    This object is applicable for an OLT and an ONU. At the
    OLT, it has a distinct value for each virtual interface.
    At the OLT, the value should be zero.
    Discontinuities of this counter can occur at
    re-initialization of the management system and at other
    times, as indicated by the value of the
    ifCounterDiscontinuityTime object of the Interfaces Group MIB
    module."
 ::= { dot3OmpEmulationStatEntry 10 }

dot3EponFecObjects OBJECT IDENTIFIER
 ::= { dot3EponObjects 3 }

dot3EponFecTable OBJECT-TYPE
SYNTAX      SEQUENCE OF Dot3EponFecEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "A table of dot3 EPON FEC management objects.
    The entries in the table are control and status objects
    and statistic counters for the FEC layer.
    Each object has a row for every virtual link denoted by
    the corresponding ifIndex.
    The LLID field, as defined in IEEE Std 802.3, is a 2-byte
    register (15-bit field and a broadcast bit) limiting the
```

```

    number of virtual links to 32768. Typically the number
    of expected virtual links in a PON is like the number of
    ONUs, which is 32-64, plus an additional entry for
    broadcast LLID."
 ::= { dot3EponFecObjects 1 }

dot3EponFecEntry OBJECT-TYPE
    SYNTAX      Dot3EponFecEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the dot3 EPON FEC table.
        Rows exist for an OLT interface and an ONU interface.
        A row in the table is denoted by the ifIndex of the link
        and it is created when the ifIndex is created.
        The rows in the table for an ONU interface are created
        at system initialization.
        The row in the table corresponding to the OLT ifIndex
        and the row corresponding to the broadcast virtual link
        are created at system initialization.
        A row in the table corresponding to the ifIndex of a
        virtual links is created when a virtual link is
        established (ONU registers) and deleted when the virtual
        link is deleted (ONU deregisters)."
```

```

    INDEX      { ifIndex }
    ::= { dot3EponFecTable 1 }

Dot3EponFecEntry ::= SEQUENCE {
    dot3EponFecPCSCodingViolation      Counter64,
    dot3EponFecAbility                 INTEGER,
    dot3EponFecMode                     INTEGER,
    dot3EponFecCorrectedBlocks          Counter64,
    dot3EponFecUncorrectableBlocks      Counter64,
    dot3EponFecBufferHeadCodingViolation Counter64
}

dot3EponFecPCSCodingViolation OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "octets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For 100 Mb/s operation it is a count of the number of events
        that cause the PHY to indicate 'Data reception with errors' on
        the MII (see IEEE Std 802.3, Table 22-2). For 1000 Mb/s operation
        it is a count of the number of events that cause the PHY to
        indicate 'Data reception error' or 'Carrier Extend Error' on the
        GMII (see IEEE Std 802.3, Table 35-1).
        This object is applicable for an OLT and an ONU in 1G-EPON. This
        object is not defined for 10G-EPON. At the OLT, it has a distinct
        value for each virtual interface. Discontinuities of this counter
        can occur at re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.1.14"
    ::= { dot3EponFecEntry 1 }
```

dot3EponFecAbility OBJECT-TYPE
SYNTAX INTEGER { unknown(1), supported(2), unsupported(3) }
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "An object that indicates the support of operation of the
 FEC sublayer in EPON.
 unknown(1) value is assigned in the initialization, for non
 FEC support state or type not yet known. unsupported(3)
 value is assigned when the sublayer is not supported.
 supported(2) value is assigned when the sublayer is
 supported.
 This object is applicable for an OLT, with the same
 value for all virtual interfaces, and for an ONU.
 The FEC counters will have a zero value when the
 interface is not supporting FEC.
 The counters:
 dot3EponFecPCSCodingViolation - not affected by FEC
 ability.
 dot3EponFecCorrectedBlocks - has a zero value when
 dot3EponFecAbility is unknown(1) and unsupported(3).
 dot3EponFecUncorrectableBlocks - has a zero value when
 dot3EponFecAbility is unknown(1) and unsupported(3).
 dot3EponFecBufferHeadCodingViolation - has a zero value
 when dot3EponFecAbility is unknown(1) and
 unsupported(3)."
REFERENCE
 "IEEE Std 802.3, 30.5.1.1.15"
 ::= { dot3EponFecEntry 2 }

dot3EponFecMode OBJECT-TYPE
SYNTAX INTEGER { unknown(1), disabled(2), enabled(3) }
MAX-ACCESS read-write
STATUS current
DESCRIPTION
 "An object that defines the mode of operation of the
 FEC sublayer in EPON, and reflects its state.
 A GET operation returns the current mode of operation
 of the PHY. A SET operation changes the mode of
 operation of the PHY to the indicated value.
 unknown(1) value is assigned in the initialization for non
 FEC support state or type not yet known.
 disabled(2) value is assigned when the FEC sublayer is
 operating in disabled mode.
 enabled(3) value is assigned when the FEC sublayer is
 operating in FEC mode.
 The write operation is not restricted in this document
 and can be done at any time. Changing dot3EponFecMode
 state can lead to disabling the Forward Error Correction
 on the respective interface, which can lead to a
 degradation of the optical link, and therefore may lead
 to an interruption of service for the users connected to
 the respective EPON interface.
 This object is applicable for an OLT and an ONU. At the
 OLT, it has a distinct value for each virtual interface.
 The counting of the FEC counters will stop when the FEC
 of the interface is disabled.
 The counters:
 dot3EponFecPCSCodingViolation - not affected by FEC mode."

dot3EponFecCorrectedBlocks - stops counting when Rx_FEC is not enabled. (unknown(1) and disabled(2)).
 dot3EponFecUncorrectableBlocks - stops counting when Rx_FEC is not enabled (unknown(1) and disabled(2)).
 dot3EponFecBufferHeadCodingViolation - stops counting when Rx_FEC is not enabled (unknown(1) and disabled(2)).

The object:

dot3EponFecAbility - indicates the FEC ability and is not affected by the dot3EponFecMode object."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.16"

DEFVAL { unknown }

::= { dot3EponFecEntry 3 }

dot3EponFecCorrectedBlocks OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For 1000BASE-PX, 10/25/40/50/100/200/400GBASE-R, 10GBASE-PR or 10/1GBASE-PRX PHYs, it is a count of corrected FEC blocks. This counter will not increment for other PHY Types. Increment the counter by one for each received block that is corrected by the FEC function in the PHY. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.17"

::= { dot3EponFecEntry 4 }

dot3EponFecUncorrectableBlocks OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For 1000BASE-PX, 10/25/40/50/100/200/400GBASE-R, 10GBASE-PR or 10/1GBASE-PRX PHYs, it is a count of uncorrectable FEC blocks. This counter will not increment for other PHY Types. Increment the counter by one for each FEC block that is determined to be uncorrectable by the FEC function in the PHY. This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. Discontinuities of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.18"

::= { dot3EponFecEntry 5 }

dot3EponFecBufferHeadCodingViolation OBJECT-TYPE

SYNTAX Counter64

```
UNITS          "octets"
MAX-ACCESS     read-only
STATUS         current
DESCRIPTION
    "For a 1000 Mb/s operation, it is a count of the number of
    invalid code-groups received directly from the link. The
    value has a meaning only in 1000 Mb/s mode and it is
    zero otherwise.
    This object is applicable for an OLT and an ONU in 1G-EPON.
    This object is not defined in 10G-EPON. At the
    OLT, it has a distinct value for each virtual interface.
    Discontinuities of this counter can occur at
    re-initialization of the management system and at other
    times, as indicated by the value of the
    ifCounterDiscontinuityTime object of the Interfaces Group MIB
    module."
 ::= { dot3EponFecEntry 6 }

dot3ExtPkgObjects OBJECT IDENTIFIER
 ::= { dot3EponObjects 4 }

dot3ExtPkgControlObjects OBJECT IDENTIFIER
 ::= { dot3ExtPkgObjects 1 }

dot3ExtPkgControlTable OBJECT-TYPE
SYNTAX         SEQUENCE OF Dot3ExtPkgControlEntry
MAX-ACCESS     not-accessible
STATUS         current
DESCRIPTION
    "A table of Extended package Control management
    objects. Entries in the table are control and status
    indication objects of an EPON interface, which are
    gathered in an extended package as an addition to the
    objects based on IEEE Std 802.3, Clause 30, attributes.
    Each object has a row for every virtual link denoted by
    the corresponding ifIndex.
    The LLID field, as defined in IEEE Std 802.3, is a 2-byte
    register (15-bit field and a broadcast bit) limiting the
    number of virtual links to 32768. Typically the number
    of expected virtual links in a PON is like the number of
    ONUs, which is 32-64, plus an additional entry for
    broadcast LLID."
 ::= { dot3ExtPkgControlObjects 1 }

dot3ExtPkgControlEntry OBJECT-TYPE
SYNTAX         Dot3ExtPkgControlEntry
MAX-ACCESS     not-accessible
STATUS         current
DESCRIPTION
    "An entry in the Extended package Control table.
    Rows exist for an OLT interface and an ONU interface.
    A row in the table is denoted by the ifIndex of the link
    and it is created when the ifIndex is created.
    The rows in the table for an ONU interface are created
    at system initialization.
    The row in the table corresponding to the OLT ifIndex
    and the row corresponding to the broadcast virtual link
    are created at system initialization.
    A row in the table corresponding to the ifIndex of a
```

```

        virtual links is created when a virtual link is
        established (ONU registers) and deleted when the virtual
        link is deleted (ONU deregisters)."
```

INDEX { ifIndex }

```
 ::= { dot3ExtPkgControlTable 1 }
```

Dot3ExtPkgControlEntry ::= SEQUENCE {

dot3ExtPkgObjectReset	INTEGER,
dot3ExtPkgObjectPowerDown	TruthValue,
dot3ExtPkgObjectNumberOfLLIDs	Unsigned32,
dot3ExtPkgObjectFecEnabled	INTEGER,
dot3ExtPkgObjectReportMaximumNumQueues	Unsigned32,
dot3ExtPkgObjectRegisterAction	INTEGER

}

dot3ExtPkgObjectReset OBJECT-TYPE

SYNTAX INTEGER { running(1), reset(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object is used to reset the EPON interface. The interface may be unavailable while the reset occurs and data may be lost.

Setting this object to running(1) will cause the interface to enter into running mode. Setting this object to reset(2) will cause the interface to go into reset mode. When getting running(1), the interface is in running mode. When getting reset(2), the interface is in reset mode.

The write operation is not restricted in this document and can be done at any time. Changing dot3ExtPkgObjectReset state can lead to a reset of the respective interface, leading to an interruption of service for the users connected to the respective EPON interface.

This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. A reset for a specific virtual interface resets only this virtual interface and not the physical interface. Thus, a virtual link that is malfunctioning can be reset without affecting the operation of other virtual interfaces.

The reset can cause discontinuities in the values of the counters of the interface, similar to re-initialization of the management system. Discontinuity should be indicated by the ifCounterDiscontinuityTime object of the Interfaces Group MIB module."

DEFVAL { running }

```
 ::= { dot3ExtPkgControlEntry 1 }
```

dot3ExtPkgObjectPowerDown OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object is used to power down the EPON interface. The interface may be unavailable while the power down occurs and data may be lost.

Setting this object to true(1) will cause the interface

to enter into power down mode. Setting this object to false(2) will cause the interface to go out of power down mode. When getting true(1), the interface is in power down mode. When getting false(2), the interface is not in power down mode.

The write operation is not restricted in this document and can be done at any time. Changing dot3ExtPkgObjectPowerDown state can lead to a power down of the respective interface, leading to an interruption of service of the users connected to the respective EPON interface.

This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. A power down/up of a specific virtual interface affects only the virtual interface and not the physical interface. Hence a virtual link, which needs a certain handling, can be powered down and then powered up without disrupting the operation of other virtual interfaces.

The object is relevant when the admin state of the interface is active as set by the dot3MpcpAdminState."

```
DEFVAL      { false }  
::= { dot3ExtPkgControlEntry 2 }
```

dot3ExtPkgObjectNumberOfLLIDs OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A read only object that indicates the number of registered LLIDs. The initialization value is 0. This object is applicable for an OLT with the same value for all virtual interfaces and for an ONU. The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID. At the ONU the number of LLIDs for an interface is one."

```
::= { dot3ExtPkgControlEntry 3 }
```

dot3ExtPkgObjectFecEnabled OBJECT-TYPE

SYNTAX INTEGER { noFecEnabled(1), fecTxEnabled(2),
fecRxEnabled(3), fecTxRxEnabled(4) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"An object defining the FEC mode of operation of the interface, and indicating its state. The modes defined in this object are extensions to the FEC modes defined in the dot3EponFecMode object.

When noFECEnabled(1), the interface does not enable FEC mode.

When fecTxEnabled(2), the interface enables the FEC transmit mode.

When fecRxEnabled(3), the interface enables the FEC receive mode.

When fecTxRxEnabled(4), the interface enables the FEC transmit and receive mode.

This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. The FEC counters are referring to the receive path. The FEC counters will stop when the FEC receive mode of the interface is disabled, as defined by fecRxEnabled(3) and fecTxRxEnabled(4) values.

The counters:

dot3EponFecPCSCodingViolation - not affected by FEC mode.

dot3EponFecCorrectedBlocks - stops counting when Rx_FEC is not enabled (noFecEnabled(1) and fecTxEnabled(2)).

dot3EponFecUncorrectableBlocks - stops counting when Rx_FEC is not enabled (noFecEnabled(1) and fecTxEnabled(2)).

dot3EponFecBufferHeadCodingViolation - stops counting when Rx_FEC is not enabled (noFecEnabled(1) and fecTxEnabled(2)).

The objects:

dot3EponFecAbility - indicates the FEC ability and is not affected by the FEC mode.

dot3EponFecMode - indicates the FEC mode for combined RX and TX.

The write operation is not restricted in this document and can be done at any time. Changing dot3ExtPkgObjectFecEnabled state can lead to disabling the Forward Error Correction on the respective interface, which can lead to a degradation of the optical link, and therefore may lead to an interruption of service for the users connected to the respective EPON interface."

DEFVAL { noFecEnabled }
::= { dot3ExtPkgControlEntry 4 }

dot3ExtPkgObjectReportMaximumNumQueues OBJECT-TYPE

SYNTAX Unsigned32 (0..7)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An object, that defines the maximal number of queues in the REPORT message as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. For further information please see the description of the queue table.

This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface."

DEFVAL { 0 }
::= { dot3ExtPkgControlEntry 5 }

dot3ExtPkgObjectRegisterAction OBJECT-TYPE

SYNTAX INTEGER { none(1), register(2), deregister(3),
reregister(4) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"An object configuring the registration state of an interface, and indicating its registration state.

Write operation changes the registration state to its new value.

Read operation returns the value of the state.

The registration state is reflected in this object and in

the dot3MpcpRegistrationState object.
 none(1) indicates an unknown state,
 register(2) indicates a registered LLID,
 deregister(3) indicates a deregistered LLID,
 reregister(4) indicates an LLID that is reregistering.
 The following list describes the operation of the interface, as specified in IEEE Std 802.3, when a write operation is setting a value.
 none(1) - not doing any action.
 register(2) - registering an LLID that has been requested for registration (The LLID is in registering mode.
 dot3MpcpRegistrationState - registering(2)).
 deregister(3) - deregisters an LLID that is registered (dot3MpcpRegistrationState - registered(3)).
 reregister(4) - reregister an LLID that is registered (dot3MpcpRegistrationState - registered(3)).
 The behavior of an ONU and OLT interfaces, at each one of the detailed operation at each state, is described in the registration state machine of figure in Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
 This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface. The write operation is not restricted in this document and can be done at any time. Changing dot3ExtPkgObjectRegisterAction state can lead to a change in the registration state of the respective interface leading to a deregistration and an interruption of service of the users connected to the respective EPON interface."

```
DEFVAL      { none }
 ::= { dot3ExtPkgControlEntry 6 }
```

dot3ExtPkgQueueTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3ExtPkgQueueEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table of the extended package objects for queue management. The IEEE Std 802.3 MPCP defines a report message of the occupancy of the transmit queues for the feedback BW request from the ONUs. These queues serve the uplink transmission of the ONU and data is gathered there until the ONU is granted for transmission.

The management table of the queues is added here mainly to control the reporting and to gather some statistics of their operation. This table is not duplicating existing management objects of bridging queues, specified in IEEE Std 802.1Q, since the existence of a dedicated transmit queuing mechanism is implied in IEEE Std 802.3, and the ONU may be a device that is not a bridge with embedded bridging queues.

The format of the REPORT message, as specified in IEEE Std 802.3, is presented below:

```
+-----+
|          Destination Address          |
+-----+
|          Source Address               |
+-----+
|          Length/Type                  |
+-----+
```

OpCode	
TimeStamp	
Number of queue Sets	
Report bitmap	/\
Queue 0 report	
Queue 1 report	repeated for
Queue 2 report	every
Queue 3 report	queue_set
Queue 4 report	
Queue 5 report	
Queue 6 report	
Queue 7 report	
Pad/reserved	\ /
FCS	

The 'Queue report' field reports the occupancy of each uplink transmission queue.

The entries in the table are control and status indication objects for managing the queues of an EPON interface that are gathered in an extended package as an addition to the objects that are based on the IEEE Std 802.3 attributes.

The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID.

```
::= { dot3ExtPkgControlObjects 2 }
```

```
dot3ExtPkgQueueEntry OBJECT-TYPE
    SYNTAX      Dot3ExtPkgQueueEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the Extended package Queue table. At the
        OLT, the rows exist for each ifIndex and dot3QueueIndex.
        At the ONU, rows exist for the single ifIndex for each
        dot3QueueIndex.
        Rows in the table are created when the ifIndex of the
        link is created. A set of rows per queue are added for
        each ifIndex, denoted by the dot3QueueIndex.
        A set of rows per queue in the table, for an ONU
        interface, are created at the system initialization.
        A set of rows per queue in the table, corresponding to
        the OLT ifIndex and a set of rows per queue
        corresponding to the broadcast virtual link, are
        created at the system initialization.
        A set of rows per queue in the table, corresponding to
        the ifIndex of a virtual link, are created when the
        virtual link is established (ONU registers), and deleted
        when the virtual link is deleted (ONU deregisters)."
```

```
INDEX      { ifIndex, dot3QueueIndex }
 ::= { dot3ExtPkgQueueTable 1 }
```

```
Dot3ExtPkgQueueEntry ::= SEQUENCE {
    dot3QueueIndex                Unsigned32,
    dot3ExtPkgObjectReportNumThreshold Unsigned32,
    dot3ExtPkgObjectReportMaximumNumThreshold Unsigned32,
    dot3ExtPkgStatTxFramesQueue   Counter64,
    dot3ExtPkgStatRxFramesQueue   Counter64,
    dot3ExtPkgStatDroppedFramesQueue Counter64
}
```

```
dot3QueueIndex OBJECT-TYPE
    SYNTAX      Unsigned32 (0..7)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An object that identifies an index for the queue table
        reflecting the queue index of the queues that are
        reported in the MPCP REPORT message as defined in
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        The number of queues is between 0 and 7, and limited by
        dot3ExtPkgObjectReportMaximumNumQueues."
```

```
 ::= { dot3ExtPkgQueueEntry 1 }
```

```
dot3ExtPkgObjectReportNumThreshold OBJECT-TYPE
    SYNTAX      Unsigned32 (0..7)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "An object that defines the number of thresholds for each
        queue in the REPORT message as defined in IEEE Std 802.3,
        Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        Each queue_set reporting will provide information on the
        queue occupancy of frames below the matching Threshold.
        Read operation reflects the number of thresholds.
        Write operation sets the number of thresholds for each
```

```
queue.
The write operation is not restricted in this document
and can be done at any time. Value cannot exceed the
maximal value defined by the
dot3ExtPkgObjectReportMaximumNumThreshold object.
Changing dot3ExtPkgObjectReportNumThreshold can lead to
a change in the reporting of the ONU interface and
therefore to a change in the bandwidth allocation of the
respective interface. This change may lead a degradation
or an interruption of service of the users connected to
the respective EPON interface.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface
and for each queue. At the ONU, it has a distinct value
for each queue."
DEFVAL      { 0 }
::= { dot3ExtPkgQueueEntry 2 }

dot3ExtPkgObjectReportMaximumNumThreshold OBJECT-TYPE
SYNTAX      Unsigned32 (0..7)
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "An object, that defines the maximal number of thresholds
    for each queue in the REPORT message as defined in
    IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
    Each queue_set reporting will provide information on the
    queue occupancy of frames below the matching Threshold.
    This object is applicable for an OLT and an ONU. At the
    OLT, it has a distinct value for each virtual interface
    and for each queue. At the ONU, it has a distinct value
    for each queue."
DEFVAL      { 0 }
::= { dot3ExtPkgQueueEntry 3 }

dot3ExtPkgStatTxFramesQueue OBJECT-TYPE
SYNTAX      Counter64
UNITS       "frames"
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "A count of the number of times a frame transmission
    occurs from the corresponding 'Queue'.
    Increment the counter by one for each frame transmitted,
    which is an output of the 'Queue'.
    The 'Queue' marking matches the REPORT MPCP message
    Queue field as defined in IEEE Std 802.3, Clause 64 for 1G-EPON
    or Clause 77 for 10G-EPON. This object is applicable for an
    OLT and an ONU. At the OLT, it has a distinct value for
    each virtual interface and for each queue. At the ONU, it
    has a distinct value for each queue.
    At the OLT the value should be zero.
    Discontinuities of this counter can occur at
    re-initialization of the management system and at other
    times, as indicated by the value of the
    ifCounterDiscontinuityTime object of the Interfaces Group MIB
    module."
::= { dot3ExtPkgQueueEntry 4 }
```

```
dot3ExtPkgStatRxFramesQueue OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times a frame reception
        occurs from the corresponding 'Queue'.
        Increment the counter by one for each frame received,
        which is an input to the corresponding 'Queue'.
        The 'Queue' marking matches the REPORT MPCP message
        Queue field as defined in IEEE Std 802.3, Clause 64 for 1G-EPON
        or Clause 77 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface
        and for each queue. At the ONU, it has a distinct value
        for each queue.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    ::= { dot3ExtPkgQueueEntry 5 }

dot3ExtPkgStatDroppedFramesQueue OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "frames"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times a frame drop
        occurs from the corresponding 'Queue'.
        Increment the counter by one for each frame dropped
        from the corresponding 'Queue'.
        The 'Queue' marking matches the REPORT MPCP message
        Queue field as defined in IEEE Std 802.3, Clause 64 for 1G-EPON
        or Clause 77 for 10G-EPON.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface
        and for each queue. At the ONU, it has a distinct value
        for each queue.
        At the OLT, the value should be zero.
        Discontinuities of this counter can occur at
        re-initialization of the management system and at other
        times, as indicated by the value of the
        ifCounterDiscontinuityTime object of the Interfaces Group MIB
        module."
    ::= { dot3ExtPkgQueueEntry 6 }

dot3ExtPkgQueueSetsTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3ExtPkgQueueSetsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of Extended package objects used for the
        management of the queue_sets. Entries are control and
        status indication objects of an EPON interface, which
        are gathered in an extended package as an addition to
        the objects based on the IEEE Std 802.3 attributes. The
```

objects in this table are specific for the queue_sets, which are reported in the MPCP REPORT message as defined in IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for 10G-EPON. The IEEE 802.3 MPCP defines a report message of the occupancy of the transmit queues for the feedback BW request from the ONUs. These queues serve the uplink transmission of the ONU and data is gathered there until the ONU is granted for transmission. The management table of the queues_sets is added here mainly to control the reporting and to gather some statistics of their operation. This table is not duplicating existing management objects of bridging queues, specified in IEEE Std 802.1Q, since the existence of a dedicated transmit queuing mechanism is implied in IEEE Std 802.3, and the ONU may be a device that is not a bridge with embedded bridging queues. The format of the REPORT message, as specified in IEEE Std 802.3, is presented below:

+-----+		
	Destination Address	
+-----+		
	Source Address	
+-----+		
	Length/Type	
+-----+		
	OpCode	
+-----+		
	TimeStamp	
+-----+		
	Number of queue Sets	
+-----+		
	Report bitmap	/\
+-----+		
	Queue 0 report	
+-----+		
	Queue 1 report	repeated for
+-----+		
	Queue 2 report	every
+-----+		
	Queue 3 report	queue_set
+-----+		
	Queue 4 report	
+-----+		
	Queue 5 report	
+-----+		
	Queue 6 report	
+-----+		
	Queue 7 report	
+-----+		
	Pad/reserved	\/\
+-----+		
	FCS	
+-----+		

As can be seen from the message format, the ONU interface reports of the status of up to 8 queues and it can report in a single MPCP REPORT message of a few sets of queues. The number of queue_sets defines the number of the

reported sets, and it can reach a value of up to 8. It means that an ONU can hold a variable number of sets between 0 and 7.

The dot3ExtPkgQueueSetsTable table has a variable queue_set size that is limited by the dot3ExtPkgObjectReportMaximumNumThreshold object as an ONU can have fewer queue_sets to report.

The 'Queue report' field reports the occupancy of each uplink transmission queue. The queue_sets can be used to report the occupancy of the queues in a few levels as to allow granting, in an accurate manner, of only part of the data available in the queues. A Threshold is defined for each queue_set to define the level of the queue that is counted for the report of the occupancy. The threshold is reflected in the queue_set table by the dot3ExtPkgObjectReportThreshold object.

For each queue set, the report bitmap defines which queues are present in the report, meaning that although the MPCP REPORT message can report of up to 8 queues in a REPORT message, the actual number is flexible.

The dot3ExtPkgQueueSetsTable table has a variable queue size that is limited by the dot3ExtPkgObjectReportMaximumNumQueues object as an ONU can have fewer queues to report.

Each object has a row for every virtual link, for each queue in the report and for each queue_set in the queue. The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID.

The number of queues is between 0 and 7 and limited by dot3ExtPkgObjectReportMaximumNumQueues.

The number of queue_sets is between 0 and 7 and limited by dot3ExtPkgObjectReportMaximumNumThreshold."

```
::= { dot3ExtPkgControlObjects 3 }
```

dot3ExtPkgQueueSetsEntry OBJECT-TYPE

SYNTAX Dot3ExtPkgQueueSetsEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the Extended package queue_set table. At the OLT, the rows exist for each ifIndex, dot3QueueSetQueueIndex and dot3QueueSetIndex. At the ONU, rows exist for the single ifIndex, for each dot3QueueSetQueueIndex and dot3QueueSetIndex. Rows in the table are created when the ifIndex of the link is created. A set of rows per queue and per queue_set are added for each ifIndex, denoted by dot3QueueSetIndex and dot3QueueSetQueueIndex. A set of rows per queue and per queue_set in the table, for an ONU interface are created at system initialization.

A set of rows per queue and per queue_Set in the table, corresponding to the OLT ifIndex and a set of rows per queue and per queue_set, corresponding to the broadcast

virtual link, are created at system initialization.
A set of rows per queue and per queue_set in the table, corresponding to the ifIndex of a virtual link are created when the virtual link is established (ONU registers) and deleted when the virtual link is deleted (ONU deregisters)."

```

INDEX      { ifIndex, dot3QueueSetQueueIndex, dot3QueueSetIndex }
 ::= { dot3ExtPkgQueueSetsTable 1 }

Dot3ExtPkgQueueSetsEntry ::= SEQUENCE {
    dot3QueueSetQueueIndex      Unsigned32,
    dot3QueueSetIndex           Unsigned32,
    dot3ExtPkgObjectReportThreshold Unsigned32
}

dot3QueueSetQueueIndex OBJECT-TYPE
    SYNTAX      Unsigned32 (0..7)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An object that identifies the queue index for the
        dot3ExtPkgQueueSetsTable table. The queues are reported
        in the MPCP REPORT message as defined in IEEE Std 802.3,
        Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        The number of queues is between 0 and 7, and limited by
        dot3ExtPkgObjectReportMaximumNumQueues.
        Value corresponds to the dot3QueueIndex of the queue
        table."
    ::= { dot3ExtPkgQueueSetsEntry 1 }

dot3QueueSetIndex OBJECT-TYPE
    SYNTAX      Unsigned32 (0..7)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An object that identifies the queue_set index for the
        dot3ExtPkgQueueSetsTable table. The queues are reported
        in the MPCP REPORT message as defined in IEEE Std 802.3,
        Clause 64 for 1G-EPON or Clause 77 for 10G-EPON.
        The number of queues_sets is between 0 and 7, and
        limited by dot3ExtPkgObjectReportMaximumNumThreshold."
    ::= { dot3ExtPkgQueueSetsEntry 2 }

dot3ExtPkgObjectReportThreshold OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS        "TQ (16 ns)"
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "An object that defines the value of a threshold report
        for each queue_set in the REPORT message as defined in
        IEEE Std 802.3, Clause 64 for 1G-EPON or Clause 77 for
        10G-EPON. The number of sets for each queue is
        dot3ExtPkgObjectReportNumThreshold.
        In the REPORT message, each queue_set reporting will
        provide information on the occupancy of the queues for
        frames below the matching threshold.
        The value returned shall be in Time quanta (TQ), which
        is 16 ns or 2 octets increments."

```


Read operation provides the threshold value. Write operation sets the value of the threshold.
The write operation is not restricted in this document and can be done at any time. Changing dot3ExtPkgObjectReportThreshold can lead to a change in the reporting of the ONU interface and therefore to a change in the bandwidth allocation of the respective interface. This change may lead a degradation or an interruption of service for the users connected to the respective EPON interface.
This object is applicable for an OLT and an ONU. At the OLT, it has a distinct value for each virtual interface, for each queue and for each queue_set. At the ONU, it has a distinct value for each queue and for each queue_set."

```
DEFVAL      { 0 }
::= { dot3ExtPkgQueueSetsEntry 3 }
```

dot3ExtPkgOptIfTable OBJECT-TYPE
SYNTAX SEQUENCE OF Dot3ExtPkgOptIfEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
 "This table defines the control and status indication objects for the optical interface of the EPON interface. Each object has a row for every virtual link denoted by the corresponding ifIndex.
 The LLID field, as defined in IEEE Std 802.3, is a 2-byte register (15-bit field and a broadcast bit) limiting the number of virtual links to 32768. Typically the number of expected virtual links in a PON is like the number of ONUs, which is 32-64, plus an additional entry for broadcast LLID.
 Although the optical interface is a physical interface, there is a row in the table for each virtual interface. The reason for having a separate row for each virtual link is that the OLT has a separate link for each one of the ONUs. For instance, ONUs could be in different distances with different link budgets and different receive powers, therefore having different power alarms. It is quite similar to a case of different physical interfaces."
 ::= { dot3ExtPkgControlObjects 5 }

dot3ExtPkgOptIfEntry OBJECT-TYPE
SYNTAX Dot3ExtPkgOptIfEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
 "An entry in the optical interface table of the EPON interface.
 Rows exist for an OLT interface and an ONU interface. A row in the table is denoted by the ifIndex of the link and it is created when the ifIndex is created.
 The rows in the table for an ONU interface are created at system initialization.
 The row in the table corresponding to the OLT ifIndex and the row corresponding to the broadcast virtual link are created at system initialization.
 A row in the table corresponding to the ifIndex of a

```

        virtual link is created when a virtual link is
        established (ONU registers) and deleted when the virtual
        link is deleted (ONU deregisters)."
```

INDEX { ifIndex }

```
 ::= { dot3ExtPkgOptIfTable 1 }
```

Dot3ExtPkgOptIfEntry ::= SEQUENCE {

dot3ExtPkgOptIfSuspectedFlag	TruthValue,
dot3ExtPkgOptIfInputPower	Integer32,
dot3ExtPkgOptIfLowInputPower	Integer32,
dot3ExtPkgOptIfHighInputPower	Integer32,
dot3ExtPkgOptIfLowerInputPowerThreshold	Integer32,
dot3ExtPkgOptIfUpperInputPowerThreshold	Integer32,
dot3ExtPkgOptIfOutputPower	Integer32,
dot3ExtPkgOptIfLowOutputPower	Integer32,
dot3ExtPkgOptIfHighOutputPower	Integer32,
dot3ExtPkgOptIfLowerOutputPowerThreshold	Integer32,
dot3ExtPkgOptIfUpperOutputPowerThreshold	Integer32,
dot3ExtPkgOptIfSignalDetect	TruthValue,
dot3ExtPkgOptIfTransmitAlarm	TruthValue,
dot3ExtPkgOptIfTransmitEnable	TruthValue

}

dot3ExtPkgOptIfSuspectedFlag OBJECT-TYPE

```

SYNTAX      TruthValue
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "This object is a reliability indication.
     If true, the data in this entry may be unreliable.
     This object is applicable for an OLT and an ONU. At the
     OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 1 }
```

dot3ExtPkgOptIfInputPower OBJECT-TYPE

```

SYNTAX      Integer32
UNITS       "0.1 dBm"
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "The optical power monitored at the input.
     This object is applicable for an OLT and an ONU. At the
     OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 2 }
```

dot3ExtPkgOptIfLowInputPower OBJECT-TYPE

```

SYNTAX      Integer32
UNITS       "0.1 dBm"
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "The lowest optical power monitored at the input during the
     current 15-minute interval.
     This object is applicable for an OLT and an ONU. At the
     OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 3 }
```

dot3ExtPkgOptIfHighInputPower OBJECT-TYPE

```

SYNTAX      Integer32
```

```

UNITS          "0.1 dBm"
MAX-ACCESS     read-only
STATUS         current
DESCRIPTION
    "The highest optical power monitored at the input during the
    current 15-minute interval.
    This object is applicable for an OLT and an ONU. At the
    OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 4 }

```

```

dot3ExtPkgOptIfLowerInputPowerThreshold OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 dBm"
    MAX-ACCESS   read-write
    STATUS      current
    DESCRIPTION
        "The lower limit threshold on input power. If
        dot3ExtPkgOptIfInputPower drops to this value or below,
        a Threshold Crossing Alert (TCA) should be sent.
        Reading will present the threshold value. Writing will
        set the value of the threshold.
        The write operation is not restricted in this document
        and can be done at any time. Changing
        dot3ExtPkgOptIfLowerInputPowerThreshold can lead to a Threshold
        Crossing Alert (TCA) being sent for the respective interface.
        This alert may lead to an interruption of service for the
        users connected to the respective EPON interface, depending on
        the system action on such an alert.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 5 }

```

```

dot3ExtPkgOptIfUpperInputPowerThreshold OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 dBm"
    MAX-ACCESS   read-write
    STATUS      current
    DESCRIPTION
        "The upper limit threshold on input power. If
        dot3ExtPkgOptIfInputPower reaches or exceeds this value,
        a Threshold Crossing Alert (TCA) should be sent.
        Reading will present the threshold value. Writing will
        set the value of the threshold.
        The write operation is not restricted in this document
        and can be done at any time. Changing
        dot3ExtPkgOptIfUpperInputPowerThreshold can lead to a Threshold
        Crossing Alert (TCA) being sent for the respective interface.
        This alert may lead to an interruption of service for the
        users connected to the respective EPON interface, depending on
        the system action on such an alert.
        This object is applicable for an OLT and an ONU. At the
        OLT, it has a distinct value for each virtual interface."
 ::= { dot3ExtPkgOptIfEntry 6 }

```

```

dot3ExtPkgOptIfOutputPower OBJECT-TYPE
    SYNTAX      Integer32
    UNITS       "0.1 dBm"
    MAX-ACCESS   read-only
    STATUS      current

```

DESCRIPTION

"The optical power monitored at the output.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."

::= { dot3ExtPkgOptIfEntry 7 }

dot3ExtPkgOptIfLowOutputPower OBJECT-TYPE

SYNTAX Integer32

UNITS "0.1 dBm"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The lowest optical power monitored at the output during the
current 15-minute interval.

This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."

::= { dot3ExtPkgOptIfEntry 8 }

dot3ExtPkgOptIfHighOutputPower OBJECT-TYPE

SYNTAX Integer32

UNITS "0.1 dBm"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The highest optical power monitored at the output during the
current 15-minute interval.

This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."

::= { dot3ExtPkgOptIfEntry 9 }

dot3ExtPkgOptIfLowerOutputPowerThreshold OBJECT-TYPE

SYNTAX Integer32

UNITS "0.1 dBm"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The lower limit threshold on output power. If
dot3ExtPkgOptIfOutputPower drops to this value or below,
a Threshold Crossing Alert (TCA) should be sent.
Reading will present the threshold value. Writing will
set the value of the threshold.
The write operation is not restricted in this document
and can be done at any time. Changing
dot3ExtPkgOptIfLowerOutputPowerThreshold can lead to a Threshold
Crossing Alert (TCA) being sent for the respective interface.
This alert may lead to an interruption of service for the
users connected to the respective EPON interface, depending on
the system action on such an alert.

This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."

::= { dot3ExtPkgOptIfEntry 10 }

dot3ExtPkgOptIfUpperOutputPowerThreshold OBJECT-TYPE

SYNTAX Integer32

UNITS "0.1 dBm"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The upper limit threshold on output power. If

dot3ExtPkgOptIfOutputPower reaches or exceeds this value,
a Threshold Crossing Alert (TCA) should be sent.
Reading will present the threshold value. Writing will
set the value of the threshold.
The write operation is not restricted in this document
and can be done at any time. Changing
dot3ExtPkgOptIfUpperOutputPowerThreshold can lead to a Threshold
Crossing Alert (TCA) being sent for the respective interface.
This alert may lead to an interruption of service of the
users connected to the respective EPON interface, depending on
the system action on such an alert.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."
::= { dot3ExtPkgOptIfEntry 11 }

dot3ExtPkgOptIfSignalDetect OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"When getting true(1), there is a valid optical signal at
the receive that is above the optical power level for
signal detection. When getting false(2) the optical
signal at the receive is below the optical power level
for signal detection.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."
DEFVAL { false }
::= { dot3ExtPkgOptIfEntry 12 }

dot3ExtPkgOptIfTransmitAlarm OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"When getting true(1) there is a non-valid optical signal
at the transmit of the interface, either a higher level
or lower level than expected. When getting false(2) the
optical signal at the transmit is valid and in the
required range.
This object is applicable for an OLT and an ONU. At the
OLT, it has a distinct value for each virtual interface."
DEFVAL { false }
::= { dot3ExtPkgOptIfEntry 13 }

dot3ExtPkgOptIfTransmitEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Setting this object to true(1) will cause the optical
interface to start transmission (according to the
control protocol specified for the logical interface).
Setting this object to false(2) will cause the
interface to stop the optical transmission.
When getting true(1), the optical interface is in
transmitting mode (obeying to the logical control
protocol).
When getting false(2), the optical interface is not in

```
transmitting mode.
The write operation is not restricted in this document
and can be done at any time. Changing
dot3ExtPkgOptIfTransmitEnable state can lead to a halt
in the optical transmission of the respective interface
leading to an interruption of service of the users
connected to the respective EPON interface.
The object is relevant when the admin state of the
interface is active as set by the dot3MpcpAdminState.
This object is applicable for an OLT and an ONU. At the
OLT it, has a distinct value for each virtual interface."
DEFVAL      { false }
::= { dot3ExtPkgOptIfEntry 14 }

dot3RecognizedMulticastIDsTable OBJECT-TYPE
SYNTAX      SEQUENCE OF Dot3RecognizedMulticastIDsEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "A table of MulticastIDs to be recognized by this device."
REFERENCE
    "IEEE Std 802.3, 30.3.5.1.25"
::= { dot3EponObjects 5 }

dot3RecognizedMulticastIDsEntry OBJECT-TYPE
SYNTAX      Dot3RecognizedMulticastIDsEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "An entry in the table of MulticastIDs to be recognized by this
    device."
INDEX       { ifIndex, dot3RecognizedMulticastIDIndex }
::= { dot3RecognizedMulticastIDsTable 1 }

Dot3RecognizedMulticastIDsEntry ::= SEQUENCE {
    dot3RecognizedMulticastIDIndex  Unsigned32,
    dot3RecognizedMulticastID       Unsigned32
}

dot3RecognizedMulticastIDIndex OBJECT-TYPE
SYNTAX      Unsigned32 (0..127)
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "An index into the table of MulticastIDs to be recognized by this
    device."
::= { dot3RecognizedMulticastIDsEntry 1 }

dot3RecognizedMulticastID OBJECT-TYPE
SYNTAX      Unsigned32
MAX-ACCESS  read-write
STATUS      current
DESCRIPTION
    "An unsigned32 representing a single MulticastID to be recognized
    by this device."
REFERENCE
    "IEEE Std 802.3, 30.3.5.1.25"
::= { dot3RecognizedMulticastIDsEntry 2 }
```

```
dot3EponConformance OBJECT IDENTIFIER
    ::= { ieee8023dot3EponMIB 2 }

dot3EponGroups OBJECT IDENTIFIER
    ::= { dot3EponConformance 1 }

dot3EponCompliances OBJECT IDENTIFIER
    ::= { dot3EponConformance 2 }

dot3MpcpGroupBase OBJECT-GROUP
    OBJECTS      { dot3MpcpOperStatus, dot3MpcpAdminState, dot3MpcpMode,
                    dot3MpcpSyncTime, dot3MpcpLinkID,
                    dot3MpcpRemoteMACAddress, dot3MpcpRegistrationState,
                    dot3MpcpMaximumPendingGrants, dot3MpcpTransmitElapsed,
                    dot3MpcpReceiveElapsed, dot3MpcpRoundTripTime }
    STATUS       current
    DESCRIPTION   "A collection of objects of dot3 Mpcp Control entity state
                    definition. Objects are per LLID."
    ::= { dot3EponGroups 1 }

dot3MpcpGroupStat OBJECT-GROUP
    OBJECTS      { dot3MpcpMACCtrlFramesTransmitted,
                    dot3MpcpMACCtrlFramesReceived,
                    dot3MpcpDiscoveryWindowsSent, dot3MpcpDiscoveryTimeout,
                    dot3MpcpTxRegRequest, dot3MpcpRxRegRequest,
                    dot3MpcpTxRegAck, dot3MpcpRxRegAck, dot3MpcpTxReport,
                    dot3MpcpRxReport, dot3MpcpTxGate, dot3MpcpRxGate,
                    dot3MpcpTxRegister, dot3MpcpRxRegister }
    STATUS       current
    DESCRIPTION   "A collection of objects of dot3 Mpcp Statistics.
                    Objects are per LLID."
    ::= { dot3EponGroups 2 }

dot3OmpGroupID OBJECT-GROUP
    OBJECTS      { dot3OmpEmulationType }
    STATUS       current
    DESCRIPTION   "A collection of objects of dot3 OMP emulation entity
                    state definition. Objects are per LLID."
    ::= { dot3EponGroups 3 }

dot3OmpGroupStat OBJECT-GROUP
    OBJECTS      { dot3OmpEmulationSLDErrors, dot3OmpEmulationCRC8Errors,
                    dot3OmpEmulationBadLLID, dot3OmpEmulationGoodLLID,
                    dot3OmpEmulationOnuPonCastLLID,
                    dot3OmpEmulationOltPonCastLLID,
                    dot3OmpEmulationBroadcastBitNotOnuLlid,
                    dot3OmpEmulationOnuLLIDNotBroadcast,
                    dot3OmpEmulationBroadcastBitPlusOnuLlid,
                    dot3OmpEmulationNotBroadcastBitNotOnuLlid }
    STATUS       current
    DESCRIPTION   "A collection of objects of dot3 OMP emulation
                    Statistics. Objects are per LLID."
    ::= { dot3EponGroups 4 }

dot3EponFecGroupAll OBJECT-GROUP
```

```
OBJECTS      { dot3EponFecPCSCodingViolation, dot3EponFecAbility,
                dot3EponFecMode, dot3EponFecCorrectedBlocks,
                dot3EponFecUncorrectableBlocks,
                dot3EponFecBufferHeadCodingViolation }
STATUS       current
DESCRIPTION   "A collection of objects of dot3 FEC group control and
                statistics. Objects are per LLID."
::= { dot3EponGroups 5 }

dot3ExtPkgGroupControl OBJECT-GROUP
OBJECTS      { dot3ExtPkgObjectReset, dot3ExtPkgObjectPowerDown,
                dot3ExtPkgObjectNumberOfLLIDs,
                dot3ExtPkgObjectFecEnabled,
                dot3ExtPkgObjectReportMaximumNumQueues,
                dot3ExtPkgObjectRegisterAction }
STATUS       current
DESCRIPTION   "A collection of objects of dot3ExtPkg control
                definition. Objects are per LLID."
::= { dot3EponGroups 6 }

dot3ExtPkgGroupQueue OBJECT-GROUP
OBJECTS      { dot3ExtPkgObjectReportNumThreshold,
                dot3ExtPkgObjectReportMaximumNumThreshold,
                dot3ExtPkgStatTxFramesQueue,
                dot3ExtPkgStatRxFramesQueue,
                dot3ExtPkgStatDroppedFramesQueue }
STATUS       current
DESCRIPTION   "A collection of objects of dot3ExtPkg Queue
                control. Objects are per LLID, per queue."
::= { dot3EponGroups 7 }

dot3ExtPkgGroupQueueSets OBJECT-GROUP
OBJECTS      { dot3ExtPkgObjectReportThreshold }
STATUS       current
DESCRIPTION   "A collection of objects of dot3ExtPkg queue_set
                control. Objects are per LLID, per queue, per
                queue_set."
::= { dot3EponGroups 8 }

dot3ExtPkgGroupOptIf OBJECT-GROUP
OBJECTS      { dot3ExtPkgOptIfSuspectedFlag,
                dot3ExtPkgOptIfInputPower,
                dot3ExtPkgOptIfLowInputPower,
                dot3ExtPkgOptIfHighInputPower,
                dot3ExtPkgOptIfLowerInputPowerThreshold,
                dot3ExtPkgOptIfUpperInputPowerThreshold,
                dot3ExtPkgOptIfOutputPower,
                dot3ExtPkgOptIfLowOutputPower,
                dot3ExtPkgOptIfHighOutputPower,
                dot3ExtPkgOptIfLowerOutputPowerThreshold,
                dot3ExtPkgOptIfUpperOutputPowerThreshold,
                dot3ExtPkgOptIfSignalDetect,
                dot3ExtPkgOptIfTransmitAlarm,
                dot3ExtPkgOptIfTransmitEnable }
STATUS       current
```



```
DESCRIPTION
    "A collection of objects of control and status indication
      of the optical interface.
      Objects are per LLID."
 ::= { dot3EponGroups 9 }

dot3EponGroupMulticastIDs OBJECT-GROUP
OBJECTS      { dot3RecognizedMulticastID }
STATUS       current
DESCRIPTION
    "One of a set of MulticastIDs recognized by an EPON interface."
 ::= { dot3EponGroups 10 }

dot3MPCPCompliance MODULE-COMPLIANCE
STATUS       current
DESCRIPTION
    "The compliance statement for MultiPoint
      Control Protocol interfaces."

MODULE       -- this module

    MANDATORY-GROUPS      { dot3MpcpGroupBase }

    GROUP      dot3MpcpGroupStat
    DESCRIPTION
        "This group is mandatory for all MPCP supporting
          interfaces for statistics collection."

 ::= { dot3EponCompliances 1 }

dot3OmpeCompliance MODULE-COMPLIANCE
STATUS       current
DESCRIPTION
    "The compliance statement for OMPEmulation
      interfaces."

MODULE       -- this module

    MANDATORY-GROUPS      { dot3OmpeGroupID }

    GROUP      dot3OmpeGroupStat
    DESCRIPTION
        "This group is mandatory for all OMPemulation
          supporting interfaces for statistics collection."

 ::= { dot3EponCompliances 2 }

dot3EponFecCompliance MODULE-COMPLIANCE
STATUS       current
DESCRIPTION
    "The compliance statement for FEC EPON interfaces.
      This group is mandatory for all FEC supporting
      interfaces for control and statistics collection."

MODULE       -- this module

    MANDATORY-GROUPS      { dot3EponFecGroupAll }

 ::= { dot3EponCompliances 3 }
```

```
dot3ExtPkgCompliance MODULE-COMPLIANCE
    STATUS          current
    DESCRIPTION
        "The compliance statement for EPON Interfaces
        using the extended package."

    MODULE          -- this module

        MANDATORY-GROUPS          { dot3ExtPkgGroupControl }

        GROUP    dot3ExtPkgGroupQueue
        DESCRIPTION
            " This group is mandatory for all EPON interfaces
            supporting REPORT queue management of the extended
            package."

        GROUP    dot3ExtPkgGroupQueueSets
        DESCRIPTION
            " This group is mandatory for all EPON interfaces
            supporting REPORT queue_sets management of the
            extended package."

        GROUP    dot3ExtPkgGroupOptIf
        DESCRIPTION
            "This group is mandatory for all EPON interfaces
            supporting optical interfaces management,
            of the extended package."

    ::= { dot3EponCompliances 4 }

dot3EponMulticastIDsCompliance MODULE-COMPLIANCE
    STATUS          current
    DESCRIPTION
        "The compliance statement for EPON Interfaces that
        support MulticastIDs."

    MODULE          -- this module

        MANDATORY-GROUPS          { dot3EponGroupMulticastIDs }

    ::= { dot3EponCompliances 5 }

END -- end of module IEEE8023-DOT3-EPON-MIB.
```

10. Ethernet-like interface MIB module

10.1 Introduction

This clause defines a portion of the MIB for use with SNMP. In particular, it defines objects for managing Ethernet-like interfaces.

10.2 Overview

Instances of these object types represent attributes of an interface to an Ethernet-like communications medium.

The definitions presented here are based on Clause 30 of IEEE Std 802.3. implementers of these MIB objects should note that IEEE Std 802.3 explicitly describes (in the form of Pascal pseudocode) when, where, and how various MAC attributes are measured. IEEE Std 802.3 also describes the effects of MAC actions that may be invoked by manipulating instances of the MIB objects defined here.

To the extent that some of the attributes defined in IEEE Std 802.3 are represented by previously defined objects in MIB-2 from IETF RFC 1213 or in the Interfaces Group MIB defined in IETF RFC 2863, such attributes are not redundantly represented by objects defined in this clause. Among the attributes represented by objects defined in other MIB module specifications are the number of octets transmitted or received on a particular interface, the number of frames transmitted or received on a particular interface, the promiscuous status of an interface, the MAC address of an interface, and multicast information associated with an interface.

10.2.1 Relation to MIB-2

This subclause applies only when this MIB is used in conjunction with the IETF RFC 1213 interface group.

The relationship between an Ethernet-like interface and an interface in the context of MIB-2 is one-to-one. As such, the value of an ifIndex object instance can be directly used to identify corresponding instances of the objects defined herein.

10.2.2 Relation to the Interfaces Group MIB

The Interfaces Group MIB defined in IETF RFC 2863 requires that any MIB that is an adjunct of the Interfaces Group MIB clarify specific areas within the Interfaces Group MIB. These areas were intentionally left vague in the Interfaces Group MIB to avoid over-constraining the MIB, thereby precluding management of certain media-types.

Section 4 of IETF RFC 2863 enumerates several areas that a media-specific MIB must (wherein the word “must” is used in accordance with the requirements of IETF RFC 2119) clarify. Each of these areas is addressed in a following subclause. The implementer is referred to IETF RFC 2863 in order to understand the general intent of these areas.

10.2.2.1 ifRcvAddressTable

This table contains all IEEE 802.3 addresses, unicast, multicast, and broadcast, for which this interface will receive packets and forward them up to a higher layer entity for local consumption. The format of the address, contained in ifRcvAddressAddress, is the same as for ifPhysAddress.

In the event that the interface is part of a MAC bridge, this table does not include unicast addresses that are accepted for possible forwarding out some other port. This table is explicitly not intended to provide a bridge address filtering mechanism.

10.2.2.2 ifType

All Ethernet-like interfaces shall return ethernetCsmacd(6) for ifType. Information on the particular port type and operating speed is available from ifSpeed in the Interfaces Group MIB, and ifMauType in the MAU-MIB module defined in Clause 13. All Ethernet-like interfaces shall also implement the MAU-MIB module defined in Clause 13.²⁴

10.2.2.3 ifXxxOctets

The Interfaces Group MIB octet counters, ifInOctets, ifOutOctets, ifHCInOctets, and ifHCOctets, include all octets in valid frames sent or received on the interface, including the MAC header and FCS, but not the preamble, start of frame delimiter, or extension octets. This corresponds to the definition of frameSize/8 in 4.2.7.1 of IEEE Std 802.3 (frameSize is defined in bits rather than in octets, and it is defined as $2 \times \text{addressSize} + \text{lengthOrTypeSize} + \text{dataSize} + \text{crcSize}$). They do not include the number of octets in collided or failed transmit attempts, since the MAC layer driver typically does not have visibility to count these octets. They also do not include octets in received invalid frames, since this information is normally not passed to the MAC layer, and since non-promiscuous MAC implementations cannot reliably determine whether an invalid frame was actually addressed to this station.

Note that these counters do include octets in valid MAC control frames sent or received on the interface, as well as octets in otherwise valid received MAC frames that are discarded by the MAC layer for some reason (insufficient buffer space, unknown protocol, etc.).

Note that the octet counters in IF-MIB do not exactly match the definition of the octet counters in IEEE Std 802.3. aOctetsTransmittedOK and aOctetsReceivedOK count only the octets in the clientData and Pad fields, whereas ifInOctets and ifOutOctets include the entire MAC frame, including MAC header and FCS. However, the IF-MIB counters can be derived from the IEEE 802.3 counters as follows in Equation (1) and Equation (2):

$$\text{ifInOctets} = \text{aOctetsReceivedOK} + (18 \times \text{aFramesReceivedOK}) \quad (1)$$

$$\text{ifOutOctets} = \text{aOctetsTransmittedOK} + (18 \times \text{aFramesTransmittedOK}) \quad (2)$$

Another difference to keep in mind between the IF-MIB counters and IEEE 802.3 counters is that, in IEEE Std 802.3, the frame counters and octet counters are incremented together. aOctetsTransmittedOK counts the number of octets in frames that were counted by aFramesTransmittedOK. aOctetsReceivedOK counts the number of octets in frames that were counted by aFramesReceivedOK. This is not the case with the IF-MIB counters. The IF-MIB octet counters count the number of octets sent to or received from the layer below this interface, whereas the packet counters count the number of packets sent to or received from the layer above. Therefore, received MAC Control frames, ifInDiscards, and ifInUnknownProtos are counted by ifInOctets, but not by ifInXcastPkts. Transmitted MAC Control frames are counted by ifOutOctets, but not by ifOutXcastPkts. ifOutDiscards and ifOutErrors are counted by ifOutXcastPkts, but not by ifOutOctets.

²⁴ There are three other interface types defined in IANAifType-MIB for Ethernet, namely, fastEther(62), fastEtherFX(69), and gigabitEthernet(117). Management applications should be prepared to receive these obsolete ifType values from older implementations.

10.2.2.4 ifXxxXcastPkts

The packet counters in the IF-MIB do not exactly match the definition of the frame counters in IEEE Std 802.3. `aFramesTransmittedOK` counts the number of frames successfully transmitted on the interface, whereas `ifOutUcastPkts`, `ifOutMulticastPkts`, and `ifOutBroadcastPkts` count the number of transmit requests made from a higher layer, whether or not the transmit attempt was successful. This means that packets counted by `ifOutErrors` or `ifOutDiscards` are also counted by `ifOutXcastPkts`, but they are not counted by `aFramesTransmittedOK`. This also means that, since MAC Control frames are generated by a sublayer internal to the interface layer rather than by a higher layer, they are not counted by `ifOutXcastPkts`, but they are counted by `aFramesTransmittedOK`:

$$\begin{aligned} \text{aFramesTransmittedOK} = & \text{ifOutUcastPkts} + \text{ifOutMulticastPkts} \\ & + \text{ifOutBroadcastPkts} + \text{dot3OutPauseFrames} \\ & - (\text{ifOutErrors} + \text{ifOutDiscards}) \end{aligned} \quad (3)$$

Similarly, `aFramesReceivedOK` counts the number of frames received successfully by the interface, whether or not they are passed to a higher layer, whereas `ifInUcastPkts`, `ifInMulticastPkts`, and `ifInBroadcastPkts` count only the number of packets passed to a higher layer. This means that packets counted by `ifInDiscards` or `ifInUnknownProtos` are also counted by `aFramesReceivedOK`, but they are not counted by `ifInXcastPkts`. This also means that, since MAC Control frames are consumed by a sublayer internal to the interface layer and not passed to a higher layer, they are not counted by `ifInXcastPkts`, but they are counted by `aFramesReceivedOK`:

$$\begin{aligned} \text{aFramesReceivedOK} = & \text{ifInUcastPkts} + \text{ifInMulticastPkts} \\ & + \text{ifInBroadcastPkts} + \text{dot3InPauseFrames} \\ & + \text{ifInDiscards} + \text{ifInUnknownProtos} \end{aligned} \quad (4)$$

This specification chooses to treat MAC control frames as being originated and consumed within the interface and not counted by the IF-MIB packet counters. MAC control frames are normally sent as multicast packets. In many network environments, MAC control frames can greatly outnumber multicast frames carrying actual data. If MAC control frames were included in the `ifInMulticastPkts` and `ifOutMulticastPkts`, the count of data-carrying multicast packets would tend to be drowned out by the count of MAC control frames, rendering those counters considerably less useful.

To better understand the issues surrounding the mapping of the IF-MIB packet and octet counters to an Ethernet interface, it is useful to refer to a Case diagram (Case and Partridge [B2]) for the IF-MIB counters, with modifications to show the proper interpretation for the Ethernet interface. This is depicted in Figure 10-1.

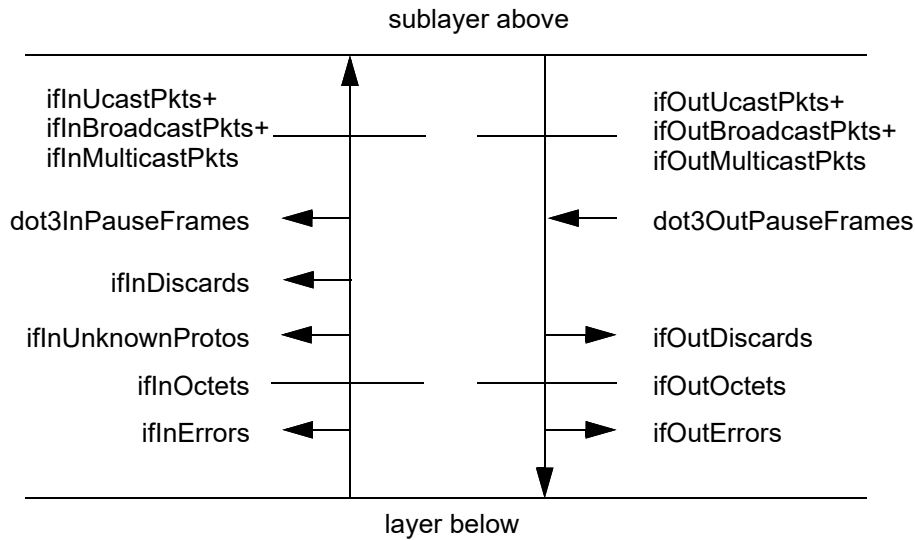


Figure 10-1—Case diagram for the IF-MIB counters

10.2.2.5 ifMtu

The defined standard MTU for Ethernet-like interfaces is 1500 octets. However, many implementations today support larger packet sizes than IEEE Std 802.3. The value of this object shall reflect the actual MTU in use on the interface, whether it matches the standard MTU or not.

This value should reflect the value seen by the MAC client interface. When a higher layer protocol, like IP, is running over Ethernet framing, this is the MTU that will be seen by that higher layer protocol. However, most Ethernet-like interfaces today run multiple protocols that use a mix of different framing types. For example, an IEEE 802.2 LLC type 1 client protocol will see an MTU of 1497 octets on an interface using the IEEE standard maximum packet size, and a protocol running over SNAP will see an MTU of 1492 octets on an interface using the IEEE standard maximum packet size. However, since the specification mandates using the MTU as seen at the MAC client interface, the value of ifMtu would be reported as 1500 octets in these cases.

10.2.2.6 ifSpeed and ifHighSpeed

For Ethernet-like interfaces operating at 1000 Megabits per second (Mb/s) or less, ifSpeed will represent the current operational speed of the interface in bits per second. For such interface types, this will be equal to 1 000 000 (1 million), 10 000 000 (10 million), 100 000 000 (100 million), or 1 000 000 000 (1 billion). ifHighSpeed will represent the current operational speed in millions of bits per second. For such Ethernet-like interfaces, this will be equal to 1, 10, 100, or 1000. If the interface implements Auto-Negotiation, Auto-Negotiation is enabled for this interface, and the interface has not yet negotiated to an operational speed, then these objects should reflect the maximum speed supported by the interface.

For Ethernet-like interfaces operating at greater than 1000 Mb/s, ifHighSpeed will represent the current operational speed of the interface in millions of bits per second. Note that for WAN implementations, this will be the payload data rate over the WAN interface sublayer. For current implementations, this will be equal to 10 000 for LAN implementations of 10 Gb/s, and 9294 for WAN implementations of the 10 Gb/s MAC over an OC-192 PHY. For these speeds, ifSpeed should report a maximum unsigned 32-bit value of 4 294 967 295 as specified in IETF RFC 2863.

These objects shall indicate the correct line speed regardless of the current duplex mode. They shall not indicate a doubled value when operating in full-duplex mode. The duplex mode of the interface may be determined by examining either the dot3StatsDuplexStatus object in this MIB module or the ifMauType MAU-MIB module object defined in Clause 13.

10.2.2.7 ifPhysAddress

This object contains the IEEE 802.3 address that is placed in the source-address field of any Ethernet, Starlan, or IEEE 802.3 frames that originate at this interface. Usually this will be kept in ROM on the interface hardware. Some systems may set this address via software.

In a system where there are several such addresses, the designer has a tougher choice. The address chosen should be the one most likely to be of use to network management (e.g., the address placed in ARP responses for systems that are primarily IP systems).

If the designer truly cannot choose, use of the factory-provided ROM address is suggested.

If the address cannot be determined, an octet string of zero length should be returned.

The address is stored in binary in this object. The address is stored in “canonical” bit order; that is, the Group Bit is positioned as the low-order bit of the first octet. Thus, the first byte of a multicast address would have the bit 0x01 set.

10.2.2.8 Specific Interfaces Group MIB objects

Table 10-1 provides specific implementation guidelines for applying the Interfaces Group objects to Ethernet-like interfaces.

Table 10-1—Implementation guidelines

Object	Guidelines
ifIndex	Each Ethernet-like interface is represented by an ifEntry. The dot3StatsTable in this MIB module is indexed by dot3StatsIndex. The interface identified by a particular value of dot3StatsIndex is the same interface as identified by the same value of ifIndex.
ifDescr	Refer to IETF RFC 2863.
ifType	Refer to 10.2.2.2.
ifMtu	Refer to 10.2.2.5.
ifSpeed	Refer to 10.2.2.6.
ifPhysAddress	Refer to 10.2.2.7.
ifAdminStatus	Write access is not required. Support for “testing” is not required.
ifOperStatus	The operational state of the interface. Support for “testing” is not required. The value “dormant” has no meaning for an Ethernet-like interface.
ifLastChange	Refer to IETF RFC 2863.

Table 10-1—Implementation guidelines (*continued*)

Object	Guidelines
ifInOctets	The number of octets in valid MAC frames received on this interface, including the MAC header and FCS. This does include the number of octets in valid MAC Control frames received on this interface. See 10.2.2.3.
ifInUcastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are consumed by the interface layer and are not passed to any higher layer protocol. See 10.2.2.4.
ifInDiscards	Refer to IETF RFC 2863.
ifInErrors	The sum for this interface of dot3StatsAlignmentErrors, dot3StatsFCSErrors, dot3StatsFrameTooLongs, and dot3StatsInternalMacReceiveErrors.
ifInUnknownProtos	Refer to IETF RFC 2863.
ifOutOctets	The number of octets transmitted in valid MAC frames on this interface, including the MAC header and FCS. This does include the number of octets in valid MAC Control frames transmitted on this interface. See 10.2.2.3.
ifOutUcastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are generated by the interface layer, and are not passed from any higher layer protocol. See 10.2.2.4.
ifOutDiscards	Refer to IETF RFC 2863.
ifOutErrors	The sum for this interface of: dot3StatsSQETestErrors, dot3StatsLateCollisions, dot3StatsExcessiveCollisions, dot3StatsInternalMacTransmitErrors and dot3StatsCarrierSenseErrors.
ifName	Locally significant textual name for the interface (e.g., lan0).
ifInMulticastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are consumed by the interface layer and are not passed to any higher layer protocol. See 10.2.2.4.
ifInBroadcastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are consumed by the interface layer, and are not passed to any higher layer protocol. See 10.2.2.4.
ifOutMulticastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are generated by the interface layer, and are not passed from any higher layer protocol. See 10.2.2.4.
ifOutBroadcastPkts	Refer to IETF RFC 2863. Note that this does not include MAC Control frames, since MAC Control frames are generated by the interface layer, and are not passed from any higher layer protocol. See 10.2.2.4.
ifHCInOctets, ifHCOctets	64-bit versions of counters. Required for Ethernet-like interfaces that are capable of operating at 20 Mb/s or faster, even if the interface is currently operating at less than 20 Mb/s.

Table 10-1—Implementation guidelines (*continued*)

Object	Guidelines
ifHCInUcastPkts, ifHCInMulticastPkts, ifHCInBroadcastPkts, ifHCOutUcastPkts, ifHCOutMulticastPkts, ifHCOutBroadcastPkts	64-bit versions of packet counters. Required for Ethernet-like interfaces that are capable of operating at 640 Mb/s or faster, even if the interface is currently operating at less than 640 Mb/s.
ifLinkUpDownTrapEnable	Refer to IETF RFC 2863. Default is “enabled”.
ifHighSpeed	Refer to 10.2.2.6.
ifPromiscuousMode	Refer to IETF RFC 2863.
ifConnectorPresent	This will normally be “true”. It will be “false” in the case where this interface uses the WAN Interface Sublayer. See Clause 12 for details.
ifAlias	Refer to IETF RFC 2863.
ifCounterDiscontinuityTime	Refer to IETF RFC 2863. Note that a discontinuity in the Interfaces Group MIB counters may also indicate a discontinuity in some or all of the counters in this MIB that are associated with that interface.
ifStackHigherLayer, ifStackLowerLayer, ifStackStatus	Refer to 11.2.1.1.
ifRcvAddressAddress, ifRcvAddressStatus, ifRcvAddressType	Refer to 10.2.2.1.

10.2.3 Relation to the IEEE 802.3 MAU-MIB module

Support for the mauModIfCompl3 compliance statement of the MAU-MIB module defined in Clause 13 is required for Ethernet-like interfaces. This MIB module is needed in order to allow applications to determine the current MAU type in use by the interface, and to control Auto-Negotiation and duplex mode for the interface. Implementing this MIB module without implementing the MAU-MIB module would leave applications with no standard way to determine the media type in use, and no standard way to control the duplex mode of the interface.

10.2.4 Mapping of IEEE 802.3 managed objects

The mapping of IEEE 802.3 managed objects to SNMP objects is shown in Table 10-2.

Table 10-2—Mapping of IEEE 802.3 managed objects

IEEE 802.3 managed object	Corresponding SNMP object
oMacEntity	.aMACID
	dot3StatsIndex or IF-MIB – ifIndex
	.aFramesTransmittedOK
	IF-MIB – ifOutUCastPkts + ifOutMulticastPkts + ifOutBroadcastPkts ^a
	.aSingleCollisionFrames
	dot3StatsSingleCollisionFrames
	.aMultipleCollisionFrames
	dot3StatsMultipleCollisionFrames
	.aFramesReceivedOK
	IF-MIB – ifInUCastPkts + ifInMulticastPkts + ifInBroadcastPkts ^a
	.aFrameCheckSequenceErrors
	dot3StatsFCSErrors
	.aAlignmentErrors
	dot3StatsAlignmentErrors
	.aOctetsTransmittedOK
	IF-MIB – ifOutOctets ^a
	.aFramesWithDeferredXmissions
	dot3StatsDeferredTransmissions
	.aLateCollisions
	dot3StatsLateCollisions
	.aFramesAbortedDueToXSColls
	dot3StatsExcessiveCollisions
	.aFramesLostDueToIntMACXmitError
	dot3StatsInternalMacTransmitErrors
	.aCarrierSenseErrors
	dot3StatsCarrierSenseErrors
	.aOctetsReceivedOK
	IF-MIB – ifInOctets ^a
	.aFramesLostDueToIntMACRcvError
	dot3StatsInternalMacReceiveErrors
	.aPromiscuousStatus
	IF-MIB – ifPromiscuousMode
	.aReadMulticastAddressList
	IF-MIB – ifRcvAddressTable
	.aMulticastFramesXmittedOK
	IF-MIB – ifOutMulticastPkts ^a
	.aBroadcastFramesXmittedOK
	IF-MIB – ifOutBroadcastPkts ^a
	.aMulticastFramesReceivedOK
	IF-MIB – ifInMulticastPkts ^a
	.aBroadcastFramesReceivedOK
	IF-MIB – ifInBroadcastPkts ^a
	.aFrameTooLongErrors
	dot3StatsFrameTooLongs
	.aReadWriteMACAddress
	IF-MIB – ifPhysAddress
	.aCollisionFrames
	dot3CollFrequencies
	.aDuplexStatus
	dot3StatsDuplexStatus
	.aRateControlAbility
	dot3StatsRateControlAbility
	.aMaxFrameLength
	dot3StatsMaxFrameLength
	.aSlowProtocolFrameLimit
	dot3SlowProtocolFrameLimit
	.aRateControlStatus
	dot3StatsRateControlStatus

Table 10-2—Mapping of IEEE 802.3 managed objects (*continued*)

IEEE 802.3 managed object		Corresponding SNMP object
	.acAddGroupAddress	IF-MIB - ifRcvAddressTable
	.acDeleteGroupAddress	IF-MIB - ifRcvAddressTable
	.acExecuteSelfTest	dot3TestLoopBack
oPHYEntity	.aPHYID	dot3StatsIndex or IF-MIB – ifIndex
	.aSQUETestErrors	dot3StatsSQUETestErrors
	.aSymbolErrorDuringCarrier	dot3StatsSymbolErrors
oMACControlEntity	.aMACControlID	dot3StatsIndex or IF-MIB – ifIndex
	.aMACControlFunctionsSupported	dot3ControlFunctionsSupported and dot3ControlFunctionsEnabled
	.aUnsupportedOpCodesReceived	dot3ControlInUnknownOpCodes
oPAUSEEntity	.aPAUSEMACCtrlFramesTransmitted	dot3OutPauseFrames
	.aPAUSEMACCtrlFramesReceived	dot3InPauseFrames

^a Note that the octet counters in IF-MIB do not exactly match the definition of the octet counters in IEEE Std 802.3. See 10.2.2.3 for details. Also note that the packet counters in the IF-MIB do not exactly match the definition of the frame counters in IEEE Std 802.3. See 10.2.2.4 for details.

10.3 Security considerations for Ethernet-like interface MIB module

There is one management object defined in this MIB that has a MAX-ACCESS clause of read-write. That object, dot3PauseAdminMode, may be used to change the flow control configuration on a network interface, which may result in dropped packets, or sending flow control packets on links where the link partner will not understand them. Either action could be detrimental to network performance.

Such objects can be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Most of the objects in this MIB module contain statistical information about particular network links. In some network environments, this information can be considered sensitive. It is thus important to control GET and/or NOTIFY access to these objects and possibly to encrypt the values of these objects when sending them over the network via SNMP.

10.4 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²⁵

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C10mib.txt

²⁵ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

10.4.1 Tree definition

```

---- ieee8023etherMIB(1.3.111.2.802.3.1.10)
+---- ieee8023etherMIBObjects(1)
| +---- dot3StatsTable(2)
| | +---- dot3StatsEntry(1) [dot3StatsIndex]
| | | +---- dot3StatsIndex(1) InterfaceIndex
| | | +---r- dot3StatsAlignmentErrors(2) Counter32
| | | +---r- dot3StatsFCSErrors(3) Counter32
| | | +---r- dot3StatsSingleCollisionFrames(4) Counter32
| | | +---r- dot3StatsMultipleCollisionFrames(5) Counter32
| | | +---r- dot3StatsSQETestErrors(6) Counter32
| | | +---r- dot3StatsDeferredTransmissions(7) Counter32
| | | +---r- dot3StatsLateCollisions(8) Counter32
| | | +---r- dot3StatsExcessiveCollisions(9) Counter32
| | | +---r- dot3StatsInternalMacTransmitErrors(10) Counter32
| | | +---r- dot3StatsCarrierSenseErrors(11) Counter32
| | | +---r- dot3StatsFrameTooLongs(13) Counter32
| | | +---r- dot3StatsInternalMacReceiveErrors(16) Counter32
| | | +---r- dot3StatsSymbolErrors(17) Counter32
| | | +---r- dot3StatsDuplexStatus(18) Enumeration
| | | +---r- dot3StatsRateControlAbility(19) TruthValue
| | | +---rw dot3StatsRateControlStatus(20) Enumeration
| | | +---r- dot3StatsMaxFrameLength(21) Enumeration
| +---- dot3CollTable(5)
| | +---- dot3CollEntry(1) [ifIndex, dot3CollCount]
| | | +---- dot3CollCount(2) Integer32
| | | +---r- dot3CollFrequencies(3) Counter32
| +---- dot3ControlTable(9)
| | +---- dot3ControlEntry(1) [dot3StatsIndex]
| | | +---r- dot3ControlFunctionsSupported(1) Bits
| | | +---r- dot3ControlInUnknownOpcodes(2) Counter32
| | | +---r- dot3HCControlInUnknownOpcodes(3) Counter64
| +---- dot3PauseTable(10)
| | +---- dot3PauseEntry(1) [dot3StatsIndex]
| | | +---rw dot3PauseAdminMode(1) Enumeration
| | | +---r- dot3PauseOperMode(2) Enumeration
| | | +---r- dot3InPauseFrames(3) Counter32
| | | +---r- dot3OutPauseFrames(4) Counter32
| | | +---r- dot3HCInPauseFrames(5) Counter64
| | | +---r- dot3HCOutPauseFrames(6) Counter64
| +---- dot3HCStatsTable(11)
| | +---- dot3HCStatsEntry(1) [dot3StatsIndex]
| | | +---r- dot3HCStatsAlignmentErrors(1) Counter64
| | | +---r- dot3HCStatsFCSErrors(2) Counter64
| | | +---r- dot3HCStatsInternalMacTransmitErrors(3) Counter64
| | | +---r- dot3HCStatsFrameTooLongs(4) Counter64
| | | +---r- dot3HCStatsInternalMacReceiveErrors(5) Counter64
| | | +---r- dot3HCStatsSymbolErrors(6) Counter64
| | | +---r- dot3HCStatsTransmitLPIMicroseconds(7) Counter64
| | | +---r- dot3HCStatsReceiveLPIMicroseconds(8) Counter64
| | | +---r- dot3HCStatsTransmitLPITransitions(9) Counter64
| | | +---r- dot3HCStatsReceiveLPITransitions(10) Counter64
| +---rw dot3SlowProtocolFrameLimit(12) Integer32
| +---- dot3ExtensionTable(13)
| | +---- dot3ExtensionEntry(1) [dot3StatsIndex]
| | | +---r- dot3HCInExtensionFrames(1) Counter64
| | | +---r- dot3HCOutExtensionFrames(2) Counter64
| | | x---r- dot3ExtensionMacCtrlStatus(3) Unsigned32

```

```

| |      +---rw dot3ExtensionMacCtrlAdmin(4)  TruthValue
| +---- dot3PFCTable(14)
|   +---- dot3PFCEnter(1) [dot3StatsIndex]
|     +---rw dot3PFCAdminMode(1)  Enumeration
|     +---r- dot3PFCOperMode(2)   Enumeration
|     +---r- dot3HCInPFCFrames(3) Counter64
|     +---r- dot3HCOuPFCFrames(4) Counter64
+---- etherConformance(2)
  +---- etherGroups(1)
    | +---- etherCollisionTableGroup(1)
    | +---- etherStatsLowSpeedGroup(2)
    | +---- etherStatsHighSpeedGroup(3)
    | +---- etherDuplexGroup(4)
    | +---- etherControlGroup(5)
    | +---- etherControlPauseGroup(6)
    | +---- etherStatsBaseGroup2(7)
    | +---- etherStatsHalfDuplexGroup(8)
    | +---- etherHCStatsGroup(9)
    | +---- etherHCControlGroup(10)
    | +---- etherHCControlPauseGroup(11)
    | +---- etherRateControlGroup(12)
    | +---- etherHCStatsLpiGroup(13)
    | +---- etherSlowProtocolsGroup(14)
    | +---- etherExtensionMacCtrlGroup(15)
    | +---- etherPfcGroup(16)
  +---- etherCompliances(2)
    +---- dot3Compliance2(1)

```

10.4.2 MIB module definition

IEEE8023-EtherLike-MIB DEFINITIONS ::= BEGIN

IMPORTS

```

    InterfaceIndex, ifIndex
    FROM IF-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
    FROM SNMPv2-CONF
    Counter32, Counter64, Integer32, MODULE-IDENTITY, OBJECT-TYPE,
    Unsigned32, org
    FROM SNMPv2-SMI
    TruthValue
    FROM SNMPv2-TC;

```

ieee8023etherMIB MODULE-IDENTITY

LAST-UPDATED "202412110000Z"

ORGANIZATION

"IEEE 802.3 Working Group"

CONTACT-INFO

" WG-URL: <http://www.ieee802.org/3/index.html>

WG-EMail: <mailto:stds-802-3-dialog@ieee.org>

Contact: IEEE 802.3 Working Group Chair

Postal: C/O IEEE 802.3 Working Group

IEEE Standards Association

445 Hoes Lane

Piscataway, NJ 08854

USA

E-mail: <mailto:stds-802-3-dialog@ieee.org>"

DESCRIPTION

```
"The MIB module to describe generic objects for
Ethernet-like network interfaces."
REVISION      "202412110000Z"
DESCRIPTION
  "Revision, based on an earlier version in IEEE Std 802.3.1-2013
  addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
  and 2022."
REVISION      "201304110000Z"
DESCRIPTION
  "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
REVISION      "201102020000Z"
DESCRIPTION
  "Initial version, based on an earlier version published
  in RFC 3635."
::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 10 }

ieee8023etherMIBObjects OBJECT IDENTIFIER
::= { ieee8023etherMIB 1 }

dot3StatsTable OBJECT-TYPE
SYNTAX        SEQUENCE OF Dot3StatsEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
  "Statistics for a collection of Ethernet-like
  interfaces attached to a particular system.
  There will be one row in this table for each
  Ethernet-like interface in the system.

  For interfaces operating at 10 Gb/s or more, 32-bit frame
  based counters can roll over in less than 5 minutes if they are
  incrementing at their maximum rate. Management stations are
  advised to use the 'HC'/64-bit versions of these counters.

  Discontinuities in the values of counters in this table
  can occur at re-initialization of the management
  system, and at other times as indicated by the
  value of ifCounterDiscontinuityTime."
::= { ieee8023etherMIBObjects 2 }

dot3StatsEntry OBJECT-TYPE
SYNTAX        Dot3StatsEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
  "Statistics for a particular interface to an
  Ethernet-like medium."
INDEX         { dot3StatsIndex }
::= { dot3StatsTable 1 }

Dot3StatsEntry ::= SEQUENCE {
    dot3StatsIndex          InterfaceIndex,
    dot3StatsAlignmentErrors Counter32,
    dot3StatsFCSErrors      Counter32,
    dot3StatsSingleCollisionFrames Counter32,
    dot3StatsMultipleCollisionFrames Counter32,
    dot3StatsSQETestErrors  Counter32,
```

```
dot3StatsDeferredTransmissions      Counter32,
dot3StatsLateCollisions             Counter32,
dot3StatsExcessiveCollisions        Counter32,
dot3StatsInternalMacTransmitErrors  Counter32,
dot3StatsCarrierSenseErrors         Counter32,
dot3StatsFrameTooLongs              Counter32,
dot3StatsInternalMacReceiveErrors   Counter32,
dot3StatsSymbolErrors               Counter32,
dot3StatsDuplexStatus               INTEGER,
dot3StatsRateControlAbility         TruthValue,
dot3StatsRateControlStatus          INTEGER,
dot3StatsMaxFrameLength              INTEGER
}

dot3StatsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndex
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index value that uniquely identifies an
         interface to an Ethernet-like medium. The
         interface identified by a particular value of
         this index is the same interface as identified
         by the same value of ifIndex."
    REFERENCE
        "IETF RFC 2863, ifIndex"
    ::= { dot3StatsEntry 1 }

dot3StatsAlignmentErrors OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of frames received on a particular
         interface that are not an integral number of
         octets in length and do not pass the FCS check.

         The count represented by an instance of this
         object is incremented when the alignmentError
         status is returned by the MAC service to the
         MAC client. Received frames for
         which multiple error conditions pertain are,
         according to the conventions of IEEE 802.3
         Layer Management, counted exclusively according
         to the error status presented to the MAC client.

         This counter does not increment for group
         encoding schemes greater than 4 bits per group."
    REFERENCE
        "IEEE Std 802.3, 30.3.1.1.7"
    ::= { dot3StatsEntry 2 }

dot3StatsFCSErrors OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of frames received on a particular
         interface that are an integral number of octets
```

in length but do not pass the FCS check. This count does not include frames received with frame-too-long or frame-too-short error.

The count represented by an instance of this object is incremented when the frameCheckError status is returned by the MAC service to the MAC client (). Received frames for which multiple error conditions pertain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the MAC client.

Note: Coding errors detected by the Physical Layer for speeds above 10 Mb/s will cause the frame to fail the FCS check."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.6"

::= { dot3StatsEntry 3 }

dot3StatsSingleCollisionFrames OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames that are involved in a single collision, and are subsequently transmitted successfully.

A frame that is counted by an instance of this object is also counted by the corresponding instance of either the ifOutUcastPkts, ifOutMulticastPkts, or ifOutBroadcastPkts, and is not counted by the corresponding instance of the dot3StatsMultipleCollisionFrames object.

This counter does not increment when the interface is operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.3"

::= { dot3StatsEntry 4 }

dot3StatsMultipleCollisionFrames OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames that are involved in more than one collision and are subsequently transmitted successfully.

A frame that is counted by an instance of this object is also counted by the corresponding instance of either the ifOutUcastPkts, ifOutMulticastPkts, or ifOutBroadcastPkts, and is not counted by the corresponding instance of the dot3StatsSingleCollisionFrames object.

This counter does not increment when the interface is operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.4"

::= { dot3StatsEntry 5 }

dot3StatsSQETestErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of times that the SQE TEST ERROR is received on a particular interface. The SQE TEST ERROR is set in accordance with the rules for verification of the SQE detection mechanism in the PLS Carrier Sense Function.

This counter does not increment on interfaces operating at speeds greater than 10 Mb/s, or on interfaces operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.4"

::= { dot3StatsEntry 6 }

dot3StatsDeferredTransmissions OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy.

The count represented by an instance of this object does not include frames involved in collisions.

This counter does not increment when the interface is operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.9"

::= { dot3StatsEntry 7 }

dot3StatsLateCollisions OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of times that a collision is detected on a particular interface later than one slotTime into the transmission of a packet.

A (late) collision included in a count represented by an instance of this object is also considered as a (generic) collision for purposes of other collision-related statistics.

This counter does not increment when the interface is operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.10"
::= { dot3StatsEntry 8 }

dot3StatsExcessiveCollisions OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames for which transmission on a particular interface fails due to excessive collisions.

This counter does not increment when the interface is operating in full-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.11"
::= { dot3StatsEntry 9 }

dot3StatsInternalMacTransmitErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of frames for which transmission on a particular interface fails due to an internal MAC sublayer transmit error. A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the dot3StatsLateCollisions object, the dot3StatsExcessiveCollisions object, or the dot3StatsCarrierSenseErrors object.

The precise meaning of the count represented by an instance of this object is implementation-specific. In particular, an instance of this object may represent a count of transmission errors on a particular interface that are not otherwise counted."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.12"
::= { dot3StatsEntry 10 }

dot3StatsCarrierSenseErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of times that the carrier sense condition was lost or never asserted when attempting to transmit a frame on a particular interface.

The count represented by an instance of this object is incremented at most once per transmission attempt, even if the carrier sense condition fluctuates during a transmission

```
        attempt.

        This counter does not increment when the
        interface is operating in full-duplex mode."
REFERENCE
    "IEEE Std 802.3, 30.3.1.1.13"
::= { dot3StatsEntry 11 }

dot3StatsFrameTooLong OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of frames received on a particular
        interface that exceed the maximum permitted
        frame size.

        The count represented by an instance of this
        object is incremented when the frameTooLong
        status is returned by the MAC service to the
        MAC client. Received frames for
        which multiple error conditions pertain are,
        according to the conventions of IEEE 802.3
        Layer Management, counted exclusively according
        to the error status presented to the MAC client."
REFERENCE
    "IEEE Std 802.3, 30.3.1.1.25"
::= { dot3StatsEntry 13 }

dot3StatsInternalMacReceiveErrors OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of frames for which reception on a
        particular interface fails due to an internal
        MAC sublayer receive error. A frame is only
        counted by an instance of this object if it is
        not counted by the corresponding instance of
        either the dot3StatsFrameTooLongs object, the
        dot3StatsAlignmentErrors object, or the
        dot3StatsFCSErrors object.

        The precise meaning of the count represented by
        an instance of this object is implementation-
        specific. In particular, an instance of this
        object may represent a count of receive errors
        on a particular interface that are not
        otherwise counted."
REFERENCE
    "IEEE Std 802.3, 30.3.1.1.15"
::= { dot3StatsEntry 16 }

dot3StatsSymbolErrors OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "For an interface operating at 100 Mb/s, the
```

number of times there was an invalid data symbol when a valid carrier was present.

For an interface operating in half-duplex mode at 1000 Mb/s, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than slotTime, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Data reception error' or 'carrier extend error' on the GMII.

For an interface operating in full-duplex mode at 1000 Mb/s, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than minFrameSize, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Data reception error' on the GMII.

For an interface operating at 10 Gb/s, 40 Gb/s, or 100 Gb/s, it is a count of the number of times the receiving media is non-idle (the time between the Start of Packet Delimiter and the End of Packet Delimiter) for a period of time equal to or greater than minFrameSize, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Receive Error' on the XGMII, the XLGMII, or the CGMII.

The count represented by an instance of this object is incremented at most once per carrier event, even if multiple symbol errors occur during the carrier event. This count does not increment if a collision is present.

This counter does not increment when the interface is operating at 10 Mb/s."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.5"
::= { dot3StatsEntry 17 }

dot3StatsDuplexStatus OBJECT-TYPE

SYNTAX INTEGER { unknown(1), halfDuplex(2), fullDuplex(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current mode of operation of the MAC entity. 'unknown' indicates that the current duplex mode could not be determined.

Management control of the duplex mode is accomplished through the MAU MIB. When an interface does not support Auto-Negotiation, or when Auto-Negotiation is not enabled, the duplex mode is controlled using ifMauDefaultType. When Auto-Negotiation is supported and enabled, duplex mode is controlled using ifMauAutoNegAdvertisedBits. In either

case, the currently operating duplex mode is reflected both in this object and in ifMauType.

Note that this object provides redundant information with ifMauType. Normally, redundant objects are discouraged. However, in this instance, it allows a management application to determine the duplex status of an interface without having to know every possible value of ifMauType. This was felt to be sufficiently valuable to justify the redundancy."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.32"

::= { dot3StatsEntry 18 }

dot3StatsRateControlAbility OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"'true' for interfaces operating at speeds above 1000 Mb/s that support Rate Control through lowering the average data rate of the MAC sublayer, with frame granularity, and 'false' otherwise."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.33"

::= { dot3StatsEntry 19 }

dot3StatsRateControlStatus OBJECT-TYPE

SYNTAX INTEGER { rateControlOff(1), rateControlOn(2),
unknown(3) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The current Rate Control mode of operation of the MAC sublayer of this interface."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.34"

::= { dot3StatsEntry 20 }

dot3StatsMaxFrameLength OBJECT-TYPE

SYNTAX INTEGER { unknown(1), baseFrame(2), qTaggedFrame(3),
envelopeFrame(4) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This indicates the MAC frame length at which the dot3StatsFrameTooLongs counter is incremented."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.37"

::= { dot3StatsEntry 21 }

dot3CollTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3CollEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A collection of collision histograms for a particular set of interfaces.

Discontinuities in the values of counters in this table can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.30"

::= { ieee8023etherMIBObjects 5 }

dot3CollEntry OBJECT-TYPE

SYNTAX Dot3CollEntry
MAX-ACCESS not-accessible
STATUS current

DESCRIPTION

"A cell in the histogram of per-frame collisions for a particular interface. An instance of this object represents the frequency of individual MAC frames for which the transmission (successful or otherwise) on a particular interface is accompanied by a particular number of media collisions."

INDEX { ifIndex, dot3CollCount }
::= { dot3CollTable 1 }

Dot3CollEntry ::= SEQUENCE {
dot3CollCount Integer32,
dot3CollFrequencies Counter32
}

dot3CollCount OBJECT-TYPE

SYNTAX Integer32 (1..16)
MAX-ACCESS not-accessible
STATUS current

DESCRIPTION

"The number of per-frame media collisions for which a particular collision histogram cell represents the frequency on a particular interface."

::= { dot3CollEntry 2 }

dot3CollFrequencies OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"A count of individual MAC frames for which the transmission (successful or otherwise) on a particular interface occurs after the frame has experienced exactly the number of collisions in the associated dot3CollCount object.

For example, a frame which is transmitted on interface 77 after experiencing exactly 4 collisions would be indicated by incrementing only dot3CollFrequencies.77.4. No other instance of dot3CollFrequencies would

be incremented in this example.

This counter does not increment when the interface is operating in full-duplex mode."

::= { dot3CollEntry 3 }

dot3ControlTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3ControlEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table of descriptive and status information about the MAC Control sublayer on the Ethernet-like interfaces attached to a particular system. There will be one row in this table for each Ethernet-like interface in the system which implements the MAC Control sublayer. If some, but not all, of the Ethernet-like interfaces in the system implement the MAC Control sublayer, there will be fewer rows in this table than in the dot3StatsTable.

For interfaces operating at 10 Gb/s or more, 32-bit frame based counters can roll over in less than 5 minutes if they are incrementing at their maximum rate. Management stations are advised to use the 'HC'/64-bit versions of these counters.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime."

::= { ieee8023etherMIBObjects 9 }

dot3ControlEntry OBJECT-TYPE

SYNTAX Dot3ControlEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the table, containing information about the MAC Control sublayer on a single Ethernet-like interface."

INDEX { dot3StatsIndex }

::= { dot3ControlTable 1 }

Dot3ControlEntry ::= SEQUENCE {

dot3ControlFunctionsSupported BITS,

dot3ControlInUnknownOpcodes Counter32,

dot3HCControlInUnknownOpcodes Counter64

}

dot3ControlFunctionsSupported OBJECT-TYPE

SYNTAX BITS { pause(0), mpcp(1), pfc(2), extension(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A list of the possible MAC Control functions implemented for this interface."

REFERENCE

"IEEE Std 802.3, 30.3.3.2"

```
::= { dot3ControlEntry 1 }

dot3ControlInUnknownOpcodes OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of MAC Control frames received on this
        interface that contain an opcode that is not
        supported by this device."
    REFERENCE
        "IEEE Std 802.3, 30.3.3.5 "
    ::= { dot3ControlEntry 2 }

dot3HCControlInUnknownOpcodes OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A 64-bit count of MAC Control frames received on this
        interface that contain an opcode that is not
        supported by this device."
    REFERENCE
        "IEEE Std 802.3, 30.3.3.5 "
    ::= { dot3ControlEntry 3 }

dot3PauseTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3PauseEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of descriptive and status information
        about the MAC Control PAUSE function on the
        Ethernet-like interfaces attached to a
        particular system. There will be one row in
        this table for each Ethernet-like interface in
        the system that supports the MAC Control PAUSE
        function (i.e., the 'pause' bit in the
        corresponding instance of
        dot3ControlFunctionsSupported is set). If some,
        but not all, of the Ethernet-like interfaces in
        the system implement the MAC Control PAUSE
        function (for example, if some interfaces only
        support half-duplex), there will be fewer rows
        in this table than in the dot3StatsTable.

        For interfaces operating at 10 Gb/s or more, 32-bit frame
        based counters can roll over in less than 5 minutes if they are
        incrementing at their maximum rate. Management stations are
        advised to use the 'HC'/64-bit versions of these counters.

        Discontinuities in the values of counters in this table
        can occur at re-initialization of the management
        system, and at other times as indicated by the
        value of ifCounterDiscontinuityTime."
    ::= { ieee8023etherMIBObjects 10 }

dot3PauseEntry OBJECT-TYPE
    SYNTAX      Dot3PauseEntry
```



```
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "An entry in the table, containing information
    about the MAC Control PAUSE function on a single
    Ethernet-like interface."
INDEX         { dot3StatsIndex }
 ::= { dot3PauseTable 1 }

Dot3PauseEntry ::= SEQUENCE {
    dot3PauseAdminMode    INTEGER,
    dot3PauseOperMode     INTEGER,
    dot3InPauseFrames     Counter32,
    dot3OutPauseFrames    Counter32,
    dot3HCInPauseFrames   Counter64,
    dot3HCOutPauseFrames  Counter64
}

dot3PauseAdminMode OBJECT-TYPE
    SYNTAX      INTEGER { disabled(1), enabledXmit(2), enabledRcv(3),
                        enabledXmitAndRcv(4) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "This object is used to configure the default
        administrative PAUSE mode for this interface.

        This object represents the
        administratively-configured PAUSE mode for this
        interface. If Auto-Negotiation is not enabled
        or is not implemented for the active MAU
        attached to this interface, the value of this
        object determines the operational PAUSE mode
        of the interface whenever it is operating in
        full-duplex mode. In this case, a set to this
        object will force the interface into the
        specified mode.

        If Auto-Negotiation is implemented and enabled
        for the MAU attached to this interface, the
        PAUSE mode for this interface is determined by
        Auto-Negotiation, and the value of this object
        denotes the mode to which the interface will
        automatically revert if/when Auto-Negotiation is
        later disabled. Note that when Auto-Negotiation
        is running, administrative control of the PAUSE
        mode may be accomplished using the
        ifMauAutoNegCapAdvertisedBits object in the
        MAU-MIB module.

        Note that the value of this object is ignored
        when the interface is not operating in
        full-duplex mode.

        An attempt to set this object to
        'enabledXmit(2)' or 'enabledRcv(3)' will fail
        on interfaces that do not support operation
        at greater than 100 Mb/s."
 ::= { dot3PauseEntry 1 }
```

```
dot3PauseOperMode OBJECT-TYPE
    SYNTAX      INTEGER { disabled(1), enabledXmit(2), enabledRcv(3),
                        enabledXmitAndRcv(4) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This object reflects the PAUSE mode currently
        in use on this interface, as determined by
        either (1) the result of the Auto-Negotiation
        function or (2) if Auto-Negotiation is not
        enabled or is not implemented for the active MAU
        attached to this interface, by the value of
        dot3PauseAdminMode. Interfaces operating at
        100 Mb/s or less will never return
        'enabledXmit(2)' or 'enabledRcv(3)'. Interfaces
        operating in half-duplex mode will return
        'disabled(1)'. Interfaces on which
        Auto-Negotiation is enabled but not yet
        completed should return the value
        'disabled(1)'."
    ::= { dot3PauseEntry 2 }

dot3InPauseFrames OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of MAC Control frames received on this
        interface with an opcode indicating the PAUSE
        operation.

        This counter does not increment when the
        interface is operating in half-duplex mode."
    REFERENCE
        "IEEE Std 802.3, 30.3.4.3"
    ::= { dot3PauseEntry 3 }

dot3OutPauseFrames OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of MAC Control frames transmitted on
        this interface with an opcode indicating the
        PAUSE operation.

        This counter does not increment when the
        interface is operating in half-duplex mode."
    REFERENCE
        "IEEE Std 802.3, 30.3.4.2"
    ::= { dot3PauseEntry 4 }

dot3HCInPauseFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A 64-bit count of MAC Control frames received on this
```

interface with an opcode indicating the PAUSE operation.

This counter does not increment when the interface is operating in half-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.4.3"

::= { dot3PauseEntry 5 }

dot3HCOutPauseFrames OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A 64-bit count of MAC Control frames transmitted on this interface with an opcode indicating the PAUSE operation.

This counter does not increment when the interface is operating in half-duplex mode."

REFERENCE

"IEEE Std 802.3, 30.3.4.2"

::= { dot3PauseEntry 6 }

dot3HCStatsTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3HCStatsEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table containing 64-bit versions of error counters from the dot3StatsTable to address counter rollover for interfaces operating at or above 10Gb/s.

For interfaces operating at 10 Gb/s or more, 32-bit frame based counters can roll over in less than 5 minutes if they are incrementing at their maximum rate. Management stations are advised to use the 'HC'/64-bit versions of these counters.

Discontinuities in the values of counters in this table can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime."

::= { ieee8023etherMIBObjects 11 }

dot3HCStatsEntry OBJECT-TYPE

SYNTAX Dot3HCStatsEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry containing 64-bit statistics for a single Ethernet-like interface."

INDEX { dot3StatsIndex }

::= { dot3HCStatsTable 1 }

Dot3HCStatsEntry ::= SEQUENCE {

dot3HCStatsAlignmentErrors Counter64,

dot3HCStatsFCSErrors Counter64,

dot3HCStatsInternalMacTransmitErrors Counter64,

dot3HCStatsFrameTooLongs Counter64,

```
dot3HCStatsInternalMacReceiveErrors      Counter64,
dot3HCStatsSymbolErrors                  Counter64,
dot3HCStatsTransmitLPIMicroseconds       Counter64,
dot3HCStatsReceiveLPIMicroseconds        Counter64,
dot3HCStatsTransmitLPITransitions        Counter64,
dot3HCStatsReceiveLPITransitions         Counter64
}
```

dot3HCStatsAlignmentErrors OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A 64-bit count of frames received on a particular interface that are not an integral number of octets in length and do not pass the FCS check.

The count represented by an instance of this object is incremented when the alignmentError status is returned by the MAC service to the MAC client (). Received frames for which multiple error conditions pertain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the MAC client.

This counter does not increment for group encoding schemes greater than 4 bits per group."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.7"

::= { dot3HCStatsEntry 1 }

dot3HCStatsFCSErrors OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A 64-bit count of frames received on a particular interface that are an integral number of octets in length but do not pass the FCS check. This count does not include frames received with frame-too-long or frame-too-short error.

The count represented by an instance of this object is incremented when the frameCheckError status is returned by the MAC service to the MAC client. Received frames for which multiple error conditions pertain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the MAC client.

Note: Coding errors detected by the Physical Layer for speeds above 10 Mb/s will cause the frame to fail the FCS check."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.6"

::= { dot3HCStatsEntry 2 }

dot3HCStatsInternalMacTransmitErrors OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "A 64-bit count of frames for which transmission on a particular interface fails due to an internal MAC sublayer transmit error. A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the dot3StatsLateCollisions object, the dot3StatsExcessiveCollisions object, or the dot3StatsCarrierSenseErrors object.

 The precise meaning of the count represented by an instance of this object is implementation-specific. In particular, an instance of this object may represent a count of transmission errors on a particular interface that are not otherwise counted."

REFERENCE
 "IEEE Std 802.3, 30.3.1.1.12"
::= { dot3HCStatsEntry 3 }

dot3HCStatsFrameTooLongs OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "A 64-bit count of frames received on a particular interface that exceed the maximum permitted frame size.

 The count represented by an instance of this object is incremented when the frameTooLong status is returned by the MAC service to the MAC client. Received frames for which multiple error conditions pertain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the MAC client."

REFERENCE
 "IEEE Std 802.3, 30.3.1.1.25"
::= { dot3HCStatsEntry 4 }

dot3HCStatsInternalMacReceiveErrors OBJECT-TYPE
SYNTAX Counter64
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "A 64-bit count of frames for which reception on a particular interface fails due to an internal MAC sublayer receive error. A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the dot3StatsFrameTooLongs object, the dot3StatsAlignmentErrors object, or the dot3StatsFCSErrors object.

The precise meaning of the count represented by an instance of this object is implementation-specific. In particular, an instance of this object may represent a count of receive errors on a particular interface that are not otherwise counted."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.15"

::= { dot3HCStatsEntry 5 }

dot3HCStatsSymbolErrors OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For an interface operating at 100 Mb/s, the number of times there was an invalid data symbol when a valid carrier was present.

For an interface operating in half-duplex mode at 1000 Mb/s, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than slotTime, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Data reception error' or 'carrier extend error' on the GMII.

For an interface operating in full-duplex mode at 1000 Mb/s, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than minFrameSize, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Data reception error' on the GMII.

For an interface operating at 10 Gb/s, 40 Gb/s, or 100 Gb/s, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than minFrameSize, and during which there was at least one occurrence of an event that causes the PHY to indicate 'Receive Error' on the XGMII, the XLGMII, or the CGMII.

The 64-bit count represented by an instance of this object is incremented at most once per carrier event, even if multiple symbol errors occur during the carrier event. This count does not increment if a collision is present."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.5"

::= { dot3HCStatsEntry 6 }

dot3HCStatsTransmitLPIMicroseconds OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count reflecting the amount of time that the

LPI_REQUEST parameter has the value ASSERT. The request is indicated to the PHY according to the requirements of the RS (see IEEE Std 802.3, 22.7, 35.4, and 46.4).

This counter has a maximum increment rate of 1 million counts per second."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.8"

::= { dot3HCStatsEntry 7 }

dot3HCStatsReceiveLPIMicroseconds OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count reflecting the amount of time that the LPI_INDICATION parameter has the value ASSERT. The indication reflects the state of the PHY according to the requirements of the RS (see IEEE Std 802.3, 22.7, 35.4, and 46.4).

This counter has a maximum increment rate of 1 million counts per second."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.9"

::= { dot3HCStatsEntry 8 }

dot3HCStatsTransmitLPITransitions OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of occurrences of the transition from state LPI_DEASSERTED to state LPI_ASSERTED of the LPI transmit state diagram is the RS. The state transition corresponds to the assertion of the LPI_REQUEST parameter. The request is indicated to the PHY according to the requirements of the RS (see IEEE Std 802.3, 22.7, 35.4, 46.4.)

This counter has a maximum increment rate of 50 thousand counts per second at 100 Mb/s; 90 thousand counts per second at 1000 Mb/s; and 230 thousand counts per second at 10 Gb/s."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.10"

::= { dot3HCStatsEntry 9 }

dot3HCStatsReceiveLPITransitions OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of occurrences of the transition from DEASSERT to ASSERT of the LPI_INDICATE parameter. The indication reflects the state of the PHY according to the requirements of the RS (see IEEE Std 802.3, 22.7, 35.4, and 46.4).

This counter has a maximum increment rate of 50 thousand counts per second at 100 Mb/s; 90 thousand counts per second at 1000 Mb/s; and 230 thousand counts per second at 10 Gb/s."

REFERENCE

"IEEE Std 802.3, 30.3.2.1.11"
::= { dot3HCStatsEntry 10 }

dot3SlowProtocolFrameLimit OBJECT-TYPE

SYNTAX Integer32
MAX-ACCESS read-write
STATUS current
DESCRIPTION

"The maximum number of Slow Protocol frames of a given subtype that can be transmitted in a one second interval. The default value is 10."

REFERENCE

"IEEE Std 802.3, 30.3.1.1.138"
DEFVAL { 10 }
::= { ieee8023etherMIBObjects 12 }

dot3ExtensionTable OBJECT-TYPE

SYNTAX SEQUENCE OF Dot3ExtensionEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"A table of status information about the Extension MAC Control frames transmitted and received on the Ethernet-like interfaces attached to a particular system. There will be one row in this table for each Ethernet-like interface in the system that supports Extension MAC Control function (i.e., the 'mpcp' bit in the corresponding instance of dot3ControlFunctionsSupported is set). If some, but not all, of the Ethernet-like interfaces in the system implement the Extension MAC Control function, there will be fewer rows in this table than in the dot3StatsTable.

For interfaces operating at 10 Gb/s or more, 32-bit frame based counters can roll over in less than 5 minutes if they are incrementing at their maximum rate. Management stations are advised to use the 'HC'/64-bit versions of these counters.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime."

::= { ieee8023etherMIBObjects 13 }

dot3ExtensionEntry OBJECT-TYPE

SYNTAX Dot3ExtensionEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"An entry in the table, containing information about the Extension MAC Control function on a single


```
        Ethernet-like interface."
INDEX      { dot3StatsIndex }
::= { dot3ExtensionTable 1 }

Dot3ExtensionEntry ::= SEQUENCE {
    dot3HCInExtensionFrames    Counter64,
    dot3HCOutExtensionFrames   Counter64,
    dot3ExtensionMacCtrlStatus Unsigned32,
    dot3ExtensionMacCtrlAdmin  TruthValue
}

dot3HCInExtensionFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of Extension MAC Control frames received on
        this interface."
    REFERENCE
        "IEEE Std 802.3, 30.3.8.2"
    ::= { dot3ExtensionEntry 1 }

dot3HCOutExtensionFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of Extension MAC Control frames transmitted on
        this interface."
    REFERENCE
        "IEEE Std 802.3, 30.3.8.1"
    ::= { dot3ExtensionEntry 2 }

dot3ExtensionMacCtrlStatus OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS   read-only
    STATUS       deprecated
    DESCRIPTION
        "The state of the EXTENSION MAC Control function as described in
        IEEE Std 802.3, 30.3.8.3."
    ::= { dot3ExtensionEntry 3 }

dot3ExtensionMacCtrlAdmin OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "The state of the EXTENSION MAC Control function as described in
        IEEE Std 802.3, 30.3.8.3."
    REFERENCE
        "IEEE Std 802.3, 30.3.8.3"
    ::= { dot3ExtensionEntry 4 }

dot3PFCTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Dot3PFCEnt
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "A table of descriptive and status information
```

about the MAC Control Priority-based Flow Control function on the Ethernet-like interfaces attached to a particular system. There will be one row in this table for each Ethernet-like interface in the system that supports the MAC Control PFC function (i.e., the 'pfc' bit in the corresponding instance of dot3ControlFunctionsSupported is set). If some, but not all, of the Ethernet-like interfaces in the system implement the MAC Control PFC function (for example, if some interfaces only support half-duplex), there will be fewer rows in this table than in the dot3StatsTable.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime."

::= { ieee8023etherMIBObjects 14 }

dot3PFCEntry OBJECT-TYPE

SYNTAX Dot3PFCEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"An entry in the table, containing information about the MAC Control PFC function on a single Ethernet-like interface."
INDEX { dot3StatsIndex }
::= { dot3PFCTable 1 }

Dot3PFCEntry ::= SEQUENCE {

dot3PFCAAdminMode INTEGER,
dot3PFCCooperMode INTEGER,
dot3HCInPFCFrames Counter64,
dot3HCOutPFCFrames Counter64

}

dot3PFCAAdminMode OBJECT-TYPE

SYNTAX INTEGER { disabled(1), enabled(2) }
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This object is used to configure the default administrative PFC mode for this interface.

This object represents the administratively-configured PFC mode for this interface. The value of this object determines the operational PFC mode of the interface. A set to this object will force the interface into the specified mode.

Note that the value of this object is ignored when the interface is not operating in full-duplex mode."
::= { dot3PFCEntry 1 }

```
dot3PFCOperMode OBJECT-TYPE
    SYNTAX      INTEGER { disabled(1), enabled(2) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "This object reflects the PFC mode currently
         in use on this interface, as determined by
         by the value of dot3PFCAdminMode."
    REFERENCE
        "IEEE Std 802.3, 30.3.3.6"
    ::= { dot3PFCEntry 2 }

dot3HCInPFCFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of MAC Control frames received on this
         interface with an opcode indicating the PFC
         operation."
    ::= { dot3PFCEntry 3 }

dot3HCOuPFCFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of MAC Control frames transmitted on
         this interface with an opcode indicating the
         PFC operation."
    ::= { dot3PFCEntry 4 }

etherConformance OBJECT IDENTIFIER
    ::= { ieee8023etherMIB 2 }

etherGroups OBJECT IDENTIFIER
    ::= { etherConformance 1 }

etherCompliances OBJECT IDENTIFIER
    ::= { etherConformance 2 }

etherCollisionTableGroup OBJECT-GROUP
    OBJECTS      { dot3CollFrequencies }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing a histogram
         of packets successfully transmitted after
         experiencing exactly N collisions."
    ::= { etherGroups 1 }

etherStatsLowSpeedGroup OBJECT-GROUP
    OBJECTS      { dot3StatsSQETestErrors }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing information
         applicable to Ethernet-like network interfaces
         capable of operating at 10 Mb/s or slower in
         half-duplex mode."
    ::= { etherGroups 2 }
```

```
etherStatsHighSpeedGroup OBJECT-GROUP
  OBJECTS      { dot3StatsSymbolErrors }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    applicable to Ethernet-like network interfaces
    capable of operating at 100 Mb/s or faster."
  ::= { etherGroups 3 }

etherDuplexGroup OBJECT-GROUP
  OBJECTS      { dot3StatsDuplexStatus }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    about the duplex mode of an Ethernet-like
    network interface."
  ::= { etherGroups 4 }

etherControlGroup OBJECT-GROUP
  OBJECTS      { dot3ControlFunctionsSupported,
                dot3ControlInUnknownOpcodes }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    about the MAC Control sublayer on Ethernet-like
    network interfaces."
  ::= { etherGroups 5 }

etherControlPauseGroup OBJECT-GROUP
  OBJECTS      { dot3PauseAdminMode, dot3PauseOperMode,
                dot3InPauseFrames, dot3OutPauseFrames }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    about and control of the MAC Control PAUSE
    function on Ethernet-like network interfaces."
  ::= { etherGroups 6 }

etherStatsBaseGroup2 OBJECT-GROUP
  OBJECTS      { dot3StatsAlignmentErrors, dot3StatsFCSErrors,
                dot3StatsInternalMacTransmitErrors,
                dot3StatsFrameTooLongs,
                dot3StatsInternalMacReceiveErrors,
                dot3StatsMaxFrameLength }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    applicable to all Ethernet-like network
    interfaces."
  ::= { etherGroups 7 }

etherStatsHalfDuplexGroup OBJECT-GROUP
  OBJECTS      { dot3StatsSingleCollisionFrames,
                dot3StatsMultipleCollisionFrames,
                dot3StatsDeferredTransmissions,
                dot3StatsLateCollisions, dot3StatsExcessiveCollisions,
                dot3StatsCarrierSenseErrors }
  STATUS       current
```

```
DESCRIPTION
    "A collection of objects providing information
    applicable only to half-duplex Ethernet-like
    network interfaces."
 ::= { etherGroups 8 }

etherHCStatsGroup OBJECT-GROUP
    OBJECTS      { dot3HCStatsAlignmentErrors, dot3HCStatsFCSErrors,
                    dot3HCStatsInternalMacTransmitErrors,
                    dot3HCStatsFrameTooLongs,
                    dot3HCStatsInternalMacReceiveErrors,
                    dot3HCStatsSymbolErrors }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing high-capacity
        statistics applicable to higher-speed
        Ethernet-like network interfaces."
 ::= { etherGroups 9 }

etherHCControlGroup OBJECT-GROUP
    OBJECTS      { dot3HCControlInUnknownOpcodes }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing high-capacity
        statistics for the MAC Control sublayer on
        higher-speed Ethernet-like network interfaces."
 ::= { etherGroups 10 }

etherHCControlPauseGroup OBJECT-GROUP
    OBJECTS      { dot3HCInPauseFrames, dot3HCCOutPauseFrames }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing high-capacity
        statistics for the MAC Control PAUSE function on
        higher-speed Ethernet-like network interfaces."
 ::= { etherGroups 11 }

etherRateControlGroup OBJECT-GROUP
    OBJECTS      { dot3StatsRateControlAbility,
                    dot3StatsRateControlStatus }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing information
        about the Rate Control function on Ethernet-like
        interfaces."
 ::= { etherGroups 12 }

etherHCStatsLpiGroup OBJECT-GROUP
    OBJECTS      { dot3HCStatsTransmitLPIMicroseconds,
                    dot3HCStatsReceiveLPIMicroseconds,
                    dot3HCStatsTransmitLPITransitions,
                    dot3HCStatsReceiveLPITransitions }
    STATUS       current
    DESCRIPTION
        "A collection of objects providing information
        about the Low Power Idle function on Ethernet-like
        interfaces."
 ::= { etherGroups 13 }
```

```
etherSlowProtocolsGroup OBJECT-GROUP
  OBJECTS      { dot3SlowProtocolFrameLimit }
  STATUS       current
  DESCRIPTION
    "An object providing control and information
    about the frame transmission rate limit for
    Slow Protocols on Ethernet-like interfaces."
  ::= { etherGroups 14 }

etherExtensionMacCtrlGroup OBJECT-GROUP
  OBJECTS      { dot3HCInExtensionFrames, dot3HCOutExtensionFrames,
                dot3ExtensionMacCtrlAdmin }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    about the Extension MAC Control function on
    Ethernet-like interfaces."
  ::= { etherGroups 15 }

etherPfcGroup OBJECT-GROUP
  OBJECTS      { dot3PFCAdminMode, dot3PFCOperMode, dot3HCInPFCFrames,
                dot3HCOutPFCFrames }
  STATUS       current
  DESCRIPTION
    "A collection of objects providing information
    about the Priority Flow Control function on
    Ethernet-like interfaces."
  ::= { etherGroups 16 }

dot3Compliance2 MODULE-COMPLIANCE
  STATUS       current
  DESCRIPTION
    "The compliance statement for managed network
    entities which have Ethernet-like network
    interfaces.

    Note that compliance with this MIB module
    requires compliance with the ifCompliance3
    MODULE-COMPLIANCE statement of the IF-MIB
    (IETF RFC 2863). In addition, compliance with this
    MIB module requires compliance with the
    mauModIfCompl3 MODULE-COMPLIANCE statement of
    the MAU-MIB module defined in Clause 13."

  MODULE      -- this module

  MANDATORY-GROUPS      { etherStatsBaseGroup2 }

  GROUP      etherDuplexGroup
  DESCRIPTION
    "This group is mandatory for all
    Ethernet-like network interfaces which are
    capable of operating in full-duplex mode.
    It is highly recommended for all
    Ethernet-like network interfaces."

  GROUP      etherRateControlGroup
  DESCRIPTION
    "This group is mandatory for all
```

Ethernet-like network interfaces which are capable of operating at speeds faster than 1000 Mb/s. It is highly recommended for all Ethernet-like network interfaces."

GROUP etherStatsLowSpeedGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces which are capable of operating at 10 Mb/s or slower in half-duplex mode."

GROUP etherStatsHighSpeedGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces which are capable of operating at 100 Mb/s or faster."

GROUP etherStatsHalfDuplexGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces which are capable of operating in half-duplex mode."

GROUP etherHCStatsGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces which are capable of operating at 10 Gb/s or faster. It is recommended for all Ethernet-like network interfaces which are capable of operating at 1000 Mb/s or faster."

GROUP etherControlGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that support the MAC Control sublayer."

GROUP etherHCControlGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that support the MAC Control sublayer and are capable of operating at 10 Gb/s or faster."

GROUP etherControlPauseGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that support the MAC Control PAUSE function."

GROUP etherHCControlPauseGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that support the MAC Control PAUSE function and are capable of operating at 10 Gb/s or faster."

GROUP etherCollisionTableGroup

DESCRIPTION

"This group is optional. It is appropriate for all Ethernet-like network interfaces which are capable of operating in half-duplex mode and have the necessary metering. Implementation in systems with such interfaces is highly recommended."

GROUP etherHCStatsLpiGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that support the Low Power Idle function."

GROUP etherSlowProtocolsGroup

DESCRIPTION

"This group is optional. It is appropriate for Ethernet-like network interfaces that implement OAM as defined in Clause 57 of IEEE Std 802.3."

GROUP etherExtensionMacCtrlGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that implement Extension MAC Control."

GROUP etherPfcGroup

DESCRIPTION

"This group is mandatory for all Ethernet-like network interfaces that implement Priority Flow Control."

::= { etherCompliances 1 }

END -- end of module IEEE8023-EtherLike-MIB.

11. Ethernet in the First Mile copper (EFMCu) interfaces MIB module

11.1 Introduction

Ethernet-like interfaces have been defined in IEEE Std 802.3 known as Ethernet in the First Mile (EFM). In particular, 2BASE-TL and 10PASS-TS physical interfaces (PHYs), defined over voice-grade copper pairs, have been specified for the long and short reach, respectively. These interfaces, collectively called EFM Copper (EFMCu), are based on single-pair high-speed digital subscriber line (SHDSL; see ITU-T G.991.2) and very high-speed digital subscriber line (VDSL; see ITU-T G.993.1) technology, supporting optional physical medium entity (PME) aggregation (a.k.a. multi-pair bonding) with variable rates.

The 2BASE-TL PHY is capable of providing at least 2 Mb/s over a 2700 m long single copper pair with a mean bit error ratio (BER) of 10^{-7} (using 5 dB target noise margin).

The 10PASS-TS PHY is capable of providing at least 10 Mb/s over a 750 m long single copper pair with a mean BER of 10^{-7} (using 6 dB target noise margin). This clause defines a MIB module for use with SNMP to manage EFMCu interfaces. In addition, a MIB module is defined describing the cross-connect capability of a stacked interface.

11.2 Relation to other MIB modules

This subclause outlines the relationship of the MIB modules defined in this clause with other MIB modules described in other clauses of this standard, or the relevant RFCs. Specifically, the Interfaces Group MIB (IF-MIB), Ethernet-like (IEEE8023-EtherLike-MIB), MAU (MAU-MIB), SHDSL (HDSL2-SHDSL-LINE-MIB), and VDSL (VDSL-LINE-EXT-MCM-MIB) modules are discussed.

11.2.1 Relation to Interfaces Group MIB module

2BASE-TL and 10PASS-TS PHYs specified in the EFM-CU-MIB module are stacked (a.k.a. aggregated or bonded) Ethernet interfaces and as such are managed using generic interface management objects defined in the IF-MIB defined in IETF RFC 2863.

The stack management (i.e., actual connection of the sublayers to the top-layer interface) is done via the ifStackTable, as defined in the IF-MIB defined in IETF RFC 2863, and its inverse ifInvStackTable, as defined in the IF-INVERTED-STACK-MIB defined in IETF RFC 2864.

The table ifCapStackTable and its inverse ifInvCapStackTable are defined in the IF-CAP-STACK-MIB module. These tables extend the stack management with an ability to describe possible connections or cross-connect capability, when a flexible cross-connect matrix is present between the interface layers. The IF-CAP-STACK-MIB module definition (Beili [B1]) can be found in:

<https://datatracker.ietf.org/doc/draft-ietf-opsawg-rfc5066bis/06/>

11.2.1.1 Layering model

An EFMCu interface can aggregate up to 32 physical medium entity (PME) sublayer devices (modems), using the so-called PME aggregation function (PAF).

A generic EFMCu device can have a number of physical coding sublayer (PCS) ports, each connected to a media access controller (MAC) via a media independent interface (MII) at the upper layer, and cross-connected to a number of underlying PMEs, with a single PCS per PME relationship. See 61.1 of IEEE Std 802.3 for more details.

Each PME in the aggregated EFMCu port is represented in the Interface table (ifTable) as a separate interface with ifType of shdsl(169) for 2BASE-TL or vdsl(97) for 10PASS-TS. The ifType values are defined in [IANAifType-MIB].

The ifSpeed for each PME shall return the actual data bitrate of the active PME (e.g., for 2BaseTL PMEs, it is a multiple of 64 kb/s). A zero value shall be returned when the PME is Initializing or Down.

The ifSpeed of the PCS is the sum of the current operating data rates of all PMEs in the aggregation group, without the 64/65-octet encapsulation overhead and PAF overhead, but accounting for the inter-frame gaps (IFGs).

When using the stated definition of ifSpeed for the PCS, there would be no frame loss in the configuration shown in Figure 11-1 (the test-sets are configured to generate 100% of back-to-back traffic, that is, minimal IFG, at 10 Mb/s or 100 Mb/s, with min and max frame sizes; the EFM interfaces are aggregated to achieve the shown speed).

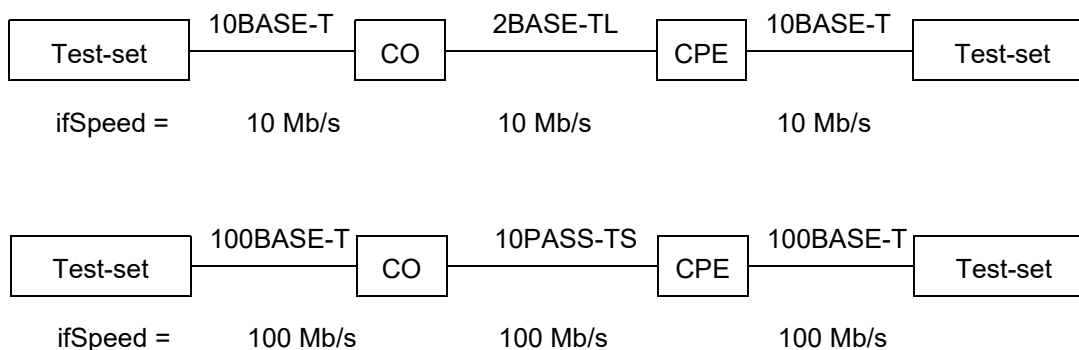
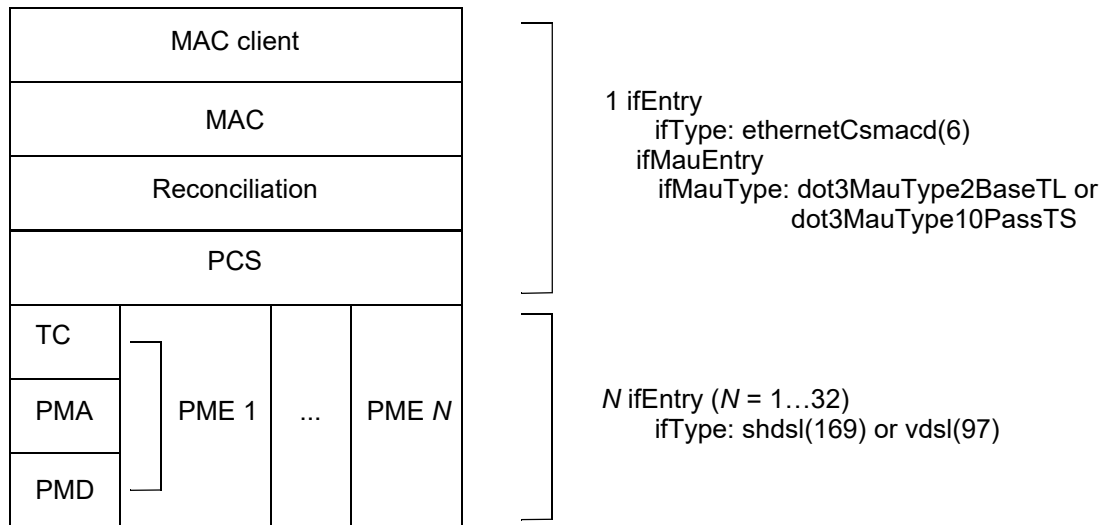


Figure 11-1—Example configuration with no frame loss

Figure 11-2 shows the IEEE 802.3 layering diagram and corresponding use of ifTable and ifMauTable.



MAC = Media Access Control
PCS = Physical Coding Sublayer
TC = Transmission Convergence

PMA = Physical Medium Attachment
PMD = Physical Medium Dependent
PME = Physical Medium Entity

Figure 11-2—Use of ifTable and ifMauTable for EFMcu ports

The ifStackTable is indexed by the ifIndex values of the aggregated EFMcu port (PCS) and the PMEs connected to it. The ifStackTable allows a Network Management application to determine which PMEs are connected to a particular PCS and change connections (if supported by the application). The ifInvStackTable, being an inverted version of the ifStackTable, provides an efficient means for a Network Management application to read a subset of the ifStackTable and thereby determine which PCS runs on top of a particular PME.

The ifCapStackTable, defined in the IF-CAP-STACK-MIB module, specifies for each higher layer interface (e.g., PCS port) a list of lower layer interfaces (e.g., PMEs), which can possibly be cross-connected to that higher layer interface, determined by the cross-connect capability of the device. This table, modeled after the ifStackTable, is read-only, reflecting the current cross-connect capability of the stacked interface, which can be dynamic in some implementations (e.g., if PMEs are located on a pluggable module and the module is pulled out). Note that PME availability per PCS, described by the ifCapStackTable, can be constrained by other parameters, for example, by the aggregation capacity of a PCS or by the PME in question being already connected to another PCS. So that a particular PME can be connected to the PCS, all respective parameters (e.g., ifCapStackTable, ifStackTable, and efmCuPAFCapacity) shall be inspected.

The ifInvCapStackTable, also defined in the IF-CAP-STACK-MIB module, describes which higher layer interfaces (e.g., PCS ports) can possibly be connected to a particular lower layer interface (e.g., PME), providing an inverted mapping of the ifCapStackTable. While it contains no additional information beyond that already contained in the ifCapStackTable, the ifInvCapStackTable has the ifIndex values in its INDEX clause in the reverse order, that is, the lower layer interface first, and the higher layer interface second, providing an efficient means for a Network Management application to read a subset of the ifCapStackTable and thereby determine which interfaces can be connected to run on top of a particular interface.

11.2.1.2 PME aggregation function (PAF)

The PME aggregation function (PAF) allows a number of PMEs to be aggregated onto a PCS port, by fragmenting the Ethernet frames, transmitting the fragments over multiple PMEs, and assembling the original frames at the remote port. PAF is optional, meaning that a device with a single PME may perform fragmentation and reassembly if this function is supported by the device. Note that the agent is required to report on the PAF capability for all EFM-Cu ports (2BASE-TL and 10PASS-TS).

The EFM-CU-MIB module allows a network management application to query the PAF capability and enable/disable it if supported. Note that enabling PAF effectively turns on fragmentation and reassembly, even on a single-PME port.

11.2.1.3 Discovery operation

The EFM-Cu ports may optionally support discovery operation, whereby PMEs, during initialization, exchange information about their respective aggregation groups (PCS). This information can then be used to detect copper misconnections or for an automatic assignment of the local PMEs into aggregation groups instead of a fixed pre-configuration.

The MIB modules defined in this clause allow a network management application to control the EFM discovery mechanism and query its results. Note that the discovery mechanism can work only if PAF is supported and enabled.

Two tables are used by the EFM discovery mechanism: ifStackTable and ifCapStackTable. The following pseudo-code gives an example of the discovery and automatic PME assignment for a generic PAF-enabled multi-PCS EFM-Cu device, located at central office (CO), using objects defined in these MIB modules and in the IF-MIB. (Note that automatic PME assignment is only shown here for the purposes of the example. Fixed PME pre-assignment, manual assignment, or auto-assignment using an alternative internal algorithm may be chosen by a particular implementation.)

```
// Go over all PCS ports in the CO device
FOREACH pcs[i] IN CO_device
{ // Perform discovery and auto-assignment only on PAF enabled ports
  // with room for more PMEs
  IF ( pcs[i].PAFSupported AND pcs[i].NumPMEs < pcs[i].PAFCapacity )
  { // Assign a unique 6-octet local discovery code to the PCS
    // e.g., MAC address
    dc = pcs[i].DiscoveryCode = MAC[i];
    // Go over all disconnected PMEs, which can
    // potentially be connected to the PCS
    FOREACH pme[j] IN ifCapStackTable[pcs[i]] AND
      NOT IN ifStackTable[pcs[i]] // not connected
    { // Try to grab the remote RT_device, by writing the value
      // of the local 6-octet discovery code to the remote
      // discovery code register (via handshake mechanism).
      // This operation is atomic Set-if-Clear action, i.e., it
      // would succeed only if the remote discovery register was
      // zero. Read the remote discovery code register via Get
      // operation to see if the RT_device, attached via the PME
      // is indeed marked as being the CO_device peer.
      pme[j].RemoteDiscoveryCode = dc; // Set-if-Clear
      r = pme[j].RemoteDiscoveryCode; // Get
      IF ( r == dc AND pcs[i].NumPMEs < pcs[i].PAFCapacity )
      { // Remote RT_device connected via PME[j] is/was a peer
        // for PCS[i] and there is room for another PME in the
        // PCS[i] aggregation group (max. PAF capacity is not
```

```
// reached yet).
// Connect this PME to the PCS (via ifStackTable,
// ifInvStackTable being inverse of ifStackTable is
// updated automatically, i.e., pcs[i] is auto-added
// to ifInvStackTable[pme[j]])
ADD pme[j] TO ifStackTable[pcs[i]];
pcs[i].NumPMEs = pcs[i].NumPMEs + 1;
// Discover all other disconnected PMEs,
// attached to the same RT_device and connect them to
// the PCS provided there is enough room for more PMEs.
FOREACH pme[k] IN ifCapStackTable[pcs[i]] AND
    NOT IN ifStackTable[pcs[i]]
{ // Get Remote Discovery Code from the PME to see if
    // it belongs to a connected RT_device "grabbed" by
    // the CO_device.
    r = pme[k].RemoteDiscoveryCode;
    IF ( r == dc AND pcs[i].NumPMEs < pcs[i].PAFCapacity)
    { // Physically connect the PME to the PCS
        // (pcs[i] is auto-added TO ifInvStackTable[pme[k]])
        ADD pme[k] TO ifStackTable[pcs[i]];
        pcs[i].NumPMEs = pcs[i].NumPMEs + 1;
    }
}
}
// At this point we have discovered all local PMEs which
// are physically connected to the same remote RT_device
// and connected them to PCS[i]. Go to the next PCS.
BREAK;
}
}
```

An SNMP Agent for an EFMcu device builds the ifCapStackTable and its inverse ifInvCapStackTable according to the information contained in the Clause 45 PME_Available_register (see 61.1.5.3 and 45.2.3.29 of IEEE Std 802.3).

Adding a PME to the ifStackTable row for a specific PCS involves actual connection of the PME to the PCS, which can be done by modifying the Clause 45 PME_Aggregate_register (see 61.1.5.3 and 45.2.3.30 of IEEE Std 802.3).

Note that the PCS port does not have to be operationally “down” for the connection to succeed. In fact, a dynamic PME addition (and removal) may be implemented with an available PME being initialized first (by setting its ifAdminStatus to “up”) and then added to an operationally “up” PCS port, by modifying a respective ifStackTable (and respective ifInvStackTable) entry.

It is recommended that a removal of the last operationally “up” PME from an operationally “up” PCS would be rejected by the implementation, as this action would completely drop the link.

11.2.1.4 EFMcu ports initialization

EFMCu ports being built on top of xDSL technology require a lengthy initialization or “training” process, before any data can pass. During this initialization, both ends of a link (peers) work cooperatively to achieve the required data rate on a particular copper pair. Sometimes, when the copper line is too long or the noise on the line is too high, that “training” process may fail to achieve a specific target rate with required characteristics.

The ifAdminStatus object from the IF-MIB controls the desired state of a PCS with all the PMEs connected to it or of an individual PME port. Setting this object to “up” instructs a particular PCS or PME to start the initialization process, which can take tens of seconds for EFMcu ports, especially if PAF is involved. The ifOperStatus object shows the operational state of an interface (extended by the ifMauMediaAvailable object from the MAU-MIB module for PCS and efmCuPmeOperStatus defined in the EFM-CU-MIB module for PME interfaces).

A disconnected PME may be initialized by changing the ifAdminState from “down” to “up”. Changing the ifAdminState to “up” on the PCS initializes all PMEs connected to that particular PCS. Note that in case of PAF, some interfaces can fail to initialize while others succeed. The PCS is considered operationally “up” if at least one PME aggregated by its PAF is operationally “up”. When all PMEs connected to the PCS are “down”, the PCS is considered operationally “lowerLayerDown”. The PCS is considered operationally “notPresent” if it is not connected to any PME. The PCS/PME interface remains operationally “down” during initialization.

The efmCuPmeOperStatus defined in the EFM-CU-MIB module expands PME’s ifOperStatus value of “down” to “downReady”, “downNotReady”, and “init” values, indicating various EFMcu PME-specific states.

11.2.1.5 Usage of ifTable

Both the PME and PCS interfaces of the EFMcu PHY are managed using interface-specific management objects defined in the EFM-CU-MIB module and generic interface objects from the ifTable of IF-MIB, with all management table entries referenced by the interface index ifIndex.

Table 11-1 summarizes EFMcu-specific interpretations for some of the ifTable objects specified in the mandatory ifGeneralInformationGroup.

Table 11-1—EFMcu interpretation of IF-MIB objects

IF-MIB object	EFMcu interpretation
ifIndex	Interface index. Each PME and each PCS in the EFMcu PHY shall have a unique index, as there are some PCS- and PME-specific attributes accessible only on the PCS or the PME level.
ifType	ethernetCsmacd(6) for PCS, shdsl(169) for 2BASE-TL PME, vdsl(97) for 10PASS-TS PME. Operating data rate for the PME. For the PCS, it is the sum of the current operating data rates of all PMEs in the aggregation group, without the 64/65-octet encapsulation overhead and PAF overhead, but accounting for the inter-frame gaps (IFGs).
ifSpeed	Setting this object to “up” instructs a particular PCS (with all PMEs connected to it) or PME to start the initialization process.
ifAdminStatus	Setting this object to “up” instructs a particular PCS (with all PMEs connected to it) or PME to start the initialization process.
ifOperStatus	efmCuPmeOperStatus supplements the “down” value of ifOperStatus for PMEs.

11.2.2 Relation to SHDSL MIB module

G.SHDSL.bis modems, similar to PMEs comprising a 2BASE-TL port, are described in the HDLSL2-SHDSL-LINE-MIB module defined in IETF RFC 4319 [B13]. Note that not all attributes of

G.SHDSL modems reflected in the HDSL2-SHDSL-LINE-MIB module have adequate management objects (Clause 30 attributes and Clause 45 registers) in IEEE Std 802.3.

Because of these differences and for the purposes of simplicity, unification of attributes common to both 2BASE-TL and 10PASS-TS PMEs, and name consistency (e.g., prefixing the 2BASE-TL PME related objects with “efmCuPme2B” instead of “hdlsl2shdsl”), it was decided not to reference HDSL2-SHDSL-LINE-MIB objects but to define all the relevant objects in the EFM-CU-MIB module.

However, if some functionality not available in the EFM-CU-MIB module is required and supported by the PME, for example performance monitoring, relevant HDSL2-SHDSL-LINE-MIB groups may be included and applied for PMEs of 2BASE-TL subtype.

11.2.3 Relation to VDSL MIB module

VDSL modems, similar to the PME(s) comprising a 10PASS-TS port, are described in the VDSL-LINE-EXT-MCM-MIB module defined in IETF RFC 4070 [B12]. Note that not all attributes of VDSL modems reflected in the VDSL-LINE-EXT-MCM-MIB module have adequate management objects (Clause 30 attributes and Clause 45 registers) in IEEE Std 802.3.

Because of these differences and for the purposes of simplicity, unification of attributes common to both 2BASE-TL and 10PASS-TS PMEs, and name consistency, it was decided not to reference VDSL-LINE-EXT-MCM-MIB objects but to define all the relevant objects in the EFM-CU-MIB module.

However, if some functionality not available in the EFM-CU-MIB module is required and supported by the PME, relevant VDSL-LINE-EXT-MCM-MIB groups may be included and applied for PMEs of 10PASS-TS subtype.

11.2.4 Relation to Ethernet-like and MAU MIB modules

An agent implementing the objects defined in this clause shall also implement the objects required by the Ethernet-like interface MIB module defined in Clause 10 and the objects required by the MAU MIB module defined in Clause 13.

Two new values of ifMauType (OBJECT-IDENTITIES of dot3MauType) and corresponding bit definitions of ifMauTypeListBits (IANAifMauTypeListBits) have been defined in the IANA-MAU-MIB module for EFMcu MAUs:

- dot3MauType2BaseTL and b2BaseTL, for 2BASE-TL MAU
- dot3MauType10PassTS and b10PassTS, for 10PASS-TS MAU

Additionally, the IANA-MAU-MIB module defines two new values of ifMauMediaAvailable, specifically for EFMcu ports: availableReduced and ready (in textual convention IANAifMauMediaAvailable). Due to the PME aggregation, the EFMcu interpretation of some possible ifMauMediaAvailable values differs from other MAUs as follows:

- unknown: the EFMcu interface (PCS with connected PMEs) is Initializing
- ready: the interface is Down, at least one PME in the aggregation group (all PMEs connected to the PCS) is ready for handshake
- available: the interface is Up, all PMEs in the aggregation group are up
- notAvailable: the interface is Down, all PMEs in the aggregation group are Down, no handshake tones are detected by any PME
- availableReduced: the interface is Up, a link fault is detected at the receive direction by one or more PMEs in the aggregation group, but at least one PME is Up
- pmdLinkFault: a link fault is detected at the receive direction by all PMEs in the aggregation group

As an Ethernet-like interface, every EFMcu port [an ifEntry representing a consolidation of LLC, MAC, and PCS (sub)layers] shall return an ifType of ethernetCsmacd(6). While most of the MAU characteristics are not applicable to the EFMcu ports (no Auto-Negotiation, false carriers, or jabber), they shall return an appropriate ifMauType (dot3MauType2BaseTL or dot3mauType10PassTS) in order to direct the management software to look in the EFM-CU-MIB module for the desired information. For example, the information on the particular EFMcu flavor that an EFMcu port is running is available from efmCuOperSubType, defined in the EFM-CU-MIB module.

Since EFMcu PMEs are not Ethernet-like interfaces, they cannot be instantiated as MAU interface objects.

11.3 MIB structure

11.3.1 EFM copper MIB overview

The main management objects defined in the EFM-CU-MIB module are split into two groups:

- efmCuPort—containing objects for configuration, capabilities, status, and notifications, common to all EFMcu PHYs.
- efmCuPme—containing objects for configuration, capabilities, status, and notifications of EFMcu PMEs.

The efmCuPme group in turn contains efmCuPme2B and efmCuPme10P groups, which define PME profiles specific to 2BASE-TL and 10PASS-TS PMEs, respectively, as well as PME-specific status information.

11.3.2 PME profiles

Since a managed node can have a large number of EFMcu PHYs, provisioning every parameter on every EFMcu PHY becomes burdensome. Moreover, most PMEs are provisioned identically with the same set of parameters. To simplify the provisioning process, the EFM-CU-MIB module makes use of configuration profiles, similar to the HDSL2-SHDSL-LINE-MIB and VDSL-LINE-EXT-MCM-MIB modules. A profile is a set of parameters, used for either configuration or representation of a PME. The same profile can be shared by multiple PME ports using the same configuration.

The PME profiles are defined in the efmCuPme2BProfileTable and efmCuPme10PProfileTable for 2BASE-TL and 10PASS-TS PMEs, respectively. There are 12 predefined standard profiles for 2BASE-TL and 22 standard profiles for 10PASS-TS, defined in IEEE Std 802.3 and dedicated for rapid provisioning of EFMcu PHYs in most scenarios. In addition, the EFM-CU-MIB defines two additional predefined profiles for “best-effort” provisioning of 2BASE-TL PMEs. An ability to define new configuration profiles is also provided to allow for EFMcu deployment tailored to specific copper environments and spectral regulations.

A specific configuration or administrative profile is assigned to a specific PME via the efmCuPmeAdminProfile object. If efmCuPmeAdminProfile is zero, then the efmCuAdminProfile object of the PCS port connected to the PME determines the configuration profile (or a list of possible profiles) for that PME. This mechanism allows specifying a common profile for all PMEs connected to the PCS port, with an ability to change individual PME profiles by setting the efmCuPmeAdminProfile object, which overwrites the profile set by efmCuAdminProfile.

A current operating PME profile is pointed to by the efmCuPmeOperProfile object. Note that this profile entry can be created automatically to reflect achieved parameters in adaptive (not fixed) initialization.

11.3.3 Mapping of IEEE 802.3 managed objects

This subclause contains the mapping between relevant managed objects (attributes) defined in Clause 30 of IEEE Std 802.3, and managed objects defined in this clause and in associated MIB modules, for example, the IF-MIB defined in IETF RFC 2863. Note that the majority of the objects defined in the EFM-CU-MIB module do not have direct counterparts in Clause 30 and instead refer to Clause 45 registers.

Table 11-2—Mapping of IEEE 802.3 managed objects

IEEE 802.3 managed object		Corresponding SNMP object
oMAU - Basic Package (Mandatory)	aMAUType	ifMauType (MAU-MIB)
	aMAUTypeList	ifMauTypeListBits (MAU-MIB)
	aMediaAvailable	ifMediaAvailable (MAU-MIB)
oPAF - Basic Package (Mandatory)	aPAFID	ifIndex (IF-MIB)
	aPhyEnd	efmCuPhySide
	aPHYCurrentStatus	efmCuStatus
	aPAFSupported	efmCuPAFSupported
oPAF - PME Aggregation Package (Optional)	aPAFAdminState	efmCuPAFAdminState
	aLocalPAFCapacity	efmCuPAFCapacity
	aLocalPMEAvailable	ifCapStackTable (IF-CAP-STACK-MIB)
	aLocalPMEAggregate	ifStackTable (IF-MIB)
	aRemotePAFSupported	efmCuRemotePAFSupported
	aRemotePAFCapacity	efmCuRemotePAFCapacity
	aRemotePMEAggregate	
oPME - 10P/2B Package (Mandatory)	aPMEID	ifIndex (IF-MIB)
	aPMEAdminState aPMEStatus	ifAdminState (IF-MIB) efmCuPmeStatus
	aPMESNRMgn	efmCuPmeSnrMgn
	aTCCodingViolations	efmCuPmeTCCodingErrors
	aTCCRCErrors	efmCuPmeTCCrcErrors
	aProfileSelect	efmCuAdminProfile, efmCuPmeAdminProfile
	aOperatingProfile	efmCuPmeOperProfile
	aPMEFECCorrectedBlocks	efmCuPme10PFECCorrectedBlocks
	aPMEFECCUncorrectableBlocks	efmCuPme10PFECUncorrectedBlocks

11.4 Security considerations for Ethernet in the First Mile copper interfaces MIB module

There are a number of managed objects defined in the EFM-CU-MIB module that have a MAX-ACCESS clause of read-write or read-create. Most objects are writeable only when the link is Down. Writing to these objects can have potentially disruptive effects on network operation, for example:

- Changing of `efmCuPmeAdminSubType` might lead to a potential locking of the link, as peer PMEs of the same subtype cannot exchange handshake messages.
- Changing of `efmCuPAFAdminState` to enabled might lead to a potential locking of the link, if the peer PHY does not support PAF.
- Changing of `efmCuPAFDiscoveryCode`, before the discovery operation, might lead to a wrongful discovery, for example, when two -O ports are connected to the same multi-PME -R port and both -O ports have the same Discovery register value.
- Changing PCS or PME configuration parameters (e.g., profile of a PCS or PME via `efmCuAdminProfile` or `efmCuPmeAdminProfile`) might lead to anything from link quality and rate degradation to a complete link initialization failure, as the ability of an EFMcu port to support a particular configuration depends on the copper environment.
- Activation of a PME can cause a severe degradation of service for another EFMcu PHY, whose PME(s) might be affected by the crosstalk from the newly activated PME.
- Removal of a PME from an operationally “up” EFMcu port, aggregating several PMEs, might cause the port’s rate degradation.

The user of the EFM-CU-MIB module should therefore be aware that support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

The readable objects in the EFM-CU-MIB module (i.e., those with MAX-ACCESS other than not-accessible) can be considered sensitive in some environments since, collectively, they provide information about the performance of network interfaces and can reveal some aspects of their configuration. In particular, since EFMcu can be carried over Unshielded Twisted Pair (UTP) voice-grade copper in a bundle with other pairs belonging to another operator/customer, it is theoretically possible to eavesdrop on an EFMcu transmission simply by “listening” to a crosstalk from the EFMcu pairs, especially if the parameters of the EFMcu link in question are known.

In such environments, it is important to control also GET and NOTIFY access to these objects and possibly to encrypt their values when sending them over the network via SNMP.

11.5 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²⁶

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C11mib.txt

²⁶ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

11.5.1 Tree hierarchy

```

----- ieee8023efmCuMIB(1.3.111.2.802.3.1.11.2)
+----- efmCuObjects(1)
| +----- efmCuPort(1)
| | +----- efmCuPortNotifications(0)
| | | +----- efmCuLowRateCrossing(1) {ifSpeed,efmCuThreshLowRate}
| | | +----- efmCuPortConfTable(1)
| | | +----- efmCuPortConfEntry(1) [ifIndex]
| | | +---rw efmCuPAFAdminState(1) Enumeration
| | | +---rw efmCuPAFDDiscoveryCode(2) PhysAddress
| | | +---rw efmCuAdminProfile(3) EfmProfileIndexList
| | | +---rw efmCuTargetDataRate(4) Unsigned32
| | | +---rw efmCuTargetSnrMgn(5) Unsigned32
| | | +---rw efmCuAdaptiveSpectra(6) TruthValue
| | | +---rw efmCuThreshLowRate(7) Unsigned32
| | | +---rw efmCuLowRateCrossingEnable(8) TruthValue
| | +----- efmCuPortCapabilityTable(2)
| | +----- efmCuPortCapabilityEntry(1) [ifIndex]
| | | +---r- efmCuPAFSupported(1) TruthValue
| | | +---r- efmCuPeerPAFSupported(2) EfmTruthValueOrUnknown
| | | +---r- efmCuPAFCapacity(3) Unsigned32
| | | +---r- efmCuPeerPAFCapacity(4) Unsigned32
| +----- efmCuPortStatusTable(3)
| | +----- efmCuPortStatusEntry(1) [ifIndex]
| | | +---r- efmCuFltStatus(1) Bits
| | | +---r- efmCuPortSide(2) Enumeration
| | | +---r- efmCuNumPMEs(3) Unsigned32
| | | +---r- efmCuPAFInErrors(4) Counter32
| | | +---r- efmCuPAFInSmallFragments(5) Counter32
| | | +---r- efmCuPAFInLargeFragments(6) Counter32
| | | +---r- efmCuPAFInBadFragments(7) Counter32
| | | +---r- efmCuPAFInLostFragments(8) Counter32
| | | +---r- efmCuPAFInLostStarts(9) Counter32
| | | +---r- efmCuPAFInLostEnds(10) Counter32
| | | +---r- efmCuPAFInOverflows(11) Counter32
| +----- efmCuPme(2)
| | +----- efmCuPmeNotifications(0)
| | +----- efmCuPmeLineAtnCrossing(1) {efmCuPmeLineAtn,efmCuPmeThreshLineAtn}

```

```

| +----- efmCuPmeSnrMgnCrossing (2) {efmCuPmeSnrMgn,efmCuPmeThreshSnrMgn}
| +----- efmCuPmeDeviceFault (3) {efmCuPmeFltStatus}
| +----- efmCuPmeConfigInitFailure (4) {efmCuPmeFltStatus,efmCuAdminProfile,efmCuPmeAdminProfile}
| +----- efmCuPmeProtocolInitFailure (5) {efmCuPmeFltStatus,efmCuPmeOperSubType}
| +----- efmCuPmeConfTable (1)
| +----- efmCuPmeConfEntry (1) [ifIndex]
| | +---rw efmCuPmeAdminSubType (1) Enumeration
| | +---rw efmCuPmeAdminProfile (2) EfmProfileIndexOrZero
| | +---rw efmCuPmeRemoteDiscoveryCode (3) PhysAddress
| | +---rw efmCuPmeThreshLineAtn (4) Integer32
| | +---rw efmCuPmeThreshSnrMgn (5) Integer32
| | +---rw efmCuPmeLineAtnCrossingEnable (6) TruthValue
| | +---rw efmCuPmeSnrMgnCrossingEnable (7) TruthValue
| | +---rw efmCuPmeDeviceFaultEnable (8) TruthValue
| | +---rw efmCuPmeConfigInitFailEnable (9) TruthValue
| | +---rw efmCuPmeProtocolInitFailEnable (10) TruthValue
| +----- efmCuPmeCapabilityTable (2)
| +----- efmCuPmeCapabilityEntry (1) [ifIndex]
| | +---r- efmCuPmeSubTypesSupported (1) Bits
| +----- efmCuPmeStatusTable (3)
| | +----- efmCuPmeStatusEntry (1) [ifIndex]
| | | +---r- efmCuPmeOperStatus (1) Enumeration
| | | +---r- efmCuPmeFltStatus (2) Bits
| | | +---r- efmCuPmeOperSubType (3) Enumeration
| | | +---r- efmCuPmeOperProfile (4) EfmProfileIndexOrZero
| | | +---r- efmCuPmeSnrMgn (5) Integer32
| | | +---r- efmCuPmePeerSnrMgn (6) Integer32
| | | +---r- efmCuPmeLineAtn (7) Integer32
| | | +---r- efmCuPmePeerLineAtn (8) Integer32
| | | +---r- efmCuPmeEquivalentLength (9) Unsigned32
| | | +---r- efmCuPmeTCCodingErrors (10) Counter32
| | | +---r- efmCuPmeTCCrcErrors (11) Counter32
| +----- efmCuPme2B (5)
| | +----- efmCuPme2BProfileTable (2)
| | | +----- efmCuPme2BProfileEntry (1) [efmCuPme2BProfileIndex]
| | | | +----- efmCuPme2BProfileIndex (1) EfmProfileIndex
| | | | +---rc efmCuPme2BProfileDescr (2) SnmpAdminString
| | | | +---rc efmCuPme2BRegion (3) Enumeration
| | | | +---rc efmCuPme2BsMode (4) EfmProfileIndexOrZero

```

```
+---rc efmCuPme2BMinDataRate (5)      Unsigned32
+---rc efmCuPme2BMaxDataRate (6)      Unsigned32
+---rc efmCuPme2BPower (7)            Unsigned32
+---rc efmCuPme2BConstellation (8)    Enumeration
+---rc efmCuPme2BProfileRowStatus (9) RowStatus
+----- efmCuPme2BsModeTable (3)
+----- efmCuPme2BsModeEntry (1) [efmCuPme2BsModeIndex]
+----- efmCuPme2BsModeIndex (1)     EfmProfileIndex
+---rc efmCuPme2BsModeDescr (2)      SnmpAdminString
+---rc efmCuPme2BsModeRowStatus (3)  RowStatus
+----- efmCuPme2BReachRateTable (4)
+----- efmCuPme2BReachRateEntry (1) [efmCuPme2BsModeIndex, efmCuPme2BReachRateIndex]
+----- efmCuPme2BReachRateIndex (1) EfmProfileIndex
+---rc efmCuPme2BEquivalentLength (2) Unsigned32
+---rc efmCuPme2BMaxDataRatePam16 (3) Unsigned32
+---rc efmCuPme2BMaxDataRatePam32 (4) Unsigned32
+---rc efmCuPme2BReachRateRowStatus (5) RowStatus
+----- efmCuPme10P (6)
+----- efmCuPme10PProfileTable (1)
+----- efmCuPme10PProfileEntry (1) [efmCuPme10PProfileIndex]
+----- efmCuPme10PProfileIndex (1)  EfmProfileIndex
+---rc efmCuPme10PProfileDescr (2)    SnmpAdminString
+---rc efmCuPme10PBandplanPSDMskProfile (3) Enumeration
+---rc efmCuPme10PUPBORReferenceProfile (4) Enumeration
+---rc efmCuPme10PBandNotchProfiles (5) Bits
+---rc efmCuPme10PPayloadDRateProfile (6) Enumeration
+---rc efmCuPme10PPayloadURateProfile (7) Enumeration
+---rc efmCuPme10PProfileRowStatus (8) RowStatus
+----- efmCuPme10PStatusTable (2)
+----- efmCuPme10PStatusEntry (1) [ifIndex]
+---r- efmCuPme10PFECCorrectedBlocks (1) Counter32
+---r- efmCuPme10PFECUncorrectedBlocks (2) Counter32
+----- efmCuConformance (2)
+----- efmCuGroups (1)
+----- efmCuBasicGroup (1)
+----- efmCuPAFGGroup (2)
+----- efmCuPAFErrorsGroup (3)
+----- efmCuPmeGroup (4)
+----- efmCuAlarmConfGroup (5)
```

```
| +---- efmCuNotificationGroup (6)
| +---- efmCuPme2BPPProfileGroup (7)
| +---- efmCuPme10PPProfileGroup (8)
| +---- efmCuPme10PStatusGroup (9)
+---- efmCuCompliances (2)
      +---- efmCuCompliance (1)
```

11.5.2 MIB module definition

```
IEEE8023-EFM-CU-MIB DEFINITIONS ::= BEGIN

IMPORTS
    ifIndex, ifSpeed
        FROM IF-MIB
    SnmpAdminString
        FROM SNMP-FRAMEWORK-MIB
    MODULE-COMPLIANCE, NOTIFICATION-GROUP, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Integer32, MODULE-IDENTITY, NOTIFICATION-TYPE,
    OBJECT-TYPE, Unsigned32, org
        FROM SNMPv2-SMI
    PhysAddress, RowStatus, TEXTUAL-CONVENTION, TruthValue
        FROM SNMPv2-TC;

ieee8023efmCuMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "The objects in this MIB module are used to manage
        the Ethernet in the First Mile (EFM) Copper (EFMCu) Interfaces
        2BASE-TL and 10PASS-TS, defined in IEEE Std 802.3.

        Of particular interest are Clause 61, 'Physical Coding
        Sublayer (PCS) and common specifications, type 10PASS-TS and
        type 2BASE-TL', Clause 30, 'Management', Clause 45,
        'Management Data Input/Output (MDIO) Interface', Annex 62A,
        'PMD profiles for 10PASS-TS' and Annex 63A, 'PMD profiles for
        2BASE-TL'."
    REVISION      "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION      "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION      "201102020000Z"
    DESCRIPTION
        "Initial version, based on an earlier version published
        as RFC 5066."

    ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) ieee8023efmcu(11) 2 }

EfmProfileIndex ::= TEXTUAL-CONVENTION
```

```
DISPLAY-HINT "d"
STATUS      current
DESCRIPTION
    "A unique value, greater than zero, for each PME configuration
    profile in the managed EFMCu port. Values should be assigned
    contiguously starting from 1. The value for each profile shall
    remain constant at least from one re-initialization of the
    entity's network management system to the next re-initialization."
SYNTAX      Unsigned32 (1..255)

EfmProfileIndexOrZero ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "d"
    STATUS      current
    DESCRIPTION
        "This textual convention is an extension of the
        EfmProfileIndex convention. The latter defines a greater than
        zero value used to identify a PME profile in the managed EFMCu
        port. This extension permits the additional value of zero.
        The value of zero is object-specific and shall therefore be
        defined as part of the description of any object that uses
        this syntax.
        Examples of the usage of zero value might include situations
        where the current operational profile is unknown."
    SYNTAX      Unsigned32 (0..255)

EfmProfileIndexList ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "ld:"
    STATUS      current
    DESCRIPTION
        "This textual convention represents a list of up to 6
        EfmProfileIndex values, any of which can be chosen for
        configuration of a PME in a managed EFMCu port.
        The EfmProfileIndex textual convention defines a greater than
        zero value used to identify a PME profile.
        The value of this object is a concatenation of zero or
        more (up to 6) octets, where each octet contains an 8-bit
        EfmProfileIndex value.
        A zero-length octet string is object-specific and shall
        therefore be defined as part of the description of any object
        that uses this syntax. Examples of the usage of a zero-length
        value might include situations where an object using this
        textual convention is irrelevant for a specific EFMCu port
        type."
    SYNTAX      OCTET STRING (SIZE(0..6))

EfmTruthValueOrUnknown ::= TEXTUAL-CONVENTION
    STATUS      current
    DESCRIPTION
        "This textual convention is an extension of the TruthValue
        convention. The latter defines a Boolean value with possible
        values of true(1) and false(2). This extension permits the
        additional value of unknown(0), which can be returned as the
        result of a GET operation when an exact true or false value
        of the object cannot be determined."
    SYNTAX      INTEGER { unknown(0), true(1), false(2) }

efmCuObjects OBJECT IDENTIFIER
    ::= { ieee8023efmCuMIB 1 }
```



```
efmCuPort OBJECT IDENTIFIER
 ::= { efmCuObjects 1 }

efmCuPortNotifications OBJECT IDENTIFIER
 ::= { efmCuPort 0 }

efmCuPortConfTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPortConfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table for Configuration of EFMCu 2BASE-TL/10PASS-TS (PCS)
        Ports. Entries in this table shall be maintained in a
        persistent manner."
    ::= { efmCuPort 1 }

efmCuPortConfEntry OBJECT-TYPE
    SYNTAX      EfmCuPortConfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the EFMCu Port Configuration table.
        Each entry represents an EFMCu port indexed by the ifIndex.
        Note that an EFMCu PCS port runs on top of a single
        or multiple PME port(s), which are also indexed by ifIndex."
    INDEX       { ifIndex }
    ::= { efmCuPortConfTable 1 }

EfmCuPortConfEntry ::= SEQUENCE {
    efmCuPAFAdminState      INTEGER,
    efmCuPAFDiscoveryCode   PhysAddress,
    efmCuAdminProfile       EfmProfileIndexList,
    efmCuTargetDataRate     Unsigned32,
    efmCuTargetSnrMgn       Unsigned32,
    efmCuAdaptiveSpectra    TruthValue,
    efmCuThreshLowRate      Unsigned32,
    efmCuLowRateCrossingEnable TruthValue
}

efmCuPAFAdminState OBJECT-TYPE
    SYNTAX      INTEGER { enabled(1), disabled(2) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "Administrative (desired) state of the PAF of the EFMCu port
        (PCS).
        When 'disabled', PME aggregation will not be performed by the
        PCS. No more than a single PME can be assigned to this PCS in
        this case.
        When 'enabled', PAF will be performed by the PCS when the link
        is Up, even on a single attached PME, if PAF is supported.

        PCS ports incapable of supporting PAF shall return a value of
        'disabled'. Attempts to 'enable' such ports shall be
        rejected.

        A PAF 'enabled' port with multiple PMEs assigned cannot be
        'disabled'. Attempts to 'disable' such port shall be
        rejected, until at most one PME is left assigned.
```

Changing PAFAdminState is a traffic-disruptive operation and as such shall be done when the link is Down. Attempts to change this object shall be rejected if the link is Up or Initializing.

This object maps to IEEE Std 802.3, Clause 30 attribute aPAFAdminState.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the PAF enable bit in the 10P/2B PCS control register.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 61.2.2, 45.2.3.28.3"

::= { efmCuPortConfEntry 1 }

efmCuPAFDiscoveryCode OBJECT-TYPE

SYNTAX PhysAddress (SIZE(0 | 6))

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"PAF Discovery Code of the EFMcu port (PCS).

A unique 6-octet code used by the Discovery function, when PAF is supported.

PCS ports incapable of supporting PAF shall return a zero-length octet string on an attempt to read this object.

An attempt to write to this object shall be rejected for such ports.

This object shall be instantiated for the -O subtype PCS before writing operations on the efmCuPAFRemoteDiscoveryCode (Set_if_Clear and Clear_if_Same) are performed by PMEs associated with the PCS.

The initial value of this object for -R subtype ports after reset is all zeros. For -R subtype ports, the value of this object cannot be changed directly. This value may be changed as a result of writing operation on the efmCuPAFRemoteDiscoveryCode object of remote PME of -O subtype, connected to one of the local PMEs associated with the PCS.

Discovery shall be performed when the link is Down.

Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

The PAF Discovery Code maps to the local Discovery code variable in PAF (note that it does not have a corresponding Clause 45 register)."

REFERENCE

"IEEE Std 802.3, 61.2.2.8.3, 61.2.2.8.4, 45.2.6.6.1, 45.2.6.8, 61A.2"

::= { efmCuPortConfEntry 2 }

efmCuAdminProfile OBJECT-TYPE

SYNTAX EfmProfileIndexList

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired configuration profile(s), common for all PME's in the EFMcu port. This object is a list of pointers to entries in either efmCuPme2BProfileTable or efmCuPme10PProfileTable, depending on the current operating SubType of the EFMcu port as indicated by efmCuPortSide.

The value of this object is a list of up to 6 indices of profiles. If this list consists of a single profile index, then all PME's assigned to this EFMcu port shall be configured according to the profile referenced by that index, unless it is overwritten by a corresponding non-zero efmCuPmeAdminProfile instance, which takes precedence over efmCuAdminProfile.

A list consisting of more than one index allows each PME in the port to be configured according to any profile specified in the list.

By default, this object has a value of 0x01, referencing the 1st entry in efmCuPme2BProfileTable or efmCuPme10PProfileTable.

This object is writeable and readable for the -O subtype (2BaseTL-O or 10PassTS-O) EFMcu ports. It is irrelevant for the -R subtype (2BaseTL-R or 10PassTS-R) ports -- a zero-length octet string shall be returned on an attempt to read this object and an attempt to change this object shall be rejected in this case.

Note that the current operational profile value is available via the efmCuPmeOperProfile object.

Any modification of this object shall be performed when the link is Down. Attempts to change this object shall be rejected, if the link is Up or Initializing. Attempts to set this object to a list with a member value that is not the value of the index for an active entry in the corresponding profile table shall be rejected.

This object maps to IEEE Std 802.3, Clause 30 attribute aProfileSelect.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.6"

DEFVAL { '01'H }

::= { efmCuPortConfEntry 3 }

efmCuTargetDataRate OBJECT-TYPE

SYNTAX Unsigned32 (1..100000 | 999999)

UNITS "Kbps"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired EFMcu port 'net' (as seen across MII) Data Rate in kb/s, to be achieved during initialization, under spectral restrictions placed on each PME via efmCuAdminProfile or efmCuPmeAdminProfile, with the desired SNR margin specified by efmCuTargetSnrMgn.

In case of PAF, this object represents a sum of individual PME

data rates, modified to compensate for fragmentation and 64/65-octet encapsulation overhead (e.g., target data rate of 10 Mb/s shall allow lossless transmission of a full-duplex 10 Mb/s Ethernet frame stream with minimal inter-frame gap).

The value is limited above by 100 Mb/s as this is the max burst rate across MII for EFMCu ports.

The value between 1 and 100000 indicates that the total data rate (ifSpeed) of the EFMCu port after initialization shall be equal to the target data rate or less, if the target data rate cannot be achieved under spectral restrictions specified by efmCuAdminProfile/efmCuPmeAdminProfile and with the desired SNR margin. In case the copper environment allows a higher total data rate to be achieved than that specified by the target, the excess capability shall be either converted to additional SNR margin or reclaimed by minimizing transmit power as controlled by efmCuAdaptiveSpectra.

The value of 999999 means that the target data rate is not fixed and shall be set to the maximum attainable rate during initialization (Best Effort), under specified spectral restrictions and with the desired SNR margin.

This object is read-write for the -O subtype EFMCu ports (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

Changing of the Target Data Rate shall be performed when the link is Down. Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

Note that the current Data Rate of the EFMCu port is represented by the ifSpeed object of IF-MIB.

This object shall be maintained in a persistent manner."
 ::= { efmCuPortConfEntry 4 }

efmCuTargetSnrMgn OBJECT-TYPE

SYNTAX Unsigned32 (0..21)

UNITS "dB"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired EFMCu port SNR margin to be achieved on all PMEs assigned to the port, during initialization. (The SNR margin is the difference between the desired SNR and the actual SNR.)"

Note that IEEE Std 802.3 recommends using a default target SNR margin of 5 dB for 2BASE-TL ports and 6 dB for 10PASS-TS ports in order to achieve a mean bit error ratio (BER) of 10^{-7} at the PMA service interface.

This object is read-write for the -O subtype EFMCu ports (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

Changing of the target SNR margin shall be performed when the link is Down. Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the

link is Up or Initializing.

Note that the current SNR margin of the PMEs comprising the EFMcu port is represented by efmCuPmeSnrmgn.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 61.1.2"

::= { efmCuPortConfEntry 5 }

efmCuAdaptiveSpectra OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Indicates how to utilize excess capacity when the copper environment allows a higher total data rate to be achieved than that specified by the efmCuTargetDataRate.

A value of true(1) indicates that the excess capability shall be reclaimed by minimizing transmit power, e.g., using higher constellations and Power Back-Off, in order to reduce interference to other copper pairs in the binder and the adverse impact to link/system performance.

A value of false(2) indicates that the excess capability shall be converted to additional SNR margin and spread evenly across all active PMEs assigned to the (PCS) port, to increase link robustness.

This object is read-write for the -O subtype EFMcu ports (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

Changing of this object shall be performed when the link is Down. Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

This object shall be maintained in a persistent manner."

::= { efmCuPortConfEntry 6 }

efmCuThreshLowRate OBJECT-TYPE

SYNTAX Unsigned32 (1..100000)

UNITS "Kbps"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This object configures the EFMcu port low-rate crossing alarm threshold. When the current value of ifSpeed for this port reaches/drops below or exceeds this threshold, an efmCuLowRateCrossing notification may be generated if enabled by efmCuLowRateCrossingEnable.

This object is read-write for the -O subtype EFMcu ports (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

This object shall be maintained in a persistent manner."

::= { efmCuPortConfEntry 7 }

```
efmCuLowRateCrossingEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "Indicates whether efmCuLowRateCrossing notifications should
        be generated for this interface.

        A value of true(1) indicates that efmCuLowRateCrossing
        notification is enabled. A value of false(2) indicates that
        the notification is disabled.

        This object is read-write for the -O subtype EFMcu ports
        (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

        This object shall be maintained in a persistent manner."
    ::= { efmCuPortConfEntry 8 }

efmCuPortCapabilityTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPortCapabilityEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "Table for Capabilities of EFMcu 2BASE-TL/10PASS-TS (PCS)
        Ports. Entries in this table shall be maintained in a
        persistent manner"
    ::= { efmCuPort 2 }

efmCuPortCapabilityEntry OBJECT-TYPE
    SYNTAX      EfmCuPortCapabilityEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "An entry in the EFMcu Port Capability table.
        Each entry represents an EFMcu port indexed by the ifIndex.
        Note that an EFMcu PCS port runs on top of a single
        or multiple PME port(s), which are also indexed by ifIndex."
    INDEX       { ifIndex }
    ::= { efmCuPortCapabilityTable 1 }

EfmCuPortCapabilityEntry ::= SEQUENCE {
    efmCuPAFSupported      TruthValue,
    efmCuPeerPAFSupported  EfmTruthValueOrUnknown,
    efmCuPAFCapacity       Unsigned32,
    efmCuPeerPAFCapacity   Unsigned32
}

efmCuPAFSupported OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "PME Aggregation Function (PAF) capability of the EFMcu port
        (PCS).
        This object has a value of true(1) when the PCS can perform
        PME aggregation on the available PMEs.
        Ports incapable of PAF shall return a value of false(2).

        This object maps to IEEE Std 802.3, Clause 30 attribute
```

aPAFSupported.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present,
then this object maps to the PAF available bit in the
10P/2B capability register."

REFERENCE

"IEEE Std 802.3, 61.2.2, 30.11.1.1.4, 45.2.3.27.1"

::= { efmCuPortCapabilityEntry 1 }

efmCuPeerPAFSupported OBJECT-TYPE

SYNTAX EfmTruthValueOrUnknown

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"PME Aggregation Function (PAF) capability of the EFMcu port
(PCS) link partner.

This object has a value of true(1) when the remote PCS can
perform PME aggregation on its available PMEs.

Ports whose peers are incapable of PAF shall return a value
of false(2).

Ports whose peers cannot be reached because of the link
state shall return a value of unknown(0).

This object maps to IEEE Std 802.3, Clause 30 attribute
aRemotePAFSupported.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present,
then this object maps to the Remote PAF supported bit in the
10P/2B capability register."

REFERENCE

"IEEE Std 802.3, 61.2.2, 30.11.1.1.9, 45.2.3.27.2"

::= { efmCuPortCapabilityEntry 2 }

efmCuPAFCapacity OBJECT-TYPE

SYNTAX Unsigned32 (1..32)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Number of PMEs that can be aggregated by the local PAF.
The number of PMEs currently assigned to a particular
EFMCu port (efmCuNumPMEs) is never greater than
efmCuPAFCapacity.

This object maps to IEEE Std 802.3, Clause 30 attribute
aLocalPAFCapacity."

REFERENCE

"IEEE Std 802.3, 61.2.2, 30.11.1.1.6"

::= { efmCuPortCapabilityEntry 3 }

efmCuPeerPAFCapacity OBJECT-TYPE

SYNTAX Unsigned32 (0..32)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Number of PMEs that can be aggregated by the PAF of the peer
PHY (PCS port).

A value of 0 is returned when peer PAF capacity is unknown
(peer cannot be reached).

This object maps to IEEE Std 802.3, Clause 30 attribute

```
aRemotePAFCapacity."
```

REFERENCE

```
"IEEE Std 802.3, 61.2.2, 30.11.1.1.10"
::= { efmCuPortCapabilityEntry 4 }
```

efmCuPortStatusTable OBJECT-TYPE

```
SYNTAX      SEQUENCE OF EfmCuPortStatusEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "This table provides overall status information of EFMCu
    2BASE-TL/10PASS-TS ports, complementing the generic status
    information from the ifTable of IF-MIB and ifMauTable of the
    MAU-MIB module. Additional status information about connected PMEs
    is available from the efmCuPmeStatusTable.

    This table contains live data from the equipment. As such,
    it is not persistent."
::= { efmCuPort 3 }
```

efmCuPortStatusEntry OBJECT-TYPE

```
SYNTAX      EfmCuPortStatusEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
    "An entry in the EFMCu Port Status table.
    Each entry represents an EFMCu port indexed by the ifIndex.
    Note that an EFMCu PCS port runs on top of a single
    or multiple PME port(s), which are also indexed by ifIndex."
INDEX       { ifIndex }
::= { efmCuPortStatusTable 1 }
```

EfmCuPortStatusEntry ::= SEQUENCE {

```
    efmCuFltStatus          BITS,
    efmCuPortSide            INTEGER,
    efmCuNumPMEs             Unsigned32,
    efmCuPAFInErrors         Counter32,
    efmCuPAFInSmallFragments Counter32,
    efmCuPAFInLargeFragments Counter32,
    efmCuPAFInBadFragments  Counter32,
    efmCuPAFInLostFragments Counter32,
    efmCuPAFInLostStarts    Counter32,
    efmCuPAFInLostEnds      Counter32,
    efmCuPAFInOverflows     Counter32
}
```

efmCuFltStatus OBJECT-TYPE

```
SYNTAX      BITS { noPeer(0), peerPowerLoss(1),
                  pmeSubTypeMismatch(2), lowRate(3) }
MAX-ACCESS  read-only
STATUS      current
DESCRIPTION
    "EFMCu (PCS) port Fault Status. This is a bitmap of possible
    conditions. The various bit positions are:
        noPeer          - the peer PHY cannot be reached (e.g.,
                        no PMEs attached, all PMEs are Down,
                        etc.). More info is available in
                        efmCuPmeFltStatus.
        peerPowerLoss    - the peer PHY has indicated impending
```


unit failure due to loss of local power ('Dying Gasp').

pmeSubTypeMismatch - local PME's in the aggregation group are not of the same subtype, e.g., some PME's in the local device are -O while others are -R subtype.

lowRate - ifSpeed of the port reached or dropped below efmCuThreshLowRate.

This object is intended to supplement the ifOperStatus object in IF-MIB and ifMauMediaAvailable in the MAU-MIB module.

Additional information is available via the efmCuPmeFltStatus object for each PME in the aggregation group (single PME if PAF is disabled)."

REFERENCE

"IF-MIB, ifOperStatus; MAU-MIB, ifMauMediaAvailable;
efmCuPmeFltStatus"

::= { efmCuPortStatusEntry 1 }

efmCuPortSide OBJECT-TYPE

SYNTAX INTEGER { subscriber(1), office(2), unknown(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"EFM port mode of operation (subtype).

The value of 'subscriber' indicates that the port is designated as '-R' subtype (all PME's assigned to this port are of subtype '-R').

The value of 'office' indicates that the port is designated as '-O' subtype (all PME's assigned to this port are of subtype '-O').

The value of 'unknown' indicates that the port has no assigned PME's yet or that the assigned PME's are not of the same side (subTypePMEMismatch).

This object partially maps to IEEE Std 802.3, Clause 30 attribute aPhyEnd."

REFERENCE

"IEEE Std 802.3, 61.1, 30.11.1.1.2"

::= { efmCuPortStatusEntry 2 }

efmCuNumPMEs OBJECT-TYPE

SYNTAX Unsigned32 (0..32)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of PME's that is currently aggregated by the local PAF (assigned to the EFMcu port using the ifStackTable).

This number is never greater than efmCuPAFCapacity.

This object shall be automatically incremented or decremented when a PME is added or deleted to/from the EFMcu port using the ifStackTable."

REFERENCE

"IEEE Std 802.3, 61.2.2, 30.11.1.1.6"

::= { efmCuPortStatusEntry 3 }

efmCuPAFInErrors OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of fragments that have been received across the gamma interface with RxErr asserted and discarded. This read-only counter is inactive (not incremented) when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF RX error register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE
"IEEE Std 802.3, 45.2.3.31"
::= { efmCuPortStatusEntry 4 }

efmCuPAFInSmallFragments OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of fragments smaller than minFragmentSize (64 bytes) that have been received across the gamma interface and discarded. This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF small fragments register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE
"IEEE Std 802.3, 45.2.3.32"
::= { efmCuPortStatusEntry 5 }

efmCuPAFInLargeFragments OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of fragments larger than maxFragmentSize (512 bytes) that have been received across the gamma interface and discarded. This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF large fragments register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.3.33"
::= { efmCuPortStatusEntry 6 }

efmCuPAFInBadFragments OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of fragments that do not fit into the sequence expected by the frame assembly function and that have been received across the gamma interface and discarded (the frame buffer is flushed to the next valid frame start). This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF bad fragments register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.3.35"
::= { efmCuPortStatusEntry 7 }

efmCuPAFInLostFragments OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of gaps in the sequence of fragments that have been received across the gamma interface (the frame buffer is flushed to the next valid frame start, when fragment/fragments expected by the frame assembly function is/are not received). This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF lost fragment register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.3.36"
::= { efmCuPortStatusEntry 8 }

efmCuPAFInLostStarts OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of missing StartOfPacket indicators expected by the frame assembly function.
This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF lost starts of fragments register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.3.37"

::= { efmCuPortStatusEntry 9 }

efmCuPAFInLostEnds OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of missing EndOfPacket indicators expected by the frame assembly function.
This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF lost ends of fragments register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.3.38"

::= { efmCuPortStatusEntry 10 }

efmCuPAFInOverflows OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of fragments, received across the gamma interface and discarded, which would have caused the frame assembly buffer to overflow.
This read-only counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the counter retains its previous value.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the 10P/2B PAF overflow register.

Discontinuities in the value of this counter can occur at

```
        re-initialization of the management system, and at other times
        as indicated by the value of ifCounterDiscontinuityTime,
        defined in IF-MIB."
REFERENCE
    "IEEE Std 802.3, 45.2.3.34"
::= { efmCuPortStatusEntry 11 }

efmCuPme OBJECT IDENTIFIER
::= { efmCuObjects 2 }

efmCuPmeNotifications OBJECT IDENTIFIER
::= { efmCuPme 0 }

efmCuPmeConfTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPmeConfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table for Configuration of common aspects for EFMcu
        2BASE-TL/10PASS-TS PME ports (modems). Configuration of
        aspects specific to 2BASE-TL or 10PASS-TS PME types is
        represented in efmCuPme2BConfTable and efmCuPme10PConfTable,
        respectively.

        Entries in this table shall be maintained in a persistent
        manner."
    ::= { efmCuPme 1 }

efmCuPmeConfEntry OBJECT-TYPE
    SYNTAX      EfmCuPmeConfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the EFMcu PME Configuration table.
        Each entry represents common aspects of an EFMcu PME port
        indexed by the ifIndex. Note that an EFMcu PME port can be
        stacked below a single PCS port, also indexed by ifIndex,
        possibly together with other PME ports if PAF is enabled."
    INDEX       { ifIndex }
    ::= { efmCuPmeConfTable 1 }

EfmCuPmeConfEntry ::= SEQUENCE {
    efmCuPmeAdminSubType      INTEGER,
    efmCuPmeAdminProfile      EfmProfileIndexOrZero,
    efmCuPAFRemoteDiscoveryCode PhysAddress,
    efmCuPmeThreshLineAtn     Integer32,
    efmCuPmeThreshSnrMgn      Integer32,
    efmCuPmeLineAtnCrossingEnable TruthValue,
    efmCuPmeSnrMgnCrossingEnable TruthValue,
    efmCuPmeDeviceFaultEnable TruthValue,
    efmCuPmeConfigInitFailEnable TruthValue,
    efmCuPmeProtocolInitFailEnable TruthValue
}

efmCuPmeAdminSubType OBJECT-TYPE
    SYNTAX      INTEGER { ieee2BaseTLO(1), ieee2BaseTLR(2),
        ieee10PassTSO(3), ieee10PassTSR(4),
        ieee2BaseTLor10PassTSR(5), ieee2BaseTLor10PassTSO(6),
        ieee10PassTSor2BaseTLO(7) }
```

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Administrative (desired) subtype of the PME.

Possible values are:

- ieee2BaseTLO - PME shall operate as 2BaseTL-O
- ieee2BaseTLR - PME shall operate as 2BaseTL-R
- ieee10PassTSO - PME shall operate as 10PassTS-O
- ieee10PassTSR - PME shall operate as 10PassTS-R
- ieee2BaseTLor10PassTSR - PME shall operate as 2BaseTL-R or 10PassTS-R. The actual value will be set by the -O link partner during initialization (handshake).
- ieee2BaseTLor10PassTSO - PME shall operate as 2BaseTL-O (preferred) or 10PassTS-O. The actual value will be set during initialization depending on the -R link partner capability (i.e., if -R is incapable of the preferred 2BaseTL mode, 10PassTS will be used).
- ieee10PassTSor2BaseTLO - PME shall operate as 10PassTS-O (preferred) or 2BaseTL-O. The actual value will be set during initialization depending on the -R link partner capability (i.e., if -R is incapable of the preferred 10PassTS mode, 2BaseTL will be used).

Changing efmCuPmeAdminSubType is a traffic-disruptive operation and as such shall be done when the link is Down. Attempts to change this object shall be rejected if the link is Up or Initializing.

Attempts to change this object to an unsupported subtype (see efmCuPmeSubTypesSupported) shall be rejected.

The current operational subtype is indicated by the efmCuPmeOperSubType variable.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this object combines values of the Port subtype select bits and the PMA/PMD type selection bits in the 10P/2B PMA/PMD control register."

REFERENCE

"IEEE Std 802.3, 61.1, 45.2.1.28.4, 45.2.1.28.7"

::= { efmCuPmeConfEntry 1 }

efmCuPmeAdminProfile OBJECT-TYPE

SYNTAX EfmProfileIndexOrZero

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired PME configuration profile. This object is a pointer to an entry in either the efmCuPme2BProfileTable or the efmCuPme10PProfileTable, depending on the current operating SubType of the PME. The value of this object is the index of the referenced profile.

The value of zero (default) indicates that the PME is

configured via the efmCuAdminProfile object for the PCS port to which this PME is assigned. That is, the profile referenced by efmCuPmeAdminProfile takes precedence over the profile(s) referenced by efmCuAdminProfile.

This object is writeable and readable for the CO subtype PMEs (2BaseTL-O or 10PassTS-O). It is irrelevant for the CPE subtype (2BaseTL-R or 10PassTS-R) -- a zero value shall be returned on an attempt to read this object and any attempt to change this object shall be rejected in this case. Note that the current operational profile value is available via efmCuPmeOperProfile object.

Any modification of this object shall be performed when the link is Down. Attempts to change this object shall be rejected, if the link is Up or Initializing.

Attempts to set this object to a value that is not the value of the index for an active entry in the corresponding profile table shall be rejected.

This object maps to IEEE Std 802.3, Clause 30 attribute aProfileSelect.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.6"

DEFVAL { 0 }

::= { efmCuPmeConfEntry 2 }

efmCuPAFRemoteDiscoveryCode OBJECT-TYPE

SYNTAX PhysAddress (SIZE(0 | 6))

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"PAF Remote Discovery Code of the PME port at the CO.

The 6-octet Discovery Code of the peer PCS connected via the PME.

Reading this object results in a Discovery Get operation.

Setting this object to all zeros results in a Discovery Clear_if_Same operation (the value of efmCuPAFDiscoveryCode at the peer PCS shall be the same as efmCuPAFDiscoveryCode of the local PCS associated with the PME for the operation to succeed).

Writing a non-zero value to this object results in a Discovery Set_if_Clear operation.

A zero-length octet string shall be returned on an attempt to read this object when PAF aggregation is not enabled.

This object is irrelevant in CPE port (-R) subtypes: in this case, a zero-length octet string shall be returned on an attempt to read this object; writing to this object shall be rejected.

Discovery shall be performed when the link is Down.

Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this object is a function of 10P/2B aggregation discovery control register, Discovery operation result bits in 10P/2B aggregation and discovery status register and 10P/2B aggregation discovery code register."

REFERENCE

"IEEE Std 802.3, 61.2.2.8.4, 45.2.6.6 to 45.2.6.8"

::= { efmCuPmeConfEntry 3 }

efmCuPmeThreshLineAtn OBJECT-TYPE

SYNTAX Integer32 (-127..128)

UNITS "dB"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired Line Attenuation threshold for the 2B/10P PME.
This object configures the line attenuation alarm threshold.
When the current value of Line Attenuation reaches or exceeds this threshold, an efmCuPmeLineAtnCrossing notification may be generated, if enabled by efmCuPmeLineAtnCrossingEnable.

This object is writeable for the CO subtype PMEs (-O).
It is read-only for the CPE subtype (-R).

Changing of the Line Attenuation threshold shall be performed when the link is Down. Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the loop attenuation threshold bits in the 10P/2B line quality thresholds register."

REFERENCE

"IEEE Std 802.3, 45.2.1.39"

::= { efmCuPmeConfEntry 4 }

efmCuPmeThreshSnrMgn OBJECT-TYPE

SYNTAX Integer32 (-127..128)

UNITS "dB"

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"Desired SNR margin threshold for the 2B/10P PME.
This object configures the SNR margin alarm threshold.
When the current value of SNR margin reaches or exceeds this threshold, an efmCuPmeSnrMgnCrossing notification may be generated, if enabled by efmCuPmeSnrMgnCrossingEnable.
This object is writeable for the CO subtype PMEs (2BaseTL-O/10PassTS-O). It is read-only for the CPE subtype (2BaseTL-R/10PassTS-R).

Changing of the SNR margin threshold shall be performed when the link is Down. Attempts to change this object shall be rejected (in case of SNMP with the error inconsistentValue), if the link is Up or Initializing.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the SNR margin threshold bits in the

10P/2B line quality thresholds register."
REFERENCE
"IEEE Std 802.3, 45.2.1.39"
::= { efmCuPmeConfEntry 5 }

efmCuPmeLineAtnCrossingEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Indicates whether efmCuPmeLineAtnCrossing notifications
should be generated for this interface.

A value of true(1) indicates that efmCuPmeLineAtnCrossing
notification is enabled. A value of false(2) indicates that
the notification is disabled."
::= { efmCuPmeConfEntry 6 }

efmCuPmeSnrMgnCrossingEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Indicates whether efmCuPmeSnrMgnCrossing notifications
should be generated for this interface.

A value of true(1) indicates that efmCuPmeSnrMgnCrossing
notification is enabled. A value of false(2) indicates that
the notification is disabled."
::= { efmCuPmeConfEntry 7 }

efmCuPmeDeviceFaultEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Indicates whether efmCuPmeDeviceFault notifications
should be generated for this interface.

A value of true(1) indicates that efmCuPmeDeviceFault
notification is enabled. A value of false(2) indicates that
the notification is disabled."
::= { efmCuPmeConfEntry 8 }

efmCuPmeConfigInitFailEnable OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"Indicates whether efmCuPmeConfigInitFailure notifications
should be generated for this interface.

A value of true(1) indicates that efmCuPmeConfigInitFailure
notification is enabled. A value of false(2) indicates that
the notification is disabled."
::= { efmCuPmeConfEntry 9 }

efmCuPmeProtocolInitFailEnable OBJECT-TYPE
SYNTAX TruthValue

```
MAX-ACCESS    read-write
STATUS        current
DESCRIPTION
    "Indicates whether efmCuPmeProtocolInitFailure notifications
    should be generated for this interface.

    A value of true(1) indicates that efmCuPmeProtocolInitFailure
    notification is enabled. A value of false(2) indicates that
    the notification is disabled."
::= { efmCuPmeConfEntry 10 }

efmCuPmeCapabilityTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPmeCapabilityEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "Table for the configuration of common aspects for EFMCu
        2BASE-TL/10PASS-TS PME ports (modems). The configuration of
        aspects specific to 2BASE-TL or 10PASS-TS PME types is
        represented in the efmCuPme2BConfTable and the
        efmCuPme10PConfTable, respectively.

        Entries in this table shall be maintained in a persistent
        manner."
    ::= { efmCuPme 2 }

efmCuPmeCapabilityEntry OBJECT-TYPE
    SYNTAX      EfmCuPmeCapabilityEntry
    MAX-ACCESS   not-accessible
    STATUS       current
    DESCRIPTION
        "An entry in the EFMCu PME Capability table.
        Each entry represents common aspects of an EFMCu PME port
        indexed by the ifIndex. Note that an EFMCu PME port can be
        stacked below a single PCS port, also indexed by ifIndex,
        possibly together with other PME ports if PAF is enabled."
    INDEX       { ifIndex }
    ::= { efmCuPmeCapabilityTable 1 }

EfmCuPmeCapabilityEntry ::= SEQUENCE {
    efmCuPmeSubTypesSupported    BITS
}

efmCuPmeSubTypesSupported OBJECT-TYPE
    SYNTAX      BITS { ieee2BaseTLO(0), ieee2BaseTLR(1),
                      ieee10PassTSO(2), ieee10PassTSR(3) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "PME supported subtypes. This is a bitmap of possible
        subtypes. The various bit positions are:
            ieee2BaseTLO    - PME is capable of operating as 2BaseTL-O
            ieee2BaseTLR    - PME is capable of operating as 2BaseTL-R
            ieee10PassTSO   - PME is capable of operating as 10PassTS-O
            ieee10PassTSR   - PME is capable of operating as 10PassTS-R

        The desired mode of operation is determined by
        efmCuPmeAdminSubType, while efmCuPmeOperSubType reflects the
        current operating mode.
```

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object combines the 10PASS-TS capable and 2BASE-TL capable bits in the PMA/PMD speed ability register and the CO supported and CPE supported bits in the 10P/2B PMA/PMD status register."

REFERENCE

"IEEE Std 802.3, 61.1, 45.2.1.4.14, 45.2.1.4.15, 45.2.1.29.2, 45.2.1.29.3"

::= { efmCuPmeCapabilityEntry 1 }

efmCuPmeStatusTable OBJECT-TYPE

SYNTAX SEQUENCE OF EfmCuPmeStatusEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table provides common status information of EFMCu 2BASE-TL/10PASS-TS PME ports. Status information specific to 10PASS-TS PME is represented in efmCuPme10PStatusTable.

This table contains live data from the equipment. As such, it is not persistent."

::= { efmCuPme 3 }

efmCuPmeStatusEntry OBJECT-TYPE

SYNTAX EfmCuPmeStatusEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the EFMCu PME Status table.

Each entry represents common aspects of an EFMCu PME port indexed by the ifIndex. Note that an EFMCu PME port can be stacked below a single PCS port, also indexed by ifIndex, possibly together with other PME ports if PAF is enabled."

INDEX { ifIndex }

::= { efmCuPmeStatusTable 1 }

EfmCuPmeStatusEntry ::= SEQUENCE {

efmCuPmeOperStatus	INTEGER,
efmCuPmeFltStatus	BITS,
efmCuPmeOperSubType	INTEGER,
efmCuPmeOperProfile	EfmProfileIndexOrZero,
efmCuPmeSnrMgn	Integer32,
efmCuPmePeerSnrMgn	Integer32,
efmCuPmeLineAtn	Integer32,
efmCuPmePeerLineAtn	Integer32,
efmCuPmeEquivalentLength	Unsigned32,
efmCuPmeTCCodingErrors	Counter32,
efmCuPmeTCCrcErrors	Counter32

}

efmCuPmeOperStatus OBJECT-TYPE

SYNTAX INTEGER { up(1), downNotReady(2), downReady(3), init(4) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Current PME link Operational Status. Possible values are:

up(1) - The link is Up and ready to pass
64/65-octet encoded frames or fragments.

- downNotReady(2) - The link is Down and the PME does not detect Handshake tones from its peer. This value indicates a possible problem with the peer PME.
- downReady(3) - The link is Down and the PME detects Handshake tones from its peer.
- init(4) - The link is Initializing, as a result of ifAdminStatus being set to 'up' for a particular PME or a PCS to which the PME is connected.

This object is intended to supplement the Down(2) state of ifOperStatus.

This object partially maps to IEEE Std 802.3, Clause 30 attribute aPMEStatus.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object partially maps to PMA/PMD link status bits in the 10P/2B PMA/PMD status register."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.3, 45.2.1.29.4"

::= { efmCuPmeStatusEntry 1 }

efmCuPmeFltStatus OBJECT-TYPE

SYNTAX BITS { lossOfFraming(0), snrMgnDefect(1),
lineAtnDefect(2), deviceFault(3),
configInitFailure(4), protocolInitFailure(5) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Current/Last PME link Fault Status. This is a bitmap of possible conditions. The various bit positions are:

- lossOfFraming - Loss of Framing for 10P or Loss of Sync word for 2B PMD or Loss of 64/65-octet framing.
- snrMgnDefect - SNR margin dropped below the threshold.
- lineAtnDefect - Line Attenuation exceeds the threshold.
- deviceFault - Indicates a vendor-dependent diagnostic or self-test fault has been detected.
- configInitFailure - Configuration initialization failure, due to inability of the PME link to support the configuration profile, requested during initialization.
- protocolInitFailure - Protocol initialization failure, due to an incompatible protocol used by the peer PME during init (that could happen if a peer PMD is a regular G.SDHSL/VDSL modem instead of a 2BASE-TL/10PASS-TS PME).

This object is intended to supplement ifOperStatus in IF-MIB.

This object holds information about the last fault. efmCuPmeFltStatus is cleared by the device restart.

In addition, `lossOfFraming`, `configInitFailure`, and `protocolInitFailure` are cleared by PME init; `deviceFault` is cleared by successful diagnostics/test; `snrMgnDefect` and `lineAtnDefect` are cleared by SNR margin and Line attenuation, respectively, returning to norm and by PME init.

This object partially maps to IEEE Std 802.3, Clause 30 attribute `apMEStatus`.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object consolidates information from various PMA/PMD registers, namely: Fault bit in PMA/PMD status 1 register, 10P/2B PMA/PMD link loss register, 10P outgoing indicator bits status register, 10P incoming indicator bits status register, 2B state defects register."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.3, 45.2.1.2.3, 45.2.1.34, 45.2.1.57, 45.2.1.58, 45.2.1.73"

::= { efmCuPmeStatusEntry 2 }

efmCuPmeOperSubType OBJECT-TYPE

SYNTAX INTEGER { ieee2BaseTLO(1), ieee2BaseTLR(2),
ieee10PassTSO(3), ieee10PassTSR(4) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Current operational subtype of the PME.

Possible values are:

ieee2BaseTLO	- PME operates as 2BaseTL-O
ieee2BaseTLR	- PME operates as 2BaseTL-R
ieee10PassTSO	- PME operates as 10PassTS-O
ieee10PassTSR	- PME operates as 10PassTS-R

The desired operational subtype of the PME can be configured via the `efmCuPmeAdminSubType` variable.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this object combines values of the Port subtype select bits, the PMA/PMD type selection bits in the 10P/2B PMA/PMD control register, and the PMA/PMD link status bits in the 10P/2B PMA/PMD status register."

REFERENCE

"IEEE Std 802.3, 61.1, 45.2.1.28.4, 45.2.1.28.7, 45.2.1.29.4"

::= { efmCuPmeStatusEntry 3 }

efmCuPmeOperProfile OBJECT-TYPE

SYNTAX EfmProfileIndexOrZero

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"PME current operating profile. This object is a pointer to an entry in either the `efmCuPme2BProfileTable` or the `efmCuPme10PProfileTable`, depending on the current operating SubType of the PME as indicated by `efmCuPmeOperSubType`. Note that a profile entry to which `efmCuPmeOperProfile` is pointing can be created automatically to reflect achieved parameters in adaptive (not fixed) initialization,

i.e., values of efmCuPmeOperProfile and efmCuAdminProfile or efmCuPmeAdminProfile may differ.
The value of zero indicates that the PME is Down or Initializing.

This object partially maps to the aOperatingProfile attribute in Clause 30."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.7"

::= { efmCuPmeStatusEntry 4 }

efmCuPmeSnrMgn OBJECT-TYPE

SYNTAX Integer32 (-127..128 | 65535)

UNITS "dB"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current signal-to-noise ratio (SNR) margin with respect to the received signal as perceived by the local PME.
The value of 65535 is returned when the PME is Down or Initializing.

This object maps to the aPMESNRMgn attribute in Clause 30.

If IEEE Std 802.3, Clause 45 MDIO Interface is present, then this object maps to the 10P/2B RX SNR margin register."

REFERENCE

"IEEE Std 802.3, 30.11.2.1.4, 45.2.1.35"

::= { efmCuPmeStatusEntry 5 }

efmCuPmePeerSnrMgn OBJECT-TYPE

SYNTAX Integer32 (-127..128 | 65535)

UNITS "dB"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current SNR margin in dB with respect to the received signal, as perceived by the remote (link partner) PME.
The value of 65535 is returned when the PME is Down or Initializing.

This object is irrelevant for the -R PME subtypes. The value of 65535 shall be returned in this case.

If IEEE Std 802.3, Clause 45 MDIO Interface is present, then this object maps to the 10P/2B link partner RX SNR margin register."

REFERENCE

"IEEE Std 802.3, 45.2.1.36"

::= { efmCuPmeStatusEntry 6 }

efmCuPmeLineAtn OBJECT-TYPE

SYNTAX Integer32 (-127..128 | 65535)

UNITS "dB"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current Line Attenuation in dB as perceived by the local PME.

The value of 65535 is returned when the PME is Down or Initializing.

If IEEE Std 802.3, Clause 45 MDIO Interface is present, then this object maps to the 10P/2B Line Attenuation register."

REFERENCE

"IEEE Std 802.3, 45.2.1.37"

::= { efmCuPmeStatusEntry 7 }

efmCuPmePeerLineAtn OBJECT-TYPE

SYNTAX Integer32 (-127..128 | 65535)

UNITS "dB"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current Line Attenuation in dB as perceived by the remote (link partner) PME.

The value of 65535 is returned when the PME is Down or Initializing.

This object is irrelevant for the -R PME subtypes. The value of 65535 shall be returned in this case.

If IEEE Std 802.3, Clause 45 MDIO Interface is present, then this object maps to the 10P/2B link partner line attenuation register."

REFERENCE

"IEEE Std 802.3, 45.2.1.38"

::= { efmCuPmeStatusEntry 8 }

efmCuPmeEquivalentLength OBJECT-TYPE

SYNTAX Unsigned32 (0..8192 | 65535)

UNITS "m"

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"An estimate of the equivalent loop's physical length in meters, as perceived by the PME after the link is established. An equivalent loop is a hypothetical 26AWG (0.4mm) loop with a perfect square root attenuation characteristic, without any bridged taps.

The value of 65535 is returned if the link is Down or Initializing or the PME is unable to estimate the equivalent length.

For a 10BASE-TL PME, If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the 10P electrical length register."

REFERENCE

"IEEE Std 802.3, 45.2.1.45"

::= { efmCuPmeStatusEntry 9 }

efmCuPmeTCCodingErrors OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of 64/65-octet encapsulation errors. This counter is incremented for each 64/65-octet encapsulation error

detected by the 64/65-octet receive function.

This object maps to aTCCodingViolations attribute in Clause 30.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME TC is present, then this object maps to the 10P/2B TPS-TC coding violations register (see IEEE Std 802.3 45.2.6.12).

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 61.3.3.1, 30.11.2.1.5, 45.2.6.12"
::= { efmCuPmeStatusEntry 10 }

efmCuPmeTCCrcErrors OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"The number of TC-CRC errors. This counter is incremented for each TC-CRC error detected by the 64/65-octet receive function (see IEEE Std 802.3 61.3.3.3 and IEEE Std 802.3 Figure 61-19).

This object maps to aTCCRCErrors attribute in Clause 30.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME TC is present, then this object maps to the 10P/2B TC CRC error register (see IEEE Std 802.3 45.2.6.11).

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 61.3.3.3, 30.11.2.1.10, 45.2.6.11"
::= { efmCuPmeStatusEntry 11 }

efmCuPme2B OBJECT IDENTIFIER

::= { efmCuPme 5 }

efmCuPme2BProfileTable OBJECT-TYPE

SYNTAX SEQUENCE OF EfmCuPme2BProfileEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION

"This table supports definitions of administrative and operating profiles for 2BASE-TL PMEs.
The first 14 entries in this table shall be defined as follows (see IEEE Std 802.3 Annex 63A):

Profile index	MinRate (kb/s)	MaxRate (kb/s)	Power (dBm)	Region	Constellation	Comment
1	5696	5696	13.5	1	32-TCPAM	default
2	3072	3072	13.5	1	32-TCPAM	

3	2048	2048	13.5	1	16-TCPAM	
4	1024	1024	13.5	1	16-TCPAM	
5	704	704	13.5	1	16-TCPAM	
6	512	512	13.5	1	16-TCPAM	
7	5696	5696	14.5	2	32-TCPAM	
8	3072	3072	14.5	2	32-TCPAM	
9	2048	2048	14.5	2	16-TCPAM	
10	1024	1024	13.5	2	16-TCPAM	
11	704	704	13.5	2	16-TCPAM	
12	512	512	13.5	2	16-TCPAM	
13	192	5696	0	1	0	best effort
14	192	5696	0	2	0	best effort
-----+-----+-----+-----+-----+-----+-----						

These default entries shall be created during agent initialization and shall not be deleted.

Entries following the first 14 can be dynamically created and deleted to provide custom administrative (configuration) profiles and automatic operating profiles.

This table shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, Annex 63A, 30.11.2.1.6"

::= { efmCuPme2B 2 }

efmCuPme2BProfileEntry OBJECT-TYPE

SYNTAX EfmCuPme2BProfileEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Each entry corresponds to a single 2BASE-TL PME profile. Each profile contains a set of parameters, used either for configuration or representation of a 2BASE-TL PME. In case a particular profile is referenced via the efmCuPmeAdminProfile object (or efmCuAdminProfile if efmCuPmeAdminProfile is zero), it represents the desired parameters for the 2BaseTL-O PME initialization. If a profile is referenced via an efmCuPmeOperProfile object, it represents the current operating parameters of an operational PME.

Profiles are created/deleted using the row creation/deletion mechanism via efmCuPme2BProfileRowStatus. If an active entry is referenced, the entry shall remain 'active' until all references are removed.

Default entries shall not be removed."

INDEX { efmCuPme2BProfileIndex }

::= { efmCuPme2BProfileTable 1 }

EfmCuPme2BProfileEntry ::= SEQUENCE {

efmCuPme2BProfileIndex	EfmProfileIndex,
efmCuPme2BProfileDescr	SnmpAdminString,
efmCuPme2BRegion	INTEGER,
efmCuPme2BsMode	EfmProfileIndexOrZero,
efmCuPme2BMinDataRate	Unsigned32,
efmCuPme2BMaxDataRate	Unsigned32,
efmCuPme2BPower	Unsigned32,
efmCuPme2BConstellation	INTEGER,

```
efmCuPme2BProfileRowStatus RowStatus
}

efmCuPme2BProfileIndex OBJECT-TYPE
    SYNTAX      EfmProfileIndex
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "2BASE-TL PME profile index.
        This object is the unique index associated with this profile.
        Entries in this table are referenced via efmCuAdminProfile or
        efmCuPmeAdminProfile objects."
    ::= { efmCuPme2BProfileEntry 1 }

efmCuPme2BProfileDescr OBJECT-TYPE
    SYNTAX      SnmpAdminString
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "A textual string containing information about a 2BASE-TL PME
        profile. The string may include information about the data
        rate and spectral limitations of this particular profile."
    ::= { efmCuPme2BProfileEntry 2 }

efmCuPme2BRegion OBJECT-TYPE
    SYNTAX      INTEGER { region1(1), region2(2) }
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "Regional settings for a 2BASE-TL PME, as specified in the
        relevant Regional Annex of ITU-T Recommendation G.991.2.
        Regional settings specify the Power Spectral Density (PSD)
        mask and the Power Back-Off (PBO) values, and place
        limitations on the max allowed data rate, power, and
        constellation.

        Possible values for this object are:
            region1      - Annexes A and F (e.g., North America)
            region2      - Annexes B and G (e.g., Europe)

        Annex A/B specify regional settings for data rates from
        192 kb/s to 2304 kb/s using 16-TCPAM encoding.
        Annex F/G specify regional settings for rates from
        2320 kb/s to 3840 kb/s using 16-TCPAM encoding and from
        768 kb/s to 5696 kb/s using 32-TCPAM encoding.

        If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present,
        then this object partially maps to the Region bits in the
        2B general parameter register."
    REFERENCE
        "IEEE Std 802.3, 45.2.1.61; ITU-T Recommendation G.991.2,
        Annexes A, B, F, and G"
    ::= { efmCuPme2BProfileEntry 3 }

efmCuPme2BsMode OBJECT-TYPE
    SYNTAX      EfmProfileIndexOrZero
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
```

"Desired custom Spectral Mode for a 2BASE-TL PME. This object is a pointer to an entry in efmCuPme2BsModeTable and a block of entries in efmCuPme2BRateReachTable, which together define (country-specific) reach-dependent rate limitations in addition to those defined by efmCuPme2BRegion.

The value of this object is the index of the referenced spectral mode.
The value of zero (default) indicates that no specific spectral mode is applicable.

Attempts to set this object to a value that is not the value of the index for an active entry in the corresponding spectral mode table shall be rejected."

REFERENCE

"efmCuPme2BsModeTable, efmCuPme2BRateReachTable"

DEFVAL { 0 }

::= { efmCuPme2BProfileEntry 4 }

efmCuPme2BMinDataRate OBJECT-TYPE

SYNTAX Unsigned32 (192..5696)

UNITS "Kbps"

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"Minimum Data Rate for the 2BASE-TL PME.

This object can take values of (n x 64)kb/s,
where n=3..60 for 16-TCPAM and n=12..89 for 32-TCPAM encoding.

The data rate of the 2BASE-TL PME is considered 'fixed' when the value of this object equals that of efmCuPme2BMaxDataRate. If efmCuPme2BMinDataRate is less than efmCuPme2BMaxDataRate in the administrative profile, the data rate is considered 'adaptive', and shall be set to the maximum attainable rate not exceeding efmCuPme2BMaxDataRate, under the spectral limitations placed by the efmCuPme2BRegion and efmCuPme2BsMode.

Note that the current operational data rate of the PME is represented by the ifSpeed object of IF-MIB.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the Min data rate1 bits in the 2B PMD parameters register.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 45.2.1.62"

::= { efmCuPme2BProfileEntry 5 }

efmCuPme2BMaxDataRate OBJECT-TYPE

SYNTAX Unsigned32 (192..5696)

UNITS "Kbps"

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"Maximum Data Rate for the 2BASE-TL PME.

This object can take values of (n x 64)kb/s,
where n=3..60 for 16-TCPAM and n=12..89 for 32-TCPAM encoding.

The data rate of the 2BASE-TL PME is considered 'fixed' when the value of this object equals that of efmCuPme2BMinDataRate. If efmCuPme2BMinDataRate is less than efmCuPme2BMaxDataRate in the administrative profile, the data rate is considered 'adaptive', and shall be set to the maximum attainable rate not exceeding efmCuPme2BMaxDataRate, under the spectral limitations placed by the efmCuPme2BRegion and efmCuPme2BsMode.

Note that the current operational data rate of the PME is represented by the ifSpeed object of IF-MIB.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the Max data ratel bits in the 2B PMD parameters register.

This object shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, 45.2.1.62"

::= { efmCuPme2BProfileEntry 6 }

efmCuPme2BPower OBJECT-TYPE

SYNTAX Unsigned32 (0 | 10..42)

UNITS "0.5 dBm"

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"Signal Transmit Power. Multiple of 0.5 dBm.

The value of 0 in the administrative profile means that the signal transmit power is not fixed and shall be set to maximize the attainable rate, under the spectral limitations placed by the efmCuPme2BRegion and efmCuPme2BsMode.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the Power1 bits in the 2B PMD parameters register."

REFERENCE

"IEEE Std 802.3, 45.2.1.62"

::= { efmCuPme2BProfileEntry 7 }

efmCuPme2BConstellation OBJECT-TYPE

SYNTAX INTEGER { adaptive(0), tcpam16(1), tcpam32(2) }

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"TCPAM Constellation of the 2BASE-TL PME.

The possible values are:

- adaptive(0) - either 16- or 32-TCPAM
- tcpam16(1) - 16-TCPAM
- tcpam32(2) - 32-TCPAM

The value of adaptive(0) in the administrative profile means that the constellation is not fixed and shall be set to maximize the attainable rate, under the spectral limitations placed by the efmCuPme2BRegion and efmCuPme2BsMode.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PME is present, then this object maps to the Constellation1 bits in the 2B PMD

```
        parameters register."
REFERENCE
    "IEEE Std 802.3, 45.2.1.62"
 ::= { efmCuPme2BProfileEntry 8 }

efmCuPme2BProfileRowStatus OBJECT-TYPE
    SYNTAX      RowStatus
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "This object controls the creation, modification, or deletion
        of the associated entry in the efmCuPme2BProfileTable per the
        semantics of RowStatus.

        If an 'active' entry is referenced via efmCuAdminProfile or
        efmCuPmeAdminProfile instance(s), the entry shall remain
        'active'.

        An 'active' entry shall not be modified. In order to modify
        an existing entry, it shall be taken out of service (by setting
        this object to 'notInService'), modified, and set 'active'
        again."
 ::= { efmCuPme2BProfileEntry 9 }

efmCuPme2BsModeTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPme2BsModeEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table, together with efmCu2BReachRateTable, supports
        definition of administrative custom spectral modes for
        2BASE-TL PMEs, describing spectral limitations in addition to
        those specified by efmCuPme2BRegion.

        In some countries, spectral regulations (e.g., UK ANFP) limit
        the length of the loops for certain data rates. This table
        allows these country-specific limitations to be specified.

        Entries in this table referenced by the efmCuPme2BsMode
        shall not be deleted until all the active references are
        removed.

        This table shall be maintained in a persistent manner."
REFERENCE
    "efmCu2BReachRateTable"
 ::= { efmCuPme2B 3 }

efmCuPme2BsModeEntry OBJECT-TYPE
    SYNTAX      EfmCuPme2BsModeEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Each entry specifies a spectral mode description and its
        index, which is used to reference corresponding entries in the
        efmCu2BReachRateTable.

        Entries are created/deleted using the row creation/
        deletion mechanism via efmCuPme2BsModeRowStatus."
    INDEX      { efmCuPme2BsModeIndex }
```

```
::= { efmCuPme2BsModeTable 1 }

EfmCuPme2BsModeEntry ::= SEQUENCE {
    efmCuPme2BsModeIndex      EfmProfileIndex,
    efmCuPme2BsModeDescr     SnmpAdminString,
    efmCuPme2BsModeRowStatus  RowStatus
}

efmCuPme2BsModeIndex OBJECT-TYPE
    SYNTAX      EfmProfileIndex
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "2BASE-TL PME Spectral Mode index.
        This object is the unique index associated with this spectral
        mode.
        Entries in this table are referenced via the efmCuPme2BsMode
        object."
    ::= { efmCuPme2BsModeEntry 1 }

efmCuPme2BsModeDescr OBJECT-TYPE
    SYNTAX      SnmpAdminString
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "A textual string containing information about a 2BASE-TL PME
        spectral mode. The string may include information about
        corresponding (country-specific) spectral regulations
        and rate/reach limitations of this particular spectral mode."
    ::= { efmCuPme2BsModeEntry 2 }

efmCuPme2BsModeRowStatus OBJECT-TYPE
    SYNTAX      RowStatus
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "This object controls creation, modification, or deletion of
        the associated entry in efmCuPme2BsModeTable per the semantics
        of RowStatus.

        If an 'active' entry is referenced via efmCuPme2BsMode
        instance(s), the entry shall remain 'active'.

        An 'active' entry shall not be modified. In order to modify
        an existing entry, it shall be taken out of service (by setting
        this object to 'notInService'), modified, and set 'active'
        again."
    ::= { efmCuPme2BsModeEntry 3 }

efmCuPme2BReachRateTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPme2BReachRateEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table supports the definition of administrative custom
        spectral modes for 2BASE-TL PMEs, providing spectral
        limitations in addition to those specified by
        efmCuPme2BRegion.
        The spectral regulations in some countries (e.g., UK ANFP)
```

limit the length of the loops for certain data rates.
This table allows these country-specific limitations to be specified.

Below is an example of this table for NICC Document ND1602:2005/08:

Equivalent Length (m)	MaxRate PAM16 (kb/s)	MaxRate PAM32 (kb/s)
975	2304	5696
1125	2304	5504
1275	2304	5120
1350	2304	4864
1425	2304	4544
1500	2304	4288
1575	2304	3968
1650	2304	3776
1725	2304	3520
1800	2304	3264
1875	2304	3072
1950	2048	2688
2100	1792	2368
2250	1536	0
2400	1408	0
2550	1280	0
2775	1152	0
2925	1152	0
3150	1088	0
3375	1024	0

Entries in this table referenced by an efmCuPme2BsMode instance shall not be deleted.

This table shall be maintained in a persistent manner."

REFERENCE

"NICC Document ND1602:2005/08"

::= { efmCuPme2B 4 }

efmCuPme2BReachRateEntry OBJECT-TYPE

SYNTAX EfmCuPme2BReachRateEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Each entry specifies maximum 2BASE-TL PME data rates allowed for a certain equivalent loop length, when using 16-TCPAM or 32-TCPAM encoding.

When a 2BASE-TL PME is initialized, its data rate shall not exceed the following limitations:

- the value of efmCuPme2BMaxDataRate
- maximum data rate allowed by efmCuPme2BRegion and efmCuPme2BPower
- maximum data rate for a given encoding specified in the efmCuPme2BsModeEntry, corresponding to the equivalent loop length, estimated by the PME

efmCuPme2BEquivalentLength values should be assigned

in increasing order, starting from the minimum value.

Entries are created/deleted using the row creation/
deletion mechanism via efmCuPme2ReachRateRowStatus."

```
INDEX      { efmCuPme2BsModeIndex, efmCuPme2BReachRateIndex }  
::= { efmCuPme2BReachRateTable 1 }
```

```
EfmCuPme2BReachRateEntry ::= SEQUENCE {  
    efmCuPme2BReachRateIndex      EfmProfileIndex,  
    efmCuPme2BEquivalentLength    Unsigned32,  
    efmCuPme2BMaxDataRatePam16    Unsigned32,  
    efmCuPme2BMaxDataRatePam32    Unsigned32,  
    efmCuPme2BReachRateRowStatus  RowStatus  
}
```

```
efmCuPme2BReachRateIndex OBJECT-TYPE  
    SYNTAX      EfmProfileIndex  
    MAX-ACCESS  not-accessible  
    STATUS      current  
    DESCRIPTION  
        "2BASE-TL custom spectral mode Reach-Rate table index.  
        This object is the unique index associated with each entry."  
    ::= { efmCuPme2BReachRateEntry 1 }
```

```
efmCuPme2BEquivalentLength OBJECT-TYPE  
    SYNTAX      Unsigned32 (0..8192)  
    UNITS       "m"  
    MAX-ACCESS  read-create  
    STATUS      current  
    DESCRIPTION  
        "Maximum allowed equivalent loop's physical length in meters  
        for the specified data rates.  
        An equivalent loop is a hypothetical 26AWG (0.4mm) loop with a  
        perfect square root attenuation characteristic, without any  
        bridged taps."  
    ::= { efmCuPme2BReachRateEntry 2 }
```

```
efmCuPme2BMaxDataRatePam16 OBJECT-TYPE  
    SYNTAX      Unsigned32 (0 | 192..5696)  
    UNITS       "Kbps"  
    MAX-ACCESS  read-create  
    STATUS      current  
    DESCRIPTION  
        "Maximum data rate for a 2BASE-TL PME at the specified  
        equivalent loop's length using TC-PAM16 encoding.  
        The value of zero means that TC-PAM16 encoding should not be  
        used at this distance."  
    ::= { efmCuPme2BReachRateEntry 3 }
```

```
efmCuPme2BMaxDataRatePam32 OBJECT-TYPE  
    SYNTAX      Unsigned32 (0 | 192..5696)  
    UNITS       "Kbps"  
    MAX-ACCESS  read-create  
    STATUS      current  
    DESCRIPTION  
        "Maximum data rate for a 2BASE-TL PME at the specified  
        equivalent loop's length using TC-PAM32 encoding.  
        The value of zero means that TC-PAM32 encoding should not be  
        used at this distance."
```



```
 ::= { efmCuPme2BReachRateEntry 4 }
```

efmCuPme2BReachRateRowStatus OBJECT-TYPE

```
SYNTAX      RowStatus
MAX-ACCESS  read-create
STATUS      current
DESCRIPTION
```

"This object controls the creation, modification, or deletion of the associated entry in the efmCuPme2BReachRateTable per the semantics of RowStatus.

If an 'active' entry is referenced via efmCuPme2BsMode instance(s), the entry shall remain 'active'.

An 'active' entry shall not be modified. In order to modify an existing entry, it shall be taken out of service (by setting this object to 'notInService'), modified, and set 'active' again."

```
 ::= { efmCuPme2BReachRateEntry 5 }
```

efmCuPme10P OBJECT IDENTIFIER

```
 ::= { efmCuPme 6 }
```

efmCuPme10PProfileTable OBJECT-TYPE

```
SYNTAX      SEQUENCE OF EfmCuPme10PProfileEntry
MAX-ACCESS  not-accessible
STATUS      current
DESCRIPTION
```

"This table supports definitions of configuration profiles for 10PASS-TS PMEs.

The first 22 entries in this table shall be defined as follows (see IEEE Std 802.3 Annex 62B.3, Table 62B-1):

Profile Index	Bandplan PSDMask#	UPBO p#	BandNotch p#	DRate p#	URate p#	Comment
1	1	3	2,6,10,11	20	20	default profile
2	13	5	0	20	20	
3	1	1	0	20	20	
4	16	0	0	100	100	
5	16	0	0	70	50	
6	6	0	0	50	10	
7	17	0	0	30	30	
8	8	0	0	30	5	
9	4	0	0	25	25	
10	4	0	0	15	15	
11	23	0	0	10	10	
12	23	0	0	5	5	
13	16	0	2,5,9,11	100	100	
14	16	0	2,5,9,11	70	50	
15	6	0	2,6,10,11	50	10	
16	17	0	2,5,9,11	30	30	
17	8	0	2,6,10,11	30	5	
18	4	0	2,6,10,11	25	25	
19	4	0	2,6,10,11	15	15	
20	23	0	2,5,9,11	10	10	
21	23	0	2,5,9,11	5	5	
22	30	0	0	200	50	

These default entries shall be created during agent initialization and shall not be deleted.

Entries following the first 22 can be dynamically created and deleted to provide custom administrative (configuration) profiles and automatic operating profiles.

This table shall be maintained in a persistent manner."

REFERENCE

"IEEE Std 802.3, Annex 62B.3, 30.11.2.1.6"

::= { efmCuPme10P 1 }

efmCuPme10PProfileEntry OBJECT-TYPE

SYNTAX EfmCuPme10PProfileEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Each entry corresponds to a single 10PASS-TS PME profile.

Each profile contains a set of parameters, used either for configuration or representation of a 10PASS-TS PME.

In case a particular profile is referenced via the efmCuPmeAdminProfile object (or efmCuAdminProfile if efmCuPmeAdminProfile is zero), it represents the desired parameters for the 10PassTS-O PME initialization.

If a profile is referenced via an efmCuPmeOperProfile object, it represents the current operating parameters of the PME.

Profiles are created/deleted using the row creation/deletion mechanism via efmCuPme10PProfileRowStatus. If an 'active' entry is referenced, the entry shall remain 'active' until all references are removed.

Default entries shall not be removed."

INDEX { efmCuPme10PProfileIndex }

::= { efmCuPme10PProfileTable 1 }

EfmCuPme10PProfileEntry ::= SEQUENCE {

efmCuPme10PProfileIndex	EfmProfileIndex,
efmCuPme10PProfileDescr	SnmpAdminString,
efmCuPme10PBandplanPSDMskProfile	INTEGER,
efmCuPme10PUPBReferenceProfile	INTEGER,
efmCuPme10PBandNotchProfiles	BITS,
efmCuPme10PPayloadDRateProfile	INTEGER,
efmCuPme10PPayloadURateProfile	INTEGER,
efmCuPme10PProfileRowStatus	RowStatus

}

efmCuPme10PProfileIndex OBJECT-TYPE

SYNTAX EfmProfileIndex

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"10PASS-TS PME profile index.

This object is the unique index associated with this profile. Entries in this table are referenced via efmCuAdminProfile or efmCuPmeAdminProfile."

::= { efmCuPme10PProfileEntry 1 }

efmCuPme10PProfileDescr OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"A textual string containing information about a 10PASS-TS PME profile. The string includes information of this particular profile, for example, data rate and spectral limitations."

::= { efmCuPme10PProfileEntry 2 }

efmCuPme10PBandplanPSDMskProfile OBJECT-TYPE

SYNTAX INTEGER { profile1(1), profile2(2), profile3(3),
profile4(4), profile5(5), profile6(6), profile7(7),
profile8(8), profile9(9), profile10(10),
profile11(11), profile12(12), profile13(13),
profile14(14), profile15(15), profile16(16),
profile17(17), profile18(18), profile19(19),
profile20(20), profile21(21), profile22(22),
profile23(23), profile24(24), profile25(25),
profile26(26), profile27(27), profile28(28),
profile29(29), profile30(30) }

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The 10PASS-TS PME Bandplan and PSD Mask Profile, as specified in IEEE Std 802.3 Annex 62A, Table 62A-1. Possible values are:

-----+-----+-----+-----+-----+-----					
Profile Name	PSD Mask			Bands	ITU-T G.993.1
				0/1/2/3/4/5	Bandplan
-----+-----+-----+-----+-----+-----					
profile1(1)	ANSI	T1.424	FTTCab.M1	x/D/U/D/U	A
profile2(2)	ANSI	T1.424	FTTEEx.M1	x/D/U/D/U	A
profile3(3)	ANSI	T1.424	FTTCab.M2	x/D/U/D/U	A
profile4(4)	ANSI	T1.424	FTTEEx.M2	x/D/U/D/U	A
profile5(5)	ANSI	T1.424	FTTCab.M1	D/D/U/D/U	A
profile6(6)	ANSI	T1.424	FTTEEx.M1	D/D/U/D/U	A
profile7(7)	ANSI	T1.424	FTTCab.M2	D/D/U/D/U	A
profile8(8)	ANSI	T1.424	FTTEEx.M2	D/D/U/D/U	A
profile9(9)	ANSI	T1.424	FTTCab.M1	U/D/U/D/x	A
profile10(10)	ANSI	T1.424	FTTEEx.M1	U/D/U/D/x	A
profile11(11)	ANSI	T1.424	FTTCab.M2	U/D/U/D/x	A
profile12(12)	ANSI	T1.424	FTTEEx.M2	U/D/U/D/x	A
profile13(13)	ETSI	TS 101	270-1 Pcab.M1.A	x/D/U/D/U	B
profile14(14)	ETSI	TS 101	270-1 Pcab.M1.B	x/D/U/D/U	B
profile15(15)	ETSI	TS 101	270-1 Pex.P1.M1	x/D/U/D/U	B
profile16(16)	ETSI	TS 101	270-1 Pex.P2.M1	x/D/U/D/U	B
profile17(17)	ETSI	TS 101	270-1 Pcab.M2	x/D/U/D/U	B
profile18(18)	ETSI	TS 101	270-1 Pex.P1.M2	x/D/U/D/U	B
profile19(19)	ETSI	TS 101	270-1 Pex.P2.M2	x/D/U/D/U	B
profile20(20)	ETSI	TS 101	270-1 Pcab.M1.A	U/D/U/D/x	B
profile21(21)	ETSI	TS 101	270-1 Pcab.M1.B	U/D/U/D/x	B
profile22(22)	ETSI	TS 101	270-1 Pex.P1.M1	U/D/U/D/x	B
profile23(23)	ETSI	TS 101	270-1 Pex.P2.M1	U/D/U/D/x	B
profile24(24)	ETSI	TS 101	270-1 Pcab.M2	U/D/U/D/x	B
profile25(25)	ETSI	TS 101	270-1 Pex.P1.M2	U/D/U/D/x	B
profile26(26)	ETSI	TS 101	270-1 Pex.P2.M2	U/D/U/D/x	B
profile27(27)	ITU-T	G.993.1	F.1.2.1	x/D/U/D/U	Annex F
profile28(28)	ITU-T	G.993.1	F.1.2.2	x/D/U/D/U	Annex F
profile29(29)	ITU-T	G.993.1	F.1.2.3	x/D/U/D/U	Annex F

REFERENCE

"IEEE Std 802.3, Annex 62A"

```
::= { efmCuPme10PProfileEntry 3 }
```

efmCuPme10PUPB0ReferenceProfile OBJECT-TYPE

```
SYNTAX      INTEGER { profile0(0), profile1(1), profile2(2),
                  profile3(3), profile4(4), profile5(5), profile6(6),
                  profile7(7), profile8(8), profile9(9) }
```

MAX-ACCESS read-create

```
STATUS      current
```

DESCRIPTION

"The 10PASS-TS PME Upstream Power Back-Off (UPBO) Reference PSD Profile, as specified in 802.3 Annex 62A, Table 62A-3. Possible values are:

Profile Name	Reference	PSD
profile0(0)	no profile	
profile1(1)	ANSI T1.424	Noise A
profile2(2)	ANSI T1.424	Noise A
profile3(3)	ANSI T1.424	Noise F
profile4(4)	ANSI T1.424	Noise F
profile5(5)	ETSI TS 101 270-1	Noise A&B
profile6(6)	ETSI TS 101 270-1	Noise C
profile7(7)	ETSI TS 101 270-1	Noise D
profile8(8)	ETSI TS 101 270-1	Noise E
profile9(9)	ETSI TS 101 270-1	Noise F

REFERENCE

"IEEE Std 802.3, Annex 62A.3.4"

```
::= { efmCuPme10PProfileEntry 4 }
```

efmCuPme10PBandNotchProfiles OBJECT-TYPE

```
SYNTAX      BITS { profile0(0), profile1(1), profile2(2),
                profile3(3), profile4(4), profile5(5), profile6(6),
                profile7(7), profile8(8), profile9(9), profile10(10),
                profile11(11) }
```

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The 10PASS-TS PME Egress Control Band Notch Profile bitmap, as specified in IEEE Std 802.3 Annex 62A, Table 62A-4. Possible values are:

Profile Name	G.991.3 table	T1.424 table	TS 101 table	270-1	StartF (MHz)	EndF (MHz)
profile0(0)	no profile					
profile1(1)	F-5 #01	-	-		1.810	1.825
profile2(2)	6-2	15-1	17		1.810	2.000
profile3(3)	F-5 #02	-	-		1.907	1.912
profile4(4)	F-5 #03	-	-		3.500	3.575
profile5(5)	6-2	-	17		3.500	3.800
profile6(6)	-	15-1	-		3.500	4.000
profile7(7)	F-5 #04	-	-		3.747	3.754
profile8(8)	F-5 #05	-	-		3.791	3.805
profile9(9)	6-2	-	17		7.000	7.100

profile10(10)	F-5	#06	15-1	-	7.000	7.300
profile11(11)	6-2		15-1	1	10.100	10.150

Any combination of profiles can be specified by ORing individual profiles, for example, a value of 0x2230 selects profiles 2, 6, 10, and 11."

REFERENCE

"IEEE Std 802.3, Annex 62A.3.5"

::= { efmCuPme10PPProfileEntry 5 }

efmCuPme10PPayloadDRateProfile OBJECT-TYPE

SYNTAX INTEGER { profile5(5), profile10(10), profile15(15),
profile20(20), profile25(25), profile30(30),
profile50(50), profile70(70), profile100(100),
profile140(140), profile200(200) }

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The 10PASS-TS PME Downstream Payload Rate Profile, as specified in IEEE Std 802.3 Annex 62A. Possible values are:

profile5(5)	- 2.5 Mb/s
profile10(10)	- 5 Mb/s
profile15(15)	- 7.5 Mb/s
profile20(20)	- 10 Mb/s
profile25(25)	- 12.5 Mb/s
profile30(30)	- 15 Mb/s
profile50(50)	- 25 Mb/s
profile70(70)	- 35 Mb/s
profile100(100)	- 50 Mb/s
profile140(140)	- 70 Mb/s
profile200(200)	- 100 Mb/s

Each value represents a target for the PME's Downstream Payload Bitrate as seen at the MII. If the payload rate of the selected profile cannot be achieved based on the loop environment, bandplan, and PSD mask, the PME initialization shall fail."

REFERENCE

"IEEE Std 802.3, Annex 62A.3.6"

::= { efmCuPme10PPProfileEntry 6 }

efmCuPme10PPayloadURateProfile OBJECT-TYPE

SYNTAX INTEGER { profile5(5), profile10(10), profile15(15),
profile20(20), profile25(25), profile30(30),
profile50(50), profile70(70), profile100(100) }

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"The 10PASS-TS PME Upstream Payload Rate Profile, as specified in 802.3 Annex 62A. Possible values are:

profile5(5)	- 2.5 Mb/s
profile10(10)	- 5 Mb/s
profile15(15)	- 7.5 Mb/s
profile20(20)	- 10 Mb/s
profile25(25)	- 12.5 Mb/s
profile30(30)	- 15 Mb/s
profile50(50)	- 25 Mb/s
profile70(70)	- 35 Mb/s

```
    profile100(100)    - 50 Mb/s
    Each value represents a target for the PME's Upstream Payload
    Bitrate as seen at the MII. If the payload rate of the
    selected profile cannot be achieved based on the loop
    environment, bandplan, and PSD mask, the PME initialization
    shall fail."
REFERENCE
    "IEEE Std 802.3, Annex 62A.3.6"
::= { efmCuPme10PProfileEntry 7 }

efmCuPme10PProfileRowStatus OBJECT-TYPE
    SYNTAX      RowStatus
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
        "This object controls creation, modification, or deletion of
        the associated entry in efmCuPme10PProfileTable per the
        semantics of RowStatus.

        If an active entry is referenced via efmCuAdminProfile or
        efmCuPmeAdminProfile, the entry shall remain 'active' until
        all references are removed.

        An 'active' entry shall not be modified. In order to modify
        an existing entry, it shall be taken out of service (by setting
        this object to 'notInService'), modified, and set 'active'
        again."
::= { efmCuPme10PProfileEntry 8 }

efmCuPme10PStatusTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EfmCuPme10PStatusEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table provides status information of EFMcu 10PASS-TS
        PMEs (modems).

        This table contains live data from the equipment. As such,
        it is not persistent."
::= { efmCuPme10P 2 }

efmCuPme10PStatusEntry OBJECT-TYPE
    SYNTAX      EfmCuPme10PStatusEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the EFMcu 10PASS-TS PME Status table."
    INDEX      { ifIndex }
::= { efmCuPme10PStatusTable 1 }

EfmCuPme10PStatusEntry ::= SEQUENCE {
    efmCuPme10PFECCorrectedBlocks    Counter32,
    efmCuPme10PFECUncorrectedBlocks  Counter32
}

efmCuPme10PFECCorrectedBlocks OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
```

DESCRIPTION

"The number of received and corrected Forward Error Correction (FEC) codewords in this 10PASS-TS PME.

This object maps to the aPMEFECCorrectedBlocks attribute in Clause 30.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this object maps to the 10P FEC correctable errors register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.1.41, 30.11.2.1.8"

::= { efmCuPme10PStatusEntry 1 }

efmCuPme10PFECUncorrectedBlocks OBJECT-TYPE

SYNTAX Counter32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The number of received uncorrectable FEC codewords in this 10PASS-TS PME.

This object maps to the aPMEFECUncorrectableBlocks attribute in Clause 30.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this object maps to the 10P FEC uncorrectable errors register.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime, defined in IF-MIB."

REFERENCE

"IEEE Std 802.3, 45.2.1.42, 30.11.2.1.9"

::= { efmCuPme10PStatusEntry 2 }

efmCuConformance OBJECT IDENTIFIER

::= { ieee8023efmCuMIB 2 }

efmCuGroups OBJECT IDENTIFIER

::= { efmCuConformance 1 }

efmCuCompliances OBJECT IDENTIFIER

::= { efmCuConformance 2 }

efmCuLowRateCrossing NOTIFICATION-TYPE

OBJECTS { ifSpeed, efmCuThreshLowRate }

STATUS current

DESCRIPTION

"This notification indicates that the EFMcu port's data rate has reached/dropped below or exceeded the low rate threshold, specified by efmCuThreshLowRate.

This notification may be sent for the -O subtype ports (2BaseTL-O/10PassTS-O) while the port is Up, on the crossing event in both directions: from normal (rate is above the threshold) to low (rate equals the threshold or below it) and from low to normal. This notification is not applicable to the -R subtypes.

A small debouncing period of 2.5 sec, between the detection of the condition and the notification, should be implemented to prevent simultaneous LinkUp/LinkDown and efmCuLowRateCrossing notifications to be sent.

The adaptive nature of the EFMCu technology allows the port to adapt itself to the changes in the copper environment, e.g., an impulse noise, alien crosstalk, or a micro-interruption that temporarily drops one or more PMEs in the aggregation group, causing a rate degradation of the aggregated EFMCu link. The dropped PMEs would then try to re-initialize, possibly at a lower rate than before, adjusting the rate to provide required target SNR margin.

Generation of this notification is controlled by the efmCuLowRateCrossingEnable object."

```
::= { efmCuPortNotifications 1 }
```

efmCuPmeLineAtnCrossing NOTIFICATION-TYPE

OBJECTS { efmCuPmeLineAtn, efmCuPmeThreshLineAtn }

STATUS current

DESCRIPTION

"This notification indicates that the loop attenuation threshold (as per the efmCuPmeThreshLineAtn value) has been reached/exceeded for the 2BASE-TL/10PASS-TS PME. This notification may be sent on the crossing event in both directions: from normal to exceeded and from exceeded to normal.

A small debouncing period of 2.5 sec, between the detection of the condition and the notification, should be implemented to prevent intermittent notifications from being sent.

Generation of this notification is controlled by the efmCuPmeLineAtnCrossingEnable object."

```
::= { efmCuPmeNotifications 1 }
```

efmCuPmeSnrMgnCrossing NOTIFICATION-TYPE

OBJECTS { efmCuPmeSnrMgn, efmCuPmeThreshSnrMgn }

STATUS current

DESCRIPTION

"This notification indicates that the SNR margin threshold (as per the efmCuPmeThreshSnrMgn value) has been reached/exceeded for the 2BASE-TL/10PASS-TS PME. This notification may be sent on the crossing event in both directions: from normal to exceeded and from exceeded to normal.

A small debouncing period of 2.5 sec, between the detection of the condition and the notification, should be implemented to prevent intermittent notifications from being sent.


```

    Generation of this notification is controlled by the
    efmCuPmeSnrMgnCrossingEnable object."
 ::= { efmCuPmeNotifications 2 }

efmCuPmeDeviceFault NOTIFICATION-TYPE
  OBJECTS      { efmCuPmeFltStatus }
  STATUS       current
  DESCRIPTION
    "This notification indicates that a fault in the PME has been
    detected by a vendor-specific diagnostic or a self-test.

    Generation of this notification is controlled by the
    efmCuPmeDeviceFaultEnable object."
 ::= { efmCuPmeNotifications 3 }

efmCuPmeConfigInitFailure NOTIFICATION-TYPE
  OBJECTS      { efmCuPmeFltStatus, efmCuAdminProfile,
                 efmCuPmeAdminProfile }
  STATUS       current
  DESCRIPTION
    "This notification indicates that PME initialization has
    failed, due to inability of the PME link to achieve the
    requested configuration profile.

    Generation of this notification is controlled by the
    efmCuPmeConfigInitFailEnable object."
 ::= { efmCuPmeNotifications 4 }

efmCuPmeProtocolInitFailure NOTIFICATION-TYPE
  OBJECTS      { efmCuPmeFltStatus, efmCuPmeOperSubType }
  STATUS       current
  DESCRIPTION
    "This notification indicates that the peer PME was using
    an incompatible protocol during initialization.

    Generation of this notification is controlled by the
    efmCuPmeProtocolInitFailEnable object."
 ::= { efmCuPmeNotifications 5 }

efmCuBasicGroup OBJECT-GROUP
  OBJECTS      { efmCuPAFSupported, efmCuAdminProfile,
                 efmCuTargetDataRate, efmCuTargetSnrMgn,
                 efmCuAdaptiveSpectra, efmCuPortSide, efmCuFltStatus }
  STATUS       current
  DESCRIPTION
    "A collection of objects representing management information
    common for all types of EFMcu ports."
 ::= { efmCuGroups 1 }

efmCuPAFGroup OBJECT-GROUP
  OBJECTS      { efmCuPeerPAFSupported, efmCuPAFCapacity,
                 efmCuPeerPAFCapacity, efmCuPAFAdminState,
                 efmCuPAFDiscoveryCode, efmCuPAFRemoteDiscoveryCode,
                 efmCuNumPMEs }
  STATUS       current
  DESCRIPTION
    "A collection of objects supporting optional PME
    Aggregation Function (PAF) and PAF discovery in EFMcu ports."
 ::= { efmCuGroups 2 }
```

efmCuPAFErrorsGroup OBJECT-GROUP

```
OBJECTS      { efmCuPAFInErrors, efmCuPAFInSmallFragments,
                efmCuPAFInLargeFragments, efmCuPAFInBadFragments,
                efmCuPAFInLostFragments, efmCuPAFInLostStarts,
                efmCuPAFInLostEnds, efmCuPAFInOverflows }

STATUS       current

DESCRIPTION
    "A collection of objects supporting optional error counters
    of PAF on EFMcu ports."

::= { efmCuGroups 3 }
```

efmCuPmeGroup OBJECT-GROUP

```
OBJECTS      { efmCuPmeAdminProfile, efmCuPmeOperStatus,
                efmCuPmeFltStatus, efmCuPmeSubTypesSupported,
                efmCuPmeAdminSubType, efmCuPmeOperSubType,
                efmCuPAFRemoteDiscoveryCode, efmCuPmeOperProfile,
                efmCuPmeSnrMgn, efmCuPmePeerSnrMgn, efmCuPmeLineAtn,
                efmCuPmePeerLineAtn, efmCuPmeEquivalentLength,
                efmCuPmeTCCodingErrors, efmCuPmeTCCrcErrors,
                efmCuPmeThreshLineAtn, efmCuPmeThreshSnrMgn }

STATUS       current

DESCRIPTION
    "A collection of objects providing information about
    a 2BASE-TL/10PASS-TS PME."

::= { efmCuGroups 4 }
```

efmCuAlarmConfGroup OBJECT-GROUP

```
OBJECTS      { efmCuThreshLowRate, efmCuLowRateCrossingEnable,
                efmCuPmeThreshLineAtn, efmCuPmeLineAtnCrossingEnable,
                efmCuPmeThreshSnrMgn, efmCuPmeSnrMgnCrossingEnable,
                efmCuPmeDeviceFaultEnable,
                efmCuPmeConfigInitFailEnable,
                efmCuPmeProtocolInitFailEnable }

STATUS       current

DESCRIPTION
    "A collection of objects supporting configuration of alarm
    thresholds and notifications in EFMcu ports."

::= { efmCuGroups 5 }
```

efmCuNotificationGroup NOTIFICATION-GROUP

```
NOTIFICATIONS { efmCuLowRateCrossing, efmCuPmeLineAtnCrossing,
                efmCuPmeSnrMgnCrossing, efmCuPmeDeviceFault,
                efmCuPmeConfigInitFailure, efmCuPmeProtocolInitFailure }

STATUS       current

DESCRIPTION
    "This group supports notifications of significant conditions
    associated with EFMcu ports."

::= { efmCuGroups 6 }
```

efmCuPme2BProfileGroup OBJECT-GROUP

```
OBJECTS      { efmCuPme2BProfileDescr, efmCuPme2BRegion,
                efmCuPme2BsMode, efmCuPme2BMinDataRate,
                efmCuPme2BMaxDataRate, efmCuPme2BPower,
                efmCuPme2BConstellation, efmCuPme2BProfileRowStatus,
                efmCuPme2BsModeDescr, efmCuPme2BsModeRowStatus,
                efmCuPme2BEquivalentLength, efmCuPme2BMaxDataRatePam16,
                efmCuPme2BMaxDataRatePam32,
                efmCuPme2BReachRateRowStatus }
```

```

STATUS      current
DESCRIPTION
    "A collection of objects that constitute a configuration
    profile for configuration of 2BASE-TL ports."
::= { efmCuGroups 7 }

efmCuPme10PProfileGroup OBJECT-GROUP
OBJECTS      { efmCuPme10PProfileDescr,
                efmCuPme10PBandplanPSDMskProfile,
                efmCuPme10PUPBReferenceProfile,
                efmCuPme10PBandNotchProfiles,
                efmCuPme10PPayloadDRateProfile,
                efmCuPme10PPayloadURateProfile,
                efmCuPme10PProfileRowStatus }
STATUS      current
DESCRIPTION
    "A collection of objects that constitute a configuration
    profile for configuration of 10PASS-TS ports."
::= { efmCuGroups 8 }

efmCuPme10PStatusGroup OBJECT-GROUP
OBJECTS      { efmCuPme10PFECCorrectedBlocks,
                efmCuPme10PFECUncorrectedBlocks }
STATUS      current
DESCRIPTION
    "A collection of objects providing status information
    specific to 10PASS-TS PMEs."
::= { efmCuGroups 9 }

efmCuCompliance MODULE-COMPLIANCE
STATUS      current
DESCRIPTION
    "The compliance statement for 2BASE-TL/10PASS-TS interfaces.
    Compliance with the following external compliance statements
    is required:

    MIB module          Compliance Statement
    -----
    IF-MIB               ifCompliance3
    IEEE8023-EtherLike-MIB dot3Compliance2
    MAU-MIB              mauModIfCompl3

    Compliance with the following external compliance statements
    is optional for implementations supporting PME Aggregation
    Function (PAF) with flexible cross-connect between the PCS
    and PME ports:

    MIB module          Compliance Statement
    -----
    IF-INVERTED-STACK-MIB ifInvCompliance
    IF-CAP-STACK-MIB     ifCapStackCompliance"

MODULE      -- this module

MANDATORY-GROUPS { efmCuBasicGroup, efmCuPmeGroup,
                    efmCuAlarmConfGroup, efmCuNotificationGroup }

GROUP      efmCuPme2BProfileGroup
DESCRIPTION

```

"Support for this group is only required for implementations supporting 2BASE-TL PHY."

GROUP efmCuPme10PProfileGroup

DESCRIPTION

"Support for this group is only required for implementations supporting 10PASS-TS PHY."

GROUP efmCuPAFGroup

DESCRIPTION

"Support for this group is only required for implementations supporting PME Aggregation Function (PAF)."

GROUP efmCuPAFErrorsGroup

DESCRIPTION

"Support for this group is optional for implementations supporting PME Aggregation Function (PAF)."

GROUP efmCuPme10PStatusGroup

DESCRIPTION

"Support for this group is optional for implementations supporting 10PASS-TS PHY."

OBJECT efmCuPmeSubTypesSupported

SYNTAX BITS { ieee2BaseTLO(0), ieee2BaseTLR(1),
ieee10PassTSO(2), ieee10PassTSR(3) }

DESCRIPTION

"Support for all subtypes is not required. However, at least one value shall be supported."

OBJECT efmCuPmeAdminSubType

MIN-ACCESS read-only

DESCRIPTION

"Write access is not required (needed only for PMEs supporting more than a single subtype, e.g., ieee2BaseTLO and ieee2BaseTLR or ieee10PassTSO and ieee10PassTSR)."

OBJECT efmCuTargetSnrMgn

MIN-ACCESS read-only

DESCRIPTION

"Write access is optional. For PHYs without write access, the target SNR margin shall be fixed at 5dB for 2BASE-TL and 6dB for 10PASS-TS."

OBJECT efmCuAdaptiveSpectra

MIN-ACCESS read-only

DESCRIPTION

"Write access is optional. For PHYs without write access, the default value should be false."

::= { efmCuCompliances 1 }

END -- end of module IEEE8023-EFM-CU-MIB.

12. Ethernet wide area network (WAN) interface sublayer (WIS) MIB module

This clause defines a portion of the MIB for use with SNMP. In particular, it defines objects for managing IEEE 802.3 WAN interface sublayers.

12.1 Overview

The objects defined in this clause are used in conjunction with objects defined in the Interfaces Group MIB in IETF RFC 2863, the SONET/SDH Interface MIB in IETF RFC 3592, and the IEEE 802.3 MAU MIB defined in Clause 13 of this document to manage the Ethernet WAN interface sublayer (WIS) defined in IEEE Std 802.3. The WIS contains functions to perform OC-192c/VC-4-64c framing and scrambling. It resides between the Physical Coding Sublayer (PCS) and the Physical Medium Attachment (PMA) sublayer within a 10GBASE-W 10 Gb/s WAN-compatible Physical Layer device (PHY) and may be used in conjunction with any of the PCS, PMA, and physical medium dependent (PMD) sublayers defined in IEEE Std 802.3 for 10GBASE-W PHYs. Three types of 10GBASE-W PHYs are defined, distinguished by the type of optics employed: 10GBASE-SW, 10GBASE-LW, and 10GBASE-EW. The objects in this clause are used to manage interfaces implemented using 10GBASE-W PHYs. They do not apply to any other kind of interface. In particular, they do not apply to so-called Ethernet line terminating equipment (ELTE) residing within a SONET network element that uses the 10GBASE-W PMA/PMD sublayers but otherwise acts as SONET line terminating equipment (LTE).

The objects presented here—along with those incorporated by reference from the Interfaces Group MIB, the SONET/SDH Interface MIB, and the IEEE 802.3 MAU MIB—are intended to provide exact representations of the mandatory attributes in the oWIS managed object class (i.e., the members of the pWISBasic package) defined in Clause 30 of IEEE Std 802.3. They are also intended to provide approximate representations of the optional attributes (i.e., the members of the pWISOptional package). Some objects with no analogs in oWIS are defined to support WIS testing features required by Clause 50 of IEEE Std 802.3.

12.1.1 Relationship to the SONET/SDH interface MIB

Since the Ethernet WAN interface sublayer was designed to be SONET-compatible, information similar to that provided by most of the members of the oWIS managed object class is available from objects defined in the SONET-MIB in IETF RFC 3592. Thus, the MIB module defined in this clause is a sparse augmentation of the SONET-MIB—in other words, every table defined here is an extension of some table in the SONET-MIB—and its compliance statement REQUIRES that an agent implementing the objects defined in this clause also implement the relevant SONET-MIB objects. That includes all objects required by sonetCompliance2 as well as some that it leaves optional.

It should be noted that some of the objects incorporated by reference from the SONET-MIB—specifically, the threshold objects and interval counter objects—provide only approximate representations of the corresponding oWIS attributes, as detailed in 12.1.6. An alternative approach would have been to define new objects to exactly match the oWIS definitions. That approach was rejected because the SONET-MIB objects are already used in deployed systems to manage the SONET sublayers of ATM over SONET and PPP over SONET interfaces, and it was deemed undesirable to use a different scheme to manage the SONET sublayers of 10 Gb/s WAN-compatible Ethernet interfaces. Note that the approach adopted by this clause requires no hardware support beyond that mandated by 50.3.11 of IEEE Std 802.3.

12.1.2 Relationship to the Ethernet-like interface MIB

An interface that includes the Ethernet WIS is, by definition, an Ethernet-like interface, and an agent implementing the objects defined in this clause shall also implement the objects required by the Ethernet-like interface MIB module defined in Clause 10.

12.1.3 Relationship to the IEEE 802.3 MAU MIB

Support for the mauModIfCompl3 compliance statement of the MAU-MIB module defined in Clause 13 is required for all Ethernet-like interfaces. The MAU-MIB module is needed in order to allow applications to control and/or determine the media type in use; this is important for devices that can support both the 10GBASE-R 10 Gb/s LAN format (which does not include the WIS) and the 10GBASE-W 10 Gb/s WAN format (which does include the WIS). The MAU-MIB module provides the means to put a device in standby mode or to reset the device. Reset can be used to re-initialize the WIS.

12.1.4 Use of the ifTable

This subclause specifies how the ifTable, as defined in IETF RFC 2863, is used for the Ethernet WIS application.

12.1.4.1 Layering model

Ethernet interfaces that employ the WIS are layered as defined in IEEE Std 802.3. The corresponding use of the ifTable defined in IETF RFC 2863 is shown in Figure 12-1.

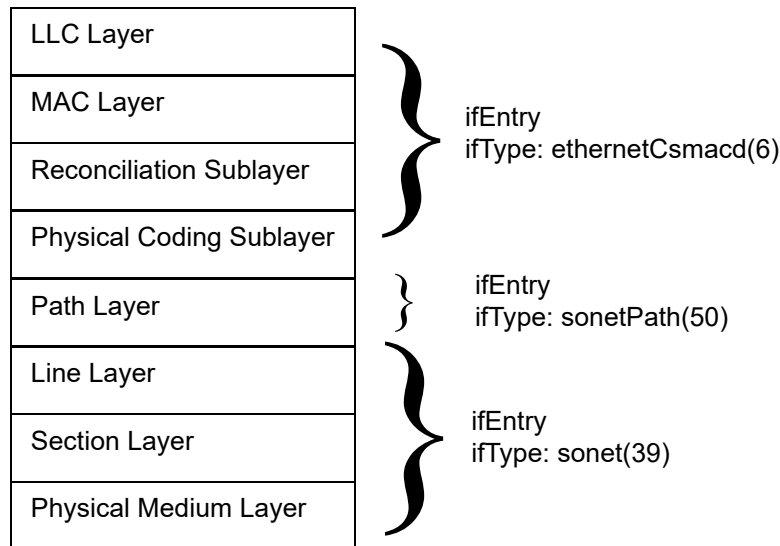


Figure 12-1—Use of ifTable for an Ethernet WIS port

The exact configuration and multiplexing of the layers is maintained in the ifStackTable in IETF RFC 2863 and in the ifInvStackTable in IETF RFC 2864.

12.1.4.2 Use of ifTable for LLC layer/MAC sublayer/reconciliation sublayer/physical coding sublayer

The ifTable shall be used as specified in Clause 10 and Clause 13 for the LLC Layer/MAC sublayer/reconciliation sublayer/physical coding sublayer.

12.1.4.3 Use of ifTable for SONET/SDH path layer

The ifTable shall be used as specified in IETF RFC 3592 for the SONET/SDH path layer. The value of ifHighSpeed is set to 9585. ifSpeed reports a value of 4294967295.

12.1.4.4 Use of ifTable for SONET/SDH medium/section/line layer

The ifTable shall be used as specified in IETF RFC 3592 for the SONET/SDH Medium/Section/Line Layer. The value of ifHighSpeed is set to 9953. ifSpeed reports a value of 4294967295.

12.1.5 SONET/SDH terminology

The SONET/SDH terminology used in IEEE Std 802.3 is mostly the same as in IETF RFC 3592, but there are a few differences. In those cases, the definitions in Clause 3 take precedence.

12.1.6 Mapping of IEEE 802.3 managed objects

Table 12-1 contains the mapping between oWIS managed objects in the pWIS Basic package defined in IEEE Std 802.3 and managed objects defined in this clause and in associated MIB modules, that is, the IF-MIB in IETF RFC 2863, the SONET-MIB in IETF RFC 3592, and the IEEE 802.3 MAU-MIB module defined in Clause 13 of this document.

Table 12-1—Mapping of IEEE 802.3 managed objects (pWIS Basic package)

IEEE 802.3 managed object		Corresponding SNMP object
oWIS - pWISBasic package	aWISID	IF-MIB - ifIndex
	aSectionStatus	SONET-MIB - sonetSectionCurrentStatus
	aLineStatus	SONET-MIB - sonetLineCurrentStatus
	aPathStatus	etherWisPathCurrentStatus
	aFarEndPathStatus	etherWisFarEndPathCurrentStatus

The Unequipped defect is not defined by IEEE Std 802.3.

Table 12-2 contains the same mapping information for the pWIS optional package.

Table 12-2—Mapping of IEEE 802.3 managed objects (pWIS optional package)

IEEE 802.3 managed object		Corresponding SNMP object
oWIS – pWISOptional package	aSectionSESThreshold	SONET-MIB – sonetSESthresholdSet
	aSectionSESs	SONET-MIB – sonetSectionCurrentSESs + sonetSectionIntervalSESs
	aSectionESs	SONET-MIB – sonetSectionCurrentESs + sonetSectionIntervalESs
	aSectionSEFSs	SONET-MIB – sonetSectionCurrentSEFSs + sonetSectionIntervalSEFSs
	aSectionCVs	SONET-MIB – sonetSectionCurrentCVs + sonetSectionIntervalCVs
	aJ0ValueTX	etherWisSectionCurrentJ0Transmitted
	aJ0ValueRX	etherWisSectionCurrentJ0Received
	aLineSESThreshold	SONET-MIB – sonetSESthresholdSet
	aLineSESs	SONET-MIB – sonetLineCurrentSESs + sonetLineIntervalSESs
	aLineESs	SONET-MIB – sonetLineCurrentESs + sonetLineIntervalESs
	aLineCVs	SONET-MIB – sonetLineCurrentCVs + sonetLineIntervalCVs
	aFarEndLineSESs	SONET-MIB – sonetFarEndLineCurrentSESs + sonetFarEndLineIntervalSESs
	aFarEndLineESs	SONET-MIB – sonetFarEndLineCurrentESs + sonetFarEndLineIntervalESs
	aFarEndLineCVs	SONET-MIB – sonetFarEndLineCurrentCVs + sonetFarEndLineIntervalCVs
	aPathSESThreshold	SONET-MIB – sonetSESthresholdSet
	aPathSESs	SONET-MIB – sonetPathCurrentSESs + sonetPathIntervalSESs
	aPathESs	SONET-MIB – sonetPathCurrentESs + sonetPathIntervalESs
	aPathCVs	SONET-MIB – sonetPathCurrentCVs + sonetPathIntervalCVs
	aJ1ValueTX	etherWisPathCurrentJ1Transmitted

Table 12-2—Mapping of IEEE 802.3 managed objects (pWIS optional package) (*continued*)

IEEE 802.3 managed object		Corresponding SNMP object
oWIS - pWISOptional package (<i>continued</i>)	aJ1ValueRX	etherWisPathCurrentJ1Received
	aFarEndPathSESSs	SONET-MIB – sonetFarEndPathCurrentSESSs + sonetFarEndPathIntervalSESSs
	aFarEndPathESs	SONET-MIB – sonetFarEndPathCurrentESs + sonetFarEndPathIntervalESs
	aFarEndPathCVs	SONET-MIB – sonetFarEndPathCurrentCVs + sonetFarEndPathIntervalCVs

The threshold and counter objects imported from the SONET-MIB are not completely equivalent to the corresponding IEEE 802.3 objects. The specific differences are presented in Table 12-3. Despite the semantic differences between the threshold objects and counter objects imported from the SONET-MIB and the corresponding IEEE 802.3 objects, the hardware support mandated by 50.3.11 of IEEE Std 802.3 suffices for both. See Annex 12A for details.

Table 12-3—IEEE 802.3 managed object and SNMP object differences

IEEE 802.3 managed object	How corresponding SNMP object differs
aSectionSESThreshold	This object is defined in IEEE Std 802.3 as an integer with one instance per interface. sonetSESThresholdSet is an enumerated value that has one instance per network element; it controls the thresholds for all layers simultaneously and allows only certain discrete values to be selected.
aSectionSESSs	This object is defined in IEEE Std 802.3 as a generalized nonresettable counter. The objects sonetSectionCurrentSESSs and sonetSectionIntervalSESSs are 15-minute interval counters.
aSectionESs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3. The objects sonetSectionCurrentESs and sonetSectionIntervalESs are 15-minute interval counters.
aSectionSEFSs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3. The objects sonetSectionCurrentSEFSs and sonetSectionIntervalSEFSs are 15-minute interval counters.
aSectionCVs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetSectionCurrentCVs and sonetSectionIntervalCVs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as severely errored seconds.
aLineSESThreshold	This object is defined in IEEE Std 802.3 as an integer with one instance per interface. sonetSESThresholdSet is an enumerated value that has one instance per network element; it controls the thresholds for all layers simultaneously and allows only certain discrete values to be selected.

Table 12-3—IEEE 802.3 managed object and SNMP object differences (*continued*)

IEEE 802.3 managed object	How corresponding SNMP object differs
aLineSESSs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetLineCurrentSESSs and sonetLineIntervalSESSs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds.
aLineESs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetLineCurrentESs and sonetLineIntervalESs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds.
aLineCVs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetLineCurrentCVs and sonetLineIntervalCVs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify either as severely errored seconds or as unavailable seconds.
aFarEndLineSESSs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndLineCurrentSESSs and sonetFarEndLineIntervalSESSs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds.
aFarEndLineESs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndLineCurrentESs and sonetFarEndLineIntervalESs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds.
aFarEndLineCVs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndLineCurrentCVs and sonetFarEndLineIntervalCVs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify either as severely errored seconds or as unavailable seconds.
aPathSESThreshold	This object is defined in IEEE Std 802.3 as an integer with one instance per interface. sonetSESThresholdSet is an enumerated value that has one instance per network element; it controls the thresholds for all layers simultaneously and allows only certain discrete values to be selected.
aPathSESSs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetPathCurrentSESSs and sonetPathIntervalSESSs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds. In addition, IEEE Std 802.3 includes PLM-P and LCD-P defects in the criteria for declaring path layer severely errored seconds, while IETF RFC 3592 does not.
aPathESs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetPathCurrentESs and sonetPathIntervalESs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds. In addition, IEEE Std 802.3 includes PLM-P and LCD-P defects in the criteria for declaring path layer errored seconds, while IETF RFC 3592 does not.
aPathCVs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetPathCurrentCVs and sonetPathIntervalCVs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify either as severely errored seconds or as unavailable seconds.

Table 12-3—IEEE 802.3 managed object and SNMP object differences (*continued*)

IEEE 802.3 managed object	How corresponding SNMP object differs
aFarEndPathSESSs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndPathCurrentSESSs and sonetFarEndPathIntervalSESSs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds. In addition, IEEE Std 802.3 includes far-end PLM-P and LCD-P defects in the criteria for declaring far-end path layer severely errored seconds, while IETF RFC 3592 does not.
aFarEndPathESs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndPathCurrentESs and sonetFarEndPathIntervalESs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify as unavailable seconds. In addition, IEEE Std 802.3 includes far-end PLM-P and LCD-P defects in the criteria for declaring far-end path layer errored seconds, while IETF RFC 3592 does not.
aFarEndPathCVs	This object is defined as a generalized nonresettable counter in IEEE Std 802.3, and it is not subject to inhibiting. The objects sonetFarEndPathCurrentCVs and sonetFarEndPathIntervalCVs are 15-minute interval counters, and they are inhibited (not incremented) during 1-second intervals that qualify either as severely errored seconds or as unavailable seconds.

12.1.7 Mapping of SNMP objects to WIS station management registers

Some of the objects defined in this clause or incorporated by reference from the SONET-MIB IETF RFC 3592, or the MAU-MIB module defined in Clause 13 require WIS-specific hardware support. Subclause 50.3.11 of IEEE Std 802.3 specifies WIS management interface requirements, including a required subset of the WIS Management Data Input/Output (MDIO) registers defined in 45.2.2 of IEEE Std 802.3. Table 12-4 provides a cross-reference between those managed objects and the WIS MDIO registers from the subset in 50.3.11 of IEEE Std 802.3 required to support them. Note that the MDIO interface is optional; however, if it is not implemented, then the capabilities of the required register subset shall be provided by other means.

Table 12-4—Cross-reference between SNMP objects and WIS MDIO registers

SNMP object	WIS MDIO register(s)
ETHER-WIS - etherWisDeviceTxTestPatternMode	10G WIS control 2
ETHER-WIS - etherWisDeviceRxTestPatternMode	10G WIS control 2
ETHER-WIS - etherWisDeviceRxTestPatternErrors	10G WIS test pattern error counter
SONET-MIB - sonetMediumType	None required
SONET-MIB - sonetMediumTimeElapsed	None required
SONET-MIB - sonetMediumValidIntervals	None required
SONET-MIB - sonetMediumLineCoding	None required
SONET-MIB - sonetMediumLineType	None required

Table 12-4—Cross-reference between SNMP objects and WIS MDIO registers (*continued*)

SNMP object	WIS MDIO register(s)
SONET-MIB - sonetMediumCircuitIdentifier	None required
SONET-MIB - sonetMediumInvalidIntervals	None required
SONET-MIB - sonetMediumLoopbackConfig	None required
SONET-MIB - sonetSESthresholdSet	None required
ETHER-WIS - etherWisSectionCurrentJ0Transmitted	10G WIS J0 transmit
ETHER-WIS - etherWisSectionCurrentJ0Received	10G WIS J0 receive
SONET-MIB - sonetSectionCurrentStatus	10G WIS status 3
SONET-MIB - sonetSectionCurrentESs	10G WIS status 3 + 10G WIS section BIP error count
SONET-MIB - sonetSectionCurrentSESs	
SONET-MIB - sonetSectionCurrentSEFSs	
SONET-MIB - sonetSectionCurrentCVs	
SONET-MIB - sonetSectionIntervalESs	
SONET-MIB - sonetSectionIntervalSESs	
SONET-MIB - sonetSectionIntervalSEFSs	
SONET-MIB - sonetSectionIntervalCVs	
SONET-MIB - sonetSectionIntervalValidData	None required
SONET-MIB - sonetLineCurrentStatus	10G WIS status 3
SONET-MIB - sonetLineCurrentESs	10G WIS status 3 + 10G WIS line BIP errors
SONET-MIB - sonetLineCurrentSESs	
SONET-MIB - sonetLineCurrentCVs	
SONET-MIB - sonetLineCurrentUASs	
SONET-MIB - sonetLineIntervalESs	
SONET-MIB - sonetLineIntervalSESs	
SONET-MIB - sonetLineIntervalCVs	
SONET-MIB - sonetLineIntervalUASs	
SONET-MIB - sonetLineIntervalValidData	None required

Table 12-4—Cross-reference between SNMP objects and WIS MDIO registers (*continued*)

SNMP object	WIS MDIO register(s)
SONET-MIB - sonetFarEndLineCurrentESs	10G WIS status 3 + 10G WIS far end line BIP errors
SONET-MIB - sonetFarEndLineCurrentSEsSs	
SONET-MIB - sonetFarEndLineCurrentCVs	
SONET-MIB - sonetFarEndLineCurrentUASs	
SONET-MIB - sonetFarEndLineIntervalESs	
SONET-MIB - sonetFarEndLineIntervalSEsSs	
SONET-MIB - sonetFarEndLineIntervalCVs	
SONET-MIB - sonetFarEndLineIntervalUASs	
SONET-MIB - sonetFarEndLineIntervalValidData	10G WIS status 3
ETHER-WIS - etherWisPathCurrentStatus	10G WIS status 3
ETHER-WIS - etherWisPathCurrentJ1Transmitted	10G WIS J1 transmit
ETHER-WIS - etherWisPathCurrentJ1Received	10G WIS J1 receive
SONET-MIB - sonetPathCurrentWidth	None required
SONET-MIB - sonetPathCurrentStatus	10G WIS status 3
SONET-MIB - sonetPathCurrentESs	10G WIS status 3 + 10G WIS path block error count
SONET-MIB - sonetPathCurrentSEsSs	
SONET-MIB - sonetPathCurrentCVs	
SONET-MIB - sonetPathCurrentUASs	
SONET-MIB - sonetPathIntervalESs	
SONET-MIB - sonetPathIntervalCVs	
SONET-MIB - sonetPathIntervalUASs	
SONET-MIB - sonetPathIntervalValidData	None required
ETHER-WIS - etherWisFarEndPathCurrentStatus	10G WIS status 3

Table 12-4—Cross-reference between SNMP objects and WIS MDIO registers (*continued*)

SNMP object	WIS MDIO register(s)
SONET-MIB - sonetFarEndPathCurrentESs	10G WIS status 3 + 10G WIS far end path block error count
SONET-MIB - sonetFarEndPathCurrentSESSs	
SONET-MIB - sonetFarEndPathCurrentCVs	
SONET-MIB - sonetFarEndPathCurrentUASs	
SONET-MIB - sonetFarEndPathIntervalESs	
SONET-MIB - sonetFarEndPathIntervalSESSs	
SONET-MIB - sonetFarEndPathIntervalCVs	
SONET-MIB - sonetFarEndPathIntervalUASs	
SONET-MIB - sonetFarEndPathIntervalValidData	
MAU-MIB - ifMauIfIndex	None required
MAU-MIB - ifMauIndex	None required
MAU-MIB - ifMauType	10G WIS control 2
MAU-MIB - ifMauStatus	WIS control 1
MAU-MIB - ifMauMediaAvailable	WIS status 1 + 10G WIS status 3
MAU-MIB - ifMauMediaAvailableStateExits	
MAU-MIB - ifMauJabberState	None required
MAU-MIB - ifMauJabberingStateEnters	None required
MAU-MIB - ifMauFalseCarriers	None required
MAU-MIB - ifMauDefaultType	10G WIS control 2
MAU-MIB - ifMauAutoNegSupported	None required
MAU-MIB - ifMauTypeListBits	10G WIS status 2

12.1.8 Structure of the MIB module

Four tables are defined in this MIB module.

12.1.8.1 etherWisDeviceTable

The purpose of this table is to define managed objects to control the WIS test pattern mode. These objects are required to support mandatory and optional WIS test features specified in 50.3.8 of IEEE Std 802.3.

The etherWisDeviceTable is a sparse augmentation of the sonetMediumTable of the SONET-MIB; in other words, for each entry in the etherWisDeviceTable, there shall be an entry in the sonetMediumTable and the same ifIndex value shall be used for both entries.

12.1.8.2 etherWisSectionCurrentTable

The purpose of this table is to define managed objects for the transmitted and received section trace messages (J0 byte).

The etherWisSectionCurrentTable is a sparse augmentation of the sonetSectionCurrentTable of the SONET-MIB; in other words, for each entry in the etherWisSectionCurrentTable, there shall be an entry in the sonetSectionCurrentTable and the same ifIndex value shall be used for both entries.

12.1.8.3 etherWisPathCurrentTable

The purpose of this table is to define managed objects for the current WIS path layer status and for the transmitted and received path trace messages (J1 byte). The path layer status object is provided because the WIS supports some near-end path status conditions that are not reported in sonetPathCurrentStatus.

The etherWisPathCurrentTable is a sparse augmentation of the sonetPathCurrentTable of the SONET-MIB; in other words, for each entry in the etherWisPathCurrentTable, there shall be an entry in the sonetPathCurrentTable and the same ifIndex value shall be used for both entries.

12.1.8.4 etherWisFarEndPathCurrentTable

The purpose of this table is to define a managed object for the current status of the far end of the path. This object is provided because the WIS supports some far-end path status conditions that are not reported in sonetPathCurrentStatus.

The etherWisFarEndPathCurrentTable is a sparse augmentation of the sonetFarEndPathCurrentTable of the SONET-MIB; in other words, for each entry in the etherWisFarEndPathCurrentTable, there shall be an entry in the sonetFarEndPathCurrentTable and the same ifIndex value shall be used for both entries.

12.2 Security considerations for Ethernet wide area network (WAN) interface sublayer (WIS) MIB module

There are five managed objects defined in this MIB module that have a MAX-ACCESS clause of read-write:

- 1) etherWisDeviceTxTestPatternMode
- 2) etherWisDeviceRxTestPatternMode
- 3) etherWisDeviceRxTestPatternErrors
- 4) etherWisSectionCurrentJ0Transmitted
- 5) etherWisPathCurrentJ1Transmitted

Writing to these objects can have the following potentially disruptive effects on network operation:

- Changing the transmit or receive test pattern mode or modifying the accumulated error count from a PRBS31 pattern test on an administratively disabled 10GBASE-W interface, which can interfere with an in-progress pattern test.
- Modifying the transmitted section trace and/or path trace message on an operational 10GBASE-W interface, which can cause connectivity alarms to be raised at the remote end of the link.

The user of this MIB module should therefore be aware that support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

The readable objects in this MIB module (i.e., those with MAX-ACCESS other than not-accessible) can be considered sensitive in some environments since, collectively, they provide information about the performance of network interfaces and can reveal some aspects of their configuration. In such environments it is important to control GET and NOTIFY access to these objects and possibly to encrypt their values when sending them over the network via SNMP.

12.3 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²⁷

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C12mib.txt

²⁷ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

12.3.1 Tree hierarchy

```

---- ieee8023etherWisMIB(1.3.111.2.802.3.1.12)
+---- etherWisObjects(1)
| +---- etherWisDevice(1)
| | +---- etherWisDeviceTable(1)
| | | +---- etherWisDeviceEntry(1) [ifIndex]
| | | +---rw etherWisDeviceTxTestPatternMode(1) Enumeration
| | | +---rw etherWisDeviceRxTestPatternMode(2) Enumeration
| | | +---rw etherWisDeviceRxTestPatternErrors(3) Gauge32
| | +---- etherWisSection(2)
| | | +---- etherWisSectionCurrentTable(1)
| | | | +---- etherWisSectionCurrentEntry(1) [ifIndex]
| | | | +---rw etherWisSectionCurrentJ0Transmitted(1) OctetString
| | | | +---r etherWisSectionCurrentJ0Received(2) OctetString
+---- etherWisObjectsPath(2)
| +---- etherWisPath(1)
| | +---- etherWisPathCurrentTable(1)
| | | +---- etherWisPathCurrentEntry(1) [ifIndex]
| | | +---r etherWisPathCurrentStatus(1) Bits
| | | +---rw etherWisPathCurrentJ1Transmitted(2) OctetString
| | | +---r etherWisPathCurrentJ1Received(3) OctetString
| | +---- etherWisFarEndPath(2)
| | | +---- etherWisFarEndPathCurrentTable(1)
| | | | +---- etherWisFarEndPathCurrentEntry(1) [ifIndex]
| | | | +---r etherWisFarEndPathCurrentStatus(1) Bits
+---- etherWisConformance(3)
+---- etherWisGroups(1)
| +---- etherWisDeviceGroupBasic(1)
| +---- etherWisDeviceGroupExtra(2)
| +---- etherWisSectionGroup(3)
| +---- etherWisPathGroup(4)
| +---- etherWisFarEndPathGroup(5)
+---- etherWisCompliances(2)
+---- etherWisCompliance(1)

```

12.3.2 MIB module definition

```

IEEE8023-ETHER-WIS-MIB DEFINITIONS ::= BEGIN

IMPORTS
    ifIndex
        FROM IF-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
        FROM SNMPv2-CONF
    Gauge32, MODULE-IDENTITY, OBJECT-TYPE, org
        FROM SNMPv2-SMI
    sonetFarEndLineStuff2, sonetFarEndPathStuff2, sonetLineStuff2,
    sonetMediumCircuitIdentifier, sonetMediumLineCoding,
    sonetMediumLineType, sonetMediumLoopbackConfig, sonetMediumStuff2,
    sonetMediumType, sonetPathCurrentWidth, sonetPathStuff2,
    sonetSESthresholdSet, sonetSectionStuff2
        FROM SNET-MIB;

ieee8023etherWisMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"

```

```
CONTACT-INFO
  " WG-URL: http://www.ieee802.org/3/index.html
    WG-EMail: mailto:stds-802-3-dialog@ieee.org
    Contact: IEEE 802.3 Working Group Chair
    Postal: C/O IEEE 802.3 Working Group
            IEEE Standards Association
            445 Hoes Lane
            Piscataway, NJ 08854
            USA
    E-mail: mailto:stds-802-3-dialog@ieee.org"

DESCRIPTION
  "The objects in this MIB module are used in conjunction
  with objects in the SONET-MIB module and the MAU-MIB module to
  manage the Ethernet WAN Interface Sublayer (WIS) defined in
  IEEE Std 802.3.

  Of particular interest are IEEE Std 802.3, Clause 50, 'WAN
  Interface Sublayer (WIS), type 10GBASE-W', Clause 30, 'Management',
  and Clause 45, 'Management Data Input/Output (MDIO) Interface'."

REVISION      "202412110000Z"
DESCRIPTION
  "Revision, based on an earlier version in IEEE Std 802.3.1-2013
  addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
  and 2022."

REVISION      "201304110000Z"
DESCRIPTION
  "Revision, based on an earlier version in IEEE Std 802.3.1-2011."

REVISION      "201102020000Z"
DESCRIPTION
  "Initial version, based on an earlier version published
  as RFC 3637."

 ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 12 }

etherWisObjects OBJECT IDENTIFIER
 ::= { ieee8023etherWisMIB 1 }

etherWisDevice OBJECT IDENTIFIER
 ::= { etherWisObjects 1 }

etherWisDeviceTable OBJECT-TYPE
 SYNTAX      SEQUENCE OF EtherWisDeviceEntry
 MAX-ACCESS  not-accessible
 STATUS      current
 DESCRIPTION
   "The table for Ethernet WIS devices"
 ::= { etherWisDevice 1 }

etherWisDeviceEntry OBJECT-TYPE
 SYNTAX      EtherWisDeviceEntry
 MAX-ACCESS  not-accessible
 STATUS      current
 DESCRIPTION
   "An entry in the Ethernet WIS device table. For each
   instance of this object there shall be a corresponding
   instance of sonetMediumEntry."
 INDEX       { ifIndex }
 ::= { etherWisDeviceTable 1 }
```

```
EtherWisDeviceEntry ::= SEQUENCE {
    etherWisDeviceTxTestPatternMode    INTEGER,
    etherWisDeviceRxTestPatternMode    INTEGER,
    etherWisDeviceRxTestPatternErrors  Gauge32
}

etherWisDeviceTxTestPatternMode OBJECT-TYPE
    SYNTAX      INTEGER { none(1), squareWave(2), prbs31(3),
                          mixedFrequency(4) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "This variable controls the transmit test pattern mode.
        The value none(1) puts the the WIS transmit path into
        the normal operating mode. The value squareWave(2) puts
        the WIS transmit path into the square wave test pattern
        mode described in IEEE Std 802.3, 50.3.8.1.
        The value prbs31(3) puts the WIS transmit path into the
        PRBS31 test pattern mode described in IEEE Std 802.3
        50.3.8.2. The value mixedFrequency(4) puts the
        WIS transmit path into the mixed frequency test pattern
        mode described in IEEE Std 802.3, 50.3.8.3.
        Any attempt to set this object to a value other than
        none(1) when the corresponding instance of ifAdminStatus
        has the value up(1) shall be rejected with the error
        inconsistentValue, and any attempt to set the corresponding
        instance of ifAdminStatus to the value up(1) when an
        instance of this object has a value other than none(1)
        shall be rejected with the error inconsistentValue."
    REFERENCE
        "IEEE Std 802.3, 50.3.8, 45.2.2.6, 45.2.2.7.2"
    ::= { etherWisDeviceEntry 1 }

etherWisDeviceRxTestPatternMode OBJECT-TYPE
    SYNTAX      INTEGER { none(1), prbs31(3), mixedFrequency(4) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "This variable controls the receive test pattern mode.
        The value none(1) puts the the WIS receive path into the
        normal operating mode. The value prbs31(3) puts the WIS
        receive path into the PRBS31 test pattern mode described
        in IEEE Std 802.3, 50.3.8.2. The value
        mixedFrequency(4) puts the WIS receive path into the mixed
        frequency test pattern mode described in IEEE Std 802.3,
        50.3.8.3. Any attempt to set this object to a
        value other than none(1) when the corresponding instance
        of ifAdminStatus has the value up(1) shall be rejected with
        the error inconsistentValue, and any attempt to set the
        corresponding instance of ifAdminStatus to the value up(1)
        when an instance of this object has a value other than
        none(1) shall be rejected with the error inconsistentValue."
    REFERENCE
        "IEEE Std 802.3, 50.3.8, 45.2.2.6, and 45.2.2.7.2"
    ::= { etherWisDeviceEntry 2 }

etherWisDeviceRxTestPatternErrors OBJECT-TYPE
    SYNTAX      Gauge32 (0..65535)
```

```
MAX-ACCESS    read-write
STATUS        current
DESCRIPTION
    "This object counts the number of errors detected when the
    WIS receive path is operating in the PRBS31 test pattern
    mode. It is reset to zero when the WIS receive path
    initially enters that mode, and it increments each time
    the PRBS pattern checker detects an error as described in
    IEEE Std 802.3, 50.3.8.2 unless its value is
    65535, in which case it remains unchanged. This object is
    writeable so that it can be reset upon explicit request
    of a command generator application while the WIS receive
    path continues to operate in PRBS31 test pattern mode."
REFERENCE
    "IEEE Std 802.3, 50.3.8, 45.2.2.7.2, and 45.2.2.8"
::= { etherWisDeviceEntry 3 }

etherWisSection OBJECT IDENTIFIER
::= { etherWisObjects 2 }

etherWisSectionCurrentTable OBJECT-TYPE
SYNTAX        SEQUENCE OF EtherWisSectionCurrentEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "The table for the current state of Ethernet WIS sections."
::= { etherWisSection 1 }

etherWisSectionCurrentEntry OBJECT-TYPE
SYNTAX        EtherWisSectionCurrentEntry
MAX-ACCESS    not-accessible
STATUS        current
DESCRIPTION
    "An entry in the etherWisSectionCurrentTable. For each
    instance of this object there shall be a corresponding
    instance of sonetSectionCurrentEntry."
INDEX        { ifIndex }
::= { etherWisSectionCurrentTable 1 }

EtherWisSectionCurrentEntry ::= SEQUENCE {
    etherWisSectionCurrentJ0Transmitted    OCTET STRING,
    etherWisSectionCurrentJ0Received       OCTET STRING
}

etherWisSectionCurrentJ0Transmitted OBJECT-TYPE
SYNTAX        OCTET STRING (SIZE(16))
MAX-ACCESS    read-write
STATUS        current
DESCRIPTION
    "This is the 16-octet section trace message that
    is transmitted in the J0 byte. The value should
    be '89'h followed by fifteen octets of '00'h
    (or some cyclic shift thereof) when the section
    trace function is not used, and the implementation
    should use that value (or a cyclic shift thereof)
    as a default if no other value has been set."
REFERENCE
    "IEEE Std 802.3, 30.8.1.1.8"
::= { etherWisSectionCurrentEntry 1 }
```

```
etherWisSectionCurrentJ0Received OBJECT-TYPE
    SYNTAX      OCTET STRING (SIZE(16))
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This is the 16-octet section trace message that
         was most recently received in the J0 byte."
    REFERENCE
        "IEEE Std 802.3, 30.8.1.1.9"
    ::= { etherWisSectionCurrentEntry 2 }

etherWisObjectsPath OBJECT IDENTIFIER
    ::= { ieee8023etherWisMIB 2 }

etherWisPath OBJECT IDENTIFIER
    ::= { etherWisObjectsPath 1 }

etherWisPathCurrentTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EtherWisPathCurrentEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The table for the current state of Ethernet WIS paths."
    ::= { etherWisPath 1 }

etherWisPathCurrentEntry OBJECT-TYPE
    SYNTAX      EtherWisPathCurrentEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the etherWisPathCurrentTable. For each
         instance of this object there shall be a corresponding
         instance of sonetPathCurrentEntry."
    INDEX       { ifIndex }
    ::= { etherWisPathCurrentTable 1 }

EtherWisPathCurrentEntry ::= SEQUENCE {
    etherWisPathCurrentStatus      BITS,
    etherWisPathCurrentJ1Transmitted OCTET STRING,
    etherWisPathCurrentJ1Received  OCTET STRING
}

etherWisPathCurrentStatus OBJECT-TYPE
    SYNTAX      BITS { etherWisPathLOP(0), etherWisPathAIS(1),
                       etherWisPathPLM(2), etherWisPathLCD(3) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This variable indicates the current status of the
         path payload with a bit map that can indicate multiple
         defects at once. The bit positions are assigned as
         follows:

         etherWisPathLOP(0)
             This bit is set to indicate that an
             LOP-P (Loss of Pointer - Path) defect
             is being experienced. When this
             bit is set, sonetPathSTSLOP shall be set
```

in the corresponding instance of
sonetPathCurrentStatus.

etherWisPathAIS(1)

This bit is set to indicate that an
AIS-P (Alarm Indication Signal - Path)
defect is being experienced. When
this bit is set, sonetPathSTSAIS shall be
set in the corresponding instance of
sonetPathCurrentStatus.

etherWisPathPLM(1)

This bit is set to indicate that a
PLM-P (Payload Label Mismatch - Path)
defect is being experienced. When
this bit is set, sonetPathSignalLabelMismatch
shall be set in the corresponding instance of
sonetPathCurrentStatus.

etherWisPathLCD(3)

This bit is set to indicate that an
LCD-P (Loss of Codegroup Delination - Path)
defect is being experienced. Since this
defect is detected by the PCS and not by
the path layer itself, there is no
corresponding bit in sonetPathCurrentStatus."

REFERENCE

"IEEE Std 802.3, 30.8.1.1.18"

::= { etherWisPathCurrentEntry 1 }

etherWisPathCurrentJ1Transmitted OBJECT-TYPE

SYNTAX OCTET STRING (SIZE(16))

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This is the 16-octet path trace message that
is transmitted in the J1 byte. The value should
be '89'h followed by fifteen octets of '00'h
(or some cyclic shift thereof) when the path
trace function is not used, and the implementation
should use that value (or a cyclic shift thereof)
as a default if no other value has been set."

REFERENCE

"IEEE Std 802.3, 30.8.1.1.23"

::= { etherWisPathCurrentEntry 2 }

etherWisPathCurrentJ1Received OBJECT-TYPE

SYNTAX OCTET STRING (SIZE(16))

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is the 16-octet path trace message that
was most recently received in the J1 byte."

REFERENCE

"IEEE Std 802.3, 30.8.1.1.24"

::= { etherWisPathCurrentEntry 3 }

etherWisFarEndPath OBJECT IDENTIFIER

::= { etherWisObjectsPath 2 }

```
etherWisFarEndPathCurrentTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF EtherWisFarEndPathCurrentEntry
    MAX-ACCESS   not-accessible
    STATUS      current
    DESCRIPTION
        "The table for the current far-end state of Ethernet WIS
        paths."
    ::= { etherWisFarEndPath 1 }

etherWisFarEndPathCurrentEntry OBJECT-TYPE
    SYNTAX      EtherWisFarEndPathCurrentEntry
    MAX-ACCESS   not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the etherWisFarEndPathCurrentTable. For each
        instance of this object there shall be a corresponding
        instance of sonetFarEndPathCurrentEntry."
    INDEX       { ifIndex }
    ::= { etherWisFarEndPathCurrentTable 1 }

EtherWisFarEndPathCurrentEntry ::= SEQUENCE {
    etherWisFarEndPathCurrentStatus    BITS
}

etherWisFarEndPathCurrentStatus OBJECT-TYPE
    SYNTAX      BITS { etherWisFarEndPayloadDefect(0),
                        etherWisFarEndServerDefect(1) }
    MAX-ACCESS   read-only
    STATUS      current
    DESCRIPTION
        "This variable indicates the current status at the
        far end of the path using a bit map that can indicate
        multiple defects at once. The bit positions are
        assigned as follows:

        etherWisFarEndPayloadDefect(0)
            A far end payload defect (i.e., far end
            PLM-P or LCD-P) is currently being signaled
            in G1 bits 5-7.

        etherWisFarEndServerDefect(1)
            A far end server defect (i.e., far end
            LOP-P or AIS-P) is currently being signaled
            in G1 bits 5-7. When this bit is set,
            sonetPathSTSRDI shall be set in the corresponding
            instance of sonetPathCurrentStatus."
    REFERENCE
        "IEEE Std 802.3, 30.8.1.1.25"
    ::= { etherWisFarEndPathCurrentEntry 1 }

etherWisConformance OBJECT IDENTIFIER
    ::= { ieee8023etherWisMIB 3 }

etherWisGroups OBJECT IDENTIFIER
    ::= { etherWisConformance 1 }

etherWisCompliances OBJECT IDENTIFIER
    ::= { etherWisConformance 2 }
```

```

etherWisDeviceGroupBasic OBJECT-GROUP
    OBJECTS      { etherWisDeviceTxTestPatternMode,
                   etherWisDeviceRxTestPatternMode }
    STATUS       current
    DESCRIPTION
        "A collection of objects that support test
         features required of all WIS devices."
    ::= { etherWisGroups 1 }

etherWisDeviceGroupExtra OBJECT-GROUP
    OBJECTS      { etherWisDeviceRxTestPatternErrors }
    STATUS       current
    DESCRIPTION
        "A collection of objects that support
         optional WIS device test features."
    ::= { etherWisGroups 2 }

etherWisSectionGroup OBJECT-GROUP
    OBJECTS      { etherWisSectionCurrentJ0Transmitted,
                   etherWisSectionCurrentJ0Received }
    STATUS       current
    DESCRIPTION
        "A collection of objects that provide
         required information about a WIS section."
    ::= { etherWisGroups 3 }

etherWisPathGroup OBJECT-GROUP
    OBJECTS      { etherWisPathCurrentStatus,
                   etherWisPathCurrentJ1Transmitted,
                   etherWisPathCurrentJ1Received }
    STATUS       current
    DESCRIPTION
        "A collection of objects that provide
         required information about a WIS path."
    ::= { etherWisGroups 4 }

etherWisFarEndPathGroup OBJECT-GROUP
    OBJECTS      { etherWisFarEndPathCurrentStatus }
    STATUS       current
    DESCRIPTION
        "A collection of objects that provide required
         information about the far end of a WIS path."
    ::= { etherWisGroups 5 }

etherWisCompliance MODULE-COMPLIANCE
    STATUS       current
    DESCRIPTION
        "The compliance statement for interfaces that include
         the Ethernet WIS. Compliance with the following
         external compliance statements is prerequisite:

        MIB module          Compliance Statement
        -----
        IF-MIB               ifCompliance3
        IF-INVERTED-STACK-MIB ifInvCompliance
        IEEE8023-EtherLike-MIB dot3Compliance2
        MAU-MIB              mauModIfCompl3"

    MODULE      -- this module

```



```
MANDATORY-GROUPS      { etherWisDeviceGroupBasic,
                        etherWisSectionGroup, etherWisPathGroup,
                        etherWisFarEndPathGroup }

GROUP    etherWisDeviceGroupExtra
DESCRIPTION
    "Implementation of this group, along with support for
    the value prbs31(3) for etherWisDeviceTxTestPatternMode
    and etherWisDeviceRxTestPatternMode, is necessary if the
    optional PRBS31 test pattern mode is to be supported."

OBJECT    etherWisDeviceTxTestPatternMode
SYNTAX    INTEGER { none(1), squareWave(2), mixedFrequency(4) }
DESCRIPTION
    "Support for values other than none(1),
    squareWave(2), and mixedFrequency(4)
    is not required."

OBJECT    etherWisDeviceRxTestPatternMode
SYNTAX    INTEGER { none(1), mixedFrequency(4) }
DESCRIPTION
    "Support for values other than none(1)
    and mixedFrequency(4) is not required."

OBJECT    etherWisDeviceRxTestPatternErrors
WRITE-SYNTAX    Gauge32 (0)
DESCRIPTION
    "An implementation is not required to
    allow values other than zero to be
    written to this object."

MODULE      SONET-MIB

MANDATORY-GROUPS      { sonetMediumStuff2, sonetSectionStuff2,
                        sonetLineStuff2, sonetFarEndLineStuff2,
                        sonetPathStuff2, sonetFarEndPathStuff2 }

OBJECT    sonetMediumType
SYNTAX    INTEGER { sonet(1) }
MIN-ACCESS    read-only
DESCRIPTION
    "Write access is not required, nor is support
    for any value other than sonet(1)."
```

```
OBJECT    sonetMediumLineCoding
SYNTAX    INTEGER { sonetMediumNRZ(4) }
MIN-ACCESS    read-only
DESCRIPTION
    "Write access is not required, nor is support
    for any value other than sonetMediumNRZ(4)."
```

```
OBJECT    sonetMediumLineType
MIN-ACCESS    read-only
DESCRIPTION
    "Write access is not required."
```

```
OBJECT    sonetMediumCircuitIdentifier
MIN-ACCESS    read-only
```

```
DESCRIPTION
    "Write access is not required."

OBJECT  sonetMediumLoopbackConfig
SYNTAX  BITS { sonetNoLoop(0), sonetFacilityLoop(1) }
MIN-ACCESS  read-only
DESCRIPTION
    "Write access is not required, nor is support for values
    other than sonetNoLoop(0) and sonetFacilityLoop(1)."
```

```
OBJECT  sonetSESthresholdSet
MIN-ACCESS  read-only
DESCRIPTION
    "Write access is not required, and only one
    of the enumerated values need be supported."
```

```
OBJECT  sonetPathCurrentWidth
SYNTAX  INTEGER { sts192cSTM64(6) }
MIN-ACCESS  read-only
DESCRIPTION
    "Write access is not required, nor is support
    for any value other than sts192cSTM64(6)."
```

```
::= { etherWisCompliances 1 }
```

```
END -- end of module IEEE8023-ETHER-WIS-MIB.
```

13. Ethernet medium attachment units (MAUs) MIB module

13.1 Introduction

This clause defines a portion of the MIB for use with SNMP. In particular, it defines objects for managing IEEE 802.3 medium attachment units (MAUs).

A previous version of this clause, IETF RFC 3636 [B10], defined a single MIB module. IETF RFC 4836 [B14] split the original MIB module into two, putting frequently updated object identities and textual conventions into a separate, IANA-maintained MIB module, in order to decrease the need of updating the basic MAU-MIB module. The MIB module defined in this clause incorporates the IANA-MAU-MIB module by reference.

13.2 Overview

Instances of these object types represent attributes of an IEEE 802.3 MAU. Several types of MAUs are defined in IEEE Std 802.3. MAUs can be connected to IEEE 802.3 repeaters or to IEEE 802.3 interfaces. For convenience, this clause refers to these devices as “repeater MAUs” and “interface MAUs”.

The definitions presented here are based on 30.5 and 30.6 of IEEE Std 802.3. This specification is intended to provide for management of all types of Ethernet/IEEE 802.3 MAUs.

13.2.1 Relationship to IETF RFC 3636 and IETF RFC 4836

The management definitions provided in this clause are intended to be a superset of those defined by IETF RFC 3636 [B10] and IETF RFC 4836 [B14].

In order to decrease the need of updating the basic MAU MIB module due to a new MAU type, Media Available state, Auto Negotiation capability, and/or Jack type introduction, all relevant object identities and textual conventions have been moved to a separate, IANA-maintained MIB module, IANA-MAU-MIB. Thus, when a new MAU type, Media Available state, Auto Negotiation capability, and/or Jack type is defined by the IEEE 802.3 Working Group, only the IANA-maintained module needs to be revised, leaving the basic MAU-MIB module defined in this clause unchanged.

The changes made in this revision are not entirely backward-compatible with MIB modules that currently import MAU-type object identity descriptors from the MAU-MIB; such modules need to be revised to import those DESCRIPTORS from the IANA-MAU-MIB. Similarly, any management applications that process the object identity definitions (e.g., to present the DESCRIPTION text to a user) need to get those definitions from the IANA-MAU-MIB instead of the MAU-MIB. While it is true that changes that require such adjustments are not strictly compliant with the SMIv2 rules governing MIB module revisions (see section 10 of IETF RFC 2578), in this case continued high maintenance costs that would result from not making these changes make the deviation from the rules justified.

13.2.2 Relationship to other MIBs

It is assumed that an agent implementing the MAU-MIB module will also implement (at least) the “system” group defined in the SNMPv2 MIB of IETF RFC 3418 [B7]. The following subclauses identify other MIBs that such an agent should implement.

13.2.2.1 Relationship to the Interfaces Group MIB

The subclauses of this clause that define interface MAU-related objects specify an extension to the Interfaces Group MIB of IETF RFC 2863. An agent implementing these interface-MAU related objects shall also implement the relevant groups of the ifCompliance3 MODULE-COMPLIANCE statement of the Interfaces Group MIB. The value of the object ifMauIfIndex is the same as the value of “ifIndex” used to instantiate the interface to which the given MAU is connected.

An agent implementing the interface-MAU related objects in the MAU-MIB module shall also fully comply with the dot3Compliance2 MODULE-COMPLIANCE statement of the Ethernet-like interface MIB defined in Clause 10. Furthermore, when the interface-MAU related objects are used to manage a 10GBASE-W PHY—i.e., when ifMauType is equal to dot3MauType10GigBaseW or any other 10GBASE-W variant—then the agent shall also support the Ethernet WAN Interface Sublayer (WIS) MIB module defined in Clause 12, and shall follow the interface layering model specified therein. In that case, the value of the object ifMauIfIndex is the same as the value of “ifIndex” for the layer at the top of the stack, that is, for the ifTable entry that has “ifType” equal to ethernetCsmacd(6). If the interface-MAU related objects are used to manage a PHY that allows the MAU type to be changed dynamically, then the agent shall create ifTable, ifStackTable, and ifInvStackTable entries that pertain to the WIS when ifMauDefaultType is changed to a 10GBASEW variant (i.e., one of dot3MauType10GigBaseW, dot3MauType10GigBaseEW, dot3MauType10GigBaseLW, or dot3MauType10GigBaseSW) from any other type, and shall destroy the WIS-related entries when ifMauDefaultType is changed to a non-10GBASE-W type. The agent shall also change the values of “ifConnectorPresent” and “ifHighSpeed” in the ifTable entry indexed by ifMauIfIndex as specified in Clause 10 and Clause 12 when ifMauDefaultType is manipulated in this way, but shall not otherwise alter that entry.

NOTE—Repeater ports are not represented as interfaces in the Interfaces Group MIB.

13.2.2.2 Relationship to the IEEE 802.3 repeater MIB module

The subclause of this clause that defines repeater MAU-related objects specifies an extension to the IEEE 802.3 repeater MIB module defined in Clause 7. An agent implementing these repeater-MAU related objects shall also comply with the snmpRptrModCompl compliance statement of the IEEE 802.3 repeater MIB module.

The values of “rpMauGroupIndex” and “rpMauPortIndex” used to instantiate a repeater MAU variable shall be the same as the values of “rptrPortGroupIndex” and “rptrPortIndex” used to instantiate the port to which the given MAU is connected.

13.2.3 Management of internal MAUs

In some situations, a MAU can be “internal”; that is, its functionality is implemented entirely within a device. For example, a managed repeater may contain an internal repeater MAU and/or an internal interface MAU through which management communications originating on one of the repeater’s external ports pass, in order to reach the management agent associated with the repeater. Such internal MAUs may be managed. If they are managed, objects describing their attributes should appear in the appropriate MIB subtree: dot3RpMauBasicGroup for internal repeater MAUs and dot3IfMauBasicGroup for internal interface MAUs.

13.2.4 Mapping of IEEE 802.3 managed objects

Table 13-1 depicts the mapping between relevant managed objects (attributes) defined in Clause 30 of IEEE Std 802.3 and managed objects defined in this clause.

Table 13-1—Mapping of IEEE 802.3 managed objects

IEEE 802.3 managed object		Corresponding SNMP object
oMAU	.aMAUID	rpMauIndex or ifMauIndex or broadMauIndex
	.aMAUType	rpMauType or ifMauType
	.aMAUTypeList	ifMauTypeListBits
	.aMediaAvailable	rpMauMediaAvailable or ifMauMediaAvailable
	.aLoseMediaCounter	rpMauMediaAvailableStateExits or ifMauMediaAvailableStateExits
	.aJabber	rpMauJabberState and rpMauJabberingStateEnters or ifMauJabberState and ifMauJabberingStateEnters
	.aMAUAdminState	rpMauStatus or ifMauStatus
	.aFalseCarriers	rpMauFalseCarriers or ifMauFalseCarriers
	.acResetMAU	rpMauStatus or ifMauStatus
	.acMAUAdminControl	rpMauStatus or ifMauStatus
	.nJabber	rpMauJabberTrap or ifMauJabberTrap

Table 13-1—Mapping of IEEE 802.3 managed objects (*continued*)

IEEE 802.3 managed object		Corresponding SNMP object
oAutoNegotiation	.aAutoNegID	ifMauIndex
	.aAutoNegAdminState	ifMauAutoNegAdminStatus
	.aAutoNegRemoteSignalling	ifMauAutoNegRemoteSignalling
	.aAutoNegAutoConfig	ifMauAutoNegConfig
	.aAutoNegLocalTechnologyAbility	ifMauAutoNegCapabilityBits
	.aAutoNegAdvertisedTechnologyAbility	ifMauAutoNegAdvertisedBits and ifMauAutoNegRemoteFaultAdvertised
	.aAutoNegReceivedTechnologyAbility	ifMauAutoNegReceivedBits and ifMauAutoNegRemoteFaultReceived
	.acAutoNegRestartAutoConfig	ifMauAutoNegRestart
	.acAutoNegAdminControl	ifMauAutoNegAdminStatus
oTimeSync	.aTimeSyncCapabilityTX	ifMauTimeSyncCapabilityTX
	.aTimeSyncCapabilityRX	ifMauTimeSyncCapabilityRX
	.aTimeSyncDelayTXmax	ifMauTimeSyncDelayTXmax
	.aTimeSyncDelayTXmin	ifMauTimeSyncDelayTXmin
	.aTimeSyncDelayRXmax	ifMauTimeSyncDelayRXmax
	.aTimeSyncDelayRXmin	ifMauTimeSyncDelayRXmin

Table 13-2 depicts the IEEE 802.3 managed objects that have not been included in the MAU-MIB module, and the reason for the exclusion.

Table 13-2—Unmapped IEEE 802.3 managed objects

IEEE 802.3 managed object		Reason for exclusion
oMAU	.aIdleErrorCount	Only useful for 100BaseT2, which is not widely implemented
oAutoNegotiation	.aAutoNegLocalSelectorAbility	Only needed for support of isoethernet (IEEE Std 802.9a-1995), which is not supported by MAU- MIB
	.aAutoNegAdvertisedSelectorAbility	
	.aAutoNegReceivedSelectorAbility	

13.2.5 Addition of new MAU types

13.2.5.1 dot3MauType OBJECT-IDENTITIES

The dot3MauType OBJECT IDENTIFIER and its OBJECT-IDENTITY definitions have been moved from the MAU-MIB module into the IANA-maintained IANA-MAU-MIB module.

When a new IEEE 802.3 MAU is defined, IANA can reissue a version of the IANA-MAU-MIB module with the new dot3MauType OBJECT-IDENTITY and its matching IANAifMauTypeListBits textual convention value and, possibly, with new IANAifMauMediaAvailable, IANAifMauAutoNegCapBits, and/or IANAifJackType values.

An Expert Review, as defined in IETF RFC 2434, is required for the addition of the new MAU, Media Available states, Auto Negotiation capabilities, and/or Jack types.

In some cases, new MAU types require additional managed objects, or have side effects on the behavior of existing managed objects. In such cases, specification in a standard is required, either in a new standard or a revision of this standard. Such specification is required to note any special properties of the MAU types that it defines—for example, side effects on the ifStackTable as noted in this document for 10GBASE-W MAUs.

13.2.5.2 IANAifMauTypeListBits TEXTUAL-CONVENTION

The syntax of ifMauTypeListBits is changed to be a textual convention, such that the enumerated integer values are now defined in the textual convention IANAifMauTypeListBits, which can be respecified (with additional values, when defined by IEEE Std 802.3) in the IANA-maintained MIB module without issuing a new version of this document.

13.2.5.3 IANAifMauMediaAvailable TEXTUAL-CONVENTION

The syntax of ifMauMediaAvailable and rpMauMediaAvailable is changed to be a textual convention, such that the enumerated integer values are now defined in the textual convention IANAifMauMediaAvailable, which can be respecified (with additional values, when defined by IEEE Std 802.3) in the IANA-maintained MIB module without issuing a new version of this document.

13.2.5.4 IANAifMauAutoNegCapBits TEXTUAL-CONVENTION

The syntax of ifMauAutoNegCapabilityBits, ifMauAutoNegCapAdvertisedBits, and ifMauAutoNegCapReceivedBits objects is changed to be a textual convention, such that the enumerated integer values are now defined in the textual convention IANAifMauAutoNegCapBits, which can be respecified (with additional values, when defined by IEEE Std 802.3) in the IANA-maintained MIB module without issuing a new version of this document.

13.2.5.5 JackType TEXTUAL-CONVENTION

The JackType Textual Convention has been deprecated in favor of the IANAifJackType defined in the IANA-maintained MIB module, so the new Jack types can be added (when defined by IEEE Std 802.3) without issuing a new version of this document.

13.3 Security considerations for Ethernet medium attachment units (MAUs) MIB module

The IANA-MAU-MIB module defines a set of textual conventions that are used by the MAU-MIB module and by other MIB modules to define management objects. Meaningful security considerations can only be written for MIB modules that define management objects.

There are a number of management objects defined in the MAU-MIB module that have a MAX-ACCESS clause of read-write. Setting these objects can have a serious effect on the operation of the network, including:

- Enabling or disabling a MAU
- Changing a MAU's default type
- Enabling, disabling, or restarting Auto-Negotiation
- Modifying the capabilities that a MAU advertises during Auto-Negotiation.

Such objects can be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Some of the readable objects in the MAU-MIB module (i.e., objects with a MAX-ACCESS other than not-accessible) can be considered sensitive or vulnerable in some network environments. In some environments, it is undesirable to allow unauthorized parties to access statistics or status information about individual links in a network. It is thus important to control GET and/or NOTIFY access to these objects and possibly to encrypt the values of these objects when sending them over the network via SNMP.

13.4 IANA considerations

It is intended that each new MAU type, Media Available state, Auto Negotiation capability, and/or Jack type defined by the IEEE 802.3 Working Group and approved for publication in a revision of IEEE Std 802.3 will be added to the IANA-maintained MIB module, provided that it is suitable for being managed by the base objects in the MAU-MIB module.

For each new MAU type added, a short description of the MAU technology and, wherever possible, a reference to a publicly available specification should be specified. An Expert Review, as defined in IETF RFC 2434, is required, for each modification.

13.5 MIB module definition

An ASCII text version of the MIB definition can be found at the following URL:²⁸

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1C13mib.txt

The IANA-MAU-MIB module can be found at the following URL:

<https://www.iana.org/assignments/ianamau-mib>

²⁸ Copyright release for MIB modules: Users of this standard may freely reproduce the MIB module contained in this subclause so that it can be used for its intended purpose.

13.5.1 Tree hierarchy

```

----- ieee8023mauMIB(1.3.111.2.802.3.1.1.13)
+----- ieee8023snmpDot3MauMgt(1)
| +----- snmpDot3MauTraps(0)
| | +----- rpMauJabberTrap(1) {rpMauJabberState}
| | +----- ifMauJabberTrap(2) {ifMauJabberState}
| +----- dot3RpMauBasicGroup(1)
| | +----- rpMauTable(1)
| | | +----- rpMauEntry(1) [rpMauGroupIndex, rpMauPortIndex, rpMauIndex]
| | | | +----- rpMauGroupIndex(1) Integer32
| | | | +----- rpMauPortIndex(2) Integer32
| | | | +----- rpMauIndex(3) Integer32
| | | | +----- rpMauType(4) AutonomousType
| | | | +----- rpMauStatus(5) Enumeration
| | | | +----- rpMauMediaAvailable(6) IANAifMauMediaAvailable
| | | | +----- rpMauMediaAvailableStateExits(7) Counter32
| | | | +----- rpMauJabberState(8) Enumeration
| | | | +----- rpMauJabberingStateEnters(9) Counter32
| | | | +----- rpMauFalseCarriers(10) Counter32
| | +----- rpJackTable(2)
| | | +----- rpJackEntry(1) [rpMauGroupIndex, rpMauPortIndex, rpMauIndex, rpJackIndex]
| | | | +----- rpJackIndex(1) Integer32
| | | | +----- rpJackType(2) IANAifJackType
| +----- dot3IfMauBasicGroup(2)
| | +----- ifMauTable(1)
| | | +----- ifMauEntry(1) [ifMauIfIndex, ifMauIndex]
| | | | +----- ifMauIfIndex(1) InterfaceIndex
| | | | +----- ifMauIndex(2) Integer32
| | | | +----- ifMauType(3) AutonomousType
| | | | +----- ifMauStatus(4) Enumeration
| | | | +----- ifMauMediaAvailable(5) IANAifMauMediaAvailable
| | | | +----- ifMauMediaAvailableStateExits(6) Counter32
| | | | +----- ifMauJabberState(7) Enumeration
| | | | +----- ifMauJabberingStateEnters(8) Counter32
| | | | +----- ifMauFalseCarriers(9) Counter32
| | | | +----- ifMauDefaultType(10) AutonomousType
| | | | +----- ifMauAutoNegSupported(11) TruthValue
| | | | +----- ifMauTypeListBits(12) IANAifMauTypeListBits

```

```

+--r- ifMauHCFalseCarriers (13) Counter64
+--r- ifMauPCSCodingViolations (14) Counter64
+--r- ifMauFECAbility (15) Enumeration
+--rw ifMauFECMode (16) Enumeration
x--r- ifMauFECCorrectedBlocks (17) Counter64
x--r- ifMauFECUncorrectableBlocks (18) Counter64
+--r- ifMausNROpMarginChn1A (19) Integer32
+--r- ifMausNROpMarginChn1B (20) Integer32
+--r- ifMausNROpMarginChn1C (21) Integer32
+--r- ifMausNROpMarginChn1D (22) Integer32
+--r- ifMauEEESupportList (23) IANAifMauTypeListBits
+--r- ifMauEEELDFastRetrainCount (24) Counter32
+--r- ifMauEEELPFastRetrainCount (25) Counter32
+--r- ifMauTimesyncCapabilityTX (26) TruthValue
+--r- ifMauTimesyncCapabilityRX (27) TruthValue
+--r- ifMauTimesyncDelayTXmax (28) Integer32
+--r- ifMauTimesyncDelayTXmin (29) Integer32
+--r- ifMauTimesyncDelayRXmax (30) Integer32
+--r- ifMauTimesyncDelayRXmin (31) Integer32
+---- ifJackTable (2)
+---- ifJackEntry (1) [ifMauIfIndex, ifMauIndex, ifJackIndex]
+---- ifJackIndex (1) Integer32
+--r- ifJackType (2) IANAifJackType
+---- ifMauPerPCSLaneStatsTable (3)
+---- ifMauPerPCSLaneStatsEntry (1) [ifMauIfIndex, ifMauIndex, ifPCSLaneIndex]
+---- ifPCSLaneIndex (1) Unsigned32
+--r- ifMauPPLFECCorrectedBlocks (2) Counter64
+--r- ifMauPPLFECUncorrectableBlocks (3) Counter64
+--r- ifMauBIPErrorCount (4) Counter32
+--r- ifMauPCStoPHYLaneMapping (5) Unsigned32
+---- dot3PlaceholderGroup (3)
+--r- dot3Placeholder (1) Enumeration
+---- dot3IfMauAutoNegGroup (5)
+---- ifMauAutoNegTable (1)
+---- ifMauAutoNegEntry (1) [ifMauIfIndex, ifMauIndex]
+--rw ifMauAutoNegAdminStatus (1) Enumeration
+--r- ifMauAutoNegRemoteSignaling (2) Enumeration
+--r- ifMauAutoNegConfig (4) Enumeration
+--rw ifMauAutoNegRestart (5) Enumeration

```

```

|   +---r- ifMauAutoNegCapabilityBits(6)   IANAifMauAutoNegCapBits
|   +---rw ifMauAutoNegCapAdvertisedBits(7) IANAifMauAutoNegCapBits
|   +---r- ifMauAutoNegCapReceivedBits(8)  IANAifMauAutoNegCapBits
|   +---rw ifMauAutoNegRemoteFaultAdvertised(9) Enumeration
|   +---r- ifMauAutoNegRemoteFaultReceived(10) Enumeration
+----- mauModConf(2)
+----- mauModCompls(1)
|   +----- mauModRpComp12(1)
|   +----- mauModIfComp13(2)
+----- mauModObjGrps(2)
|   +----- mauRpGrpBasic(1)
|   +----- mauRpGrp100Mbps(2)
|   +----- mauRpGrpJack(3)
|   +----- mauIfGrpBasic(4)
|   +----- mauIfGrpJack(5)
|   +----- mauIfGrpHighCapacity(6)
|   +----- mauIfGrpAutoNeg2(7)
|   +----- mauIfGrpAutoNeg1000Mbps(8)
|   +----- mauIfGrpHCStats(9)
|   +----- mauIfGrpFEC(10)
|   +----- mauIfGrpSNR(11)
|   +----- mauIfGrpEEE(12)
|   +----- mauIfGrpTimeSync(13)
|   +----- mauIfGrpPerPCSLaneStats(14)
+----- mauModNotGrps(3)
+----- rpMauNotifications(1)
+----- ifMauNotifications(2)

```

13.5.2 MIB module definition

```
IEEE8023-MAU-MIB DEFINITIONS ::= BEGIN

IMPORTS
    IANAifJackType, IANAifMauAutoNegCapBits, IANAifMauMediaAvailable,
    IANAifMauTypeListBits
        FROM IANA-MAU-MIB
    InterfaceIndex
        FROM IF-MIB
    MODULE-COMPLIANCE, NOTIFICATION-GROUP, OBJECT-GROUP
        FROM SNMPv2-CONF
    Counter32, Counter64, Integer32, MODULE-IDENTITY, NOTIFICATION-TYPE,
    OBJECT-TYPE, Unsigned32, org
        FROM SNMPv2-SMI
    AutonomousType, TruthValue
        FROM SNMPv2-TC;

ieee8023mauMIB MODULE-IDENTITY
    LAST-UPDATED "202412110000Z"
    ORGANIZATION
        "IEEE 802.3 Working Group"
    CONTACT-INFO
        " WG-URL: http://www.ieee802.org/3/index.html
        WG-EMail: mailto:stds-802-3-dialog@ieee.org
        Contact: IEEE 802.3 Working Group Chair
        Postal: C/O IEEE 802.3 Working Group
                IEEE Standards Association
                445 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: mailto:stds-802-3-dialog@ieee.org"
    DESCRIPTION
        "Management information for 802.3 MAUs."
    REVISION      "202412110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2013
        addressing changes from IEEE Std 802.3 revisions 2012, 2015, 2018,
        and 2022."
    REVISION      "201304110000Z"
    DESCRIPTION
        "Revision, based on an earlier version in IEEE Std 802.3.1-2011."
    REVISION      "201102020000Z"
    DESCRIPTION
        "Initial version, based on an earlier version published
        as RFC 4836."
    ::= { org ieee(111) standards-association-numbers-series-standards(2) lan-
    man-stds(802) ieee802dot3(3) ieee802dot3dot1mibs(1) 13 }

ieee8023snmpDot3MauMgt OBJECT IDENTIFIER
    ::= { ieee8023mauMIB 1 }

snmpDot3MauTraps OBJECT IDENTIFIER
    ::= { ieee8023snmpDot3MauMgt 0 }

dot3RpMauBasicGroup OBJECT IDENTIFIER
    ::= { ieee8023snmpDot3MauMgt 1 }
```

rpMauTable OBJECT-TYPE

SYNTAX SEQUENCE OF RpMauEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"Table of descriptive and status information
about the MAU(s) attached to the ports of a
repeater."
::= { dot3RpMauBasicGroup 1 }

rpMauEntry OBJECT-TYPE

SYNTAX RpMauEntry
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"An entry in the table, containing information
about a single MAU."
INDEX { rpMauGroupIndex, rpMauPortIndex, rpMauIndex }
::= { rpMauTable 1 }

RpMauEntry ::= SEQUENCE {

rpMauGroupIndex	Integer32,
rpMauPortIndex	Integer32,
rpMauIndex	Integer32,
rpMauType	AutonomousType,
rpMauStatus	INTEGER,
rpMauMediaAvailable	IANAIfMauMediaAvailable,
rpMauMediaAvailableStateExits	Counter32,
rpMauJabberState	INTEGER,
rpMauJabberingStateEnters	Counter32,
rpMauFalseCarriers	Counter32

}

rpMauGroupIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current
DESCRIPTION
"This variable uniquely identifies the group
containing the port to which the MAU described
by this entry is connected.

Note: In practice, a group will generally be
a field-replaceable unit (i.e., module, card,
or board) that can fit in the physical system
enclosure, and the group number will correspond
to a number marked on the physical enclosure.

The group denoted by a particular value of this
object is the same as the group denoted by the
same value of rpPtrGroupIndex."
REFERENCE
"RFC 2108, rpPtrGroupIndex."
::= { rpMauEntry 1 }

rpMauPortIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)
MAX-ACCESS not-accessible
STATUS current

DESCRIPTION

"This variable uniquely identifies the repeater port within group rpMauGroupIndex to which the MAU described by this entry is connected."

REFERENCE

"RFC 2108, rpPtrPortIndex."

::= { rpMauEntry 2 }

rpMauIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This variable uniquely identifies the MAU described by this entry from among other MAUs connected to the same port (rpMauPortIndex)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.1"

::= { rpMauEntry 3 }

rpMauType OBJECT-TYPE

SYNTAX AutonomousType

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object identifies the MAU type. Values for standard IEEE 802.3 MAU types are defined in the IANA maintained IANA-MAU-MIB module, as OBJECT-IDENTITIES of dot3MauType. If the MAU type is unknown, the object identifier zeroDotZero is returned."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.2"

::= { rpMauEntry 4 }

rpMauStatus OBJECT-TYPE

SYNTAX INTEGER { other(1), unknown(2), operational(3), standby(4), shutdown(5), reset(6) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The current state of the MAU. This object may be implemented as a read-only object by those agents and MAUs that do not implement software control of the MAU state. Agents may support a subset of the enumerated values."

The value other(1) is returned if the MAU is in a state other than one of the states 2 through 6.

The value unknown(2) is returned when the MAU's true state is unknown; for example, when it is being initialized.

A MAU in the operational(3) state is fully functional; it operates, and passes signals to its attached DTE or repeater port in accordance to its specification.

A MAU in standby(4) state forces DI and CI to idle, and the media transmitter to idle or fault, if supported. Standby(4) mode only applies to link type MAUs. The state of rpMauMediaAvailable is unaffected.

A MAU in shutdown(5) state assumes the same condition on DI, CI, and the media transmitter, as though it were powered down or not connected. The MAU may return other(1) value for the rpMauJabberState and rpMauMediaAvailable objects when it is in this state. For an AUI, this state will remove power from the AUI.

Setting this variable to the value reset(6) resets the MAU in the same manner as a power-off, power-on cycle of at least one-half second would. The agent is not required to return the value reset(6).

Setting this variable to the value operational(3), standby(4), or shutdown(5) causes the MAU to assume the respective state, except that setting a mixing-type MAU or an AUI to standby(4) will cause the MAU to enter the shutdown state."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.7, 30.5.1.2.2, and 30.5.1.2.1"
::= { rpMauEntry 5 }

rpMauMediaAvailable OBJECT-TYPE

SYNTAX IANAIfMauMediaAvailable
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"This object identifies Media Available state of the MAU, complementary to the rpMauStatus. Values for the standard IEEE 802.3 Media Available states are defined in the IANA maintained IANA-MAU-MIB module, as IANAIfMauMediaAvailable TC."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.4, aMediaAvailable."
::= { rpMauEntry 6 }

rpMauMediaAvailableStateExits OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION

"A count of the number of times that rpMauMediaAvailable for this MAU instance leaves the state available(3).

Discontinuities in the value of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of rpPtrMonitorPortLastChange."

REFERENCE

```
"IEEE Std 802.3, 30.5.1.1.5
RFC 2108, rpPtrMonitorPortLastChange"
::= { rpMauEntry 7 }

rpMauJabberState OBJECT-TYPE
    SYNTAX      INTEGER { other(1), unknown(2), noJabber(3),
                        jabbering(4) }
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "The value other(1) is returned if the jabber
        state is not 2, 3, or 4. The agent shall
        return other(1) for MAU type dot3MauTypeAUI.

        The value unknown(2) is returned when the MAU's
        true state is unknown; for example, when it is
        being initialized.

        If the MAU is not jabbering the agent returns
        noJabber(3). This is the 'normal' state.

        If the MAU is in jabber state the agent returns
        the jabbering(4) value."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.1.6"
    ::= { rpMauEntry 8 }

rpMauJabberingStateEnters OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of the number of times that
        mauJabberState for this MAU instance enters the
        state jabbering(4). For MAUs of type
        dot3MauTypeAUI, dot3MauType100BaseT4,
        dot3MauType100BaseTX, dot3MauType100BaseFX, and
        all 1000 Mb/s types, this counter will
        indicate zero.

        Discontinuities in the value of this counter can
        occur at re-initialization of the management
        system and at other times, as indicated by the
        value of rpPtrMonitorPortLastChange."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.1.6,
        RFC 2108, rpPtrMonitorPortLastChange"
    ::= { rpMauEntry 9 }

rpMauFalseCarriers OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "A count of the number of false carrier events
        during IDLE in 100BASE-X links. This counter
        does not increment at the symbol rate. It can
        increment after a valid carrier completion at a
        maximum rate of once per 100 ms until the next
```


carrier event.

This counter increments only for MAUs of type
dot3MauType100BaseT4, dot3MauType100BaseTX,
dot3MauType100BaseFX, and all 1000 Mb/s types.

For all other MAU types, this counter will
indicate zero.

The approximate minimum time for rollover of
this counter is 7.4 hours.

Discontinuities in the value of this counter can
occur at re-initialization of the management
system and at other times, as indicated by the
value of rpPtrMonitorPortLastChange."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.10,
RFC 2108, rpPtrMonitorPortLastChange"

::= { rpMauEntry 10 }

rpJackTable OBJECT-TYPE

SYNTAX SEQUENCE OF RpJackEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Information about the external jacks attached
to MAUs attached to the ports of a repeater."

::= { dot3RpMauBasicGroup 2 }

rpJackEntry OBJECT-TYPE

SYNTAX RpJackEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry in the table, containing information
about a particular jack."

INDEX { rpMauGroupIndex, rpMauPortIndex, rpMauIndex,
rpJackIndex }

::= { rpJackTable 1 }

RpJackEntry ::= SEQUENCE {

rpJackIndex Integer32,

rpJackType IANAifJackType

}

rpJackIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This variable uniquely identifies the jack
described by this entry from among other jacks
attached to the same MAU (rpMauIndex)."

::= { rpJackEntry 1 }

rpJackType OBJECT-TYPE

SYNTAX IANAifJackType

MAX-ACCESS read-only

```

STATUS      current
DESCRIPTION
    "The jack connector type, as it appears on the
    outside of the system."
::= { rpJackEntry 2 }

dot3IfMauBasicGroup OBJECT IDENTIFIER
::= { ieee8023snmpDot3MauMgt 2 }

ifMauTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF IfMauEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of descriptive and status information
        about MAU(s) attached to an interface."
    ::= { dot3IfMauBasicGroup 1 }

ifMauEntry OBJECT-TYPE
    SYNTAX      IfMauEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
        about a single MAU."
    INDEX       { ifMauIfIndex, ifMauIndex }
    ::= { ifMauTable 1 }

IfMauEntry ::= SEQUENCE {
    ifMauIfIndex      InterfaceIndex,
    ifMauIndex        Integer32,
    ifMauType         AutonomousType,
    ifMauStatus       INTEGER,
    ifMauMediaAvailable IANAifMauMediaAvailable,
    ifMauMediaAvailableStateExits Counter32,
    ifMauJabberState  INTEGER,
    ifMauJabberingStateEnters Counter32,
    ifMauFalseCarriers Counter32,
    ifMauDefaultType  AutonomousType,
    ifMauAutoNegSupported TruthValue,
    ifMauTypeListBits IANAifMauTypeListBits,
    ifMauHCFALSECarriers Counter64,
    ifMauPCSCodingViolations Counter64,
    ifMauFECAbility   INTEGER,
    ifMauFECMode      INTEGER,
    ifMauFECCorrectedBlocks Counter64,
    ifMauFECUnCorrectableBlocks Counter64,
    ifMauSNROpMarginChn1A Integer32,
    ifMauSNROpMarginChn1B Integer32,
    ifMauSNROpMarginChn1C Integer32,
    ifMauSNROpMarginChn1D Integer32,
    ifMauEEESupportList IANAifMauTypeListBits,
    ifMauEEELDFastRetrainCount Counter32,
    ifMauEEELPFastRetrainCount Counter32,
    ifMauTimeSyncCapabilityTX TruthValue,
    ifMauTimeSyncCapabilityRX TruthValue,
    ifMauTimeSyncDelayTXmax Integer32,
    ifMauTimeSyncDelayTXmin Integer32,
    ifMauTimeSyncDelayRXmax Integer32,

```

```
    ifMauTimeSyncDelayRXmin          Integer32
}

ifMauIfIndex OBJECT-TYPE
    SYNTAX          InterfaceIndex
    MAX-ACCESS      not-accessible
    STATUS          current
    DESCRIPTION
        "This variable uniquely identifies the interface
        to which the MAU described by this entry is
        connected."
    REFERENCE
        "RFC 2863, ifIndex"
    ::= { ifMauEntry 1 }

ifMauIndex OBJECT-TYPE
    SYNTAX          Integer32 (1..2147483647)
    MAX-ACCESS      not-accessible
    STATUS          current
    DESCRIPTION
        "This variable uniquely identifies the MAU
        described by this entry from among other MAUs
        connected to the same interface (ifMauIfIndex)."
```

agents and MAUs that do not implement software control of the MAU state. Agents may support a subset of the enumerated values.

The value other(1) is returned if the MAU is in a state other than one of the states 2 through 6.

The value unknown(2) is returned when the MAU's true state is unknown; for example, when it is being initialized.

A MAU in the operational(3) state is fully functional; it operates, and passes signals to its attached DTE or repeater port in accordance to its specification.

A MAU in standby(4) state forces DI and CI to idle and the media transmitter to idle or fault, if supported. Standby(4) mode only applies to link type MAUs. The state of ifMauMediaAvailable is unaffected.

A MAU in shutdown(5) state assumes the same condition on DI, CI, and the media transmitter, as though it were powered down or not connected. The MAU may return other(1) value for the ifMauJabberState and ifMauMediaAvailable objects when it is in this state. For an AUI, this state will remove power from the AUI.

Setting this variable to the value reset(6) resets the MAU in the same manner as a power-off, power-on cycle of at least one-half second would. The agent is not required to return the value reset(6).

Setting this variable to the value operational(3), standby(4), or shutdown(5) causes the MAU to assume the respective state, except that setting a mixing-type MAU or an AUI to standby(4) will cause the MAU to enter the shutdown state."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.7, 30.5.1.2.2, and 30.5.1.2.1"
::= { ifMauEntry 4 }

ifMauMediaAvailable OBJECT-TYPE

SYNTAX IANAifMauMediaAvailable

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object identifies Media Available state of the MAU, complementary to the ifMauStatus. Values for the standard IEEE 802.3 Media Available states are defined in the IANA maintained IANA-MAU-MIB module, as IANAifMauMediaAvailable TC."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.4"

```
::= { ifMauEntry 5 }

ifMauMediaAvailableStateExits OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times that
         ifMauMediaAvailable for this MAU instance leaves
         the state available(3).

         Discontinuities in the value of this counter can
         occur at re-initialization of the management
         system and at other times, as indicated by the
         value of ifCounterDiscontinuityTime."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.1.5,
         RFC 2863, ifCounterDiscontinuityTime."
    ::= { ifMauEntry 6 }

ifMauJabberState OBJECT-TYPE
    SYNTAX      INTEGER { other(1), unknown(2), noJabber(3),
                          jabbering(4) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value other(1) is returned if the jabber
         state is not 2, 3, or 4. The agent shall
         return other(1) for MAU type dot3MauTypeAUI.

         The value unknown(2) is returned when the MAU's
         true state is unknown; for example, when it is
         being initialized.

         If the MAU is not jabbering the agent returns
         noJabber(3). This is the 'normal' state.

         If the MAU is in jabber state the agent returns
         the jabbering(4) value."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.1.6"
    ::= { ifMauEntry 7 }

ifMauJabberingStateEnters OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A count of the number of times that
         mauJabberState for this MAU instance enters the
         state jabbering(4). This counter will
         indicate zero for MAUs of type dot3MauTypeAUI
         and those of speeds above 10 Mb/s.

         Discontinuities in the value of this counter can
         occur at re-initialization of the management
         system and at other times, as indicated by the
         value of ifCounterDiscontinuityTime."
    REFERENCE
```

```
"IEEE Std 802.3, 30.5.1.1.6,  
RFC 2863, ifCounterDiscontinuityTime."  
::= { ifMauEntry 8 }
```

ifMauFalseCarriers OBJECT-TYPE
SYNTAX Counter32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"A count of the number of false carrier events
during IDLE in 100BASE-X and 1000BASE-X links.

For all other MAU types, this counter will
indicate zero. This counter does not
increment at the symbol rate.

It can increment after a valid carrier
completion at a maximum rate of once per 100 ms
for 100BASE-X and once per 10 us for 1000BASE-X
until the next CarrierEvent.

This counter can roll over very quickly. A
management station is advised to poll the
ifMauHCFALSECarriers instead of this counter in
order to avoid loss of information.

Discontinuities in the value of this counter can
occur at re-initialization of the management
system and at other times, as indicated by the
value of ifCounterDiscontinuityTime."
REFERENCE
"IEEE Std 802.3, 30.5.1.1.10,
RFC 2863, ifCounterDiscontinuityTime."
::= { ifMauEntry 9 }

ifMauDefaultType OBJECT-TYPE
SYNTAX AutonomousType
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This object identifies the default
administrative baseband MAU type to be used in
conjunction with the operational MAU type
denoted by ifMauType.

The set of possible values for this object is
the same as the set defined for the ifMauType
object.

This object represents the
administratively-configured type of the MAU. If
Auto-Negotiation is not enabled or is not
implemented for this MAU, the value of this
object determines the operational type of the
MAU. In this case, a set to this object will
force the MAU into the specified operating mode.

If Auto-Negotiation is implemented and enabled
for this MAU, the operational type of the MAU

is determined by Auto-Negotiation, and the value of this object denotes the type to which the MAU will automatically revert if/when Auto-Negotiation is later disabled.

In particular, when ifMauAutoNegAdminStatus transitions from enabled to disabled, the agent implementation shall verify that the operational type of the MAU (as reported by ifMauType) correctly transitions to the value specified by this object, rather than continuing to operate at the value earlier determined by the Auto-Negotiation function."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.1, and 22.2.4.1.4"

::= { ifMauEntry 10 }

ifMauAutoNegSupported OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This object indicates whether or not Auto-Negotiation is supported on this MAU."

::= { ifMauEntry 11 }

ifMauTypeListBits OBJECT-TYPE

SYNTAX IANAifMauTypeListBits

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A value that uniquely identifies the set of possible IEEE 802.3 types that the MAU could be. If Auto-Negotiation is present on this MAU, this object maps to ifMauAutoNegCapabilityBits.

To indicate that this MAU is capable of operating as a MAU type that is beyond the scope of this MIB, return the bit value bOther in addition to the bit values set for standard capabilities that are listed in the IANAifMauTypeListBits TC."

::= { ifMauEntry 12 }

ifMauHCFALSECarriers OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A count of the number of false carrier events during IDLE in 100BASE-X and 1000BASE-X links.

For all other MAU types, this counter will indicate zero. This counter does not increment at the symbol rate.

This counter is a 64-bit version of ifMauFalseCarriers. Since the 32-bit version of this counter can roll over very quickly, management stations are advised to poll the 64-bit version instead, in order to avoid loss

of information.

Discontinuities in the value of this counter can occur at re-initialization of the management system and at other times, as indicated by the value of ifCounterDiscontinuityTime."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.10,
RFC 2863, ifCounterDiscontinuityTime."

::= { ifMauEntry 13 }

ifMauPCSCodingViolations OBJECT-TYPE

SYNTAX Counter64

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"Generalized nonresettable counter. This counter has a maximum increment rate of 25 000 000 counts per second for 100 Mb/s implementations and 125 000 000 counts per second for 1000 Mb/s implementations.

For 100 Mb/s operation it is a count of the number of events that cause the PHY to indicate 'Data reception with errors' on the MII (see IEEE Std 802.3, Table 22-2).

For 1000 Mb/s operation it is a count of the number of events that cause the PHY to indicate 'Data reception error' or 'Carrier Extend Error' on the GMII (see IEEE Std 802.3, Table 35-2). The contents of this attribute is undefined when FEC is operating."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.14"

::= { ifMauEntry 14 }

ifMauFECAbility OBJECT-TYPE

SYNTAX INTEGER { unknown(1), supported(2), notsupported(3) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A read-only value that indicates if the PHY supports an optional FEC sublayer for forward error correction (see IEEE Std 802.3, 65.2 and IEEE Std 802.3, Clause 74, Clause 91, and Clause 108).

If an IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this attribute will map to the FEC capability register (see IEEE Std 802.3, 45.2.10.2 or 45.2.1.107)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.15"

::= { ifMauEntry 15 }

ifMauFECMode OBJECT-TYPE

SYNTAX INTEGER { unknown(1), disabled(2), enabled(3),
baseREnabled(4), rsFecEnabled(5) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"A read-write value that indicates the mode of operation of the optional FEC sublayer for forward error correction (see IEEE Std 802.3, 65.2 and IEEE Std 802.3, Clause 74, Clause 91, and clause 108).

A GET operation returns the current mode of operation of the PHY. A SET operation changes the mode of operation of the PHY to the indicated value. The enumerations 'baseREnabled' and 'rsFecEnabled' are only used for 25GBASE-CR, 25GBASE-CR-S, 25GBASE-KR, and 25GBASE-KR-S PHYs where operation in the no-FEC mode maps to the enumeration 'disabled', operation in the BASE-R FEC mode maps to the enumeration 'baseREnabled', and operation in the RS-FEC mode maps to the enumeration 'rsFecEnabled' (see IEEE Std 802.3, 110.6 and 111.6).

If an IEEE Std 802.3, Clause 45 MDIO Interface is present, this attribute maps to the FEC enable bit or to the RS-FEC enable bit in the appropriate FEC control register based upon the PHY type and the FEC operating mode (see IEEE Std 802.3, 45.2.10.3, 45.2.1.108, and 45.2.1.116)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.16"

::= { ifMauEntry 16 }

ifMauFECCorrectedBlocks OBJECT-TYPE

SYNTAX Counter64
MAX-ACCESS read-only
STATUS deprecated

DESCRIPTION

"***** THIS OBJECT IS DEPRECATED *****"

Generalized nonresettable counter. This counter has a maximum increment rate of 1 200 000 counts per second for 1000 Mb/s implementations, and 5 000 000 counts per second for 10 Gb/s implementations.

For 1000BASE-PX PHYs or 10GBASE-R PHYs, a count of corrected FEC blocks. This counter will not increment for other PHY types. Increment the counter by one for each received block that is corrected by the FEC function in the PHY. If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the FEC corrected blocks counter (see IEEE Std 802.3, 45.2.10.5 and 45.2.1.109)"

REFERENCE

"IEEE Std 802.3, 30.5.1.1.17"

::= { ifMauEntry 17 }

ifMauFECUnCorrectableBlocks OBJECT-TYPE

SYNTAX Counter64
MAX-ACCESS read-only
STATUS deprecated

DESCRIPTION

"***** THIS OBJECT IS DEPRECATED *****"

Generalized nonresettable counter. This counter has a maximum increment rate of 1 200 000 counts per second for 1000 Mb/s implementations, and 5 000 000 counts per second for 10 Gb/s implementations.

For 1000BASE-PX, 10/25/40/50/100/200/400GBASE-R, 100GBASE-P, 10GBASE-PR, or 10/1GBASE-PRX PHYs, a count of uncorrectable FEC blocks. This counter will not increment for other PHY types. Increment the counter by one for each received block that is determined to be uncorrectable by the FEC function in the PHY.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the FEC uncorrectable blocks counter (see IEEE Std 802.3, 45.2.10.6 and 45.2.1.110)"

REFERENCE

"IEEE Std 802.3, 30.5.1.1.18"

::= { ifMauEntry 18 }

ifMauSNROpMarginChnlA OBJECT-TYPE

SYNTAX Integer32 (-127..127)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current SNR operating margin measured at the slicer input for channel A for the 10GBASE-T PMA. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of -12.7 dB to 12.7 dB. If an IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this attribute maps to the SNR operating margin channel A register (see IEEE Std 802.3, 45.2.1.81)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.19"

::= { ifMauEntry 19 }

ifMauSNROpMarginChnlB OBJECT-TYPE

SYNTAX Integer32 (-127..127)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The current SNR operating margin measured at the slicer input for channel B for the 10GBASE-T PMA. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of -12.7 dB to 12.7 dB. If an IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this attribute maps to the SNR operating margin channel B register (see IEEE Std 802.3, 45.2.1.82)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.20"

::= { ifMauEntry 20 }

ifMauSNROpMarginChnlC OBJECT-TYPE

SYNTAX Integer32 (-127..127)

MAX-ACCESS read-only

```
STATUS          current
DESCRIPTION
    "The current SNR operating margin measured at the
    slicer input for channel C for the 10GBASE-T PMA.
    It is reported in units of 0.1 dB to an accuracy of
    0.5 dB within the range of -12.7 dB to 12.7 dB.
    If an IEEE Std 802.3, Clause 45 MDIO Interface to the
    PMA/PMD is present, then this attribute maps to the SNR
    operating margin channel C register
    (see IEEE Std 802.3, 45.2.1.83)."
```

REFERENCE

```
    "IEEE Std 802.3, 30.5.1.1.21"
 ::= { ifMauEntry 21 }
```

ifMauSNROpMarginChn1D OBJECT-TYPE

```
SYNTAX          Integer32 (-127..127)
MAX-ACCESS      read-only
STATUS          current
DESCRIPTION
    "The current SNR operating margin measured at the
    slicer input for channel D for the 10GBASE-T PMA.
    It is reported in units of 0.1 dB to an accuracy of
    0.5 dB within the range of -12.7 dB to 12.7 dB.
    If an IEEE Std 802.3, Clause 45 MDIO Interface to the
    PMA/PMD is present, then this attribute maps to the SNR
    operating margin channel D register
    (see IEEE Std 802.3, 45.2.1.84)."
```

REFERENCE

```
    "IEEE Std 802.3, 30.5.1.1.22"
 ::= { ifMauEntry 22 }
```

ifMauEEESupportList OBJECT-TYPE

```
SYNTAX          IANAifMauTypeListBits
MAX-ACCESS      read-only
STATUS          current
DESCRIPTION
    "A read-only list of the possible PHY types for which
    the underlying system supports Energy-Efficient Ethernet
    (EEE) as defined in IEEE Std 802.3, Clause 78.
    If IEEE Std 802.3, Clause 28 or Clause 73 Auto-Negotiation
    Is present, then this attribute maps to the local
    technology ability or advertised ability of the local
    device "
```

REFERENCE

```
    "IEEE Std 802.3, 30.5.1.1.23"
 ::= { ifMauEntry 23 }
```

ifMauEEELDFastRetrainCount OBJECT-TYPE

```
SYNTAX          Counter32
MAX-ACCESS      read-only
STATUS          current
DESCRIPTION
    "A count of the number of fast retrains initiated by the
    local device. This counter can be derived from
    fr_tx_counter (see IEEE Std 802.3, 55.4.5.4, 113.4.5.4,
    and 126.4.5.4). If IEEE Std 802.3, Clause 45 MDIO
    Interface to the PMA/PMD is present, then this attribute
    Can be derived from the LD fast retrain count register (see
    IEEE Std 802.3, 45.2.1.94.2)."
```

REFERENCE

"IEEE Std 802.3, 30.5.1.1.24"
::= { ifMauEntry 24 }

ifMauEEEELPFastRetrainCount OBJECT-TYPE

SYNTAX Counter32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"A count of the number of fast retrains initiated by the link partner. This counter can be derived from fr_rx_counter (see IEEE Std 802.3, 55.4.5.4, 113.4.5.4, and 126.4.5.4). If IEEE Std 802.3, Clause 45 MDIO Interface to the PMA/PMD is present, then this attribute Can be derived from the LP fast retrain count register (see IEEE Std 802.3, 45.2.1.94.1)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.25"
::= { ifMauEntry 25 }

ifMauTimeSyncCapabilityTX OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This object indicates whether or not transmit Time Sync is supported on this MAU."

REFERENCE

"IEEE Std 802.3, 30.13.1.1"
::= { ifMauEntry 26 }

ifMauTimeSyncCapabilityRX OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"This object indicates whether or not receive Time Sync is supported on this MAU."

REFERENCE

"IEEE Std 802.3, 30.13.1.2"
::= { ifMauEntry 27 }

ifMauTimeSyncDelayTXmax OBJECT-TYPE

SYNTAX Integer32
MAX-ACCESS read-only
STATUS current

DESCRIPTION

"The maximum data delay as specified in IEEE Std 802.3, 90.7, expressed in units of ns."

If an IEEE Std 802.3, Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum transmit path data delay values, consisting of the sum of the values of the registers in the instantiated sublayers (for each MMD, in case of multiple instances)"

REFERENCE

"IEEE Std 802.3, 30.13.1.3"

```
::= { ifMauEntry 28 }

ifMauTimeSyncDelayTXmin OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The minimum data delay as specified in IEEE Std 802.3,
        90.7, expressed in units of ns.

        If an IEEE Std 802.3, Clause 45 MDIO Interface to
        PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is
        present, then the value stored in this attribute
        represents the minimum transmit path data delay
        values, consisting of the sum of the values of the
        registers in the instantiated sublayers (for each MMD,
        in case of multiple instances)"
    REFERENCE
        "IEEE Std 802.3, 30.13.1.4"
::= { ifMauEntry 29 }

ifMauTimeSyncDelayRXmax OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The maximum data delay as specified in IEEE Std 802.3,
        90.7, expressed in units of ns.

        If an IEEE Std 802.3, Clause 45 MDIO Interface to
        PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is
        present, then the value stored in this attribute
        represents the maximum receive path data delay
        values, consisting of the sum of the values of the
        registers in the instantiated sublayers (for each MMD,
        in case of multiple instances)"
    REFERENCE
        "IEEE Std 802.3, 30.13.1.5"
::= { ifMauEntry 30 }

ifMauTimeSyncDelayRXmin OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The minimum data delay as specified in IEEE Std 802.3,
        90.7, expressed in units of ns.

        If an IEEE Std 802.3, Clause 45 MDIO Interface to
        PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is
        present, then the value stored in this attribute
        represents the minimum receive path data delay
        values, consisting of the sum of the values of the
        registers in the instantiated sublayers (for each MMD,
        in case of multiple instances)"
    REFERENCE
        "IEEE Std 802.3, 30.13.1.6"
::= { ifMauEntry 31 }
```

```
ifJackTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF IfJackEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Information about the external jacks attached
         to MAUs attached to an interface."
    ::= { dot3IfMauBasicGroup 2 }

ifJackEntry OBJECT-TYPE
    SYNTAX      IfJackEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
         about a particular jack."
    INDEX       { ifMauIfIndex, ifMauIndex, ifJackIndex }
    ::= { ifJackTable 1 }

IfJackEntry ::= SEQUENCE {
    ifJackIndex      Integer32,
    ifJackType       IANAifJackType
}

ifJackIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This variable uniquely identifies the jack
         described by this entry from among other jacks
         attached to the same MAU."
    ::= { ifJackEntry 1 }

ifJackType OBJECT-TYPE
    SYNTAX      IANAifJackType
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The jack connector type, as it appears on the
         outside of the system."
    ::= { ifJackEntry 2 }

ifMauPerPCSLaneStatsTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF IfMauPerPCSLaneStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Table of Per-PCS lane status information
         about MAUs attached to an interface."
    ::= { dot3IfMauBasicGroup 3 }

ifMauPerPCSLaneStatsEntry OBJECT-TYPE
    SYNTAX      IfMauPerPCSLaneStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing information
         about a single PCS lane."
```

```
INDEX      { ifMauIfIndex, ifMauIndex, ifPCSLaneIndex }
 ::= { ifMauPerPCSLaneStatsTable 1 }
```

IfMauPerPCSLaneStatsEntry ::= SEQUENCE {
 ifPCSLaneIndex Unsigned32,
 ifMauPPLFECCorrectedBlocks Counter64,
 ifMauPPLFECUncorrectableBlocks Counter64,
 ifMauBIPErrorsCount Counter32,
 ifMauPCStoPHYLaneMapping Unsigned32
}

ifPCSLaneIndex OBJECT-TYPE
 SYNTAX Unsigned32 (0..255)
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "This object provides the identification of the
 PCS lane for which this ifMauPerPCSLaneStatsEntry
 is applicable. This object can hold an integer value
 from 0 to N-1, where N is the total number of PCS
 lanes supported by the given PCS. "
 ::= { ifMauPerPCSLaneStatsTable 1 }

ifMauPPLFECCorrectedBlocks OBJECT-TYPE
 SYNTAX Counter64
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "Generalized nonresettable counter. This counter has a
 maximum increment rate of 1 200 000 counts per second
 for 1000 Mb/s implementations, 5 000 000 counts per
 second for 10 Gb/s and 40 Gb/s implementations, and
 2 500 000 counts per second for 100 Gb/s implementations.

 For 1000BASE-PX, 10/25/40/50/100/200/400GBASE-R,
 100GBASE-P, 10GBASE-PR, or 10/1GBASE-PRX PHYs that
 support FEC across the MDI, an array of corrected FEC
 block counters. The counters do not increment for other
 PHY types. The indices of this array (0 to N - 1) denote
 the FEC sublayer instance number where N is the number
 of FEC sublayer instances in use.

 The number of FEC sublayer instances in use is set to one
 for PHYs that do not use PCS lanes or use a single FEC
 instance for all lanes. Each element of this array
 contains a count of corrected FEC blocks for that FEC
 sublayer instance.

 Increment the counter by one for each FEC block received
 across the MDI that is corrected by the FEC function in
 the PHY for the corresponding lane or FEC sublayer
 instance.

 If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS
 is present, then this object maps to the FEC corrected
 blocks counter for PCS lane number n, identified by
 the ifPCSLaneIndex object
 (see IEEE Std 802.3, 45.2.10.5 and 45.2.1.109
 for 10GBASE-R, 45.2.3.41 for 10GBASE-PR and

```
10/1GBASE-PRX, 45.2.1.131 for BASE-R, 45.2.1.118 for
RS-FEC, 45.2.3.62 for PCS FEC, and 45.2.1.227 for
SC-FEC)."
REFERENCE
    "IEEE Std 802.3, 30.5.1.1.17"
::= { ifMauPerPCSLaneStatsEntry 2 }

ifMauPPLFECUncorrectableBlocks OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "Generalized nonresettable counter. This counter has a
        maximum increment rate of 1 200 000 counts per second
        for 1000 Mb/s implementations, 5 000 000 counts
        per second for 10 Gb/s and 40 Gb/s implementations,
        and 2 500 000 counts per second for 100 Gb/s
        implementations.

        For 1000BASE-PX, 10/25/40/50/100/200/400GBASE-R,
        100GBASE-P, 10GBASE-PR, or 10/1GBASE-PRX PHYs that
        support FEC across the MDI, an array of uncorrectable FEC
        block counters. The counters do not increment for other
        PHY types. The indices of this array (0 to N - 1) denote
        the FEC sublayer instance number where N is the number
        of FEC sublayer instances in use.

        The number of FEC sublayer instances in use is set to one
        for PHYs that do not use PCS lanes or use a single FEC
        instance for all lanes. Each element of this array
        contains a count of uncorrectable FEC blocks for that FEC
        sublayer instance.

        Increment the counter by one for each FEC block that
        is determined to be uncorrectable by the FEC function
        in the PHY for the corresponding lane or FEC sublayer
        instance.

        If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS
        is present, then this object maps to the FEC uncorrectable
        blocks counter for PSC lane number n, identified by
        the ifPCSLaneIndex object
        (see IEEE Std 802.3, 45.2.10.6 and 45.2.1.110 for 10GBASE-R,
        45.2.3.42 for 10GBASE-PR and 10/1GBASE-PRX, 45.2.1.149 for
        BASE-R, 45.2.1.119 for RS-FEC, 45.2.3.63 for PCS FEC, and
        45.2.1.228 for SC-FEC)."
```

```
REFERENCE
    "IEEE Std 802.3, 30.5.1.1.18"
::= { ifMauPerPCSLaneStatsEntry 3 }

ifMauBIPErrorCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS   read-only
    STATUS       current
    DESCRIPTION
        "Generalized nonresettable counter. This counter
        has a maximum increment rate of 10 000 counts per
        second for 40 Gb/s and 50 Gb/s implementations and
        5 000 counts per second for 100 Gb/ s implementations."
```


For 40/50/100GBASE-R PHYs and 100GBASE-P PHYs, an array of BIP error counters. The counters do not increment for other PHY types. The indices of this array (0 to N - 1) denote the PCS lane number where N is the number of PCS lanes in use. Each element of this array contains a count of BIP errors for that PCS lane.

Increment the counter by one for each BIP error detected during alignment marker removal in the PCS identified by the ifPCSLaneIndex object.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the BIP error counter for PCS lane number n, identified by the ifPCSLaneIndex object
(see IEEE Std 802.3, 45.2.3.47 and 45.2.3.48)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.11"

::= { ifMauPerPCSLaneStatsEntry 4 }

ifMauPCStoPHYLaneMapping OBJECT-TYPE

SYNTAX Unsigned32

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For 40/50/100/200/400GBASE-R PHYs and 100GBASE-P PHYs, an array of PCS lane identifiers. The indices of this array (0 to N - 1) denote the service interface lane number where N is the number of PCS lanes in use. Each element of this array contains the PCS lane number for the PCS lane that has been detected in the corresponding service interface lane.

If IEEE Std 802.3, Clause 45 MDIO Interface to the PCS is present, then this object maps to the Lane mapping register for PCS lane number n, identified by the ifPCSLaneIndex object
(see IEEE Std 802.3, 45.2.3.49 and 45.2.3.50)."

REFERENCE

"IEEE Std 802.3, 30.5.1.1.12"

::= { ifMauPerPCSLaneStatsEntry 5 }

dot3PlaceholderGroup OBJECT IDENTIFIER

::= { ieee8023snmpDot3MauMgt 3 }

dot3Placeholder OBJECT-TYPE

SYNTAX INTEGER { placeholder(1) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A placeholder object to preserve the assignments that follow in the module. The assignment was given to the object broadMauBasicTable in earlier versions of this module. Preserving the assignments that follow is considered important because they are used for the IANA-MAU-MIB to assign as MAU type values."

REFERENCE

```
"none"
 ::= { dot3PlaceholderGroup 1 }

dot3IfMauAutoNegGroup OBJECT IDENTIFIER
 ::= { ieee8023snmpDot3MauMgt 5 }

ifMauAutoNegTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF IfMauAutoNegEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Configuration and status objects for the
         Auto-Negotiation function of MAUs attached to
         interfaces.

         The ifMauAutoNegTable applies to systems in
         which Auto-Negotiation is supported on one or
         more MAUs attached to interfaces. Note that if
         Auto-Negotiation is present and enabled, the
         ifMauType object reflects the result of the
         Auto-Negotiation function."
    ::= { dot3IfMauAutoNegGroup 1 }

ifMauAutoNegEntry OBJECT-TYPE
    SYNTAX      IfMauAutoNegEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry in the table, containing configuration
         and status information for the Auto-Negotiation
         function of a particular MAU."
    INDEX       { ifMauIfIndex, ifMauIndex }
    ::= { ifMauAutoNegTable 1 }

IfMauAutoNegEntry ::= SEQUENCE {
    ifMauAutoNegAdminStatus      INTEGER,
    ifMauAutoNegRemoteSignaling  INTEGER,
    ifMauAutoNegConfig           INTEGER,
    ifMauAutoNegRestart          INTEGER,
    ifMauAutoNegCapabilityBits   IANAifMauAutoNegCapBits,
    ifMauAutoNegCapAdvertisedBits IANAifMauAutoNegCapBits,
    ifMauAutoNegCapReceivedBits  IANAifMauAutoNegCapBits,
    ifMauAutoNegRemoteFaultAdvertised INTEGER,
    ifMauAutoNegRemoteFaultReceived INTEGER
}

ifMauAutoNegAdminStatus OBJECT-TYPE
    SYNTAX      INTEGER { enabled(1), disabled(2) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "Setting this object to enabled(1) will cause
         the interface that has the Auto-Negotiation
         signaling ability to be enabled.

         If the value of this object is disabled(2) then
         the interface will act as it would if it had no
         Auto-Negotiation signaling. Under these
         conditions, an IEEE 802.3 MAU will immediately
```

be forced to the state indicated by the value of the object ifMauDefaultType.

When ifMauAutoNegAdminStatus transitions from enabled to disabled, the agent implementation shall verify that the operational type of the MAU (as reported by ifMauType) correctly transitions to the value specified by the ifMauDefaultType object, rather than continuing to operate at the value earlier determined by the Auto-Negotiation function."

REFERENCE

"IEEE Std 802.3, 30.6.1.1.2, and 30.6.1.2.2"

::= { ifMauAutoNegEntry 1 }

ifMauAutoNegRemoteSignaling OBJECT-TYPE

SYNTAX INTEGER { detected(1), notdetected(2) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A value indicating whether the remote end of the link is using Auto-Negotiation signaling. It takes the value detected(1) if and only if, during the previous link negotiation, FLP Bursts were received."

REFERENCE

"IEEE Std 802.3, 30.6.1.1.3"

::= { ifMauAutoNegEntry 2 }

ifMauAutoNegConfig OBJECT-TYPE

SYNTAX INTEGER { other(1), configuring(2), complete(3), disabled(4), parallelDetectFail(5) }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A value indicating the current status of the Auto-Negotiation process. The enumeration parallelDetectFail(5) maps to a failure in parallel detection as defined in IEEE Std 802.3, 28.2.3.1."

REFERENCE

"IEEE Std 802.3, 30.6.1.1.4"

::= { ifMauAutoNegEntry 4 }

ifMauAutoNegRestart OBJECT-TYPE

SYNTAX INTEGER { restart(1), norestart(2) }

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"If the value of this object is set to restart(1) then this will force Auto-Negotiation to begin link renegotiation. If Auto-Negotiation signaling is disabled, a write to this object has no effect. Setting the value of this object to norestart(2) has no effect."

REFERENCE

"IEEE Std 802.3, 30.6.1.2.1"

::= { ifMauAutoNegEntry 5 }

```
ifMauAutoNegCapabilityBits OBJECT-TYPE
    SYNTAX      IANAifMauAutoNegCapBits
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A value that uniquely identifies the set of
        capabilities of the local Auto-Negotiation
        entity.

        Note that the local Auto-Negotiation entity may
        support some capabilities beyond the scope of
        this MIB. This is indicated by returning the
        bit value bOther in addition to any bit values
        for standard capabilities that are listed in the
        IANAifMauAutoNegCapBits TC."
    REFERENCE
        "IEEE Std 802.3, 30.6.1.1.5
        ."
    ::= { ifMauAutoNegEntry 6 }

ifMauAutoNegCapAdvertisedBits OBJECT-TYPE
    SYNTAX      IANAifMauAutoNegCapBits
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "A value that uniquely identifies the set of
        capabilities advertised by the local
        Auto-Negotiation entity.

        Capabilities in this object that are not
        available in ifMauAutoNegCapabilityBits cannot
        be enabled.

        Note that the local Auto-Negotiation entity may
        advertise some capabilities beyond the scope of
        this MIB. This is indicated by returning the
        bit value bOther in addition to any bit values
        for standard capabilities that are listed in the
        IANAifMauAutoNegCapBits TC."
    REFERENCE
        "IEEE Std 802.3, 30.6.1.1.6"
    ::= { ifMauAutoNegEntry 7 }

ifMauAutoNegCapReceivedBits OBJECT-TYPE
    SYNTAX      IANAifMauAutoNegCapBits
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A value that uniquely identifies the set of
        capabilities received from the remote
        Auto-Negotiation entity.

        To indicate that an interface or interfaces that
        support this MIB are attached to remote Auto-Negotiation
        entities, return the bit value bOther in addition to the
        bit values set for standard capabilities that are listed
        in the IANAifMauAutoNegCapBits TC."
    REFERENCE
        "IEEE Std 802.3, 30.6.1.1.7"
```

```
::= { ifMauAutoNegEntry 8 }

ifMauAutoNegRemoteFaultAdvertised OBJECT-TYPE
    SYNTAX      INTEGER { noError(1), offline(2), linkFailure(3),
                          autoNegError(4) }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "A value that identifies any local fault
         indications that this MAU has detected and will
         advertise at the next Auto-Negotiation
         interaction for 1000 Mb/s MAUs."
    REFERENCE
        "IEEE Std 802.3, 30.6.1.1.6"
    ::= { ifMauAutoNegEntry 9 }

ifMauAutoNegRemoteFaultReceived OBJECT-TYPE
    SYNTAX      INTEGER { noError(1), offline(2), linkFailure(3),
                          autoNegError(4) }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A value that identifies any fault indications
         received from the far end of a link by the
         local Auto-Negotiation entity for 1000 Mb/s
         MAUs."
    REFERENCE
        "IEEE Std 802.3, 30.6.1.1.7"
    ::= { ifMauAutoNegEntry 10 }

mauModConf OBJECT IDENTIFIER
    ::= { ieee8023mauMIB 2 }

mauModCompls OBJECT IDENTIFIER
    ::= { mauModConf 1 }

mauModObjGrps OBJECT IDENTIFIER
    ::= { mauModConf 2 }

mauModNotGrps OBJECT IDENTIFIER
    ::= { mauModConf 3 }

rpMauJabberTrap NOTIFICATION-TYPE
    OBJECTS      { rpMauJabberState }
    STATUS      current
    DESCRIPTION
        "This trap is sent whenever a managed repeater
         MAU enters the jabber state.

         The agent shall limit the generation of
         consecutive rpMauJabberTraps so that there is at
         least a five-second gap between them."
    REFERENCE
        "IEEE Std 802.3, 30.5.1.3.1"
    ::= { snmpDot3MauTraps 1 }

ifMauJabberTrap NOTIFICATION-TYPE
    OBJECTS      { ifMauJabberState }
    STATUS      current
```

DESCRIPTION

"This trap is sent whenever a managed interface MAU enters the jabber state.

The agent shall limit the generation of consecutive ifMauJabberTraps so that there is at least a five-second gap between them."

REFERENCE

"IEEE Std 802.3, 30.5.1.3.1"

::= { snmpDot3MauTraps 2 }

mauRpGrpBasic OBJECT-GROUP

OBJECTS { rpMauType, rpMauStatus, rpMauMediaAvailable,
rpMauMediaAvailableStateExits, rpMauJabberState,
rpMauJabberingStateEnters }

STATUS current

DESCRIPTION

"Basic conformance group for MAUs attached to repeater ports. This group is also the conformance specification for RFC 1515 implementations."

::= { mauModObjGrps 1 }

mauRpGrp100Mbs OBJECT-GROUP

OBJECTS { rpMauFalseCarriers }

STATUS current

DESCRIPTION

"Conformance group for MAUs attached to repeater ports with 100 Mb/s or greater capability."

::= { mauModObjGrps 2 }

mauRpGrpJack OBJECT-GROUP

OBJECTS { rpJackType }

STATUS current

DESCRIPTION

"Conformance group for MAUs attached to repeater ports with managed jacks."

::= { mauModObjGrps 3 }

mauIfGrpBasic OBJECT-GROUP

OBJECTS { ifMauType, ifMauStatus, ifMauMediaAvailable,
ifMauMediaAvailableStateExits, ifMauJabberState,
ifMauJabberingStateEnters, dot3Placeholder }

STATUS current

DESCRIPTION

"Basic conformance group for MAUs attached to interfaces. This group also provides a conformance specification for RFC 1515 implementations."

::= { mauModObjGrps 4 }

mauIfGrpJack OBJECT-GROUP

OBJECTS { ifJackType }

STATUS current

DESCRIPTION

"Conformance group for MAUs attached to interfaces with managed jacks."

::= { mauModObjGrps 5 }

```
mauIfGrpHighCapacity OBJECT-GROUP
  OBJECTS      { ifMauFalseCarriers, ifMauTypeListBits,
                  ifMauDefaultType, ifMauAutoNegSupported }
  STATUS       current
  DESCRIPTION
    "Conformance group for MAUs attached to
    interfaces with 100 Mb/s or greater capability."
  ::= { mauModObjGrps 6 }

mauIfGrpAutoNeg2 OBJECT-GROUP
  OBJECTS      { ifMauAutoNegAdminStatus, ifMauAutoNegRemoteSignaling,
                  ifMauAutoNegConfig, ifMauAutoNegCapabilityBits,
                  ifMauAutoNegCapAdvertisedBits,
                  ifMauAutoNegCapReceivedBits, ifMauAutoNegRestart }
  STATUS       current
  DESCRIPTION
    "Conformance group for MAUs attached to
    interfaces with managed Auto-Negotiation."
  ::= { mauModObjGrps 7 }

mauIfGrpAutoNeg1000Mbps OBJECT-GROUP
  OBJECTS      { ifMauAutoNegRemoteFaultAdvertised,
                  ifMauAutoNegRemoteFaultReceived }
  STATUS       current
  DESCRIPTION
    "Conformance group for 1000 Mb/s MAUs attached to
    interfaces with managed Auto-Negotiation."
  ::= { mauModObjGrps 8 }

mauIfGrpHCStats OBJECT-GROUP
  OBJECTS      { ifMauHCFALSECarriers, ifMauPCSCodingViolations }
  STATUS       current
  DESCRIPTION
    "Conformance for high capacity statistics for
    MAUs attached to interfaces."
  ::= { mauModObjGrps 9 }

mauIfGrpFEC OBJECT-GROUP
  OBJECTS      { ifMauFECAbility, ifMauFECMode }
  STATUS       current
  DESCRIPTION
    "Conformance for FEC capable
    MAUs attached to interfaces."
  ::= { mauModObjGrps 10 }

mauIfGrpSNR OBJECT-GROUP
  OBJECTS      { ifMauSNROpMarginChnlA, ifMauSNROpMarginChnlB,
                  ifMauSNROpMarginChnlC, ifMauSNROpMarginChnlD }
  STATUS       current
  DESCRIPTION
    "Conformance for SNR operating margin reporting
    MAUs attached to interfaces."
  ::= { mauModObjGrps 11 }

mauIfGrpEEE OBJECT-GROUP
  OBJECTS      { ifMauEEESupportList, ifMauEEELDFastRetrainCount,
                  ifMauEEELPFastRetrainCount }
  STATUS       current
```

```
DESCRIPTION
    "Conformance EEE support and Fast Retrain count
    reporting MAUs attached to interfaces."
 ::= { mauModObjGrps 12 }

mauIfGrpTimeSync OBJECT-GROUP
    OBJECTS      { ifMauTimeSyncCapabilityTX, ifMauTimeSyncCapabilityRX,
                  ifMauTimeSyncDelayTXmax, ifMauTimeSyncDelayTXmin,
                  ifMauTimeSyncDelayRXmax, ifMauTimeSyncDelayRXmin }
    STATUS       current
    DESCRIPTION
        "Conformance Time Sync support and delay
        reporting MAUs attached to interfaces."
 ::= { mauModObjGrps 13 }

mauIfGrpPerPCSLaneStats OBJECT-GROUP
    OBJECTS      { ifMauPPLFECCorrectedBlocks,
                  ifMauPPLFECUncorrectableBlocks, ifMauBIPErrorCount,
                  ifMauPCStoPHYLaneMapping }
    STATUS       current
    DESCRIPTION
        "Conformance Per-PCS lane statistics
        reporting MAUs attached to interfaces."
 ::= { mauModObjGrps 14 }

rpMauNotifications NOTIFICATION-GROUP
    NOTIFICATIONS { rpMauJabberTrap }
    STATUS       current
    DESCRIPTION
        "Notifications for repeater MAUs."
 ::= { mauModNotGrps 1 }

ifMauNotifications NOTIFICATION-GROUP
    NOTIFICATIONS { ifMauJabberTrap }
    STATUS       current
    DESCRIPTION
        "Notifications for interface MAUs."
 ::= { mauModNotGrps 2 }

mauModRpCompl2 MODULE-COMPLIANCE
    STATUS       current
    DESCRIPTION
        "Compliance for MAUs attached to repeater
        ports.

        Note that compliance with this compliance
        statement requires compliance with the
        snmpRpPtrModCompl MODULE-COMPLIANCE statement of
        the IEEE8023-SNMP-REPEATER-MIB defined in Clause 7."

    MODULE      -- this module

    MANDATORY-GROUPS      { mauRpGrpBasic }

    GROUP      mauRpGrp100Mbs
    DESCRIPTION
        "Implementation of this optional group is
        recommended for MAUs that have 100 Mb/s or
        greater capability."
```



```
GROUP    mauRpGrpJack
DESCRIPTION
    "Implementation of this optional group is
    recommended for MAUs that have one or more
    external jacks."

GROUP    rpMauNotifications
DESCRIPTION
    "Implementation of this group is recommended
    for MAUs attached to repeater ports."

OBJECT   rpMauStatus
MIN-ACCESS read-only
DESCRIPTION
    "Write access is not required."

::= { mauModCompls 1 }

mauModIfCompl3 MODULE-COMPLIANCE
STATUS    current
DESCRIPTION
    "Compliance for MAUs attached to interfaces.

    Note that compliance with this compliance
    statement requires compliance with the
    ifCompliance3 MODULE-COMPLIANCE statement of the
    IF-MIB (RFC 2863) and the dot3Compliance2
    MODULE-COMPLIANCE statement of the
    IEEE8023-EtherLike-MIB defined in Clause 10."

MODULE    -- this module

MANDATORY-GROUPS    { mauIfGrpBasic }

GROUP    mauIfGrpHighCapacity
DESCRIPTION
    "Implementation of this optional group is
    recommended for MAUs that have 100 Mb/s
    or greater capability."

GROUP    mauIfGrpHCStats
DESCRIPTION
    "Implementation of this group is mandatory
    for MAUs that have 1000 Mb/s capacity, and
    is recommended for MAUs that have 100 Mb/s
    capacity."

GROUP    mauIfGrpJack
DESCRIPTION
    "Implementation of this optional group is
    recommended for MAUs that have one or more
    external jacks."

GROUP    mauIfGrpAutoNeg2
DESCRIPTION
    "Implementation of this group is mandatory
    for MAUs that support managed
    Auto-Negotiation."
```

```
GROUP    mauIfGrpAutoNeg1000Mbps
DESCRIPTION
    "Implementation of this group is mandatory
    for MAUs that have 1000 Mb/s or greater
    capability and support managed
    Auto-Negotiation."

GROUP    ifMauNotifications
DESCRIPTION
    "Implementation of this group is recommended
    for MAUs attached to interfaces."

GROUP    mauIfGrpFEC
DESCRIPTION
    "Implementation of this optional group is
    recommended for MAUs that incorporate FEC."

GROUP    mauIfGrpSNR
DESCRIPTION
    "Implementation of this optional group is
    recommended for MAUs that report SNR operating
    margin."

GROUP    mauIfGrpEEE
DESCRIPTION
    "Implementation of this group is
    mandatory for MAUs that support EEE."

GROUP    mauIfGrpTimeSync
DESCRIPTION
    "Implementation of this group is
    mandatory for MAUs that support Time Sync"

GROUP    mauIfGrpPerPCSLaneStats
DESCRIPTION
    "Implementation of this group is
    mandatory for MAUs that report per-PCS lane
    statistics."

OBJECT   ifMauStatus
MIN-ACCESS read-only
DESCRIPTION
    "Write access is not required."

 ::= { mauModCompls 2 }

END -- end of module IEEE8023-MAU-MIB.
```

Annex A

(informative)

Bibliography

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[B5] IEEE Std 802.3™-2008, IEEE Standard for Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications.

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²⁹ This document is available at the following URL: <https://datatracker.ietf.org/doc/draft-ietf-opsawg-rfc5066bis/06/>.

³⁰ The tutorial is available at <https://standards.ieee.org/regauth>. Follow the tutorials link and search for the tutorial title.

³¹ IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

³² The IEEE standards or products referred to in Annex A are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

- [B14] IETF RFC 4836, Definitions of Managed Objects for IEEE 802.3 Medium Attachment Units (MAUs), E. Beili, Apr. 2007.
- [B15] IETF RFC 4837, Managed Objects of Ethernet Passive Optical Networks (EPON), L. Kharmosh, Jul. 2007.
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Annex B

(normative)

Branch and leaf assignments for IEEE 802.3 and IEEE 802.3.1 managed objects

This annex formally defines the branch and leaf assignments for the IEEE 802.3 and IEEE 802.3.1 managed objects. The branch and leaf assignments currently specified in this annex supersede any object identifiers (OIDs) formerly specified in this annex.

One use for these branch and leaf assignments can be found in Clause 57 of IEEE Std 802.3, which defines OAM, for example the variable descriptor format found in 57.6.1 of IEEE Std 802.3.

B.1 Branch and leaf table

An ASCII machine readable extract of Table B-1 can be obtained at the following URL:³³

https://www.ieee802.org/3/1/public/mib_modules/20241211/802dot3dot1AB.txt

Table B-1 contains the branch and leaf assignments for Ethernet managed objects. The branch and leaf assignments are provided for use in the variable descriptors found in Clause 57 of IEEE Std 802.3.

Table B-1—Branch and leaf assignments for managed objects

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aResourceTypeIDName	ATTRIBUTE	GET	7	1
aResourceInfo	ATTRIBUTE	GET	7	2
oMACEntity	OBJECT	GET	3	1
aMACID	ATTRIBUTE	GET	7	1
aFramesTransmittedOK	ATTRIBUTE	GET	7	2
aSingleCollisionFrames	ATTRIBUTE	GET	7	3
aMultipleCollisionFrames	ATTRIBUTE	GET	7	4
aFramesReceivedOK	ATTRIBUTE	GET	7	5
aFrameCheckSequenceErrors	ATTRIBUTE	GET	7	6
aAlignmentErrors	ATTRIBUTE	GET	7	7
aOctetsTransmittedOK	ATTRIBUTE	GET	7	8

³³ Copyright release for branch and leaf table: Users of this standard may freely reproduce the branch and leaf table contained in this subclause so that it can be used for its intended purpose.

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aFramesWithDeferredXmissions	ATTRIBUTE	GET	7	9
aLateCollisions	ATTRIBUTE	GET	7	10
aFramesAbortedDueToXSColls	ATTRIBUTE	GET	7	11
aFramesLostDueToIntMACXmitError	ATTRIBUTE	GET	7	12
aCarrierSenseErrors	ATTRIBUTE	GET	7	13
aOctetsReceivedOK	ATTRIBUTE	GET	7	14
aFramesLostDueToIntMACRcvError	ATTRIBUTE	GET	7	15
aPromiscuousStatus	ATTRIBUTE	GET-SET	7	16
aReadMulticastAddressList	ATTRIBUTE	GET	7	17
aMaxFrameLength	ATTRIBUTE	GET	7	357
aSlowProtocolFrameLimit	ATTRIBUTE	GET	7	426
aMulticastFramesXmittedOK	ATTRIBUTE	GET	7	18
aBroadcastFramesXmittedOK	ATTRIBUTE	GET	7	19
aFramesWithExcessiveDeferral	ATTRIBUTE	GET	7	20
aMulticastFramesReceivedOK	ATTRIBUTE	GET	7	21
aBroadcastFramesReceivedOK	ATTRIBUTE	GET	7	22
aInRangeLengthErrors	ATTRIBUTE	GET	7	23
aOutOfRangeLengthField	ATTRIBUTE	GET	7	24
aFrameTooLongErrors	ATTRIBUTE	GET	7	25
aMACEnableStatus	ATTRIBUTE	GET-SET	7	26
aTransmitEnableStatus	ATTRIBUTE	GET-SET	7	27
aMulticastReceiveStatus	ATTRIBUTE	GET-SET	7	28
aReadWriteMACAddress	ATTRIBUTE	GET-SET	7	29
aCollisionFrames	ATTRIBUTE	GET	7	30
aMACCapabilities	ATTRIBUTE	GET	7	89
aDuplexStatus	ATTRIBUTE	GET-SET	7	90
aRateControlAbility	ATTRIBUTE	GET	7	179

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aRateControlStatus	ATTRIBUTE	GET-SET	7	180
aDeferControlAbility	ATTRIBUTE	GET	7	311
aDeferControlStatus	ATTRIBUTE	GET-SET	7	312
acInitializeMAC	ACTION	—	9	1
acAddGroupAddress	ACTION	—	9	2
acDeleteGroupAddress	ACTION	—	9	3
acExecuteSelfTest	ACTION	—	9	4
oPHYEntity	OBJECT	GET	3	2
aPHYID	ATTRIBUTE	GET	7	31
aPHYType	ATTRIBUTE	GET	7	32
aPHYTypeList	ATTRIBUTE	GET	7	33
aSQETestErrors	ATTRIBUTE	GET	7	34
aSymbolErrorDuringCarrier	ATTRIBUTE	GET	7	35
aMIIDetect	ATTRIBUTE	GET	7	36
aPHYAdminState	ATTRIBUTE	GET	7	37
acPHYAdminControl	ACTION	—	9	5
oMACControlEntity	OBJECT	GET	3	8
aMACControlIID	ATTRIBUTE	GET	7	92
aMACControlFunctionsSupported	ATTRIBUTE	GET-SET	7	93
aMACControlFramesTransmitted	ATTRIBUTE	GET	7	94
aMACControlFramesReceived	ATTRIBUTE	GET	7	95
aUnsupportedOpcodesReceived	ATTRIBUTE	GET	7	96
aPFCEnableStatus	ATTRIBUTE	GET	7	415
oMACControlFunctionEntity	OBJECT	GET	3	9
aPAUSELinkDelayAllowance	ATTRIBUTE	GET-SET	7	97
aPAUSEMACCtrlFramesTransmitted	ATTRIBUTE	GET	7	98
aPAUSEMACCtrlFramesReceived	ATTRIBUTE	GET	7	99

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
oMPCP	OBJECT	GET	3	21
aMPCPID	ATTRIBUTE	GET	7	351
aMPCPAdminState	ATTRIBUTE	GET	7	278
aMPCPMode	ATTRIBUTE	GET	7	279
aMPCPLinkID	ATTRIBUTE	GET	7	282
aMPCPRemoteMACAddress	ATTRIBUTE	GET	7	283
aMPCPRegistrationState	ATTRIBUTE	GET	7	284
aMPCPMACCtrlFramesTransmitted	ATTRIBUTE	GET	7	280
aMPCPMACCtrlFramesReceived	ATTRIBUTE	GET	7	281
aMPCPTxGate	ATTRIBUTE	GET	7	315
aMPCPTxRegAck	ATTRIBUTE	GET	7	316
aMPCPTxRegister	ATTRIBUTE	GET	7	317
aMPCPTxRegRequest	ATTRIBUTE	GET	7	318
aMPCPTxReport	ATTRIBUTE	GET	7	319
aMPCPRxGate	ATTRIBUTE	GET	7	320
aMPCPRxRegAck	ATTRIBUTE	GET	7	321
aMPCPRxRegister	ATTRIBUTE	GET	7	322
aMPCPRxRegRequest	ATTRIBUTE	GET	7	318
aMPCPRxReport	ATTRIBUTE	GET	7	324
aMPCPTransmitElapsed	ATTRIBUTE	GET	7	285
aMPCPReceiveElapsed	ATTRIBUTE	GET	7	286
aMPCPRoundTripTime	ATTRIBUTE	GET	7	287
aMPCPDiscoveryWindowsSent	ATTRIBUTE	GET	7	288
aMPCPDiscoveryTimeout	ATTRIBUTE	GET	7	290
aMPCPMaximumPendingGrants	ATTRIBUTE	GET	7	291
acMPCPAdminControl	ACTION	—	9	16
oOAM	OBJECT	GET	3	20

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aOAMID	ATTRIBUTE	GET	7	236
aOAMAdminState	ATTRIBUTE	GET	7	237
aOAMMode	ATTRIBUTE	GET-SET	7	238
aOAMDiscoveryState	ATTRIBUTE	GET	7	333
aOAMRemoteMACAddress	ATTRIBUTE	GET	7	239
aOAMLocalConfiguration	ATTRIBUTE	GET	7	334
aOAMRemoteConfiguration	ATTRIBUTE	GET	7	240
aOAMLocalPDUConfiguration	ATTRIBUTE	GET	7	335
aOAMRemotePDUConfiguration	ATTRIBUTE	GET	7	241
aOAMLocalFlagsField	ATTRIBUTE	GET	7	242
aOAMRemoteFlagsField	ATTRIBUTE	GET	7	243
aOAMLocalRevision	ATTRIBUTE	GET	7	336
aOAMRemoteRevision	ATTRIBUTE	GET	7	244
aOAMLocalState	ATTRIBUTE	GET	7	337
aOAMRemoteState	ATTRIBUTE	GET	7	245
aOAMRemoteVendorOUI	ATTRIBUTE	GET	7	246
aOAMRemoteVendorSpecificInfo	ATTRIBUTE	GET	7	247
aOAMUnsupportedCodesTx	ATTRIBUTE	GET	7	338
aOAMUnsupportedCodesRx	ATTRIBUTE	GET	7	250
aOAMInformationTx	ATTRIBUTE	GET	7	251
aOAMInformationRx	ATTRIBUTE	GET	7	252
aOAMUniqueEventNotificationTx	ATTRIBUTE	GET	7	339
aOAMDuplicateEventNotificationTx	ATTRIBUTE	GET	7	340
aOAMUniqueEventNotificationRx	ATTRIBUTE	GET	7	254
aOAMDuplicateEventNotificationRx	ATTRIBUTE	GET	7	255
aOAMLoopbackControlTx	ATTRIBUTE	GET	7	256
aOAMLoopbackControlRx	ATTRIBUTE	GET	7	257

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aOAMVariableRequestTx	ATTRIBUTE	GET	7	258
aOAMVariableRequestRx	ATTRIBUTE	GET	7	259
aOAMVariableResponseTx	ATTRIBUTE	GET	7	260
aOAMVariableResponseRx	ATTRIBUTE	GET	7	261
aOAMOrganizationSpecificTx	ATTRIBUTE	GET	7	262
aOAMOrganizationSpecificRx	ATTRIBUTE	GET	7	263
aOAMLocalErrSymPeriodConfig	ATTRIBUTE	GET	7	264
aOAMLocalErrSymPeriodEvent	ATTRIBUTE	GET	7	265
aOAMLocalErrFrameConfig	ATTRIBUTE	GET	7	266
aOAMLocalErrFrameEvent	ATTRIBUTE	GET	7	267
aOAMLocalErrFramePeriodConfig	ATTRIBUTE	GET	7	268
aOAMLocalErrFramePeriodEvent	ATTRIBUTE	GET	7	269
aOAMLocalErrFrameSecsSummaryConfig	ATTRIBUTE	GET	7	270
aOAMLocalErrFrameSecsSummaryEvent	ATTRIBUTE	GET	7	271
aOAMRemoteErrSymPeriodEvent	ATTRIBUTE	GET	7	272
aOAMRemoteErrFrameEvent	ATTRIBUTE	GET	7	273
aOAMRemoteErrFramePeriodEvent	ATTRIBUTE	GET	7	274
aOAMRemoteErrFrameSecsSummaryEvent	ATTRIBUTE	GET	7	275
aFramesLostDueToOAMError	ATTRIBUTE	GET	7	276
acOAMAdminControl	ACTION	—	9	15
oOMPEmulation	OBJECT	GET	3	19
aOMPEmulationID	ATTRIBUTE	GET	7	231
aOMPEmulationType	ATTRIBUTE	GET	7	232
aSLDErrors	ATTRIBUTE	GET	7	233
aCRC8Errors	ATTRIBUTE	GET	7	234
aGoodLLID	ATTRIBUTE	GET	7	341
aONUPONcastLLID	ATTRIBUTE	GET	7	342

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aOLTPONcastLLID	ATTRIBUTE	GET	7	343
aBadLLID	ATTRIBUTE	GET	7	235
oRepeater	OBJECT	GET	3	3
aRepeaterID	ATTRIBUTE	GET	7	38
aRepeaterType	ATTRIBUTE	GET	7	39
aRepeaterGroupCapacity	ATTRIBUTE	GET	7	40
aGroupMap	ATTRIBUTE	GET	7	41
aRepeaterHealthState	ATTRIBUTE	GET	7	42
aRepeaterHealthText	ATTRIBUTE	GET	7	43
aRepeaterHealthData	ATTRIBUTE	GET	7	44
aTransmitCollisions	ATTRIBUTE	GET	7	45
acResetRepeater	ACTION	—	9	6
acExecuteNonDisruptiveSelfTest	ACTION	—	9	7
nRepeaterHealth	NOTIFICATION	—	10	1
nRepeaterReset	NOTIFICATION	—	10	2
nGroupMapChange	NOTIFICATION	—	10	3
oGroup	OBJECT	GET	3	4
aGroupID	ATTRIBUTE	GET	7	46
aGroupPortCapacity	ATTRIBUTE	GET	7	47
aPortMap	ATTRIBUTE	GET	7	48
nPortMapChange	NOTIFICATION	—	10	4
oRepeaterPort	OBJECT	GET	3	5
aPortID	ATTRIBUTE	GET	7	49
aPortAdminState	ATTRIBUTE	GET	7	50
aAutoPartitionState	ATTRIBUTE	GET	7	51
aReadableFrames	ATTRIBUTE	GET	7	52
aReadableOctets	ATTRIBUTE	GET	7	53

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aFrameCheckSequenceErrors	ATTRIBUTE	GET	7	54
aAlignmentErrors	ATTRIBUTE	GET	7	55
aFramesTooLong	ATTRIBUTE	GET	7	56
aShortEvents	ATTRIBUTE	GET	7	57
aRunts	ATTRIBUTE	GET	7	58
aCollisions	ATTRIBUTE	GET	7	59
aLateEvents	ATTRIBUTE	GET	7	60
aVeryLongEvents	ATTRIBUTE	GET	7	61
aDataRateMismatches	ATTRIBUTE	GET	7	62
aAutoPartitions	ATTRIBUTE	GET	7	63
aIsolates	ATTRIBUTE	GET	7	64
aSymbolErrorDuringPacket	ATTRIBUTE	GET	7	65
aLastSourceAddress	ATTRIBUTE	GET	7	66
aSourceAddressChanges	ATTRIBUTE	GET	7	67
aBursts	ATTRIBUTE	GET	7	100
acPortAdminControl	ACTION	—	9	8
oMAU	OBJECT	GET	3	6
aMAUID	ATTRIBUTE	GET	7	68
aMAUType	ATTRIBUTE	GET-SET	7	69
aMAUTypeList	ATTRIBUTE	GET	7	70
aMediaAvailable	ATTRIBUTE	GET	7	71
aLoseMediaCounter	ATTRIBUTE	GET	7	72
aJabber	ATTRIBUTE	GET	7	73
aMAUAdminState	ATTRIBUTE	GET	7	74
aBbMAUXmitRcvSplitType	ATTRIBUTE	GET	7	75
aBroadbandFrequencies	ATTRIBUTE	GET	7	76
aFalseCarriers	ATTRIBUTE	GET	7	77

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aIdleErrorCount	ATTRIBUTE	GET	7	91
aSNROpMarginChnlA	ATTRIBUTE	GET	7	353
aSNROpMarginChnlB	ATTRIBUTE	GET	7	354
aSNROpMarginChnlC	ATTRIBUTE	GET	7	355
aSNROpMarginChnlD	ATTRIBUTE	GET	7	356
aPCSCodingViolation	ATTRIBUTE	GET	7	292
aFECAbility	ATTRIBUTE	GET	7	313
aFECmode	ATTRIBUTE	GET-SET	7	314
aFECCorrectedBlocks	ATTRIBUTE	GET	7	293
aFECUncorrectableBlocks	ATTRIBUTE	GET	7	294
acResetMAU	ACTION	—	9	9
acMAUAdminControl	ACTION	—	9	10
nJabber	NOTIFICATION	—	10	5
oAutoNegotiation	OBJECT	GET	3	7
aAutoNegID	ATTRIBUTE	GET	7	78
aAutoNegAdminState	ATTRIBUTE	GET	7	79
aAutoNegRemoteSignaling	ATTRIBUTE	GET	7	80
aAutoNegAutoConfig	ATTRIBUTE	GET-SET	7	81
aAutoNegLocalTechnologyAbility	ATTRIBUTE	GET	7	82
aAutoNegAdvertisedTechnologyAbility	ATTRIBUTE	GET-SET	7	83
aAutoNegReceivedTechnologyAbility	ATTRIBUTE	GET	7	84
aAutoNegLocalSelectorAbility	ATTRIBUTE	GET	7	85
aAutoNegAdvertisedSelectorAbility	ATTRIBUTE	GET-SET	7	86
aAutoNegReceivedSelectorAbility	ATTRIBUTE	GET	7	87
acAutoNegRestartAutoConfig	ACTION	—	9	11
acAutoNegAdminControl	ACTION	—	9	12
oAggregator	OBJECT	GET	3	10

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aAggID	ATTRIBUTE	GET	7	101
aAggDescription	ATTRIBUTE	GET	7	102
aAggName	ATTRIBUTE	GET-SET	7	103
aAggActorSystemID	ATTRIBUTE	GET-SET	7	104
aAggActorSystemPriority	ATTRIBUTE	GET-SET	7	105
aAggAggregateOrIndividual	ATTRIBUTE	GET	7	106
aAggActorAdminKey	ATTRIBUTE	GET-SET	7	107
aAggActorOperKey	ATTRIBUTE	GET	7	108
aAggMACAddress	ATTRIBUTE	GET	7	109
aAggPartnerSystemID	ATTRIBUTE	GET	7	110
aAggPartnerSystemPriority	ATTRIBUTE	GET	7	111
aAggPartnerOperKey	ATTRIBUTE	GET	7	112
aAggAdminState	ATTRIBUTE	GET-SET	7	113
aAggOperState	ATTRIBUTE	GET	7	114
aAggTimeOfLastOperChange	ATTRIBUTE	GET	7	115
aAggDataRate	ATTRIBUTE	GET	7	116
aAggOctetsTxOK	ATTRIBUTE	GET	7	117
aAggOctetsRxOK	ATTRIBUTE	GET	7	118
aAggFramesTxOK	ATTRIBUTE	GET	7	119
aAggFramesRxOK	ATTRIBUTE	GET	7	120
aAggMulticastFramesTxOK	ATTRIBUTE	GET	7	121
aAggMulticastFramesRxOK	ATTRIBUTE	GET	7	122
aAggBroadcastFramesTxOK	ATTRIBUTE	GET	7	123
aAggBroadcastFramesRxOK	ATTRIBUTE	GET	7	124
aAggFramesDiscardedOnTx	ATTRIBUTE	GET	7	125
aAggFramesDiscardedOnRx	ATTRIBUTE	GET	7	126
aAggFramesWithTxErrors	ATTRIBUTE	GET	7	127

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aAggFramesWithRxErrors	ATTRIBUTE	GET	7	128
aAggUnknownProtocolFrames	ATTRIBUTE	GET	7	129
aAggLinkUpDownNotificationEnable	ATTRIBUTE	GET-SET	7	130
nAggLinkUpNotification	NOTIFICATION	—	10	6
nAggLinkDownNotification	NOTIFICATION	—	10	7
aAggPortList	ATTRIBUTE	GET	7	131
aAggCollectorMaxDelay	ATTRIBUTE	GET-SET	7	132
oAggregationPort	OBJECT	GET	3	11
aAggPortID	ATTRIBUTE	GET	7	133
aAggPortActorSystemPriority	ATTRIBUTE	GET-SET	7	134
aAggPortActorSystemID	ATTRIBUTE	GET	7	135
aAggPortActorAdminKey	ATTRIBUTE	GET-SET	7	136
aAggPortActorOperKey	ATTRIBUTE	GET	7	137
aAggPortPartnerAdminSystemPriority	ATTRIBUTE	GET-SET	7	138
aAggPortPartnerOperSystemPriority	ATTRIBUTE	GET	7	139
aAggPortPartnerAdminSystemID	ATTRIBUTE	GET-SET	7	140
aAggPortPartnerOperSystemID	ATTRIBUTE	GET	7	141
aAggPortPartnerAdminKey	ATTRIBUTE	GET-SET	7	142
aAggPortPartnerOperKey	ATTRIBUTE	GET	7	143
aAggPortSelectedAggID	ATTRIBUTE	GET	7	144
aAggPortAttachedAggID	ATTRIBUTE	GET	7	145
aAggPortActorPort	ATTRIBUTE	GET	7	146
aAggPortActorPortPriority	ATTRIBUTE	GET-SET	7	147
aAggPortPartnerAdminPort	ATTRIBUTE	GET-SET	7	148
aAggPortPartnerOperPort	ATTRIBUTE	GET	7	149
aAggPortPartnerAdminPortPriority	ATTRIBUTE	GET-SET	7	150
aAggPortPartnerOperPortPriority	ATTRIBUTE	GET	7	151

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aAggPortActorAdminState	ATTRIBUTE	GET-SET	7	152
aAggPortActorOperState	ATTRIBUTE	GET	7	153
aAggPortPartnerAdminState	ATTRIBUTE	GET-SET	7	154
aAggPortPartnerOperState	ATTRIBUTE	GET	7	155
aAggPortAggregateOrIndividual	ATTRIBUTE	GET	7	156
oAggPortStats	OBJECT	GET	3	12
aAggPortStatsID	ATTRIBUTE	GET	7	157
aAggPortStatsLACPDUrx	ATTRIBUTE	GET	7	158
aAggPortStatsMarkerPDUsRx	ATTRIBUTE	GET	7	159
aAggPortStatsMarkerResponsePDUsRx	ATTRIBUTE	GET	7	160
aAggPortStatsUnknownRx	ATTRIBUTE	GET	7	161
aAggPortStatsIllegalRx	ATTRIBUTE	GET	7	162
aAggPortStatsLACPDUstx	ATTRIBUTE	GET	7	163
aAggPortStatsMarkerPDUsTx	ATTRIBUTE	GET	7	164
aAggPortStatsMarkerResponsePDUsTx	ATTRIBUTE	GET	7	165
oAggPortDebugInformation	OBJECT	GET	3	13
aAggPortDebugInformationID	ATTRIBUTE	GET	7	166
aAggPortDebugRxState	ATTRIBUTE	GET	7	167
aAggPortDebugLastRxTime	ATTRIBUTE	GET	7	168
aAggPortDebugMuxState	ATTRIBUTE	GET	7	169
aAggPortDebugMuxReason	ATTRIBUTE	GET	7	170
aAggPortDebugActorChurnState	ATTRIBUTE	GET	7	171
aAggPortDebugPartnerChurnState	ATTRIBUTE	GET	7	172
aAggPortDebugActorChurnCount	ATTRIBUTE	GET	7	173
aAggPortDebugPartnerChurnCount	ATTRIBUTE	GET	7	174
aAggPortDebugActorSyncTransitionCount	ATTRIBUTE	GET	7	175
aAggPortDebugPartnerSyncTransitionCount	ATTRIBUTE	GET	7	176

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aAggPortDebugActorChangeCount	ATTRIBUTE	GET	7	177
aAggPortDebugPartnerChangeCount	ATTRIBUTE	GET	7	178
oWIS	OBJECT	GET	3	14
aWISID	ATTRIBUTE	GET	7	181
aSectionStatus	ATTRIBUTE	GET	7	182
aSectionSESThreshold	ATTRIBUTE	GET-SET	7	183
aSectionSESSs	ATTRIBUTE	GET	7	184
aSectionESs	ATTRIBUTE	GET	7	185
aSectionSEFSs	ATTRIBUTE	GET	7	186
aSectionCVs	ATTRIBUTE	GET	7	187
aJ0ValueTX	ATTRIBUTE	GET-SET	7	188
aJ0ValueRX	ATTRIBUTE	GET	7	189
aLineStatus	ATTRIBUTE	GET	7	190
aLineSESThreshold	ATTRIBUTE	GET-SET	7	191
aLineSESSs	ATTRIBUTE	GET	7	192
aLineESs	ATTRIBUTE	GET	7	193
aLineCVs	ATTRIBUTE	GET	7	194
aFarEndLineSESSs	ATTRIBUTE	GET	7	195
aFarEndLineESs	ATTRIBUTE	GET	7	196
aFarEndLineCVs	ATTRIBUTE	GET	7	197
aPathStatus	ATTRIBUTE	GET	7	198
aPathSESThreshold	ATTRIBUTE	GET-SET	7	199
aPathSESSs	ATTRIBUTE	GET	7	200
aPathESs	ATTRIBUTE	GET	7	201
aPathCVs	ATTRIBUTE	GET	7	202
aJ1ValueTX	ATTRIBUTE	GET-SET	7	203
aJ1ValueRX	ATTRIBUTE	GET	7	204

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aFarEndPathStatus	ATTRIBUTE	GET	7	205
aFarEndPathSESs	ATTRIBUTE	GET	7	206
aFarEndPathESs	ATTRIBUTE	GET	7	207
aFarEndPathCVs	ATTRIBUTE	GET	7	208
oPSE	OBJECT	GET	3	15
aPSEID	ATTRIBUTE	GET	7	209
aPSEAdminState	ATTRIBUTE	GET	7	210
aPSEPowerPairsControlAbility	ATTRIBUTE	GET	7	211
aPSEPowerPairs	ATTRIBUTE	GET-SET	7	212
aPSEPowerDetectionStatus	ATTRIBUTE	GET	7	214
aPSEPowerClassification	ATTRIBUTE	GET	7	215
aPSEInvalidSignatureCounter	ATTRIBUTE	GET	7	227
aPSEPowerDeniedCounter	ATTRIBUTE	GET	7	228
aPSEOverLoadCounter	ATTRIBUTE	GET	7	229
aPSEShortCounter	ATTRIBUTE	GET	7	230
aPSEMPSAbsentCounter	ATTRIBUTE	GET	7	217
acPSEAdminControl	ACTION	—	9	13
aPSEActualPower	ATTRIBUTE	GET	7	427
aPSEPowerAccuracy	ATTRIBUTE	GET	7	428
aPSECumulativeEnergy	ATTRIBUTE	GET	7	429
oMidSpan	OBJECT	GET	3	17
aMidSpanID	ATTRIBUTE	GET	7	221
aMidSpanPSEGroupCapacity	ATTRIBUTE	GET	7	222
aMidSpanPSEGroupMap	ATTRIBUTE	GET	7	223
nMidSpanPSEGroupMapChange	NOTIFICATION	—	10	8
oPSEGroup	OBJECT	GET	3	18
aPSEGroupID	ATTRIBUTE	GET	7	224

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aPSECapacity	ATTRIBUTE	GET	7	225
aPSEMap	ATTRIBUTE	GET	7	226
nPSEMapChange	NOTIFICATION	—	10	9
oPAF	OBJECT	GET	3	24
aPAFID	ATTRIBUTE	GET	7	344
aPhyEnd	ATTRIBUTE	GET	7	326
aPHYCurrentStatus	ATTRIBUTE	GET	7	296
aPAFSupported	ATTRIBUTE	GET	7	304
aPAFAdminState	ATTRIBUTE	GET-SET	7	305
aLocalPAFCapacity	ATTRIBUTE	GET	7	327
aLocalPMEAvailable	ATTRIBUTE	GET	7	306
aLocalPMEAggregate	ATTRIBUTE	GET	7	307
aRemotePAFSupported	ATTRIBUTE	GET	7	328
aRemotePAFCapacity	ATTRIBUTE	GET	7	329
aRemotePMEAggregate	ATTRIBUTE	GET	7	310
oPME	OBJECT	GET	3	25
aPMEID	ATTRIBUTE	GET	7	330
aPMEAdminState	ATTRIBUTE	GET-SET	7	345
aPMEStatus	ATTRIBUTE	GET	7	346
aPMESNRMgn	ATTRIBUTE	GET	7	331
aTCCodingViolations	ATTRIBUTE	GET	7	332
aProfileSelect	ATTRIBUTE	GET-SET	7	347
aOperatingProfile	ATTRIBUTE	GET	7	348
aPMEFECCorrectedBlocks	ATTRIBUTE	GET	7	349
aPMEFECCUncorrectableBlocks	ATTRIBUTE	GET	7	350
aTCCRCErrors	ATTRIBUTE	GET	7	352
aCMCounter	ATTRIBUTE	GET	7	88

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
oMACEntity	OBJECT	GET	—	—
Basic Package (Mandatory)	PACKAGE	—	—	—
Mandatory Package (Mandatory)	PACKAGE	—	4	1
Recommended Package (Conditional)	PACKAGE	—	4	2
Conditional Package (Conditional)	PACKAGE	—	4	3
Array Package (Conditional)	PACKAGE	—	4	4
ExcessiveDeferral Package (Conditional)	PACKAGE	—	4	5
PHYRecommended Package (Conditional)	PACKAGE	—	4	6
oPHYEntity	OBJECT	GET	—	—
Basic Package (Mandatory)	PACKAGE	—	—	—
MultiplePHY Package (Conditional)	PACKAGE	—	4	7
100MbsMonitor Capability (Conditional)	PACKAGE	—	4	8
oMACControlEntity	OBJECT	GET	—	—
Mandatory Package (Mandatory)	PACKAGE	—	—	—
Recommended Package (Conditional)	PACKAGE	—	4	17
oMACControlFunctionEntity	OBJECT	GET	—	—
Mandatory Package (Mandatory)	PACKAGE	—	—	—
Recommended Package (Conditional)	PACKAGE	—	4	25
oAggregator	OBJECT	GET	—	—
AggregatorBasic Package (Mandatory)	PACKAGE	—	—	—
AggregatorMandatory Package (Mandatory)	PACKAGE	—	4	19
AggregatorRecommended Package (Conditional)	PACKAGE	—	4	20
AggregatorConditional Package (Conditional)	PACKAGE	—	4	21
oAggregationPort	OBJECT	GET	—	—
AggregationPortBasic Package (Mandatory)	PACKAGE	—	—	—
AggregationPortMandatory Package (Mandatory)	PACKAGE	—	4	22
oAggPortStats	OBJECT	GET	—	—

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
AggPortStatsPackage (Conditional)	PACKAGE	—	4	23
oAggPortDebugInformation	OBJECT	GET	—	—
AggPortDebugInformation Package (Conditional)	PACKAGE	—	4	24
oOMP	OBJECT	GET	—	—
Optical Multipoint Emulation Package (Conditional)	PACKAGE	—	—	—
Optical Multipoint Emulation Error Package (Conditional)	PACKAGE	—	4	37
oRepeater	OBJECT	GET	—	—
RepeaterBasicControl Package (Mandatory)	PACKAGE	—	—	—
RepeaterPerfMonitor Package (Conditional)	PACKAGE	—	4	9
oGroup	OBJECT	GET	—	—
GroupBasicControl Package (Mandatory)	PACKAGE	—	—	—
oRepeaterPort	OBJECT	GET	—	—
PortBasicControl Package (Mandatory)	PACKAGE	—	—	—
PortPerfMonitor Package (Conditional)	PACKAGE	—	4	10
PortAddrTracking Package (Conditional)	PACKAGE	—	4	11
100MbpsMonitor Package (Conditional)	PACKAGE	—	4	12
Burst Package (Conditional)	PACKAGE	—	4	18
oMAU	OBJECT	GET	—	—
Basic Package (Mandatory)	PACKAGE	—	—	—
MAUControl Package (Conditional)	PACKAGE	—	4	13
MediaLossTracking Package (Conditional)	PACKAGE	—	4	14
BroadbandDTEMAU Package (Conditional)	PACKAGE	—	4	15
100MbpsMonitor Capability (Conditional)	PACKAGE	—	4	16
10GBASE-T Operating Margin package (Conditional)	PACKAGE	—	4	39
PCS Code Error Monitor Capability (Optional)	PACKAGE	—	4	35
Forward Error Correction Capability (Conditional)	PACKAGE	—	4	30
oWIS	OBJECT	GET	—	—

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
WIS Basic Package (Mandatory)	PACKAGE	—	—	—
WIS Recommended Package (Conditional)	PACKAGE	—	4	26
oAutoNegotiation	OBJECT	GET	—	—
Auto Negotiation Package (Mandatory)	PACKAGE	—	—	—
oPSE	OBJECT	GET	—	—
PSEBasic Package (Mandatory)	PACKAGE	—	—	—
PSERecommended Package (Conditional)	PACKAGE	—	4	27
oMidSpan	OBJECT	GET	—	—
MidSpanBasic (Mandatory)	PACKAGE	—	—	—
oPAF	OBJECT	GET	—	—
Basic Package (Mandatory)	PACKAGE	—	—	—
PME Aggregation Package (Optional)	PACKAGE	—	4	38
oPME	OBJECT	GET	—	—
10P/2B Package (Mandatory)	PACKAGE	—	—	—
oResourceTypeID	OBJECT	GET	—	—
MII Package (Conditional)	PACKAGE	—	—	—
oTimeSync	OBJECT	GET	—	—
Support for Time Sync (Mandatory)	PACKAGE	—	4	40
aLdpXdot3LocPowerType	ATTRIBUTE	GET	7	358
aLdpXdot3LocPowerSource	ATTRIBUTE	GET	7	359
aLdpXdot3LocPowerPriority	ATTRIBUTE	GET	7	360
aLdpXdot3LocPDRrequestedPowerValue	ATTRIBUTE	GET	7	361
aLdpXdot3LocPSEAllocatedPowerValue	ATTRIBUTE	GET	7	362
aLdpXdot3LocResponseTime	ATTRIBUTE	GET	7	363
aLdpXdot3LocReady	ATTRIBUTE	GET	7	364
aLdpXdot3LocReducedOperationPowerValue	ATTRIBUTE	GET	7	365
aLdpXdot3RemPowerType	ATTRIBUTE	GET	7	366

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aLdpXdot3RemPowerSource	ATTRIBUTE	GET	7	367
aLdpXdot3RemPowerPriority	ATTRIBUTE	GET	7	368
aLdpXdot3RemPDRRequestedPowerValue	ATTRIBUTE	GET	7	369
aLdpXdot3RemPSEAllocatedPowerValue	ATTRIBUTE	GET	7	370
aTransmitLPIMicroseconds	ATTRIBUTE	GET	7	371
aReceiveLPIMicroseconds	ATTRIBUTE	GET	7	372
aTransmitLPITransitions	ATTRIBUTE	GET	7	373
aReceiveLPITransitions	ATTRIBUTE	GET	7	374
aLDFastRetrainCount	ATTRIBUTE	GET	7	375
aLPFastRetrainCount	ATTRIBUTE	GET	7	376
aEEESupportList	ATTRIBUTE	GET	7	377
oLdpXdot3Config	OBJECT	GET	3	27
LLDP Basic Package (mandatory)	PACKAGE	—	4	41
oLdpXdot3LocSystemsGroup	OBJECT	GET	3	28
LLDP MAC/PHY Configuration/Status Local Package (conditional)	PACKAGE	—	4	42
LLDP Power via MDI Local Package (conditional)	PACKAGE	—	4	44
LLDP Link Aggregation Local Package (conditional)	PACKAGE	—	4	46
LLDP Maximum Frame Size Local Package (conditional)	PACKAGE	—	4	48
LLDP EEE Local Package (optional)	PACKAGE	—	4	50
oLdpXdot3RemSystemsGroup	OBJECT	GET	3	29
LLDP MAC/PHY Configuration/Status Remote Package (conditional)	PACKAGE	—	4	43
LLDP Power via MDI Remote Package (conditional)	PACKAGE	—	4	45
LLDP Link Aggregation Remote Package (conditional)	PACKAGE	—	4	47
LLDP Maximum Frame Size Remote Package (conditional)	PACKAGE	—	4	49
LLDP EEE Remote Package (optional)	PACKAGE	—	4	51
aLdpXdot3LocTxTwSys	ATTRIBUTE	GET	7	378

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aLdpXdot3LocTxTwSysEcho	ATTRIBUTE	GET	7	379
aLdpXdot3LocRxTwSys	ATTRIBUTE	GET	7	380
aLdpXdot3LocRxTwSysEcho	ATTRIBUTE	GET	7	381
aLdpXdot3LocFbTwSys	ATTRIBUTE	GET	7	382
aLdpXdot3TxDllReady	ATTRIBUTE	GET	7	383
aLdpXdot3RxDllReady	ATTRIBUTE	GET	7	384
aLdpXdot3LocDllEnabled	ATTRIBUTE	GET	7	385
aLdpXdot3RemTxTwSys	ATTRIBUTE	GET	7	386
aLdpXdot3RemTxTwSysEcho	ATTRIBUTE	GET	7	387
aLdpXdot3RemRxTwSys	ATTRIBUTE	GET	7	388
aLdpXdot3RemRxTwSysEcho	ATTRIBUTE	GET	7	389
aLdpXdot3RemFbTwSys	ATTRIBUTE	GET	7	390
aLdpXdot3PortConfigTLVsTxEnable	ATTRIBUTE	GET	7	391
aLdpXdot3LocPortAutoNegSupported	ATTRIBUTE	GET	7	392
aLdpXdot3LocPortAutoNegEnabled	ATTRIBUTE	GET	7	393
aLdpXdot3LocPortAutoNegAdvertisedCap	ATTRIBUTE	GET	7	394
aLdpXdot3LocPortOperMauType	ATTRIBUTE	GET	7	395
aLdpXdot3LocPowerPortClass	ATTRIBUTE	GET	7	396
aLdpXdot3LocPowerMDISupported	ATTRIBUTE	GET	7	397
aLdpXdot3LocPowerMDIEnabled	ATTRIBUTE	GET	7	398
aLdpXdot3LocPowerPairControlable	ATTRIBUTE	GET	7	399
aLdpXdot3LocPowerPairs	ATTRIBUTE	GET	7	400
aLdpXdot3LocPowerClass	ATTRIBUTE	GET	7	401
aLdpXdot3LocMaxFrameSize	ATTRIBUTE	GET	7	402
aLdpXdot3RemPortAutoNegSupported	ATTRIBUTE	GET	7	403
aLdpXdot3RemPortAutoNegEnabled	ATTRIBUTE	GET	7	404
aLdpXdot3RemPortAutoNegAdvertisedCap	ATTRIBUTE	GET	7	405

Table B-1—Branch and leaf assignments for managed objects (*continued*)

IEEE Std 802.3 Clause 30 object name	Type	Access	BRANCH	LEAF
aLdpXdot3RemPortOperMauType	ATTRIBUTE	GET	7	406
aLdpXdot3RemPowerPortClass	ATTRIBUTE	GET	7	407
aLdpXdot3RemPowerMDISupported	ATTRIBUTE	GET	7	408
aLdpXdot3RemPowerMDIEnabled	ATTRIBUTE	GET	7	409
aLdpXdot3RemPowerPairControlable	ATTRIBUTE	GET	7	410
aLdpXdot3RemPowerPairs	ATTRIBUTE	GET	7	411
aLdpXdot3RemPowerClass	ATTRIBUTE	GET	7	412
aLdpXdot3RemMaxFrameSize	ATTRIBUTE	GET	7	413
oTimeSync	OBJECT	GET	3	26
aTimeSyncCapabilityTX	ATTRIBUTE	GET	7	416
aTimeSyncCapabilityRX	ATTRIBUTE	GET	7	417
aTimeSyncDelayTXmax	ATTRIBUTE	GET	7	418
aTimeSyncDelayTXmin	ATTRIBUTE	GET	7	419
aTimeSyncDelayRXmax	ATTRIBUTE	GET	7	420
aTimeSyncDelayRXmin	ATTRIBUTE	GET	7	421
oEXTENSION	OBJECT	GET	3	30
aEXTENSIONMACCtrlStatus	ATTRIBUTE	GET	7	422
aEXTENSIONMACCtrlFramesTransmitted	ATTRIBUTE	GET	7	423
aEXTENSIONMACCtrlFramesReceived	ATTRIBUTE	GET	7	424
aMPCPRecognizedMulticastIDs	ATTRIBUTE	GET	7	425
ifMauPerPCSLaneStatsTable	ATTRIBUTE	GET	7	430
ifMauPerPCSLaneStatsEntry	ATTRIBUTE	GET	7	431
ifPCSLaneIndex	ATTRIBUTE	GET	7	432
ifMauPPLFECCorrectedBlocks	ATTRIBUTE	GET	7	433
ifMauPPLFECUncorrectableBlocks	ATTRIBUTE	GET	7	434
ifMauBIPErrorsCount	ATTRIBUTE	GET	7	435
ifMauPCStoPHYLaneMapping	ATTRIBUTE	GET	7	436

Annex 12A

(informative)

Collection of performance data using WIS MDIO registers

The purpose of this annex is to illustrate how the WIS MDIO registers specified in 45.2.2 of IEEE Std 802.3 (and more specifically the subset required by 50.3.11 of IEEE Std 802.3) can be used to collect performance data either according to the conventions adopted by this document or according to the conventions specified in Clause 30 of IEEE Std 802.3.

For an agent implementing the SNMP managed objects required by this document, the first step in collecting WIS performance data would be to poll the 10G WIS status 3 register and the various error count registers (10G WIS section BIP error count, 10G WIS line BIP errors, 10G WIS far end line BIP errors, 10G WIS path block error count, and 10G WIS far end path block error count) once per second. The 10G WIS status 3 register bits are all latched until read and so would indicate whether a given defect occurred any time during the previous second. The error count registers roll over modulo 2^{16} or 2^{32} , and so to find the number of errors within the previous second, the agent would need to subtract (modulo 2^{16} or 2^{32}) the current reading from the reading taken 1 second ago. Armed with that information, the agent could determine for any layer whether the 1-second interval was an errored second, a severely errored second (that requires comparison with a threshold unless a defect is present), or a severely errored frame second. Determining whether a given second is or is not part of unavailable time requires additional logic; the most straightforward and accurate method is the delay-line approach outlined in Appendix A of IETF RFC 3592. With that information available, the agent would be able to determine by how much each current count should be incremented (including effects of inhibiting). Implementations that conform to ANSI T1.231-1997 would end each 15-minute interval on time-of-day clock 1/4 hour boundaries; if the delay-line approach is used, then a time-of-day timestamp would accompany the 1-second statistics. At the end of each interval, the current registers would be pushed onto the history stack and then would be cleared. The `xyxIntervalValidData` flags would be set to `False(2)` if the number of samples was not between 890 and 910 or, in the case of far-end counts, if a near-end defect occurred during the just-completed interval (see section 9.1.2.2 of ANSI T1.231-1997 for details).

An agent implementing the oWIS objects of Clause 30 of IEEE Std 802.3 could also start by polling the 10G WIS status 3 register and the various error count registers to find the defects and error counts for the previous second, and it could determine the number of errors and whether the second was an errored second, a severely errored second, or a severely errored frame second in the same manner as above. The rest of the process would simply be to increment the generalized non-resettable counters without consideration of any inhibiting rules.

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